

The Observation Theory Encyclopedia

Companion to *Data Mining as Observation*, 241 entries

The observation theory program, ahb-sjsu

Built 2026-09-07 from commit f6d1fe2

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Preface

This is the encyclopedia of the observation theory program, the companion to the textbook *Data Mining as Observation*. It has 241 entries, one for every headword of the book's glossary and three that were filled by hand before the generator existed. Of the entries, 159 are concepts, 71 are instruments, 7 are results, 3 are corrections, and 1 is a reference.

Where the entries come from. An entry is built from the program's records rather than written beside them. The claims ledger in the repository `geometric-observation` is the authority for what a result is worth. A campaign track file is the authority for what was measured. The book's sources tables are the authority for every number the book states. The generator, `encyclopedia/generate.py`, gathers what those records say about one thing, keyed by their identifiers, so that a correction in a record propagates to the entry on the next rebuild and never has to be chased by hand. The curated parts of an entry, its definition, its conditions, and the list of related entries, live in `encyclopedia/entries.toml`.

How to read an entry. Every entry carries the same headings in the same order. A heading with nothing to say keeps the word `none`, so that a missing field and an empty field cannot be confused. A record joins an entry by a typed edge. An equation defines the entry when the book's paragraph around it names the entry's terms. A ledger row proves, measures, or refutes and corrects the entry, by its class, when its claim names the entry, and a sources-table row measures the entry when its claim names it. Records that were cited beside an entry without naming it are not printed in the body. They are listed under the entry's `see-also` heading, and the back matter lists every ledger row with the entries it bears on. A result or correction entry trusts every record chosen for it.

The status line at the end of each entry gives the date it was generated and the book commit, and the provenance section gives the commit of every other record.

Field	What it holds	Source of truth
id	the file name without extension, lowercase, hyphenated, never reused	this directory
kind	one of concept, instrument, result, correction	this directory
definition	one or two sentences a reader can act on, with the symbols the equation uses	the volume or paper that first stated it
equation	the displayed form, with the book equation number where the book states it	the volume, the book
ledger	every claims-ledger row that bears on the entry, with its class in brackets, [proved] [demonstrated] [replicated] [predicted] [exploratory] [refuted]	geometric-observation/claims/LEDGER.md
first stated	where the idea first appears in the record, with a DOI or commit	papers, volumes
measurements	a table of the numbers, each with the file and line range and the commit it was read at	campaign tracks, experiment results
failures and corrections	every retraction, refuted bar, vacuous pass, and erratum that touches the entry, dated, at the same prominence as the measurements	ledger, ERRATA.md files
conditions	the assumptions under which the theorem or law holds, stated so a reader can tell when it does not apply	the volume, the book's revision notes
machine checked	the Lean file and theorem names, or none	observation-data-mining/lean/, ot-lean
used in	papers by DOI, book chapters by number, instruments by repository	the book's sources tables and index
related	other entry ids	this directory
status	the date the entry was last rebuilt or checked, and the commits of the records it was read from	this directory

Conventions. A ledger row carries its class in brackets, proved, demonstrated, replicated, predicted, exploratory, refuted, missed, or void, and a refuted row sits in the entry at the same size as the results it bounds. A ledger row is printed once in full in the ledger itself. An entry carries the row's first sentence, its class, and a link to the line. A number names its repository, file, line range, and commit, exactly as the book's sources tables do, and a number without a row was removed. The machine-checked heading names the Lean file and theorems that check the entry's core against Mathlib, built with no sorry and the standard axioms only, and the book's appendix C says what each check does and does not cover. Book equation numbers, chapter numbers, and section numbers refer to *Data Mining as Observation*, draft 0.2.

Prior art. Where an idea has a name outside the program, the entry says so under the heading known as. The read operator is, in the scalar Euclidean case, the active-subspace matrix of Constantine and Gleich, and consumer-relative compression sits beside the information bottleneck and task-based quantization. What the program adds is the output-metric pullback, the observer triple, the budget, and the audit discipline.

How to cite. Cite the repository `ahb-sjsu/observation-theory-campaigns` at commit `f6d1fe2`, the entry by its id, and, for a number, the source row the entry gives. The provenance section at the end names the commit of every record the entries were read from.

Cross-references. Each entry ends with its related entries, and each reference carries the page it is found on. The back matter lists the entries by kind, by chapter of the book, by the Lean file that checks them, and by the ledger row that bears on them.

From TSK to the encyclopedia

A student reading Tan, Steinbach, Karpatne, and Kumar, *Introduction to Data Mining*, second edition, meets a term and does not know which entries to read. This table gives, for each TSK term, the entries to read in order and the chapter of *Data Mining as Observation* that takes the term up.

TSK term	TSK	Entries to read	Book chapter
attribute types, nominal, ordinal, interval, ratio	2.1	attribute type (p. 28); standardization (p. 348); discretization (p. 112)	2
data quality, noise and outliers	2.2	outlier (p. 250); detector (p. 109); skewness (p. 335)	2
missing values	2.2	imputation (p. 182); leakage (p. 210)	2
aggregation and sampling	2.3	aggregation (p. 13); sampling (p. 322); Simpson's paradox (p. 333)	2
dimensionality reduction	2.3, appendix B	principal component analysis (p. 269); truncation (p. 369); explained variance (p. 137); the flip (p. 147)	4
feature subset selection	2.3	read subspace (p. 293); nuisance (p. 240); importance (p. 181)	4
discretization and binarization	2.3	discretization (p. 112); quotient (p. 279)	2
variable transformation, standardization	2.3	standardization (p. 348); whitening (p. 381)	2
Euclidean distance, Minkowski distance	2.4	Euclidean distance (p. 133); metric (p. 224); identity reader (p. 180)	3
cosine similarity	2.4	cosine (p. 87); dot product (p. 118); read direction (p. 288); quotient (p. 279)	3
Jaccard coefficient, simple matching	2.4	Jaccard (p. 191); quotient (p. 279)	3
correlation	2.4	correlation (p. 86); Spearman correlation (p. 339); Kendall correlation (p. 194)	3
Mahalanobis distance	2.4	Mahalanobis distance (p. 219); whitening (p. 381); covariance matrix (p. 91)	10
decision tree induction	3.3	decision tree (p. 102); importance (p. 181); classifier (p. 65)	6
model overfitting	3.4	capacity (p. 57); harness (p. 168); early stopping (p. 122)	6
model selection, validation set	3.5, 3.6	split (p. 346); cross-validation (p. 96); leakage (p. 210)	8
model evaluation, holdout, cross-validation, bootstrap	3.6	cross-validation (p. 96); bootstrap (p. 49); Nadeau and Bengio correction (p. 233); seed (p. 326)	8
confusion matrix, accuracy, precision, recall, F-measure	3.2, 4.11	precision, recall (p. 266); F1 (p. 140); balanced accuracy (p. 37); threshold (p. 362)	6
ROC curve	4.11	ROC curve (p. 314); AUROC (p. 32); Youden index (p. 387); Monotone Invariance Theorem (p. 227)	5
rule-based classifier	4.2	formula search (p. 154); safe pruning (p. 320)	6
nearest neighbor classifier	4.3	nearest neighbour (p. 238); neighbourhood graph (p. 239); hub (p. 175)	6

TSK term	TSK	Entries to read	Book chapter
naive Bayes classifier	4.4	naive Bayes (p. 235); Hessian (p. 172)	6
logistic regression	4.6	logistic regression (p. 217); linear classifier (p. 216); decision boundary (p. 101)	6
artificial neural network, deep learning	4.7, 4.8	encoder (p. 127); embedding (p. 126); attention (p. 26); language model (p. 206)	11
support vector machine, margin	4.9	margin (p. 221); linear classifier (p. 216); decision boundary (p. 101)	6
ensemble methods, bagging, boosting, random forest	4.10	ensemble (p. 129); bagging (p. 35); boosting (p. 48); random forest (p. 282)	7
class imbalance	4.11	balanced accuracy (p. 37); chance level (p. 62); stratification (p. 350)	6
itemset, support, frequent itemset	5.1, 5.2	itemset (p. 188); support (p. 353); transaction (p. 366)	5
Apriori principle, candidate pruning	5.2	Apriori principle (p. 25); anti-monotonicity (p. 24); safe pruning (p. 320)	5
association rule, confidence	5.3	confidence (p. 75); lift (p. 214); cross-support ratio (p. 95)	5
FP-growth, closed and maximal itemsets	5.4, 5.5	itemset (p. 188); anti-monotonicity (p. 24)	5
objective interestingness measures	5.6	lift (p. 214); output metric (p. 251); Simpson's paradox (p. 333)	5
k-means	7.2	k-means (p. 198); sum of squared errors (p. 351); identity reader (p. 180)	9
agglomerative hierarchical clustering, linkage	7.3	hierarchical clustering (p. 173); metric (p. 224)	9
DBSCAN, density-based clustering	7.4	DBSCAN (p. 100); density (p. 106); degree (p. 105)	9
cluster validity, silhouette, SSE	7.5	validity index (p. 372); silhouette (p. 332); sum of squared errors (p. 351); vacuity threshold (p. 371)	9
spectral clustering, graph-based clustering	8.5	spectral clustering (p. 340); spectral embedding (p. 342); Laplacian (p. 207); commute time (p. 72)	3
curse of dimensionality	8.1	distance concentration (p. 115); intrinsic dimension (p. 184); manifold (p. 220)	3
statistical anomaly detection	9.3	outlier (p. 250); Gaussian (p. 160); Mahalanobis distance (p. 219)	10
proximity-based and density-based anomaly detection	9.4, 9.5	detector (p. 109); density (p. 106); anti-hub (p. 20)	10
clustering-based and reconstruction-based anomaly detection	9.6, 9.7	detector (p. 109); reconstruction error (p. 298)	10
evaluation of anomaly detection	9.10	min-over-strata (p. 226); abstention (p. 12); anti-hub recall (p. 22)	10
hypothesis testing, p-value, null hypothesis	10.2	p-value (p. 273); null model (p. 242); control (p. 83); bar (p. 39)	8
multiple hypothesis testing, Bonferroni, false discovery rate	10.3	multiple comparisons (p. 229); harness (p. 168)	8

TSK term	TSK	Entries to read	Book chapter
pitfalls, p-hacking, data dredging	10.4	preregistration (p. 267); sealed (p. 324); registered (p. 305); declaration (p. 103)	8
confidence interval, standard error	appendix C, 10.2	confidence interval (p. 76); standard error (p. 347); paired (p. 253)	8
eigenvalues and eigenvectors, matrix rank	appendix A	eigenvalue, eigenvector (p. 125); spectrum (p. 344); rank (p. 283); positive semidefinite (p. 265)	0
singular value decomposition, PCA	appendix B	principal component analysis (p. 269); latent semantic analysis (p. 209); effective rank (p. 123)	4
covariance matrix, variance	appendix A, C	covariance matrix (p. 91); variance (p. 374); trace (p. 365); anisotropic (p. 18)	0
gradient, derivative	appendix E	sensitivity (p. 328); finite difference (p. 146); curvature (p. 98); Jacobian (p. 192)	0
TF-IDF, document vectors	2.4, 8.1	TF-IDF (p. 360); bag of words (p. 36); latent semantic analysis (p. 209)	12
vector quantization, compression	7.2, 2.3	quantization (p. 275); codebook (p. 66); product quantization (p. 271); bit (p. 43); matched bits (p. 223)	11
scalability, indexing, nearest-neighbour search	4.3, 8.4	inverted file (p. 185); recall at k (p. 295); shard (p. 331); rerank (p. 309); rank certificate (p. 285)	11

A

abstention

concept. Entry id abstention.

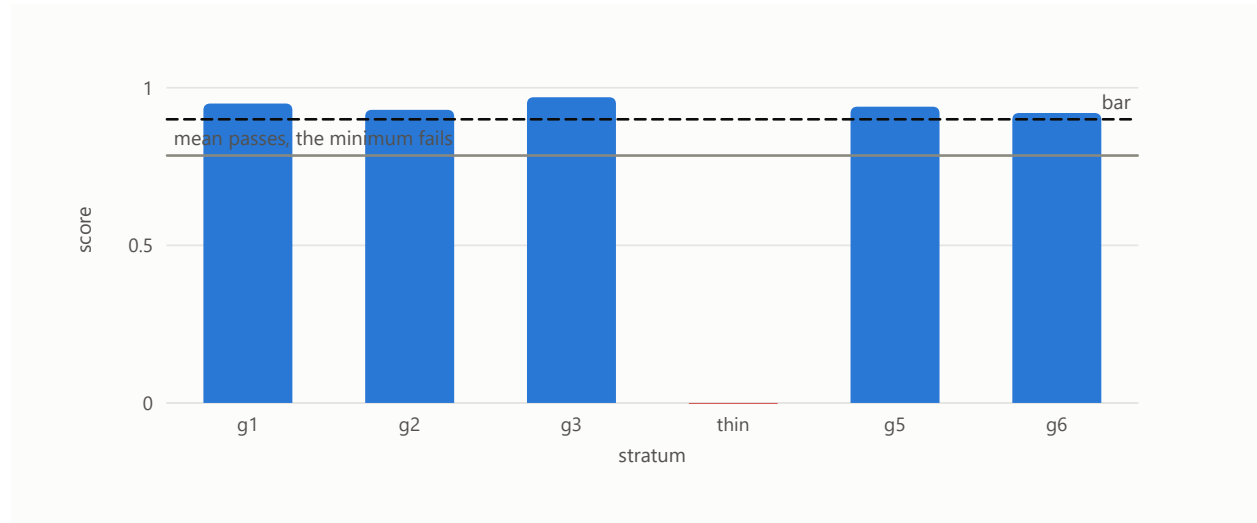


Figure 1: A verdict per group, with the group too thin to score reported as an abstention rather than dropped.

DEFINITION

The verdict a group receives when it has too few rows or too few queries to score. It is reported as a verdict, not dropped. Chapter 10.

Example. A group with 12 rows against a minimum of 50 is reported as abstained rather than scored, and the count of abstentions is printed beside the verdict.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

CONDITIONS

- A stratum with too few rows or too few queries to score receives the verdict abstain, which is reported under a registered cause and excluded from the minimum. Abstain is not a pass, and when no stratum is eligible the verdict itself is abstain.
- Raising the eligibility bar can only remove strata from the minimum and so can only raise the verdict, which is why every abstention is printed beside the verdict rather than dropped.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

The strata design in `turboquant-pro`, `turboquant-pro/docs/STRATA_RFC.md:24-98`, and chapter 10 section 10.4 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.3	abstain below 2.5 k	turboquant-pro/docs/HUBNESS_PRIMER.md:140-160
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro/docs/STRATA_RFC.md:24-98

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Abstention.lean](#), theorems `abstain_not_passes`, `verdict_eq_none_iff`, `passes_verdict_iff`, `inf_le_of_subset`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 10, 11, 14.

RELATED

[min-over-strata](#) (p. 226); [anti-hub](#) (p. 20); [certificate](#) (p. 59); [coverage](#) (p. 92).

SEE ALSO

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.4.

aggregation

concept. Entry id aggregation.

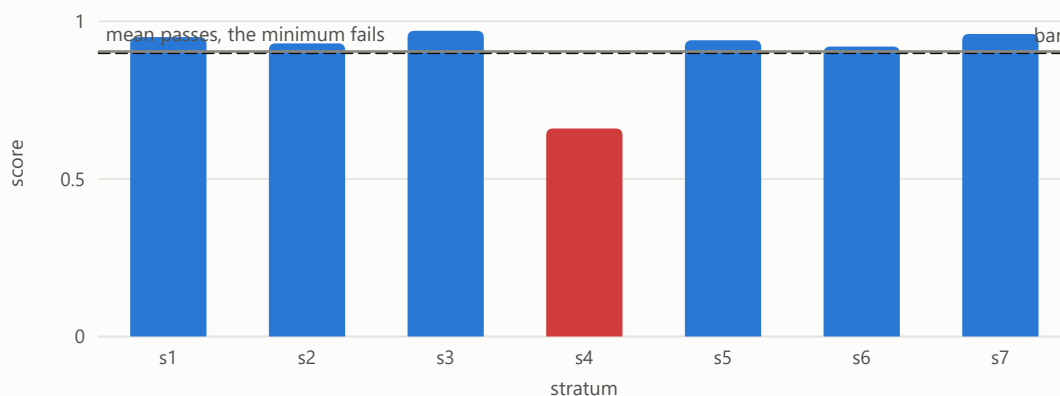


Figure 2: An aggregate over strata passes a bar that one stratum fails.

DEFINITION

Combining rows or groups into one number. An average lies between its members, and an aggregate over strata can pass a bar that one stratum fails and can reverse the sign every group shows. Chapter 2 section 2.7 and

chapter 10 section 10.4.

Example. Strata scoring 0.95, 0.93, 0.97, and 0.66 average 0.878 and pass a bar of 0.85, while the minimum, 0.66, fails it.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

CONDITIONS

- Combining rows or groups into one number, a mean, a sum, or a rate. An average lies between the smallest and largest member, a weighted aggregate over strata can pass a bar while one stratum fails it, and an aggregate can reverse the sign every group shows.
- Aggregate recall barely moves while anti-hubs fail, the aggregate staleness rate describes neither reader, and the report takes the worst group and counts the groups too thin to score.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 2 section 2.7 and chapter 10 section 10.4 of *Data Mining as Observation*, with the aggregate staleness rate of chapter 13 section 13.3.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.7	the aggregate staleness rate describing neither reader	chapter 13 of this book, section 13.3
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Ensemble.lean](#), theorems `average_between`, `sq_average_le`, `average_const`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Simpson.lean](#), theorems `pooled_eq_weighted`, `reversal`, `no_reversal_of_equal_sizes`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/RecallAtK.lean](#), theorems `recallAtK_mem_unit`, `recallAtK_eq_one_iff`, `aggregate_le_of_failing`, `aggregate_example`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2, 5, 8, 10, 11, 12, 13.

RELATED

[min-over-strata](#) (p. 226); [Simpson's paradox](#) (p. 333); [stratification](#) (p. 350); [anti-hub recall](#) (p. 22); [sampling](#) (p. 322).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 14.5, 8.2.

Ledger rows that cite the entry's records without naming it: NEG-14, GO-B-Llama, GO-B-Llama-rematch.

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.4, chapter 11 section 11.6.

alignment

concept. Entry id alignment.

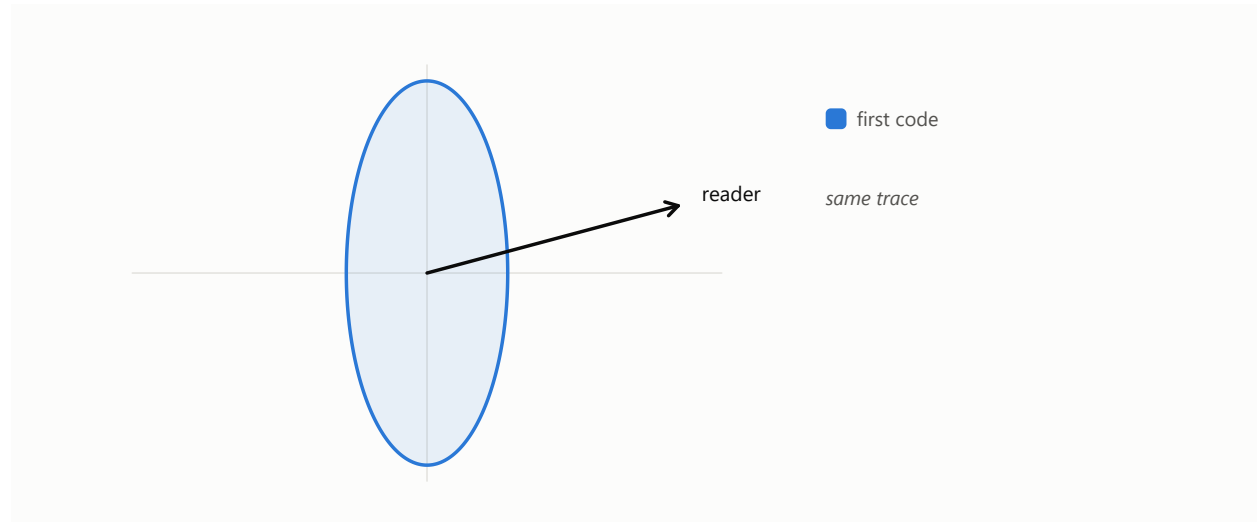


Figure 3: The overlap between the consumer's read direction and the data's covariance.

DEFINITION

The normalized overlap between a consumer's average read operator and the covariance of the data on the read subspace, written kappa, between zero and one. Near one the read subspace and the high-variance subspace coincide and no flip is possible. Chapter 4, equation 4.6.

Example. A read operator that lies wholly in one direction and a covariance with half its variance in that direction have alignment one half.

EQUATION

Book equation 4.6.

$$\kappa = \text{tr}(\bar{P}_C \bar{\Sigma}_x) \in [0, 1], \quad \kappa \rightarrow 1 \text{ is the coupling null (no flip).}$$

CONDITIONS

- Defined as the normalized overlap between the averaged read operator and the source covariance on the read subspace, between zero and one, and so a statement about one workload.
- The alignment law that relates it to the size of the flip is a retrospective fit with a single prospective point and is carried as exploratory, not as a theorem.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 12, [geometric-observation/chapters/ch12_failure_taxonomy_and_kappa.md:56-82](https://doi.org/10.5281/zenodo.21776291), DOI 10.5281/zenodo.21776291, where the alignment law is a retrospective fit carried as exploratory with one prospective point.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.4	alignment law, retrospective fit exploratory, one prospective point	geometric-observation/chapters/ch12_failure_taxonomy_and_kappa.md:56-82

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Alignment.lean](#), theorems `overlap_sq_le`, `alignment_le_one`, `alignment_nonneg`, `alignment_eq_one_of_proportional`, `coupling_null`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2, 4, 7, 11, 14.

RELATED

[the flip](#) (p. 147); [read distortion](#) (p. 290); [coupling null](#) (p. 89); [water-filling](#) (p. 378).

SEE ALSO

Ledger rows that cite the entry's records without naming it: GO-4.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.4.

allocation

instrument. Entry id allocation.

DEFINITION

The assignment of a bit budget across directions. Water-filling gives directions below the water no bits, and the allocation report warns below effective rank two. Chapter 4 section 4.2.

Example. With directions of variance 4, 1, and 0.1 and a water level of 0.5, the first two directions get bits and the third gets none.

EQUATION

none

CONDITIONS

- The assignment of a bit budget across directions. Water-filling gives directions below the water no bits, the allocation report states its gain over uniform and warns below effective rank two, and a spectrum with half its mass in one direction has effective rank at most four.
- The fragile-first allocation read 0.7118 against 0.7251 and was redesigned, and the budget inversion of GO-4 shows an allocation at fixed m rising while the matched- m allocation collapses.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

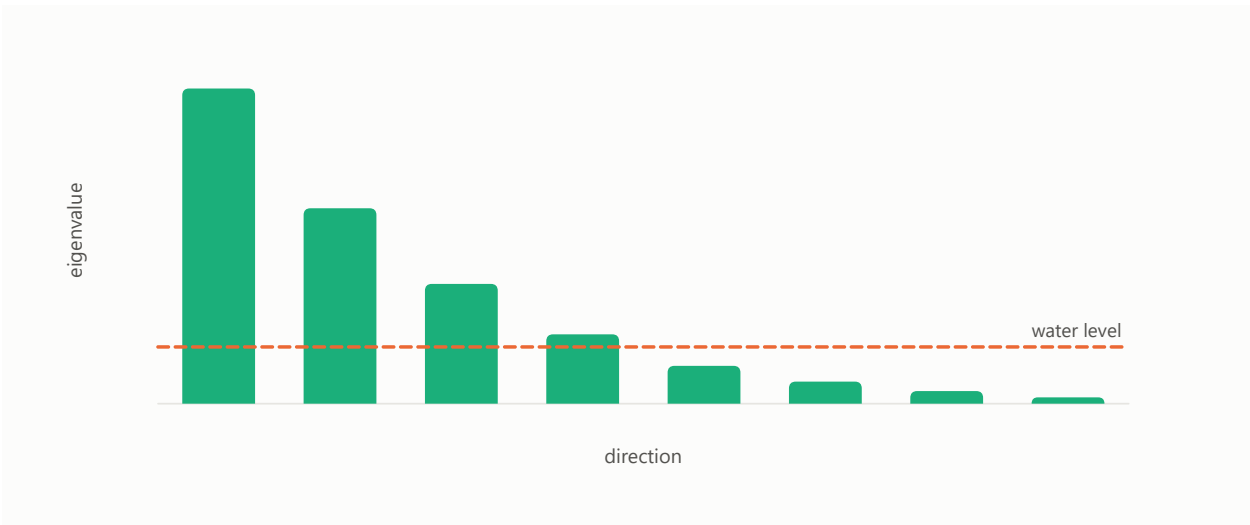


Figure 4: Bits go to the directions above the water level and none to those below it.

FIRST STATED

Chapter 4 section 4.2 of *Data Mining as Observation*, with the allocation report in `turboquant-pro/turboquant_pro/read_allocation.py:244-307`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 4 section 4.2	allocation report, gain over uniform, concentration caution below effective rank 2	turboquant-pro/turboquant_pro/read_allocation.py:244-307
chapter 10 section 10.5	fragile-first allocation 0.7118 vs 0.7251, targets up 0.004 to 0.034, redesigned allocator passed	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:110-125

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/WaterFilling.lean](#), theorems `contribution_eq_water`, `terms_mul`, `two_sqrt_le`, `distortion2_ge`, `distortion2_eq_of_equal`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

[lean/DataMiningAsObservation/EffectiveRank.lean](#), theorems `sq_sum_le`, `effRank_le`, `sum_sq_le_sq_sum`, `one_le_effRank`, `effRank_const`, `effRank_single`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 10, 11.

RELATED

[water-filling](#) (p. 378); [budget](#) (p. 51); [effective rank](#) (p. 123); [concentrated](#) (p. 73); [bit](#) (p. 43).

SEE ALSO

Book equations stated beside the entry’s terms, not defining it: 4.2, 0.7, 4.6.

Ledger rows that cite the entry's records without naming it: GO-4, GO-6.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.2.

anisotropic

concept. Entry id anisotropic.

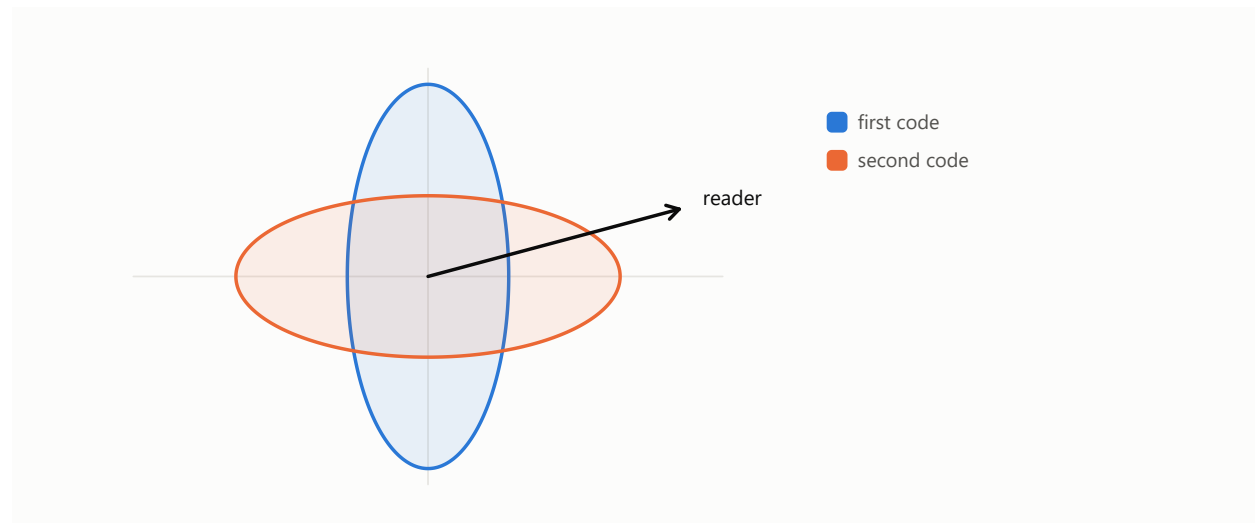


Figure 5: Unequal variances across directions, so a reader at an angle sees a different variance from another.

DEFINITION

Of a covariance, having different variances in different directions. Chapter 0.

Example. Covariance $\text{diag}(0.3, 1.7)$ has variance 0.3 along the first axis and 1.7 along the second, so it is anisotropic.

EQUATION

none

CONDITIONS

- Of a covariance, having different variances in different directions, which is to say unequal eigenvalues. Two readers at different angles to an anisotropic covariance read different variances, and the flip between two codes exists exactly when the covariance is anisotropic along the directions that separate them.
- Centring an embedding raised a ranker from 0.79 to 0.84 on the legal corpus, which is the condition under which a learned quotient and a variance quotient disagree most, and an isotropic control collapses the flip.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.4 of *Data Mining as Observation*, with the program's centring measurement in `geometric-observation/claims/LEDGER.md` row GO-B-legal and the anisotropy channel in `1ebse/README.md:40-75`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.8	0.765 to 0.971 with interval 0.190 to 0.223, 0.545 to 0.562 with interval 0.004 to 0.031, v1 0.340 with interval negative 0.214 to negative 0.081, anisotropy 0.570 to 0.259	lebse/README.md:40-75; lebse\PAPER.md:1-15,60-66; lebse/MODEL_CARD.md:48-52

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Isotropy.lean](#), theorems isotropic_reads_same, isotropic_no_flip, anisotropic_readers_differ, flip_iff_anisotropic, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 9, 11, 12.

RELATED

[isotropic](#) (p. 187); [covariance matrix](#) (p. 91); [spectrum](#) (p. 344); [whitening](#) (p. 381).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.5, 3.1, 4.6.

Ledger rows that cite the entry's records without naming it: GO-2 (neg. half: not reconstruction), GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 12 section 12.3, chapter 12 section 12.4.

anti arm

instrument. Entry id anti-arm.

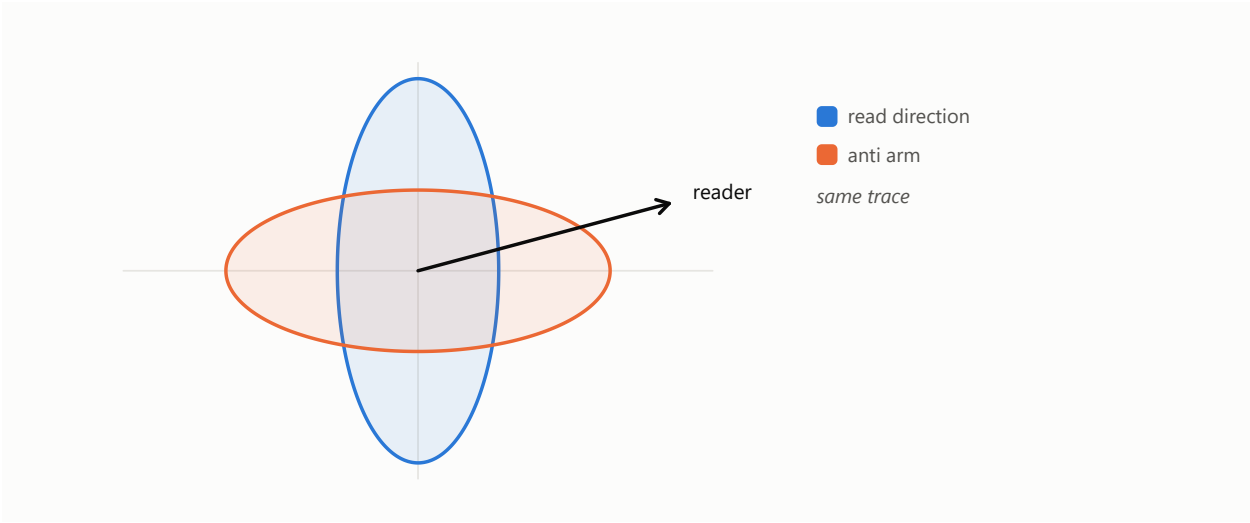


Figure 6: The control that puts the error on the read direction, confirmed worst.

DEFINITION

In a flip comparison, a third code built to destroy the consumer's read subspace at the same bits. It is expected to score worst, and a comparison in which it does not is not a flip. Chapter 4.

Example. Put the whole error of 0.4 on the direction the reader reads and the reader sees 0.4; put it on the other direction and the reader sees zero.

EQUATION

Book equation 4.5.

$$\text{flip} : \quad \text{task}(O) > \text{task}(R) \text{ and } \text{tr } M_\delta^O > \text{tr } M_\delta^R, \quad \text{task}(\text{anti}) < \text{task}(R), \quad \text{bits}(O) = \text{bits}(R).$$

CONDITIONS

- The anti arm is matched on bits with the other two codes, and its purpose is to show that the read subspace is what the comparison is about, since destroying it should be worst.
- A comparison in which the anti arm is not worst has not shown a flip, whatever the other two arms did.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 8, `geometric-observation/chapters/ch08_value.md:1-30`, DOI 10.5281/zenodo.21776291, as the third arm of every registered flip.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/AntiArm.lean](#), theorems `readDiag_le_max`, `readDiag_le_total`, `anti_arm_worst`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 4, 6, 7, 12.

RELATED

[the flip](#) (p. 147); [read distortion](#) (p. 290); [read operator](#) (p. 291).

SEE ALSO

Ledger rows that cite the entry's records without naming it: GO-2 (neg. half: not reconstruction), NEG-16 (KV serving, end-task), GO-B-whale (038), GO-B-blind (041).

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 4 section 4.3, chapter 4 section 4.4, chapter 6 section 6.2.

anti-hub

concept. Entry id `anti-hub`.

DEFINITION

A row that is never or almost never retrieved as anyone's nearest neighbour. Chapter 10 names five kinds and shows one is manufactured by the query budget. Chapters 3 and 10.

Example. With 1000 queries at `k` equal to 10 there are 10000 slots, so in a corpus of 100000 rows at least 90

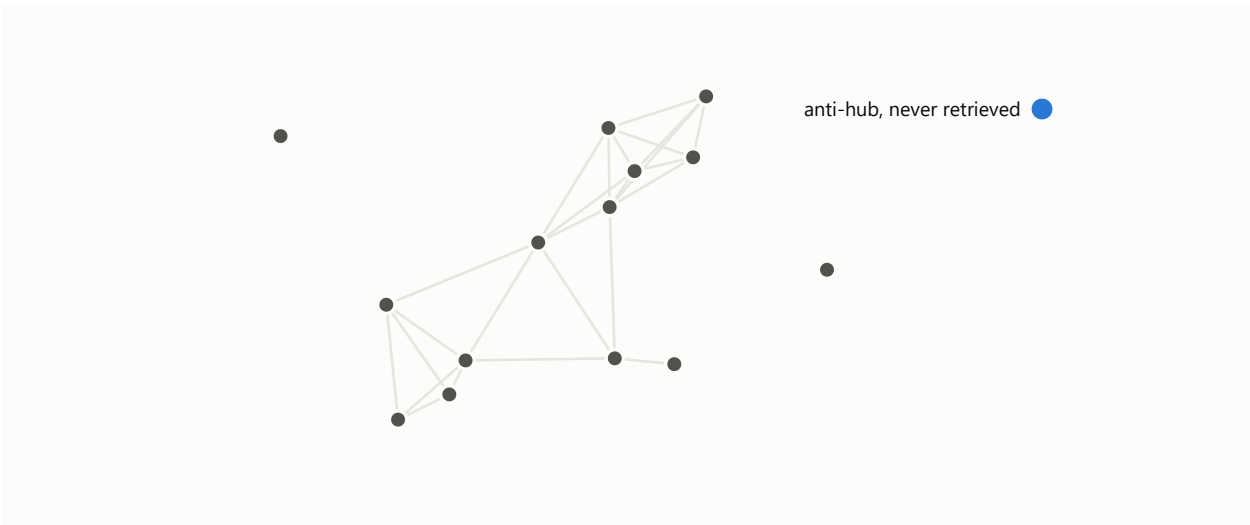


Figure 7: A row that no query reaches.

percent are never retrieved.

EQUATION

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

CONDITIONS

- An anti-hub is a row no query retrieves, and chapter 10 names five kinds, of which one is manufactured by the query budget rather than by the corpus.
- Anti-hubs are where compressed indexes fail first, since quantization rounds away the fine distinctions by which they are found, while aggregate recall barely moves, which is why recall is reported per stratum with the anti-hub stratum named.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Radovanović, Nanopoulos, and Ivanović, hubs in space, 2010, as chapter 3 cites it, with the program’s five kinds in `turboquant-pro`, `turboquant-pro/docs/HUBNESS_PRIMER.md:86-131`, and chapter 10 section 10.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131
chapter 11 section 11.6	count of ten, hubs and anti-hubs, max 78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, abstain below 2.5k, centering vs mutual-proximity rescaling	<code>turboquant-pro/docs/HUBNESS_PRIMER.md:1-60, 60-170</code>

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Hubness.lean](#), theorems `sum_count`, `sum_count_eq`, `count_congr`, `antiHub_iff`, at observation-data-mining 95af30c; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 3, 10, 11, 12.

RELATED

[hubness](#) (p. 176); [Poisson ceiling](#) (p. 262); [min-over-strata](#) (p. 226); [rank certificate](#) (p. 285).

SEE ALSO

Book equations stated beside the entry’s terms, not defining it: 10.7.

Ledger rows that cite the entry’s records without naming it: NEG-11, NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.5, chapter 8 section 8.10, chapter 10 section 10.3, chapter 10 section 10.4, chapter 10 section 10.5, chapter 11 section 11.6, chapter 12 section 12.4.

anti-hub recall

instrument. Entry id anti-hub-recall.

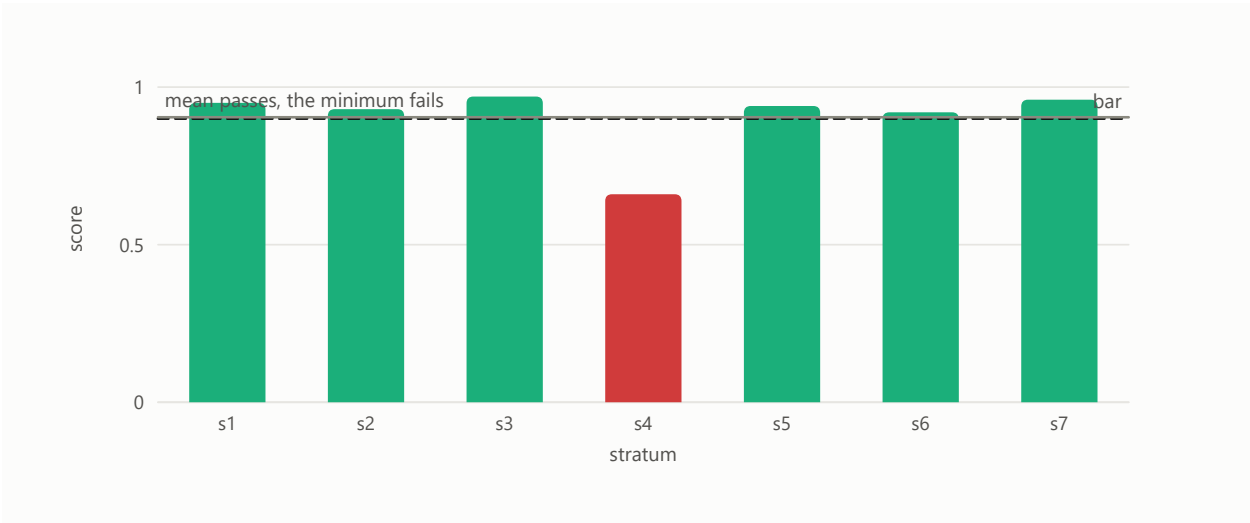


Figure 8: Recall on the rarely retrieved rows, taken as the minimum over strata.

DEFINITION

Recall at *k* restricted to the rows that are rarely retrieved, the observer’s outliers, reported as the minimum over strata so that a mean cannot hide the tail. Chapter 10.

Example. Seven strata of rarely retrieved rows scored between 0.62 and 0.69, so the minimum, 0.663, failed the bar of 0.90.

EQUATION

none

CONDITIONS

- Recall at k restricted to the rows that are rarely retrieved, the observer's outliers, reported as the minimum over strata so that a mean cannot hide the tail. It lies in the unit interval and is one exactly when every anti-hub's true neighbours are returned.
- Compressed indexes fail on anti-hubs first while aggregate recall barely moves, and the remedy that promotes isolated rows is the remedy that compression damages first, which is why the gate is a build gate and not a report line.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 10 section 10.2 of *Data Mining as Observation*, with the program's gate in `turboquant-pro/docs/HUBNESS_PRIMER.md:86-131` and the strata result in `turboquant-pro/docs/RESULTS_strata_phase23_gates.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/RecallAtK.lean](#), theorems `recallAtK_mem_unit`, `recallAtK_eq_one_iff`, `aggregate_le_of_failing`, `aggregate_example`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Hubness.lean](#), theorems `sum_count`, `sum_count_eq`, `count_congr`, `antiHub_iff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 10, 11, 12.

RELATED

[anti-hub](#) (p. 20); [recall at k](#) (p. 295); [min-over-strata](#) (p. 226); [hub](#) (p. 175); [stratification](#) (p. 350).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 10.7, 11.3, 0.35.

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.2, chapter 10 section 10.4, chapter 10 section 10.5, chapter 12 section 12.4.

anti-monotonicity

concept. Entry id anti-monotonicity.

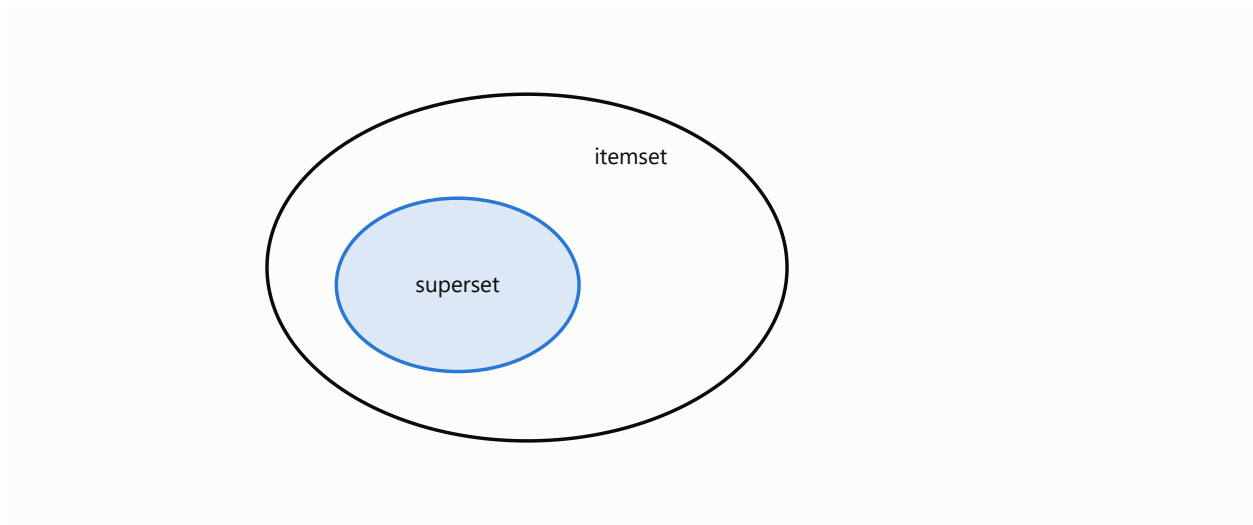


Figure 9: Support cannot rise as an itemset grows, so a superset of an infrequent itemset is pruned.

DEFINITION

The property that support cannot increase when an itemset grows, which licenses Apriori to prune every superset of an infrequent itemset without loss. Chapter 5.

Example. If bread and butter appear together in 30 baskets, then bread, butter, and jam appear together in at most 30.

EQUATION

none

CONDITIONS

- The property that support cannot increase when an itemset grows, since every transaction containing the larger set contains the smaller. It licenses Apriori to prune every superset of an infrequent itemset without loss, because no frequent itemset can lie above an infrequent one.
- The license is a statement about the measure and not about the data, and the book's sidebar carries it to thresholds outside itemsets, where the analogous license is the ceiling on a score that a Youden bound supplies.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Agrawal and Srikant, fast algorithms for mining association rules, 1994, as chapter 5 section 5.3 of *Data Mining as Observation* reads it.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/SafePruning.lean](#), theorems `support_anti`, `apriori`, `subset_of_frequent`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/ItemSet.lean](#), theorems `supportFrac_mem_unit`, `supportFrac_empty`, `supportFrac_anti`, `supportFrac_union_le`, `closed_of_maximal`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 5.

RELATED

[support](#) (p. 353); [safe pruning](#) (p. 320); [itemset](#) (p. 188).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 5.1, 5.3.

Sources-table rows that share a record with the entry without naming it: chapter 5 section 5.1 to 5.3, 5.5, chapter 5 section 5.3.

Apriori principle

result. Entry id `apriori`.

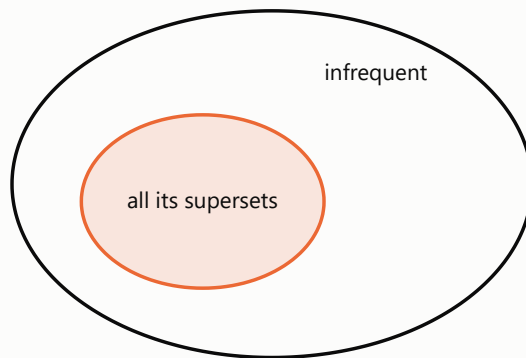


Figure 10: Every superset of an infrequent itemset is infrequent.

DEFINITION

The statement that every subset of a frequent itemset is frequent, the contrapositive of anti-monotonicity. Chapter 5.

Example. If the pair bread and jam is infrequent, every superset such as bread, jam, and milk is skipped without being counted.

EQUATION

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

CONDITIONS

- Support cannot increase when an itemset grows, so an infrequent itemset has only infrequent supersets and every subset of a frequent itemset is frequent. This holds for any finite collection of transactions.
- The principle licenses pruning the lattice above an infrequent itemset without counting, and the count of candidates rather than the count of transactions decides whether the run finishes.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Agrawal and Srikant, fast algorithms for mining association rules, 1994, as TSK chapter 5 presents it, and chapter 5 section 5.3 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/SafePruning.lean](#), theorems `support_anti`, `apriori`, `subset_of_frequent`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 5.

RELATED

[safe pruning](#) (p. 320); [Monotone Invariance Theorem](#) (p. 227).

SEE ALSO

none

attention

concept. Entry id attention.

DEFINITION

The operation inside a language model that lets each token look at earlier ones by comparing its query to their keys and averaging their values. Chapter 0 section 0.11.

Example. Scores 2, 1, and 0 against three keys give softmax weights 0.665, 0.245, and 0.090.

EQUATION

Book equation 0.23.

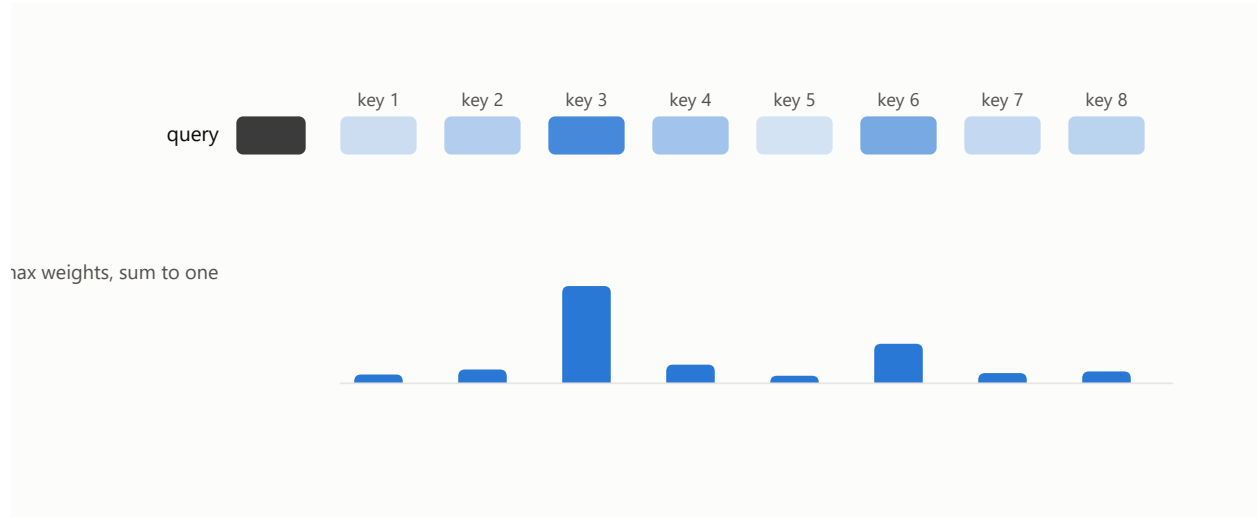


Figure 11: A query against every key, and a softmax that turns the scores into weights.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

CONDITIONS

- A head weights each earlier token's value by the softmax of its query-key score and sums. For scalar values the output lies between the smallest and largest value, and the head reads the keys only through their scores against the query, so a key change the query does not read leaves the output unchanged however large it is.
- That is the head's read subspace, a few query-weighted directions of each key, and the reason a key reconstruction at cosine 0.995 raised the perplexity by three orders of magnitude.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.11 and chapter 11 section 11.1 of *Data Mining as Observation*, with the program's head-level measurements in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49` and the serving-stack ledger row.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Attention.lean](#), theorems `output_le_max`, `min_le_output`, `output_congr`, `output_nuisance`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 8, 11, 13.

RELATED

[softmax](#) (p. 336); [read subspace](#) (p. 293); [nuisance](#) (p. 240); [the flip](#) (p. 147).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.1.

Ledger rows that cite the entry's records without naming it: GO-2/GO-12/GO-13 operational (KV serving, 077), NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 8 section 8.2, chapter 11 section 11.1, chapter 11 section 11.2.

attribute type

concept. Entry id attribute-type.



rescale $\subset$ affine $\subset$ increasing $\subset$ injective

Figure 12: The four scales and the transformations each permits, nested from ratio to nominal.

DEFINITION

The declaration of which transformations of a column keep its meaning, any injective relabelling for nominal, a strictly increasing map for ordinal, an affine map for interval, and a positive rescaling for ratio, nested in that order. Chapter 2 section 2.1.

Example. Temperature in Celsius is an interval scale, so a ratio of two temperatures means nothing, while 4 metres is twice 2 metres on the ratio scale of length.

EQUATION

none

CONDITIONS

- The declaration of which transformations of a column do not change its meaning. A ratio scale permits a positive rescaling, an interval scale an affine map with positive slope, an ordinal scale a strictly increasing map, and a nominal scale any injective relabelling, and the four classes are nested in that order.
- A rescaling preserves ratios and an affine map preserves ratios of differences, so a consumer that reads a ratio of two interval-scale columns reads the origin the scale said was arbitrary.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Stevens, on the theory of scales of measurement, 1946, as chapter 2 section 2.1 of *Data Mining as Observation* reads it after TSK section 2.1.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.1	the four attribute types and permitted transformations	TSK 2e section 2.1

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/AttributeType.lean](#), theorems `scale_affine`, `affine_strictMono`, `strictMono_injective`, `scale_preserves_ratio`, `affine_preserves_difference_ratio`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2, 13.

RELATED

[standardization](#) (p. 348); [discretization](#) (p. 112); [quotient](#) (p. 279).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 2.1, 0.12a.

attribution

instrument. Entry id attribution.

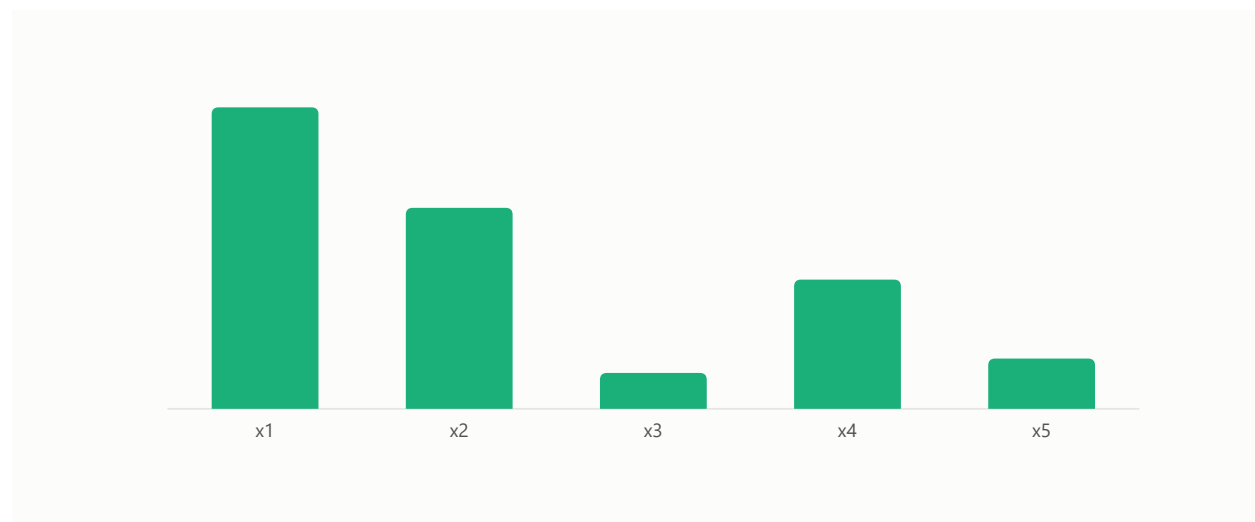


Figure 13: A share of one prediction per feature, and the averaged squared gradient that estimates the read operator.

DEFINITION

An assignment to each feature of a share of one prediction. The raw gradient is the sensitivity at a row, gradient times input is the share of the change in output, and the averaged squared gradient is a diagonal entry of the read operator, and the three are kept apart. Shapley, permutation, and partial-dependence methods measure other things and are named for them. Chapter 14.

Example. For the affine consumer $3x_1 + 2x_2$, moving from $(0, 0)$ to $(1, 1)$ gives attributions 3 and 2, which sum to the change in output, 5.

EQUATION

none

CONDITIONS

- An assignment to each feature of a share of one prediction. Three gradient quantities are kept apart. The raw gradient is the sensitivity at a row. Gradient times input, the sensitivity times the move along a coordinate, sums exactly to the change in output for an affine consumer and is zero on an unread coordinate. The weighted mean of the squared sensitivity is a diagonal entry of the read operator, so an averaged squared gradient estimates the read operator and a single-row attribution does not.
- Shapley, permutation, and partial-dependence methods measure other things and are named for them, and for a tree the sensitivity is zero almost everywhere, so importances that count splits are the right reading and finite differences the wrong one.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Lundberg and Lee, a unified approach to interpreting model predictions, 2017, as chapter 14 section 14.5 of *Data Mining as Observation* reads it, with the program's contraction formula in `gtc-prototype/docs/SPECTRUM_FINDINGS.md:90-99`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Attribution.lean](#), theorems `attr_sum_affine`, `attr_unread`, `sq_sensitivity_eq_readOp_diag`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 8, 14.

RELATED

[sensitivity](#) (p. 328); [read operator](#) (p. 291); [finite difference](#) (p. 146); [consumer](#) (p. 79).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.8, 0.9, 14.5.

Ledger rows that cite the entry's records without naming it: GO-1, GO-EC-3.

Sources-table rows that share a record with the entry without naming it: chapter 11 section 11.7, chapter 14 section 14.5.

audit

instrument. Entry id audit.

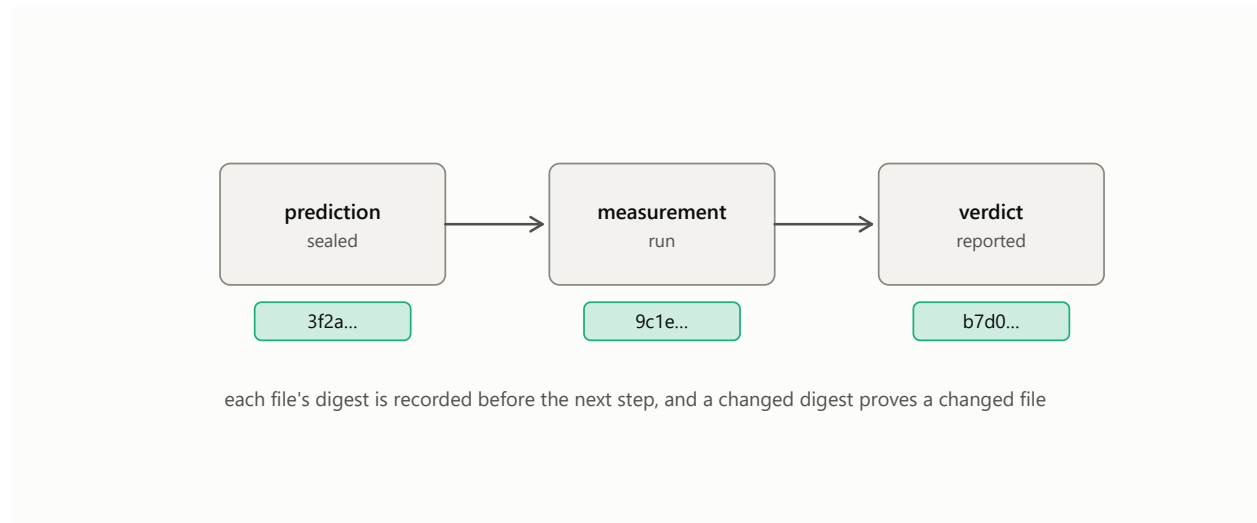


Figure 14: A reader who did not produce the number checks it against the sealed record.

DEFINITION

A check of a number or system against its record by a reader who did not produce it, bound to a sealed file. Chapter 8 section 8.9 and chapter 14 section 14.1.

Example. The audit of the measurement notes found twenty-seven errors, four of them in the checking tools themselves.

EQUATION

none

CONDITIONS

- A check of a published number or system against its record, by a reader who did not produce it. An audit binds a verdict to a sealed file whose digest proves it unchanged, and a witness's disagreement with a certificate is the audit's finding.
- The audit of the measurement notes found twenty-seven errors, four in the checking tools, and the constraint-first Bell audit reached a maximum of 2.00000 across 72 configurations.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.9 and chapter 14 section 14.1 of *Data Mining as Observation*, with the audit binding in `xbse/README.md:195-219` and the Bell audit in `geometric-observation/articles/2026-08-03`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.6	Bell audit, max S 2.00000 across 72 configurations, d 3 to 128, correlation negative 0.036, post-selected 2.7308, controls 2.748, 2.386, 3.174, seed 20260817, sealed at 6e825d8	geometric-observation/articles/2026-08-03-hubness-does-not-weaken-bell.md:1-60 ; geometric-observation/claims/LEDGER.md row NEG-15; geometric-observation/results/GO-bell-geometry-audit.json
chapter 14 section 14.1	the three questions, the trust beats, the audit binding	gtc-prototype/README.md:10-58 ; xbse/README.md:195-219

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Seal.lean](#), theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Certificate.lean](#), theorems `falseClear_mul_coverage`, `coverage_empty`, `falseClear_mem_unit`, `minOverStrata_passes_iff`, `minOverStrata_le_weighted_mean`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 2, 3, 4, 8, 12, 14.

RELATED

[sealed](#) (p. 324); [witness](#) (p. 382); [certificate](#) (p. 59); [false-clear rate](#) (p. 141); [registered](#) (p. 305).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.1, 14.2, 13.2.

Ledger rows that cite the entry's records without naming it: NEG-15 (Bell boundary), GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.9.

AUROC

concept. Entry id `auroc`.

DEFINITION

The area under the receiver operating characteristic curve, equal to the probability that a random positive scores above a random negative, ties counted as one half. A strictly increasing transform of the score leaves it unchanged and a strictly decreasing one sends it to one minus itself. Equation 0.16.

Example. Positives scoring 0.9 and 0.7 against negatives scoring 0.8 and 0.2 win three of the four pairs, so the AUROC is 0.75.

EQUATION

Book equation 0.16.

$$\text{AUROC} = \Pr[s^+ > s^-] + \frac{1}{2} \Pr[s^+ = s^-].$$

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

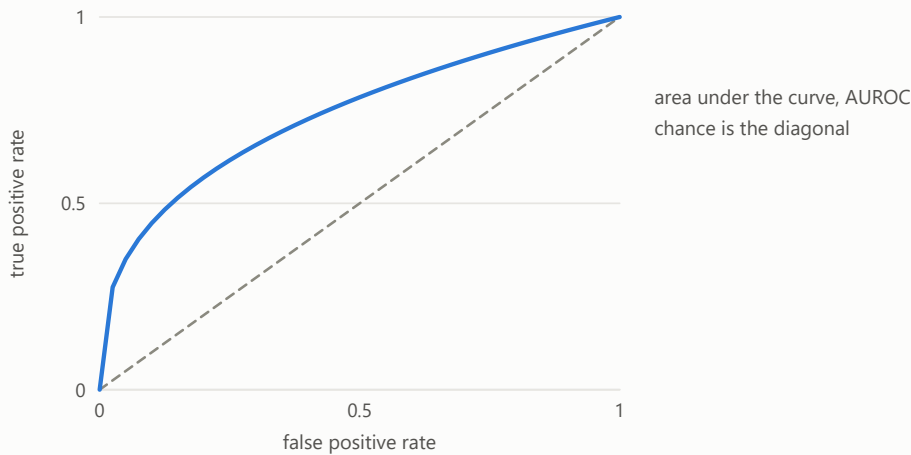


Figure 15: The area under the curve is the chance that a random positive outscores a random negative.

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

CONDITIONS

- The area under the ROC curve, the probability that a random positive scores above a random negative with ties counted as one half. It lies in the unit interval, is one half for a constant score, one for a perfect ranking, and zero for a perfectly reversed one.
- A strictly increasing transform of the score leaves it unchanged. A strictly decreasing transform reverses every ranking and sends A to $1 - A$, so the AUROC is not free of orientation, and an authority that should ignore orientation reads the larger of A and $1 - A$.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.8 of *Data Mining as Observation*, the pairwise definition with ties counted one half, and the held-out AUROC of every encoder gate and flip in the program.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Auoc.lean](#), theorems `pair_nonneg`, `auroc_nonneg`, `auroc_le_one`, `auroc_perfect`, `auroc_reversed`, `auroc_chance`, `auroc_monotone_invariant`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 5, 6, 8, 11, 12, 14.

RELATED

[Monotone Invariance Theorem](#) (p. 227); [reliability weight](#) (p. 306); [chance level](#) (p. 62); [Youden F1 bound](#) (p. 385); [cross-corpus gate](#) (p. 94).

SEE ALSO

Ledger rows that cite the entry's records without naming it: GO-B-legal (035→036), GO-B-whale (038).

B

bagging

instrument. Entry id bagging.



Figure 16: Resamples of the same rows, each fit and then averaged.

DEFINITION

Fitting the same model on bootstrap resamples and averaging. The average's squared error is at most the members' mean squared error, which is how it reduces variance. Chapter 7.

Example. Ten trees fit on ten bootstrap resamples and averaged have at most the mean squared error of the ten members.

EQUATION

none

CONDITIONS

- Fitting the same model on bootstrap resamples of the rows and averaging. The average's squared error is at most the mean of the members' squared errors, which is the arithmetic by which bagging reduces variance, and the reduction is largest when the members covary least.
- Bagging reduces variance and boosting reduces bias, and neither changes what the base model reads.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Breiman, bagging predictors, 1996, as chapter 7 section 7.1 of *Data Mining as Observation* reads it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.1	bagging variance, out-of-bag estimation, random forests, importances, AdaBoost	Hastie, Tibshirani, Friedman, ESL 2e chapters 15 and 10.1; TSK 2e 4.10

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Ensemble.lean](#), theorems `average_between`, `sq_average_le`, `average_const`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Bootstrap.lean](#), theorems `mean_sub`, `var_sub`, `paired_lt_iff`, `cov_comm`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 7.

RELATED

[ensemble](#) (p. 129); [bootstrap](#) (p. 49); [random forest](#) (p. 282); [standard error](#) (p. 347); [variance](#) (p. 374).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.18, 8.2.

bag of words

concept. Entry id bag-of-words.

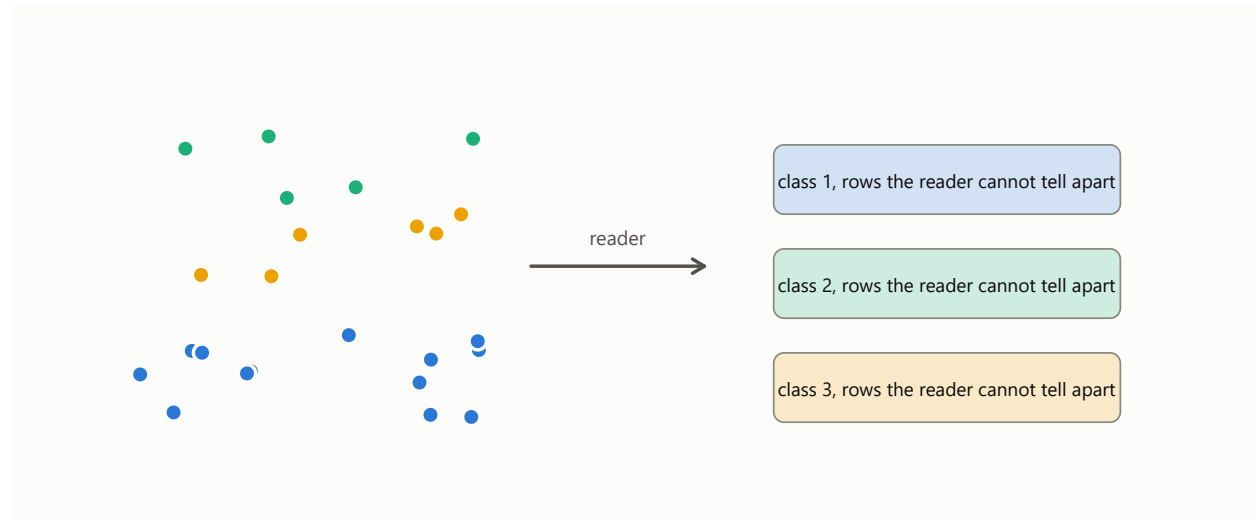


Figure 17: Documents with the same counts are the same document to the reader, whatever the order.

DEFINITION

A representation of a document as a vector with one coordinate per vocabulary term holding that term's count. Chapter 12.

Example. The texts 'the reader reads the row' and 'the row reads the reader' have the same counts and are one document to a bag of words.

EQUATION

Book equation 0.36.

$$w_{t,d} = \text{tf}_{t,d} \cdot \ln \frac{N}{\text{df}_t}, \quad \text{tf}_{t,d} = \frac{\text{count of } t \text{ in } d}{\text{length of } d}, \quad \text{df}_t = \text{documents containing } t.$$

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

CONDITIONS

- A document as the vector of its term counts. Two documents that are rearrangements of each other have the same bag, so the bag is a quotient that declares word order irrelevant, the counts are nonnegative, and they sum to the document's length.
- As a null model it reads only counts, and an encoder is validated only when its held-out AUROC clears the bag's by the preregistered margin. The scorecard's bag scores lie between 0.46 and 0.54, near chance.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 12 section 12.1 of *Data Mining as Observation*, with the program's bag-of-words null in `xbse/README.md`: 104-130 and the chapter 14 scorecard.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/BagOfWords.lean](#), theorems `bag_perm`, `bag_example`, `bag_sum`, `bag_absent`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 12, 14.

RELATED

[TF-IDF](#) (p. 360); [cross-corpus gate](#) (p. 94); [quotient](#) (p. 279); [nuisance](#) (p. 240).

SEE ALSO

none

balanced accuracy

concept. Entry id `balanced-accuracy`.

DEFINITION

The mean of the per-class recalls, so that a large class cannot hide a small one. Equation 0.34.

Example. Recalls of 0.90 on class A and 0.30 on class B give balanced accuracy 0.60, whatever the class sizes.

EQUATION

Book equation 0.34.

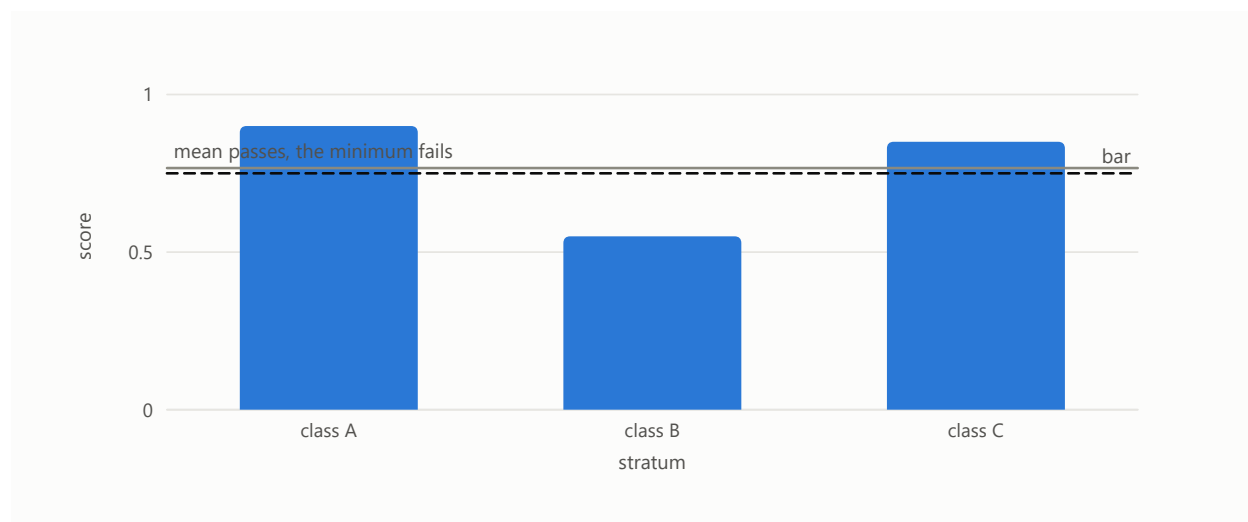


Figure 18: The mean of the per-class recalls, so a large class cannot hide a small one.

$$BA = \frac{1}{K} \sum_{k=1}^K \frac{\text{correct in class } k}{\text{rows in class } k}.$$

CONDITIONS

- The mean of the per-class recalls, between the worst class and the best. Plain accuracy is the size-weighted mean of the same recalls, so a class of 990 rows recalled perfectly and a class of 10 never recalled give accuracy 0.99 and balanced accuracy 0.5.
- The book prefers the min over classes to the mean where a verdict is at stake, since a mean can still hide one failing class among many.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.14 of *Data Mining as Observation*, with the program's use in the anti-hub category table, `openvector-bench/results/R13_STAGE1_RESULTS.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.2	five kinds, balanced accuracy 0.611, 0.718, 0.684, the category table, per-category 0.57 to 0.82, the sweep, pigeonhole floor	<code>openvector-bench/results/R13_STAGE1_RESULT.md:40-75</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/BalancedAccuracy.lean](#), `theorems min_le_balanced`, `balanced_le_max`, `hidden_class`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 10.

RELATED

[min-over-strata](#) (p. 226); [anti-hub](#) (p. 20); [abstention](#) (p. 12); [harness](#) (p. 168).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 10.7.

Ledger rows that cite the entry's records without naming it: NEG-11.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.5, chapter 10 section 10.2, chapter 11 section 11.6.

bar

instrument. Entry id bar.

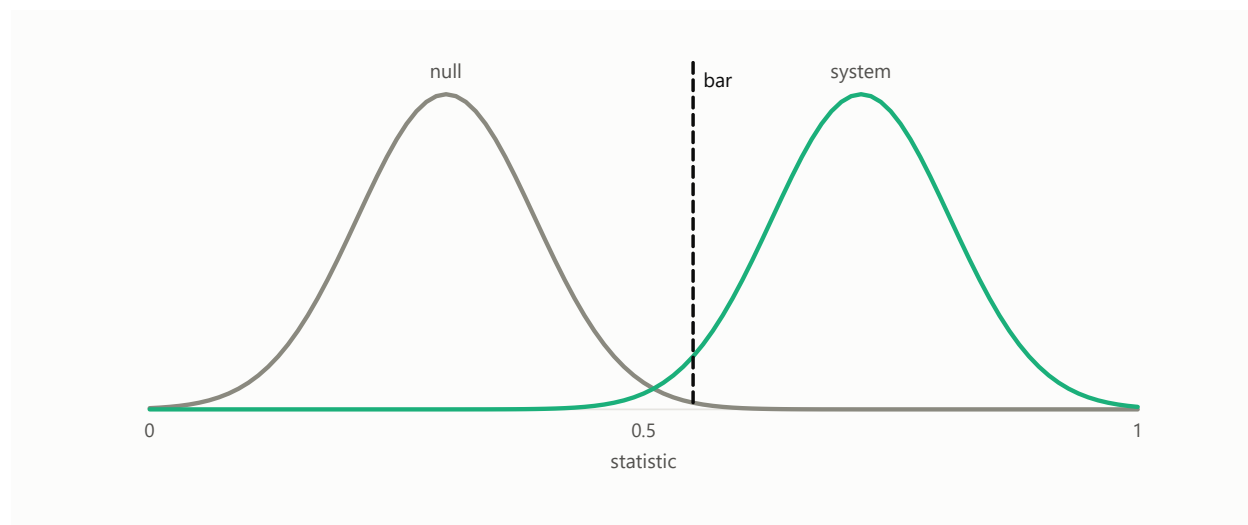


Figure 19: A bar discriminates when the null fails it and the system passes it.

DEFINITION

A threshold a statistic must reach, written down before the measurement. A bar discriminates only when the null fails it and the real system passes it, and a bar the null passes is vacuous. Chapter 8 section 8.3.

Example. A null scoring 0.30 and a system scoring 0.72 are separated by a bar at 0.55, and a bar at 0.25 is vacuous, since the null passes it.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

CONDITIONS

- A threshold a statistic must reach, written down before the measurement. Passing a higher bar passes every lower one, a larger statistic passes every bar a smaller one passes, a bar discriminates exactly when it sits strictly above the null and at or below the real system, no bar discriminates when the null scores at least as high, and a bar the null passes is vacuous.

- A pass on eight cells at six parts in ten million against a bar of one in a hundred thousand was vacuous because the null cleared the bar too, which is where the anti-vacuity bar came from.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.3 of *Data Mining as Observation*, with the anti-vacuity bar quoted from `observation-theory-campaigns/experiments/PREREG-PF5-002.md:79-86`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.3	anti-vacuity bar, quoted	observation-theory-campaigns/experiments/PREREG-PF5-002.md:79-86
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	<code>the-angular-observer/PREREG_RECOGNIZER_BATTERY.md:1-80</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Bar.lean](#), `theorems passes_anti, passes_mono, discriminates_iff, no_bar_of_null_ge, exists_bar_of_lt, vacuous_of_null_passes`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[gate](#) (p. 159); [vacuity threshold](#) (p. 371); [null model](#) (p. 242); [preregistration](#) (p. 267); [verdict](#) (p. 375).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.3, 8.1.

Ledger rows that cite the entry's records without naming it: OT-11, NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.3.

baseline

instrument. Entry id baseline.

DEFINITION

The simplest scorer a claim must beat under the same protocol, chance, a constant, an untrained encoder, or a single model. A score below it is a finding. Chapter 6 section 6.3 and chapter 7 section 7.4.

Example. An untrained encoder scored 0.50 on the rights corpus and the trained one 0.467, below the baseline.

EQUATION

none

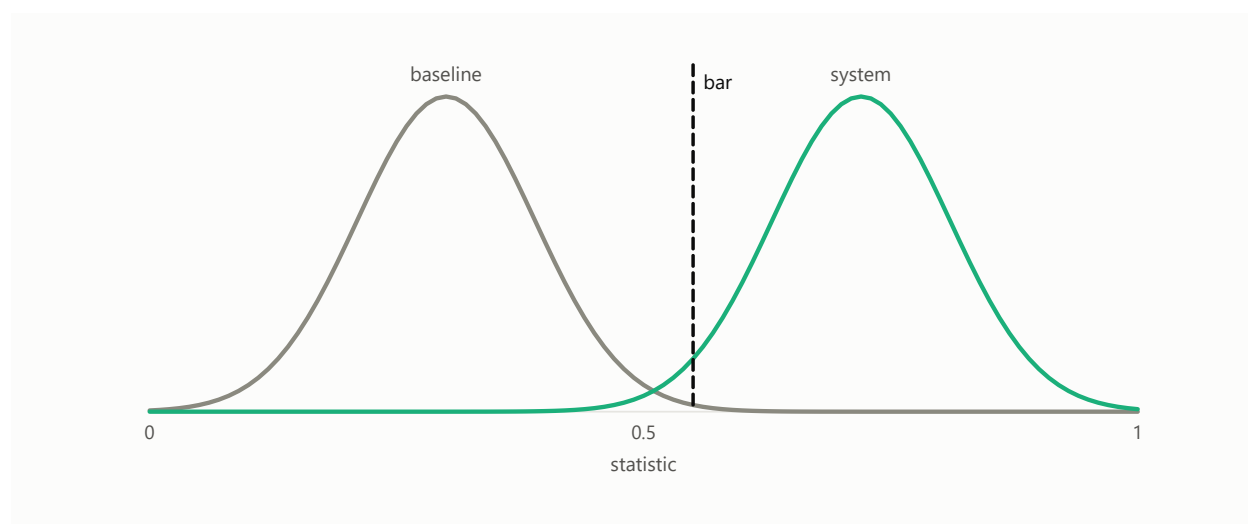


Figure 20: The simplest scorer the claim must beat, under the same protocol.

CONDITIONS

- The simplest scorer a claim must beat, chance, a constant, an untrained encoder, or a single model, run under the same protocol. A baseline that passes the bar makes the bar vacuous, and a score below the baseline is a finding.
- The rights encoder read 0.467, below the untrained baseline, and the formula's comparison with the ensemble is fair only against the single-model baseline inside one fold protocol.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 6 section 6.3 and chapter 7 section 7.4 of *Data Mining as Observation*, with the single-model baseline in `constraint-gap/review/INDETERMINATES.md:1-40`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.6	rights encoder 0.467 below untrained baseline, the AUROC lesson, cross-corpus same-sign fix	<code>xbse/README.md:160-172</code> ; <code>xbse/experiments/rights_r6_summary.json</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ChanceLevel.lean](#), theorems `aurocNum_const`, `auroc_const`, `weight_const`, `tau_const`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Bar.lean](#), theorems `passes_anti`, `passes_mono`, `discriminates_iff`, `no_bar_of_null_ge`, `exists_bar_of_lt`, `vacuous_of_null_passes`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 5, 6, 7, 8, 10, 11, 12, 13, 14.

RELATED

[chance level](#) (p. 62); [null model](#) (p. 242); [bar](#) (p. 39); [ensemble](#) (p. 129).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.2, 0.28, 6.2.

Ledger rows that cite the entry's records without naming it: NEG-4, GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.3, chapter 7 section 7.3, chapter 7 section 7.4.

bi-Lipschitz

concept. Entry id bi-lipschitz.

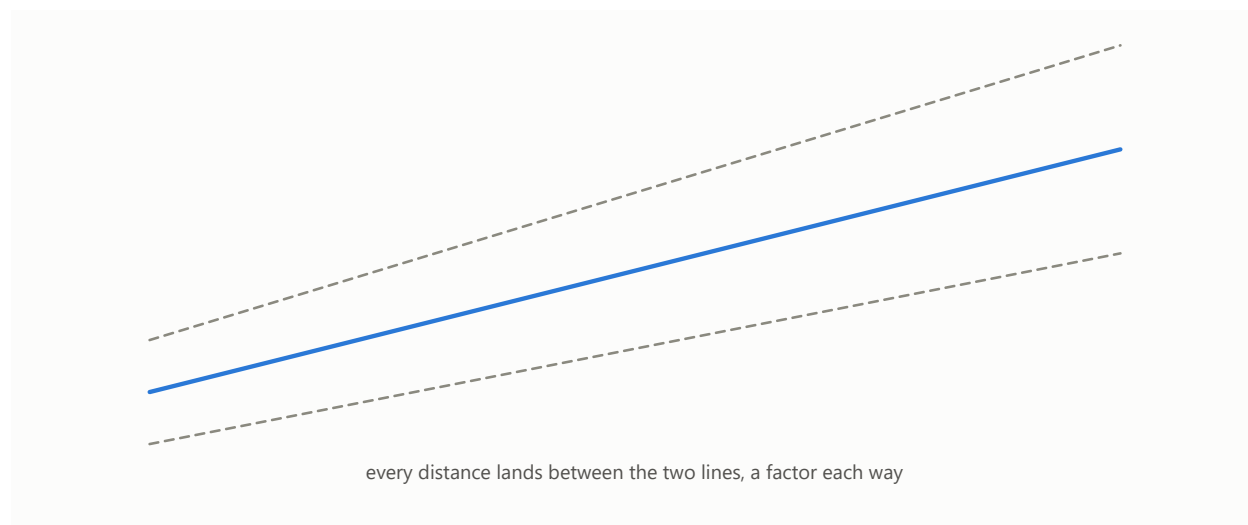


Figure 21: Every distance stretched or shrunk by at most a fixed factor.

DEFINITION

Of a map between two spaces, neither stretching nor shrinking any distance by more than a fixed factor. Equation 0.21.

Example. A map with factor 2 sends a distance of 1 to something between 0.5 and 2.

EQUATION

Book equation 0.21.

$$\frac{1}{K} d(x, y) \leq d'(f(x), f(y)) \leq K d(x, y) \quad \text{for all } x, y.$$

CONDITIONS

- A map is bi-Lipschitz with factor K when it neither stretches nor shrinks any distance by more than K . The factor is at least one, compositions multiply the factors, and the map preserves a nearest neighbour whenever the runner-up is more than K squared times as far, the rank certificate's per-pair fact.
- It does not preserve every ordering, so it is not rank-faithful, and the program's fixed-scale uniform bi-Lipschitz claim for the commute filter is refuted and carried in the ledger.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects*. NEG-1 [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [geometric-observation/claims/LEDGER.md:94](#).

FIRST STATED

Chapter 0 section 0.10 of *Data Mining as Observation*, with the refuted uniform bi-Lipschitz claim for the commute filter in Volume 14's honest negatives, ledger row NEG-1.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [geometric-observation/claims/LEDGER.md:94](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/BiLipschitz.lean](#), theorems `one_le_factor`, `within_comp`, `nn_preserved`, `not_rank_faithful`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3.

RELATED

[rank-faithful](#) (p. 286); [rank certificate](#) (p. 285); [recognizer](#) (p. 296); [quotient](#) (p. 279).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.2, chapter 3 section 3.3, chapter 9 section 9.1.

bit

concept. Entry id bit.

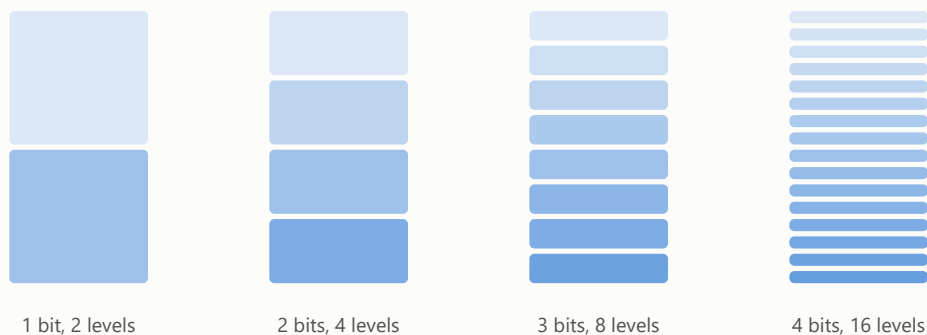


Figure 22: Each bit doubles the levels and halves the step.

DEFINITION

The unit of a budget. A direction quantized with b bits is stored at one of two to the power b levels. Chapter 0 section 0.7.

Example. Four bits give 16 levels; a fifth bit gives 32, halving the step and quartering the squared error.

EQUATION

Book equation 0.13.

$$D(b) = \sum_i s_i v_i 4^{-b_i}, \quad b_i = \max\left(0, \frac{1}{2} \log_2 \frac{s_i v_i}{\theta}\right), \quad \sum_i b_i = B.$$

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

CONDITIONS

- The unit of a budget. A direction quantized with b bits is stored at one of two to the b levels, so one more bit doubles the levels, halves a uniform quantizer's step, and quarters the squared error of the high-rate model, and bits add across directions since the levels multiply.
- Every flip comparison is at matched bits, since a code that spends more bits is a different budget and not a better reader.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-7 [replicated]. A stored description's description rate and its conditional Landauer reset content are operationally separate resources: the same finite- n code index needing $\hat{R} \approx 0.67$ bits/symbol to describe is fully recoverable from retained ... [geometric-observation/claims/LEDGER.md:69](#).

FIRST STATED

Chapter 0 section 0.7 of *Data Mining as Observation*, with the program's matched-bits comparisons in Volume 14 and `turboquant-pro`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Bit.lean](#), theorems `levels_succ`, `levels_add`, `step_succ`, `sqError_succ`, `sqError_antitone`, `levels_zero`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 10, 11, 12, 13, 14.

RELATED

[budget](#) (p. 51); [water-filling](#) (p. 378); [the flip](#) (p. 147); [Landauer's principle](#) (p. 203).

SEE ALSO

Ledger rows that cite the entry's records without naming it: GO-4.

blind probe

instrument. Entry id `blind-probe`.

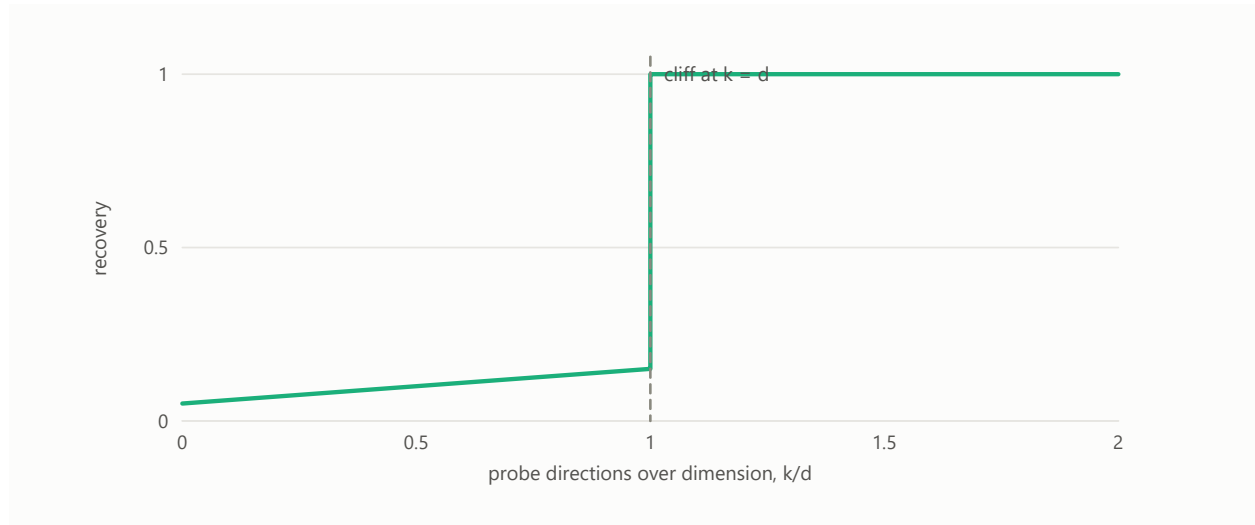


Figure 23: Recovery of the read operator against the probe’s budget, with a cliff at k equal to d .

DEFINITION

An instrument that recovers a consumer’s read operator from calls to the consumer alone, without access to its gradients or its code. Chapters 11 and 12.

Example. Probing a 16-dimensional consumer with 16 directions costs 32 calls and resolves the operator, and 15 directions do not.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

CONDITIONS

- The probe recovers the read operator at one operating point from calls to the consumer alone, two calls per input dimension for a central difference.
- Its budget law is a theorem for subspace-confined designs at an operating point, which is what its per-point estimators are, and does not cover allocations of calls across many operating points.
- The probe refuses when samples do not exceed the dimension and warns below a stated margin, since the identifiability miss of chapter 2 came from reading it past that point.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* GO-B-Llama [predicted]. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — blind probe on real post-RoPE keys [geometric-observation/claims/LEDGER.md:115](#).
- *measures.* GO-EC-3 [predicted]. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe ... [geometric-observation/claims/LEDGER.md:162](#).
- *refutes or corrects.* NEG-12 [refuted]. (Gate B, prospective, real LLM) On a trained frontier attention layer the blind probe recovers the read operator above the sealed bar *and* projection beats reconstruction. [geometric-observation/claims/LEDGER.md:105](#).

FIRST STATED

readscope, the blind probe as a specified instrument, PyPI readscope, readscope/PRINCIPLES.md and readscope/SPEC.md, and Volume 14 chapter 10.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- NEG-12, [refuted]. (Gate B, prospective, real LLM) On a trained frontier attention layer the blind probe recovers the read operator above the sealed bar *and* projection beats reconstruction. [geometric-observation/claims/LEDGER.md:105](#).
- [readscope/README.md:102-116](#) at c8d0289. What none of this changes: **the budget law, in its proven scope**. The cliff at $k = d$ is a property of consumer calls, not FLOPs — a faster backend buys speed, never admission. The theorem behind it ([PRINCIPLES.md](#), P3; OT-3) covers **subspace-confined directional designs at an operating point**, which is what this probe's per-point estimators are; it does not cover every allocation of calls across many operating points, and the sketch expectation $(1+1/k) \cdot S + \text{tr}(S)/k \cdot I$ shares S 's eigenspaces at every k — so whether many cheap points can average their way back to the population operator was a **sample-complexity question, not a proven impossibility** — and C-15 has now measured it: at equal total consumer calls, sub-dimensional budgets do not catch up, at any graded rank, within $8\times$ the full-dimension spend ([SPEC.md](#), C-15). The cliff is a property of total calls in the measured range; only the far asymptotic regime remains open.

MACHINE CHECKED

[lean/DataMiningAsObservation/ProbeCliff.lean](#), theorems `centralDiff_affine`, `centralDiff_basis`, `exists_blind_direction`, `indistinguishable`, `budget_cliff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 7, 11, 12, 14.

RELATED

[read operator](#) (p. 291); [budget cliff](#) (p. 52); [read distortion](#) (p. 290); [the flip](#) (p. 147).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.9, 11.4.

Ledger rows that cite the entry's records without naming it: OT-3, OT-6, GO-B-Llama-rematch, GO-B-Llama-rematch, GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.3, chapter 2 section 2.4, chapter 2 section 2.6, chapter 3 section 3.4, chapter 4 section 4.2, chapter 6 section 6.1, chapter 8 section 8.3, chapter 8 section 8.4, chapter 11 section 11.7, chapter 11 section 11.9.

block error rate

concept. Entry id `block-error-rate`.

DEFINITION

The fraction of transmitted radio blocks the receiver cannot decode, abbreviated BLER. Chapter 13.

Example. Two of ten transmitted blocks undecodable is a block error rate of 0.2.

EQUATION

none

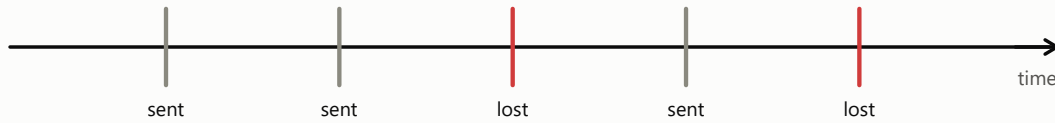


Figure 24: Blocks a receiver cannot decode, as a fraction of those sent in a window.

CONDITIONS

- The fraction of transmitted radio blocks the receiver cannot decode, in the unit interval. A block of n symbols each lost independently with probability q fails with probability one minus one minus q to the n , at most nq , never falling as the block grows, and equal to q for a single symbol.
- The measured rates of the radio track and the refuted refresh law fit against them are the ledger's numbers, with the sealed sweep's calibrated baseline as the correction.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 13 section 13.4 of *Data Mining as Observation*, with the radio freshness track, `observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35`, and the sealed sweep `observation-theory-campaigns/analysis/csi/PREREG-XPROTO-CSI-SWEEP2.md`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- [observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-50](#) at `f6d1fe2`. **The age horizon** (`analysis/csi/`, XPROTO-CSI-SWEEP2 sealed 2026-08-27, graded PASS): at the calibrated 0.10 budget the largest compliant report period is a few TTI at 10 Hz and 1 TTI at 50 Hz and above, so there is no proportionality law to fit. The earlier `CSI-refreshfloor.*` claim of a floor $\approx 0.177 \cdot T_{\text{coh}}$ ($R^2=0.915$) was an **unsealed exploration** measured at a relaxed 0.15 threshold against the 0.10 target, with a fresh baseline that never met the budget. It is refuted; the record is kept, not cited. Separately, the optimal linear predictor cannot beat the one-coherence-time wall (Gaussian fading $\rightarrow$ Wiener optimal), and that wall sits well outside the budget horizon. The **RAN governor** (`analysis/ran`) is the reference O-RAN rApp core: observe $\rightarrow$ measure false-clear $\rightarrow$ refresh at the floor $\rightarrow$ escalate (diversity / re-route).

MACHINE CHECKED

`lean/DataMiningAsObservation/BlockErrorRate.lean`, theorems `bler_mem_unit`, `blockFail_le`, `blockFail_mono`, `blockFail_one`, `blockFail_zero`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

none

RELATED

[coherence time](#) (p. 67); [refresh floor](#) (p. 299); [freshness](#) (p. 156); [false-clear rate](#) (p. 141).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 13.2, 0.25.

Ledger rows that cite the entry's records without naming it: OT-4, OT-11.

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.3, chapter 13 section 13.4.

boosting

instrument. Entry id boosting.

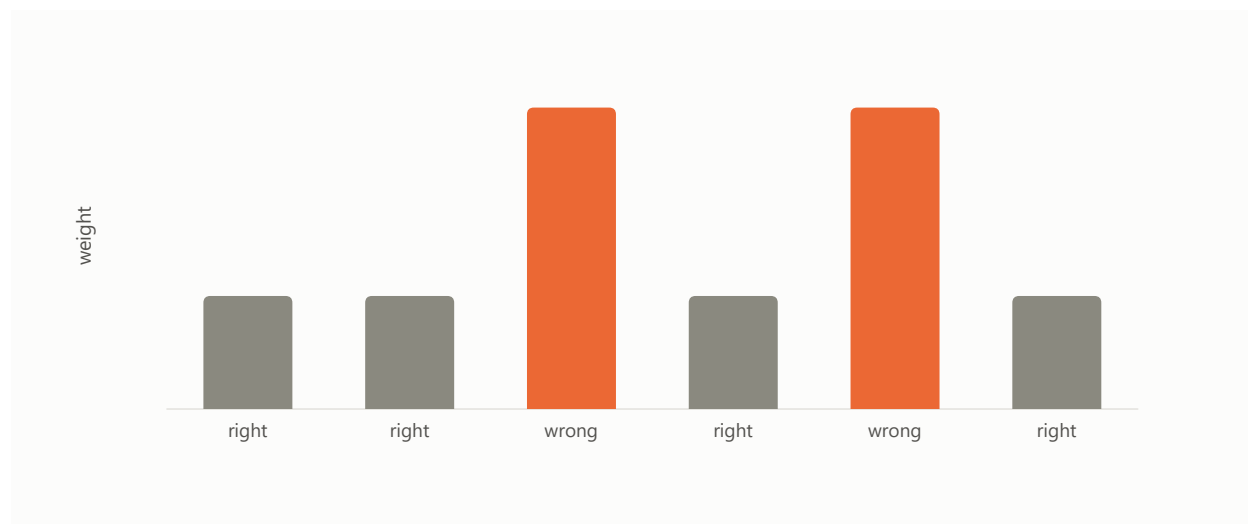


Figure 25: Rows the last scorer got wrong are reweighted so that its error becomes one half.

DEFINITION

Fitting a sequence of weak scorers, each on the rows the previous ones got wrong, and summing them. Its step size is positive exactly when the weak scorer beats chance, and the reweighting makes the last scorer's weighted error one half. Chapter 7.

Example. A weak scorer with weighted error 0.25 gets step size one half of the log of 3, about 0.549, and after reweighting its error is one half.

EQUATION

none

CONDITIONS

- Fitting a sequence of weak scorers, each on the rows the previous ones got wrong, and summing them with a step size set by the weighted error. The step size is positive exactly when the weak scorer beats chance, and after the rows are reweighted the weighted error of the scorer just fitted is one half, so the next scorer must find something new.
- Gradient boosting reads the loss through its curvature, so the optimizer is a consumer whose read operator is the Hessian, and on the author's five-dataset benchmark the ensemble wins when the boundary needs many features and a formula wins when it needs few.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Freund and Schapire, a decision-theoretic generalization of on-line learning, 1997, as chapter 7 section 7.1 of *Data Mining as Observation* reads it, with the curvature reader in `geometric-observation/claims/LEDGER.md` row GO-B-optim-D4.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.1	bagging variance, out-of-bag estimation, random forests, importances, AdaBoost	Hastie, Tibshirani, Friedman, ESL 2e chapters 15 and 10.1; TSK 2e 4.10

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Boosting.lean](#), theorems `alpha_pos_iff`, `alpha_half`, `reweighted_error_half`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 6, 7, 8.

RELATED

[ensemble](#) (p. 129); [curvature](#) (p. 98); [decision tree](#) (p. 102); [importance](#) (p. 181).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 6.1, 0.8, 0.9.

Ledger rows that cite the entry's records without naming it: GO-B-optim-D4 (034 · D4).

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.3, chapter 7 section 7.2, chapter 7 section 7.3.

bootstrap

instrument. Entry id bootstrap.

DEFINITION

An estimate of a confidence interval made by resampling the data with replacement many times and recomputing the statistic each time. A paired bootstrap resamples the same rows for two methods at once so that shared sampling variation cancels. Chapter 0 section 0.9.

Example. Resample 100 rows with replacement 1000 times, and the spread of the 1000 means is the bootstrap standard error.

EQUATION

none

CONDITIONS

- An interval estimated by resampling the data with replacement and recomputing the statistic each time. A paired bootstrap resamples the same rows for two methods at once, and the variance of the difference is the sum of the two variances minus twice their covariance, so pairing narrows the interval exactly when



Figure 26: Resampling the rows with replacement to see how far the estimate moves.

the two scores covary positively across rows.

- Resampling rows does not undo dependence between folds, which is the correction's job, and the bootstrap's interval is only as wide as the sample it resamples.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Efron, bootstrap methods, 1979, as chapter 0 section 0.9 states it, with the program's paired form in the flip comparisons of Volume 14 and the constraint-gap review.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Bootstrap.lean](#), theorems `mean_sub`, `var_sub`, `paired_lt_iff`, `cov_comm`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 6, 7, 11, 12, 14.

RELATED

[confidence interval](#) (p. 76); [standard error](#) (p. 347); [Nadeau and Bengio correction](#) (p. 233); [harness](#) (p. 168).

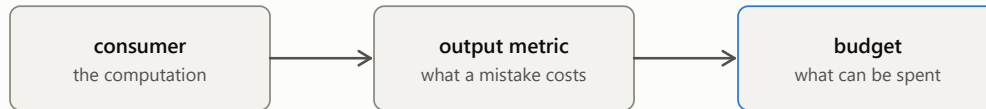
SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.2, 0.18.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.4, chapter 8 section 8.5.

budget

concept. Entry id budget.



the read operator is what the triple induces, and its kernel is the nuisance

Figure 27: The third element of the observer, what the consumer can spend.

DEFINITION

The third element of an observer. The bits, rows, calls, seconds, or dollars a consumer may spend. Chapter 1.

Example. A probe with $2d$ calls at d equal to 16 spends 32 evaluations for one operating point.

EQUATION

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

CONDITIONS

- The budget is the bits, rows, calls, seconds, or dollars a consumer may spend. It bounds what of the read operator can be measured and used, and changes the operator only when it changes the consumer.
- Comparisons are made at matched budget. A verdict at a fixed budget can invert under budget matching, and the ledger carries the case.
- For a probe the budget is consumer calls, and recovery of the read operator is a cliff at the full dimension of the space.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* GO-4 [replicated]. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [geometric-observation/claims/LEDGER.md:66](#).

FIRST STATED

Volume 14, chapter 4 for the triple and chapter 7 for cost, [geometric-observation/chapters/ch07_cost.md](#), DOI 10.5281/zenodo.21776291.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation/claims/LEDGER.md row GO-4

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ProbeCliff.lean](#), theorems `centralDiff_affine`, `centralDiff_basis`, `exists_blind_direction`, `indistinguishable`, `budget_cliff`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13.

RELATED

[observer](#) (p. 244); [water-filling](#) (p. 378); [budget cliff](#) (p. 52); [coverage](#) (p. 92).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 1.1, 0.13.

Ledger rows that cite the entry's records without naming it: OT-3, GO-12.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.2, chapter 11 section 11.7, chapter 13 section 13.6.

budget cliff

result. Entry id `budget-cliff`.

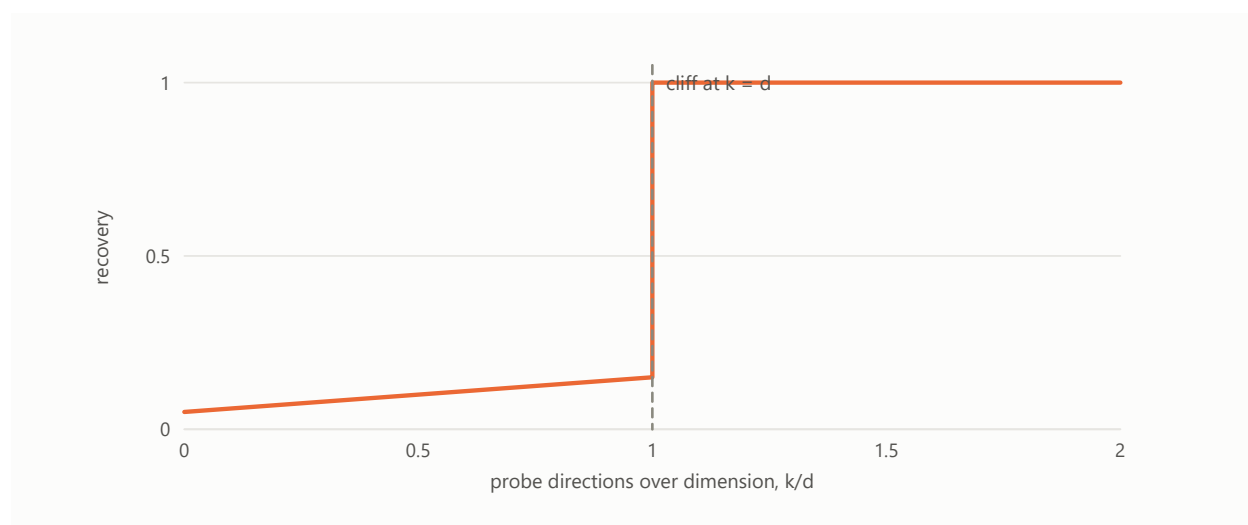


Figure 28: Recovery jumps at k equal to d and does not soften below it.

DEFINITION

The finding that recovery of a read operator by a probe is a cliff at the full dimension of the space rather than a slope, and that the cliff does not move with the operator's rank. Chapter 11.

Example. At half the dimension the probe recovered 0.366 and at the full dimension 1.000, with nothing in between reaching 0.90.

EQUATION

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

CONDITIONS

- The reader receives responses confined to the k directions it chose, the read operator is planted in a fixed subspace, and noise enters as the noisy theorem models it. Under those conditions the cliff at k equal to the dimension is a theorem and is rank-independent.
- Known side information about a k_0 -dimensional exclusion moves the cliff to d minus k_0 and never softens it.
- Whether many cheap points spread across operating points can average their way back to the population operator was measured within a budget range and is not a theorem.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* OT-3 [demonstrated]. Under subspace-confined second-order transcripts, fewer than d directions cannot identify a hidden leading eigenspace (theorem, adaptive to $d-2$ / oblivious to $d-1$); a known k_0 -dim exclusion moves the cliff to exactly $d-k_0$ and never softens ... [geometric-observation/claims/LEDGER.md:31](#).
- *measures.* OT-10 [**[refuted-as-sealed]** warning]. The noisy cliff: noise floors accuracy at a derived level; the cliff's location never moves. [geometric-observation/claims/LEDGER.md:45](#).

FIRST STATED

readscope, predicted then measured, and proved as the confinement theorem in Volume 14, [geometric-observation/crucible/OT3-THEOREM.md](#) and [geometric-observation/crucible/OT3-NOISY-THEOREM.md](#), DOI 10.5281/zenodo.21776291.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.7	confinement theorem, side information moves the cliff to d minus k_0 , noisy cliff proved then measured	readscope/PRINCIPLES.md:112-142 ; geometric-observation/crucible/OT3-THEOREM.md ; geometric-observation/crucible/OT3-NOISY-THEOREM.md

FAILURES AND CORRECTIONS

- [readscope/README.md:102-116](#) at c8d0289. What none of this changes: **the budget law, in its proven scope.** The cliff at $k = d$ is a property of consumer calls, not FLOPs — a faster backend buys speed, never admission. The theorem behind it ([PRINCIPLES.md](#), P3; OT-3) covers **subspace-confined directional designs at an operating point**, which is what this probe's per-point estimators are; it does not cover every allocation of calls across many operating points, and the sketch expectation $(1+1/k) \cdot S + \text{tr}(S)/k \cdot I$ shares S 's eigenspaces at every k — so whether many cheap points can average their way back to the population operator was a **sample-complexity question, not a proven impossibility** — and C-15 has now measured it: at equal total consumer calls, sub-dimensional budgets do not catch up, at any graded rank, within $8\times$ the full-dimension spend ([SPEC.md](#), C-15). The cliff is a property of total calls in the measured range; only the far asymptotic regime remains open.

MACHINE CHECKED

[lean/DataMiningAsObservation/ProbeCliff.lean](#), theorems `centralDiff_affine`, `centralDiff_basis`, `exists_blind_direction`, `indistinguishable`, `budget_cliff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 4, 11.

RELATED

[blind probe](#) (p. 45); [read operator](#) (p. 291).

SEE ALSO

none

C

cache

concept. Entry id cache.

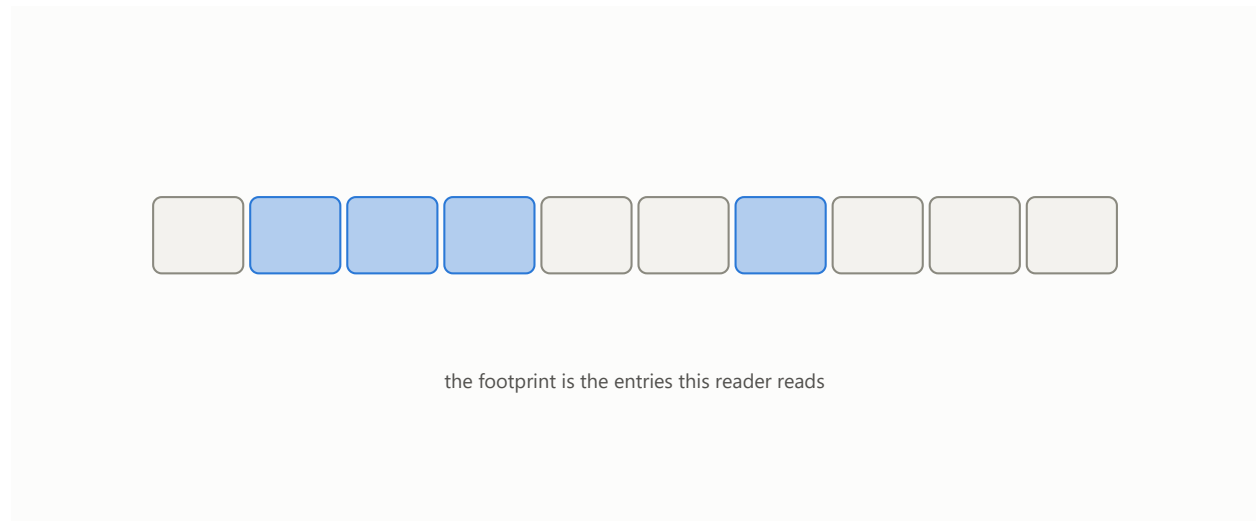


Figure 29: Recently used entries kept so they need not be fetched again.

DEFINITION

A store of recently used data kept so that it need not be fetched again. A cold measurement is taken before the cache holds anything useful, a warm one after. Chapter 13.

Example. A cache holding 4 of a reader's 10 entries hits 40 percent of that reader's reads.

EQUATION

none

CONDITIONS

- A store of recently used data kept so that it need not be fetched again. A cache reader reads the entries in its footprint and nothing else, so its hit rate over a stream of requests lies in the unit interval, is zero cold and one warm, never falls when the cache holds more, and is unchanged by entries outside the requests.
- A cold and a warm measurement of the same system are two observers of it. The ZooKeeper substrate's false-clear rate was 0.99 hot and 0.01 cold, which is the case that a cache's number is the requests' number.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 13 section 13.5 of *Data Mining as Observation*, with the cache reader row of chapter 1's consumer table and the hot-and-cold substrate measurements of the freshness track.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Cache.lean](#), theorems `hitRate_mem_unit`, `hitRate_cold`, `hitRate_warm`, `hitRate_mono`, `hitRate_footprint`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[KV cache](#) (p. 200); [eviction](#) (p. 134); [freshness](#) (p. 156); [replica](#) (p. 307); [consumer](#) (p. 79).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 13.1, 0.26.

Ledger rows that cite the entry's records without naming it: OT-11, GO-12.

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.9.

calibration

concept. Entry id calibration.

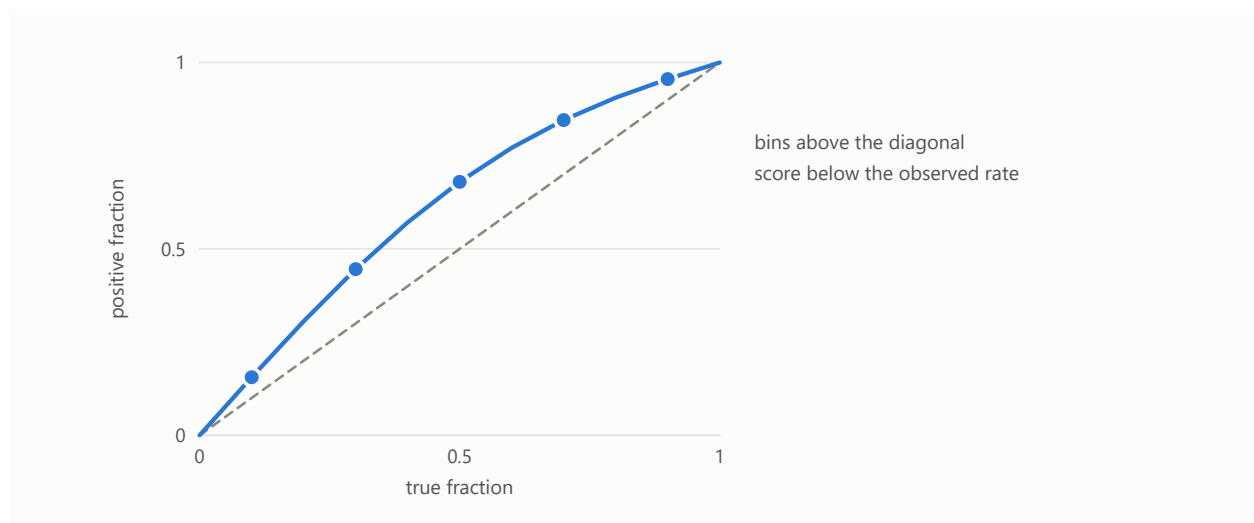


Figure 30: Score bins against the fraction of positives in each.

DEFINITION

The property of a score that a value of 0.8 means the row is positive eighty percent of the time. Measured by expected calibration error, equation 0.29.

Example. If the rows scored near 0.8 are positive 80 percent of the time, the score is calibrated at 0.8.

EQUATION

Book equation 0.29.

$$\text{ECE} = \sum_b \frac{n_b}{n} |\bar{s}_b - \bar{y}_b|.$$

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

CONDITIONS

- A score is calibrated when a value of 0.8 means the row is positive eighty percent of the time. The expected calibration error, the bin-weighted mean absolute gap between mean score and fraction of positives, lies in the unit interval and is zero exactly when every weighted bin is calibrated.
- Calibration is a property of the score values and not of their ranking, so a recalibration can change it while leaving AUROC, the reliability weight, and every threshold-swept metric unchanged. The two are reported side by side and neither stands in for the other.
- Recalibrating a reconstruction toward the input improved the reconstruction and worsened the consumer, the standing negative that keeps calibration off the list of acceptance metrics for a code.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects*. NEG-4 [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [geometric-observation/claims/LEDGER.md:97](#).

FIRST STATED

Chapter 0 section 0.14 of *Data Mining as Observation*, with the program's calibrated authority in `gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63` and the recalibration negative of ledger row NEG-4.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.5	ECE 0.018 to 0.101 vs raw up to 0.223, reliability weight, audit binding	xbse/README.md:175-195 ; gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63
chapter 14 section 14.2	reliability weights and calibration errors per axis, the collapsed family's mean weight 0.559 against the general valence channel's own 0.735, the three design rules, 0.048 to 0.049 and 0.089 to 0.101 at 696 pairs	gtc-prototype/docs/CALIBRATED_AUTHORITY.md:1-65

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [geometric-observation/claims/LEDGER.md:97](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/Calibration.lean](#), theorems `ece_nonneg`, `ece_le_one`, `ece_eq_zero_iff`, `ece_of_calibrated`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[reliability weight](#) (p. 306); [cross-corpus gate](#) (p. 94); [Monotone Invariance Theorem](#) (p. 227); [identity reader](#) (p. 180).

SEE ALSO

none

capacity

concept. Entry id capacity.

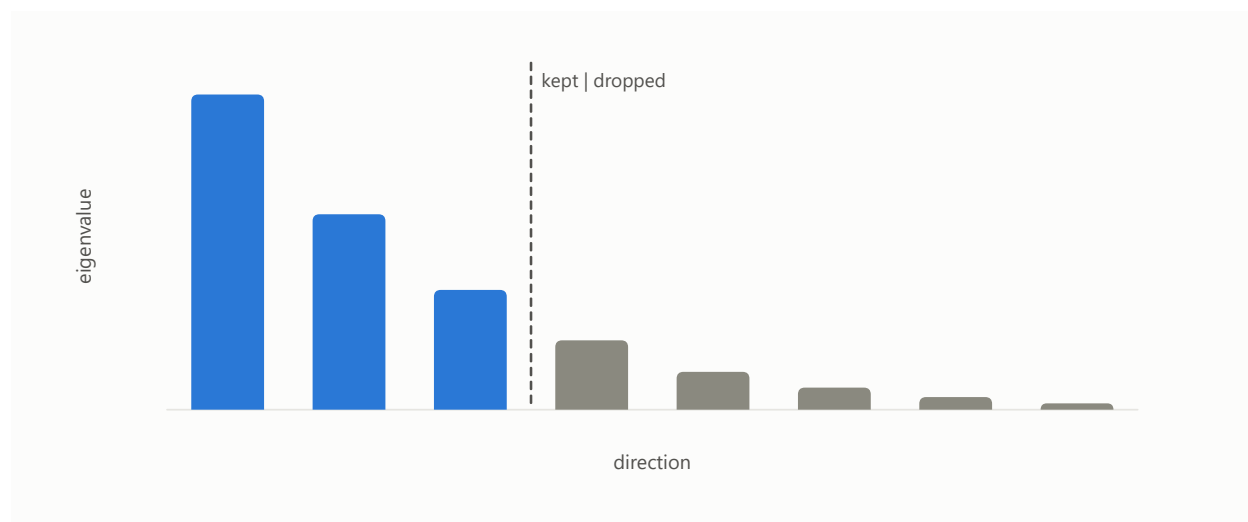


Figure 31: The distinctions a model class can draw, the model-side half of the budget.

DEFINITION

The number of distinctions a model class can draw, the model-side half of the budget. Chapter 6.

Example. A formula of depth three over 18 operations can draw far fewer boundaries than a forest of 500 trees.

EQUATION

none

CONDITIONS

- The number of distinctions a model class can draw, the budget's model-side half. A read operator's rank is at most the dimension, and a probe with fewer directions than the dimension cannot resolve the operator, which is the cliff.
- A formula of depth three over eighteen operations has a capacity the ensemble exceeds, and the ensemble wins exactly when the boundary needs that excess.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the budget inversion in `geometric-observation/claims/LEDGER.md` row GO-4.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ProbeCliff.lean](#), theorems `centralDiff_affine`, `centralDiff_basis`, `exists_blind_direction`, `indistinguishable`, `budget_cliff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Rank.lean](#), theorems `rank_le_width`, `rank_le_height`, `rank_outer_le_one`, `rank_mul_le`, `rank_zero`, `rank_transpose`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's

appendix C.

USED IN

Data Mining as Observation chapters 6, 7.

RELATED

[budget](#) (p. 51); [budget cliff](#) (p. 52); [rank](#) (p. 283); [ensemble](#) (p. 129).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 1.1, 6.1, 11.4.

Ledger rows that cite the entry's records without naming it: GO-4.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.4, chapter 6 section 6.3.

certificate

concept. Entry id certificate.

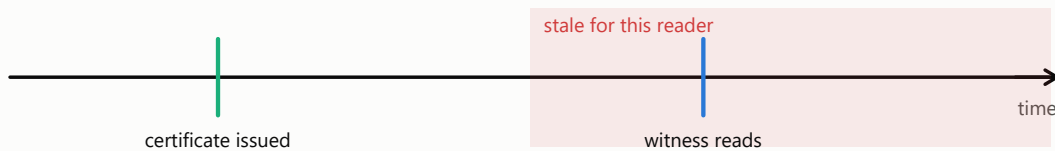


Figure 32: A claim about a read at a time, and a witness that checks it later.

DEFINITION

A claim, made at a time, that something is safe to act on. A clustering, a benchmark score, a freshness guarantee, and a rank bound are all certificates. Chapters 9 and 13.

Example. A certificate cleared 1000 decisions and the witness disagreed on 206 of them, a false-clear rate of 0.206.

EQUATION

Book equation 0.26.

$$\text{FC} = \Pr [W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E} \left[\max_{N-1} \mathcal{N}(0, 1) \right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous.}$$

Book equation 11.2.

$$r = \frac{d_{\text{compressed}}}{d_{\text{exact}}}, \quad \kappa_{\text{strict}} = \frac{\max r}{\min r}, \quad \tau \geq 1 - 2\hat{\mu}(\kappa_{\text{strict}}), \quad \rho_S \geq 1 - 3\hat{\mu}(\kappa_{\text{strict}}),$$

$$\kappa_{97.5/2.5} = \frac{q_{97.5}(r)}{q_{2.5}(r)} \text{ gives the same two expressions as estimates, not floors.}$$

Book equation 13.1.

$$\text{FC} = \Pr[W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr}(P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

CONDITIONS

- A certificate is graded only where a witness exists, and where none exists it is unassessed rather than safe.
- A certificate is bound to what it certified, the bytes, the anchor sample, the footprint, and the time, and carries none of it to another corpus, workload, or moment.
- A vacuous certificate proves nothing about the object. It does not prove that no code preserves rank or that no structure exists, only that this test could not certify one.
- The rank certificate's floor is a guarantee in its strict setting and an estimate in its percentile setting.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-3 [demonstrated]. The certificate's vacuity threshold predicts where single-stage retrieval dies. [geometric-observation/claims/LEDGER.md:65](#).

FIRST STATED

Volume 14, chapter 19 for the certificate that ages and chapter 9 of *Data Mining as Observation* for the clustering certificate. The rank certificate is specified in `turboquant-pro/docs/CERTIFICATE_SPEC.md`, DOI 10.5281/zenodo.20660087.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.3	the margin certificate, mu crit as the expected maximum of N minus 1 standard normals, rho, death at 0.948 within 6 percent, Spearman 0.991 vs 0.873, fourteen corpora, six gates, the v1 to v3 path, the standing correction	geometric-observation/experiments/GO3-certificate-vacuity-v3-NOTES.md:1-60 ; geometric-observation/claims/LEDGER.md row GO-3
chapter 13 section 13.2	the grammar, certificate, witness, refresh floor, false-clear rate, vacuity, the witness table	geometric-observation/chapters/ch19_the_certificate_that_ages.md:1-95 ; observation-theory-campaigns/experiments/FRESHNESS-PROGRAM.md:1-40

FAILURES AND CORRECTIONS

- [observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:33-39](#) at f6d1fe2. warning **XPROTO-AICSI scope correction (2026-08-25)**. The v1 seal stands for what it tested, but the reconstruction-vs-consumer dissociation does **not** survive the community-standard substrate. On real 3GPP CDL-C with a CsiNet-class codec the NMSE-optimal codec reconstructs near-perfectly (NMSE $\approx$ 0.03) and false-clears 0.0 on all seeds; the pre-registered kill fired. See `analysis/aicsi/PREREG-XPROTO-AICSI-v2.md` (REFUTED AT SHAKEDOWN, NOT SEALED, kept negative). This is a scope correction, not a retraction. Do not headline the AICSI row; WCNC §III-C is dropped.

MACHINE CHECKED

[lean/DataMiningAsObservation/Certificate.lean](#), theorems `falseClear_mul_coverage`, `coverage_empty`, `falseClear_mem_unit`, `minOverStrata_passes_iff`, `minOverStrata_le_weighted_mean`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 9, 11, 12, 13, 14.

RELATED

[witness](#) (p. 382); [false-clear rate](#) (p. 141); [coverage](#) (p. 92); [rank certificate](#) (p. 285); [vacuity threshold](#) (p. 371); [coherence time](#) (p. 67).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 11 section 11.3, chapter 12 section 12.6.

challenge set

instrument. Entry id `challenge-set`.

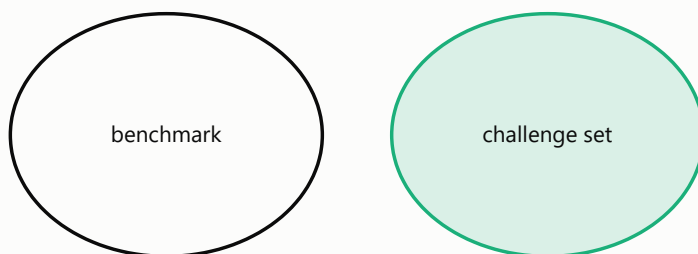


Figure 33: A collection on which a benchmark-passing shortcut fails, outside the benchmark.

DEFINITION

A test collection built so that a shortcut which passes the benchmark fails on it. Chapter 12.

Example. The identity-attack set scored 0.237, with an interval of 0.20 to 0.28 on 289 items, while the benchmark scored 0.909.

EQUATION

none

CONDITIONS

- A test collection built so that a shortcut which passes the benchmark fails on it. Every finite benchmark admits a shortcut, the lookup that returns the stored label for every benchmark row, and two scorers that agree on every benchmark row have the same benchmark error whatever they do elsewhere.
- A challenge set is any collection on which the shortcut and the intended rule disagree, which can only lie outside the benchmark, and on it the shortcut's error is positive while its benchmark error is zero.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 12 section 12.6 and chapter 14 section 14.3 of *Data Mining as Observation*, with the program's identity-attack and euphemism sets in `gtc-prototype/docs/SPECTRUM_FINDINGS.md`: 66-99.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ChallengeSet.lean](#), theorems `shortcut_passes`, `shortcut_fails`, `benchmark_blind`, `challenge_outside`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 9, 11, 12.

RELATED

[leakage](#) (p. 210); [harness](#) (p. 168); [deployment mismatch](#) (p. 108); [preregistration](#) (p. 267).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 14.2, 12.3.

Ledger rows that cite the entry's records without naming it: NEG-15 (Bell boundary).

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.4, chapter 7 section 7.3, chapter 7 section 7.4, chapter 8 section 8.5.

chance level

concept. Entry id `chance-level`.

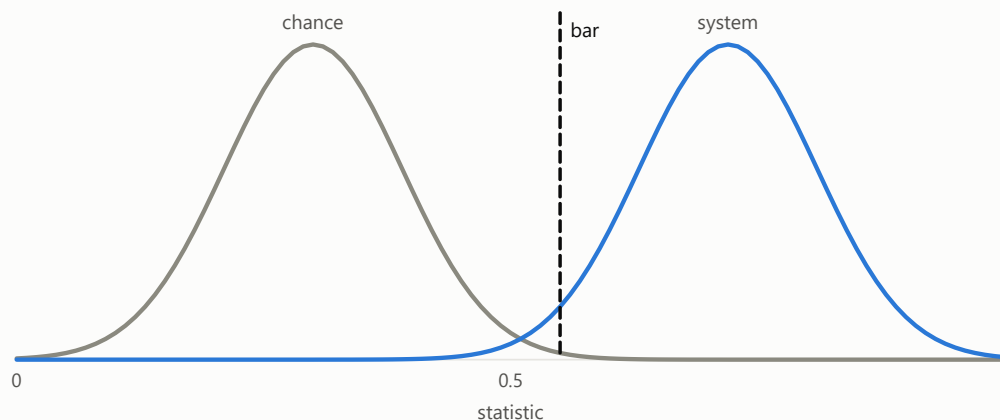


Figure 34: The score a null scorer reaches, which a bar must clear.

DEFINITION

The value a rank statistic reaches on average under a random ranking. Zero for a correlation, one half for AUROC. Chapter 0 section 0.8.

Example. A constant score has AUROC 0.5, and recall at 10 from 100000 rows at random is 0.0001.

EQUATION

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

CONDITIONS

- The value a rank statistic reaches under a ranking that carries no information. A constant score ties every pair, so its AUROC is one half when both classes are present, its Kendall correlation with any score is zero, and its reliability weight is zero.
- Every reading the probe reports comes with its chance overlap beside it, and a gate subtracts the chance level before it counts a margin.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.8 of *Data Mining as Observation*, with the program's chance-overlap reporting in `readscope/readscope/metrics.py:31-99`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ChanceLevel.lean](#), theorems `auROCNum_const`, `auROC_const`, `weight_const`, `tau_const`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 6, 11.

RELATED

[reliability weight](#) (p. 306); [cross-corpus gate](#) (p. 94); [Kendall correlation](#) (p. 194); [harness](#) (p. 168).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.16, 0.15.

chunk

concept. Entry id chunk.

DEFINITION

A piece of a document, the unit a retrieval pipeline encodes and indexes. Chunking is a discrete stage with no Jacobian. Chapter 12.

Example. A 3000-word document cut into 500-word pieces yields six chunks, each encoded on its own.

EQUATION

none

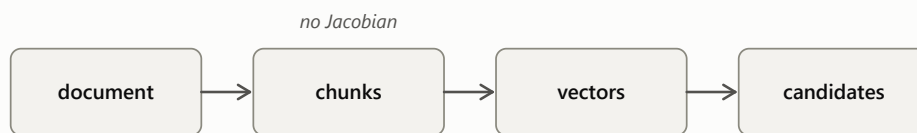


Figure 35: A document goes to pieces at a stage that has no Jacobian.

CONDITIONS

- A piece of a document, the unit a retrieval pipeline encodes and indexes. Chunking is a discrete stage with no Jacobian, and a stage that identifies two inputs identifies them for every stage after it, so what a chunker merges no later stage can separate.
- A chunk's vector is read by the index against a query vector, and the probe depth, cell, and cross-article share of the candidates are properties of the chunks and the query set together.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 12 section 12.2 of *Data Mining as Observation*, with the pullback composition in `geometric-observation/chapters/ch06_mathematical_preliminaries.md:10-27`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Pipeline.lean](#), theorems `quotient_inherited`, `quotient_inherited_chain`, `rank_comp_le_first`, `rank_comp_le_second`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 12.

RELATED

[retrieval-augmented pipeline](#) (p. 312); [Jacobian](#) (p. 192); [encoder](#) (p. 127); [inverted file](#) (p. 185).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 12.3, 0.36.

Ledger rows that cite the entry's records without naming it: `GO-B-legal (035→036)`.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 11 section 11.4, chapter 12 section 12.2.

classifier

concept. Entry id classifier.

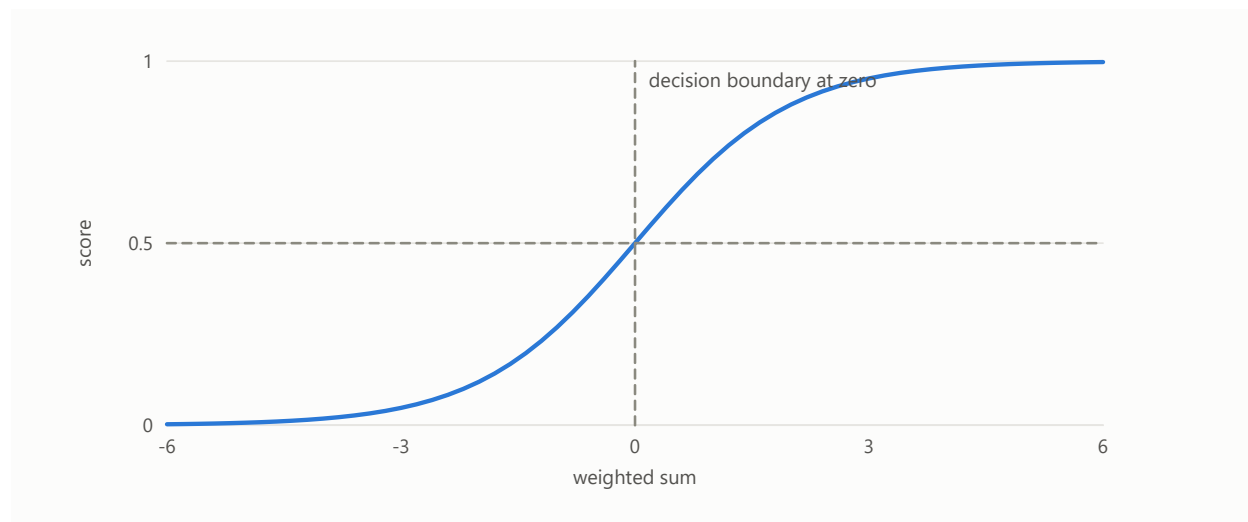


Figure 36: A score per row and a threshold that turns it into a decision.

DEFINITION

A consumer that produces a score per row and turns it into a decision with a threshold. Chapter 6.

Example. A score of 0.7 against a threshold of 0.5 is a positive decision, and raising the threshold to 0.8 flips it.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

CONDITIONS

- A consumer that produces a score per row and turns it into a decision with a threshold. For a linear classifier the sensitivity is the link's slope times the weight direction, so the read operator is the workload mean of the squared slope times the outer product of the weights, sends every vector to a multiple of the weights, and reads nothing orthogonal to them.
- The score is affine, so chapter 0's exact quotient applies before the link. The whale clan classifier is the flip's classifier case, held-out AUROC 0.934 against 0.883 for a code that reconstructs twice as well.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* GO-B-whale (038) [predicted]. Sperm-whale coda dialect (DSWP/Sharma 2024), Clan classifier — cetacean communication; promotes the exploratory (A2) verdict to a sealed flip [geometric-observation/claims/LEDGER.md:120](#).

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the classifier row of Volume 14's consumer table and the whale clan classifier of ledger row GO-B-whale.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Classifier.lean](#), theorems `readOp_classifier`, `readOp_classifier_mulVec`, `readOp_classifier_orth`, `score_orth`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 5, 6, 7, 8, 10, 11, 12, 14.

RELATED

[consumer](#) (p. 79); [read operator](#) (p. 291); [threshold](#) (p. 362); [decision boundary](#) (p. 101); [margin](#) (p. 221).

SEE ALSO

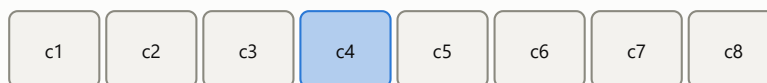
Book equations stated beside the entry's terms, not defining it: 1.1.

Ledger rows that cite the entry's records without naming it: GO-1.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.3, chapter 6 section 6.1.

codebook

concept. Entry id codebook.



each number rounds to its nearest code

Figure 37: The small set of values a quantizer replaces each number with.

DEFINITION

The small set of allowed values a quantizer replaces each number with. Chapter 0 section 0.11.

Example. A codebook of four values, 0, 1, 2, and 3, replaces 1.4 with 1 and 2.6 with 3.

EQUATION

none

CONDITIONS

- The small set of allowed values a quantizer replaces each input with, under the nearest-codeword rule. The distortion of an input is its distance to the nearest codeword, nonnegative, zero when the input is a codeword, achieved by some codeword, and never larger for a larger codebook.
- Two arms compared at unequal codebooks are not a flip comparison, which is the twenty-five percent codebook confound the ledger carries, and the reset content of a stored description is separate from its rate, which is row GO-7's finding across six codebooks.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.11 and chapter 4 section 4.2 of *Data Mining as Observation*, with the six codebooks of ledger row GO-7 and the codebook confound of the honest negatives.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.5	GO-7, 0.67 bits per symbol, bin rate 0.26 equals 0.39, error 1.00 without side information, two families, six codebooks	geometric-observation/claims/LEDGER.md row GO-7

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Codebook.lean](#), theorems `distortion_nonneg`, `distortion_codeword`, `distortion_anti`, `exists_nearest`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 4, 8, 11, 13.

RELATED

[quantization](#) (p. 275); [product quantization](#) (p. 271); [confound](#) (p. 77); [the flip](#) (p. 147).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 4.2, 0.13.

Ledger rows that cite the entry's records without naming it: GO-7, NEG-4.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.2, chapter 4 section 4.4, chapter 6 section 6.2, chapter 7 section 7.2, chapter 9 section 9.3, chapter 11 section 11.7, chapter 12 section 12.2, chapter 12 section 12.3, chapter 13 section 13.5, chapter 13 section 13.6.

coherence time

concept. Entry id coherence-time.

DEFINITION

The interval over which a changing quantity stays correlated with itself. A value measured longer ago than the coherence time is a guess. Equation 0.25.

Example. A channel whose fading repeats every 20 milliseconds has a coherence time of 20 milliseconds.

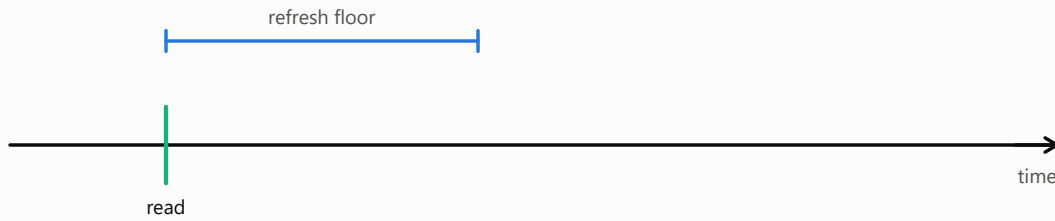


Figure 38: How long a quantity stays what it was.

EQUATION

Book equation 0.25.

$$T_{\text{coh}} = \frac{0.423}{f_D}.$$

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz, } 1 \text{ TTI at } \geq 50 \text{ Hz}.$

CONDITIONS

- Clarke's formula gives the coherence time from the Doppler frequency alone and assumes the fading model behind it.
- A value measured longer ago than the coherence time is a guess, and a certificate issued that long ago has decorrelated from the state it certified.
- The proportional refresh-floor law that was fit to it was refuted and is not cited. Its sealed replacement calibrated the baseline first and found floors at a few transmission intervals.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Clarke's model of mobile-radio reception for the formula, and Volume 14 chapter 19 for its role as the interval past which a certificate has decorrelated.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- [observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-50](#) at f6d1fe2. **The age horizon** (`analysis/csi/`, XPROTO-CSI-SWEEP2 sealed 2026-08-27, graded PASS): at the calibrated 0.10 budget the

largest compliant report period is a few TTI at 10 Hz and 1 TTI at 50 Hz and above, so there is no proportionality law to fit. The earlier `CSI-refreshfloor.*` claim of a floor $\approx 0.177 \cdot T_{\text{coh}}$ ($R^2=0.915$) was an **unsealed exploration** measured at a relaxed 0.15 threshold against the 0.10 target, with a fresh baseline that never met the budget. It is refuted; the record is kept, not cited. Separately, the optimal linear predictor cannot beat the one-coherence-time wall (Gaussian fading $\rightarrow$ Wiener optimal), and that wall sits well outside the budget horizon. The **RAN governor** (`analysis/ran`) is the reference O-RAN rApp core: observe $\rightarrow$ measure false-clear $\rightarrow$ refresh at the floor $\rightarrow$ escalate (diversity / re-route).

MACHINE CHECKED

[lean/DataMiningAsObservation/CoherenceTime.lean](#), theorems `clarke_to_three_decimals`, `examples`, `refuted_law_exceeds_sealed`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 13.

RELATED

[certificate](#) (p. 59); [false-clear rate](#) (p. 141); [refresh floor](#) (p. 299); [witness](#) (p. 382).

SEE ALSO

Ledger rows that cite the entry's records without naming it: OT-4.

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.3, chapter 13 section 13.4.

cold, warm

concept. Entry id cold-warm.

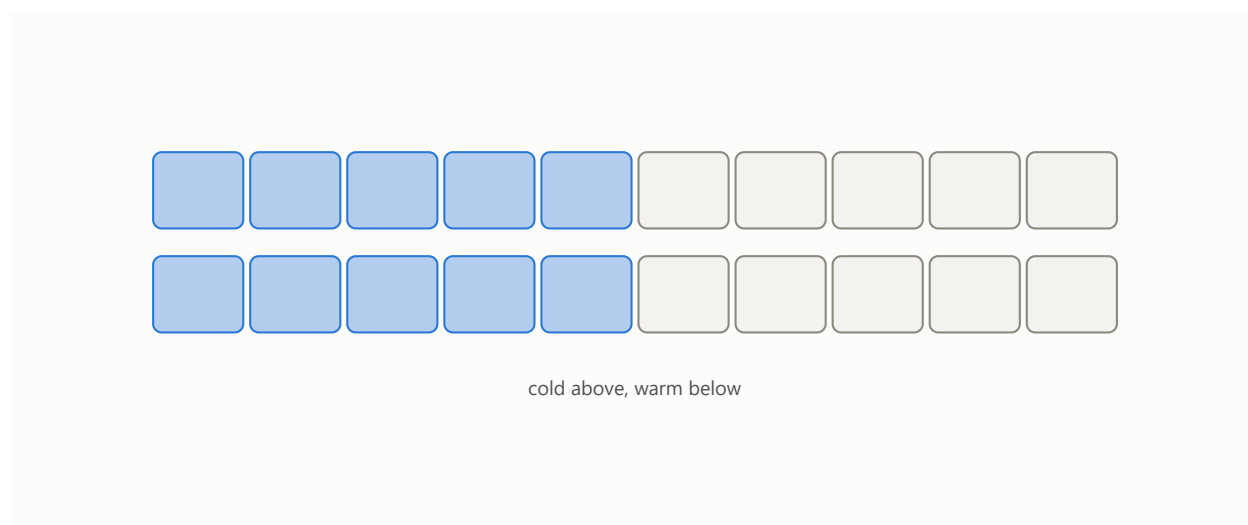


Figure 39: Before the cache holds anything useful, and after.

DEFINITION

A cold measurement is taken before the cache holds anything useful, a warm one after. A cost figure without one of the two words beside it is not a cost figure. Chapter 0 section 0.13 and chapter 13.

Example. The first search at eight probes took 22.30 seconds cold and 0.835 seconds warm.

EQUATION

none

CONDITIONS

- A cold measurement is taken before the cache holds anything useful, a warm one after. The hit rate is zero on an empty cache, one when the cache holds the whole footprint, grows as the cache grows, and does not move when entries outside the footprint are added.
- A single cold measurement at eight probes read 22.30 seconds against 0.835 seconds warm, and every previously reported placement number was warm, so a cost figure without the word warm or cold beside it is not a cost figure.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.13 of *Data Mining as Observation*, with the placement study's cold measurement in `turboquant-pro/docs/RESEARCH_ROADMAP.md:133-160`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.1	22.30 s cold vs 0.835 s warm at eight probes, every prior number warm, fragmentation 63.7 bracketing 42 to 47, 64 shards, routing sparsity computed and discarded	turboquant-pro/docs/RESEARCH_ROADMAP.md:133-160

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Cache.lean](#), `theorems hitRate_mem_unit, hitRate_cold, hitRate_warm, hitRate_mono, hitRate_footprint`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[cache](#) (p. 55); [eviction](#) (p. 134); [replica](#) (p. 307); [budget](#) (p. 51); [certificate](#) (p. 59).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.26, 13.2.

Ledger rows that cite the entry's records without naming it: OT-11, GO-12.

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.4, chapter 11 section 11.4.

commit hash

instrument. Entry id `commit-hash`.

DEFINITION

The fingerprint git assigns to a snapshot of a repository and its history, which identifies the content exactly and does not by itself prove when it existed. Chronology comes from a public push, a signed tag, or an archival

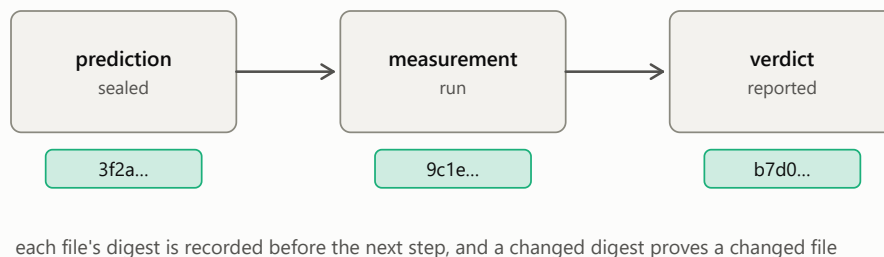


Figure 40: A fingerprint of a snapshot and its history, proving content and not time.

deposit. Chapter 0 section 0.12.

Example. Commit 3172e00 identifies one snapshot of the book's repository and every file in it, and says nothing about the clock.

EQUATION

none

CONDITIONS

- The fingerprint git assigns to a snapshot of a repository together with its history. It identifies exact content, so a changed hash proves a changed file and an equal hash is strong evidence of an unchanged one, but it does not by itself prove when that content existed, since git's author and committer dates are supplied fields.
- Chronology comes from a public repository receipt, a signed tag, a transparency log, or an archival deposit such as a Zenodo DOI, and a sealed prediction names one of those beside its hash.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.12 of *Data Mining as Observation*, with the program's seal ledger, `observation-theory-campaigns/experiments/SEALS.md:1-10`, where every seal is a registration id, a sealing commit, and a hash.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Seal.lean](#), theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[hash](#) (p. 169); [sealed](#) (p. 324); [sources table](#) (p. 338); [preregistration](#) (p. 267).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.3.

Ledger rows that cite the entry's records without naming it: OT-7.

commute time

concept. Entry id commute-time.

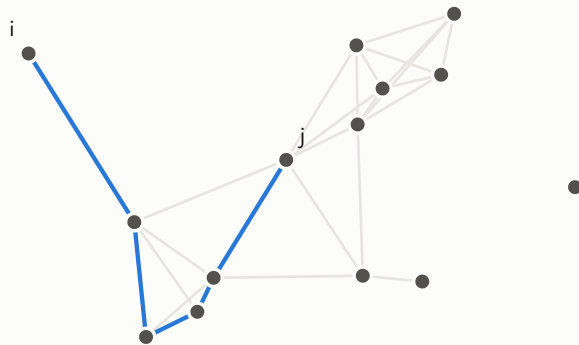


Figure 41: Steps there and back for a random walk, whose scaling by the volume is the resistance.

DEFINITION

The expected number of steps a random walk on a graph needs to go from one node to another and back. Divided by the graph's volume it is the resistance, which a particular scaling of the spectral embedding makes equal to the squared Euclidean distance. Equation 0.20.

Example. On a path of three nodes the commute time between the ends is 8 steps, and dividing by the volume, 4, gives the resistance, 2.

EQUATION

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

CONDITIONS

- The expected number of steps a random walk needs to go from one node to another and back, the hitting time there plus the hitting time back, so it is symmetric and zero from a node to itself. The resistance is the commute time divided by the volume, and the squared Euclidean distance in the embedding of equation 0.20 equals the resistance, not the commute time itself.
- On a large neighbourhood graph of a shape of dimension three or more the resistance converges to one over each degree, uniformly over pairs, so it depends on two nodes only through their degrees and carries no geometry, which is why the magnitude of the spectral embedding is degree noise. The program's first claim of a commute-metric form was self-refuted, NEG-1.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.10 of *Data Mining as Observation*, equation 0.20, with von Luxburg, Radl, and Hein, 2014, read in chapter 3 section 3.3 and the program's collapse proof in `the-angular-observer/theorem.md:110-146`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	resistance converges to $1/\sqrt{d_i} + 1/\sqrt{d_j}$, simplex argument, radius converges to $1/\sqrt{d_i}$	<code>the-angular-observer/theorem.md:110-146</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

`lean/DataMiningAsObservation/CommuteTime.lean`, theorems `commute_symm`, `commute_self`, `resistance_symm`, `commute_eq_resistance_mul`, `collapsed_congr`, `collapsed_const`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3.

RELATED

[spectral embedding](#) (p. 342); [Laplacian](#) (p. 207); [geodesic distance](#) (p. 162); [degree](#) (p. 105); [graph](#) (p. 166).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.19, 3.3.

Ledger rows that cite the entry's records without naming it: NEG-1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.3, chapter 9 section 9.1.

concentrated

concept. Entry id concentrated.

DEFINITION

Of a spectrum, having a few eigenvalues that carry most of the total. Chapter 0 section 0.4.

Example. A spectrum of 1.0, 0.12, 0.06, and 0.02 puts 83 percent of its total in one direction.

EQUATION

none

CONDITIONS

- Of a spectrum, having a few eigenvalues that carry most of the total. If one eigenvalue carries a fraction f of the total, the effective rank is at most one over f squared, so half the mass in one direction gives effective rank at most four.
- The allocation report warns below effective rank two, which is the same statement read from the rank side, and an allocation over a concentrated spectrum spends almost everything on one direction.

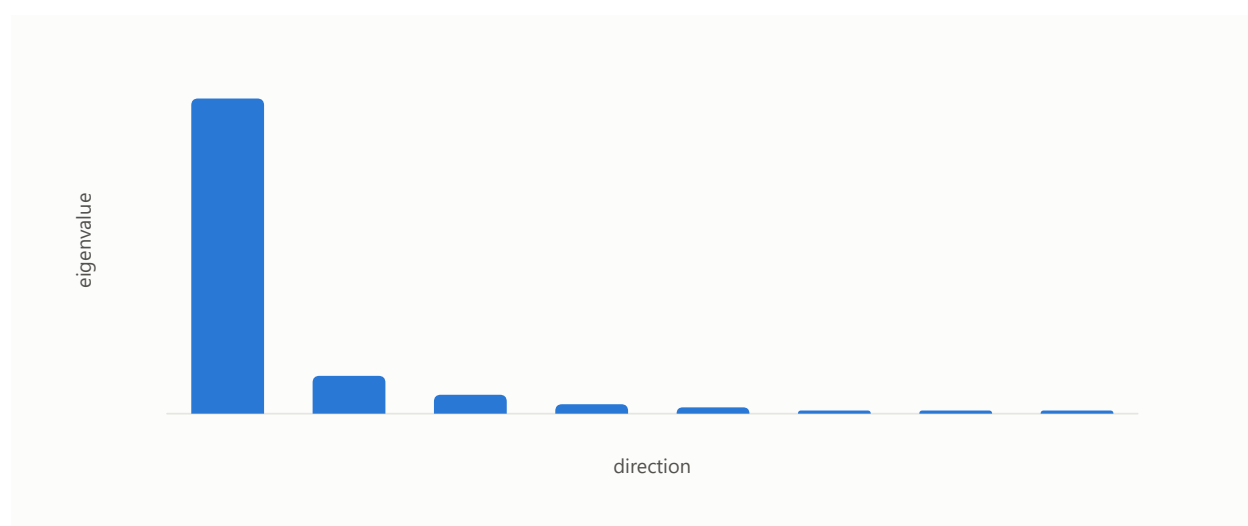


Figure 42: A few eigenvalues carry most of the total.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.4 of *Data Mining as Observation*, with the allocation report's concentration caution in `turboquant-pro/turboquant_pro/read_allocation.py:244-307`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	allocation report, gain over uniform, concentration caution below effective rank 2	turboquant-pro/turboquant_pro/read_allocation.py:244-307

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Concentrated.lean](#), theorems `sq_le_sum_sq`, `effRank_le_of_fraction`, `effRank_le_four`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 4, 6, 10, 11, 13.

RELATED

[spectrum](#) (p. 344); [effective rank](#) (p. 123); [water-filling](#) (p. 378); [isotropic](#) (p. 187).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.7, 0.5.

Ledger rows that cite the entry's records without naming it: OT-7.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.2.

confidence

concept. Entry id confidence.

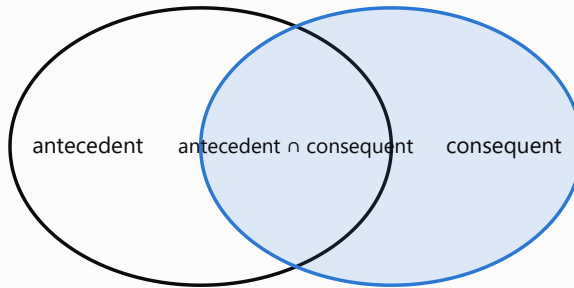


Figure 43: The fraction of transactions with the antecedent that also hold the consequent.

DEFINITION

Of a rule, the support of the rule's itemset divided by the support of its antecedent. It is not symmetric and it misleads when the consequent is common. Chapter 5.

Example. If 30 baskets hold bread and 24 of them also hold butter, the confidence of bread implies butter is 0.8.

EQUATION

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

Book equation 5.2.

$$\text{lift}(X \rightarrow Y) = \frac{c(X \rightarrow Y)}{s(Y)} = \frac{s(X \cup Y)}{s(X) s(Y)}.$$

CONDITIONS

- The support of a rule's itemset over the support of its antecedent, in the unit interval and not symmetric. Under independence the confidence equals the consequent's support whatever the antecedent, so a consequent in nine of ten transactions gives every rule into it confidence 0.9 with no association at all.
- Lift is confidence over the consequent's support and is one in that case, which is why chapter 5 sends rule evaluation to chapter 8's nulls.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

TSK chapter 5, as chapter 5 section 5.1 of *Data Mining as Observation* states it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Confidence.lean](#), theorems `confidence_mem_unit`, `confidence_asymm`, `confidence_indep`, `common_consequent`, `lift_of_indep`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 5, 6, 8, 9, 11, 13, 14.

RELATED

[lift](#) (p. 214); [support](#) (p. 353); [itemset](#) (p. 188); [multiple comparisons](#) (p. 229).

SEE ALSO

none

confidence interval

concept. Entry id confidence-interval.

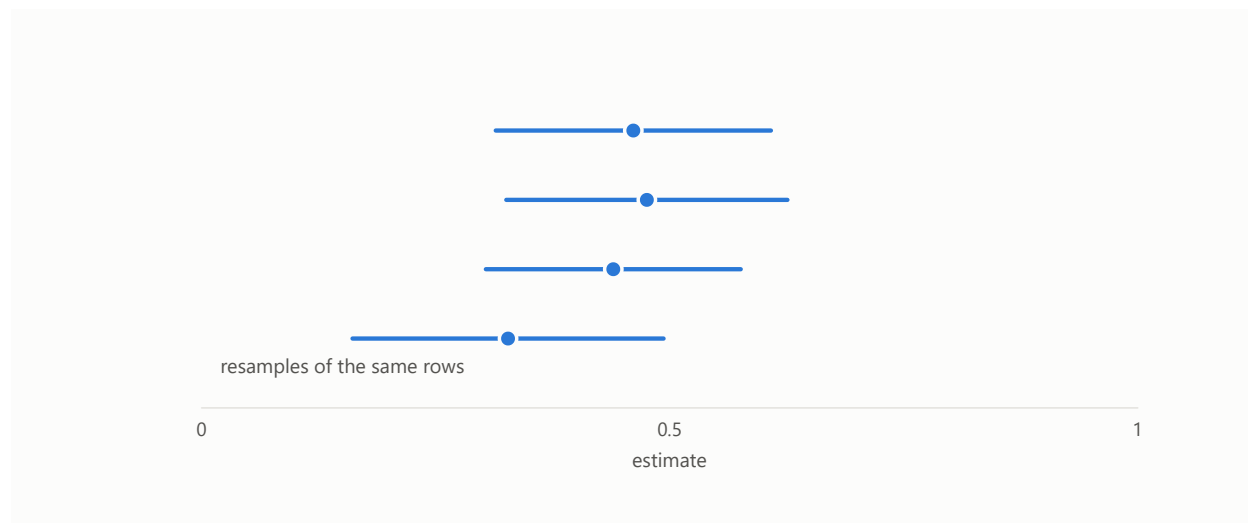


Figure 44: A range the estimate falls in with a stated frequency over repetitions.

DEFINITION

A range that would contain the true value in a stated fraction of repeated samples, usually 95 percent. Chapter 0 section 0.9.

Example. A mean of 0.62 with standard error 0.04 has a 95 percent interval of about 0.54 to 0.70.

EQUATION

none

CONDITIONS

- A range that would contain the true value in a stated fraction of repeated samples. An interval with coverage one minus alpha misses in a fraction alpha, over m independent intervals the expected misses are m alpha and the chance that all cover is one minus alpha to the m , and the chance that at least one misses is at most m alpha.
- Nine intervals at ninety-five percent have a chance near seven percent of two or more misses, which is how the re-gate table's seven of nine inside the interval is read.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.9 of *Data Mining as Observation*, with the program's intervals in the re-gate table of chapter 14 and the hubness intervals of chapter 11.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ConfidenceInterval.lean](#), theorems `missRate_eq`, `some_miss_le`, `nine_intervals`, `nine_expected`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 6, 8, 14.

RELATED

[bootstrap](#) (p. 49); [standard error](#) (p. 347); [p-value](#) (p. 273); [preregistration](#) (p. 267).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.3, 8.2.

confound

concept. Entry id `confound`.

DEFINITION

A variable that moves with both the treatment and the outcome so that a measured difference cannot be attributed. Chapter 8 requires controls before a claim.

Example. A codebook change that arrived with a compression change moved 25 percent of the score on its own, so the comparison was confounded.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

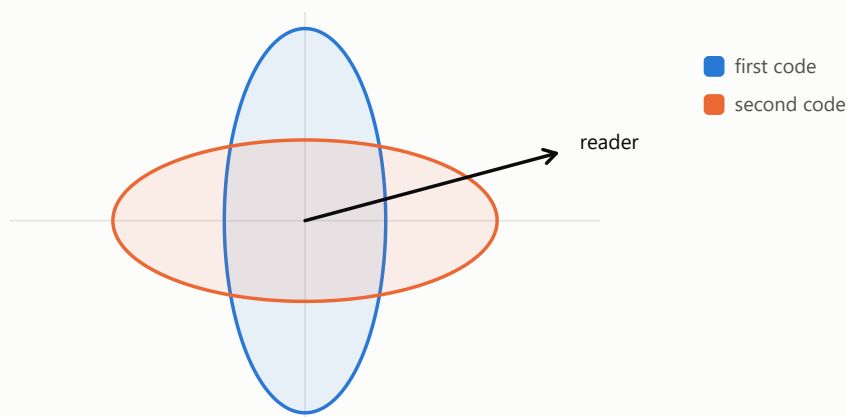


Figure 45: A variable that moves both arms of a comparison.

CONDITIONS

- A variable that moves with both the treatment and the outcome. In the linear case the naive difference of group means is the treatment effect plus the confound's effect times the difference of the confound's means between the groups, so it equals the effect exactly when the confound is balanced or has no effect, and the bias can have either sign and any size.
- Controls come before claims. The twenty-five percent codebook confound in the flip's early arms and the smooth-perturbation control that was withdrawn are the program's cases, each carried in the ledger.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects*. NEG-6 [refuted]. Relative per-channel-demeaned error norm is the quotient-tangential quantity that controls softmax-KL. [geometric-observation/claims/LEDGER.md:99](#).

FIRST STATED

Chapter 8 section 8.6 of *Data Mining as Observation*, with the program's codebook confound in Volume 14's honest negatives and the negative controls of the constraint-gap review.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- NEG-6, [refuted]. Relative per-channel-demeaned error norm is the quotient-tangential quantity that controls softmax-KL. [geometric-observation/claims/LEDGER.md:99](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/Confound.lean](#), theorems `mean_outcome`, `naive_diff`, `naive_diff_of_balanced`, `naive_diff_of_no_effect`, `bias_unbounded`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2, 3, 4, 6, 7, 8, 9, 12, 13, 14.

RELATED

[harness](#) (p. 168); [null model](#) (p. 242); [Simpson's paradox](#) (p. 333); [preregistration](#) (p. 267).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.2.

Ledger rows that cite the entry's records without naming it: NEG-4, NEG-5.

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.4, chapter 8 section 8.5.

consumer

concept. Entry id consumer.



the read operator is what the triple induces, and its kernel is the nuisance

Figure 46: The computation that reads the vector, the first element of the observer.

DEFINITION

The function that acts on a vector, the first element of an observer. A softmax, a classifier, an index, a person reading a report. Chapter 1.

Example. A logistic classifier reading a 30-column table is a consumer; so is a cosine ranker, and so is a person reading a summary.

EQUATION

Book equation 1.1.

$$O = (C, G, B).$$

CONDITIONS

- A consumer is any computation that takes a vector in and gives an output out, a softmax, a classifier, an index, a person reading a report. The book's quantities are functions of the consumer, and two consumers with the same read operator are not thereby the same consumer.
- A selection consumer, a tree or a threshold, has zero sensitivity almost everywhere and is read by counting where it selects rather than by finite differences.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* GO-1 [predicted]. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [geometric-observation/claims/LEDGER.md:62](#).

- *measures*. GO-EC-3 [predicted]. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe ...[geometric-observation/claims/LEDGER.md:162](#).

FIRST STATED

Volume 14, chapter 4, [geometric-observation/chapters/ch04_the_observer_triple.md:9-60](#) and its consumer table at lines 60-135, DOI 10.5281/zenodo.21776291.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	readscope/readscope/regimes.py:1-60
chapter 6 section 6.1	the classifier row of the consumer table, the output metric makes a different observer	geometric-observation/chapters/ch04_the_observer_triple.md:60-135
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	readscope/readscope/regimes.py:1-60

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at [observation-data-mining 95af30c](#); what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[observer](#) (p. 244); [read operator](#) (p. 291); [sensitivity](#) (p. 328); [budget](#) (p. 51).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.8.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.2, chapter 6 section 6.2, chapter 11 section 11.7.

contraction

instrument. Entry id contraction.

DEFINITION

A formula that reduces several axes to one verdict. The instrument's contraction from ten axes is fairness minus the general valence component. Chapter 14 section 14.5.

Example. Ten axis scores contracted to fairness minus the general component is one number a person can read.

EQUATION

none

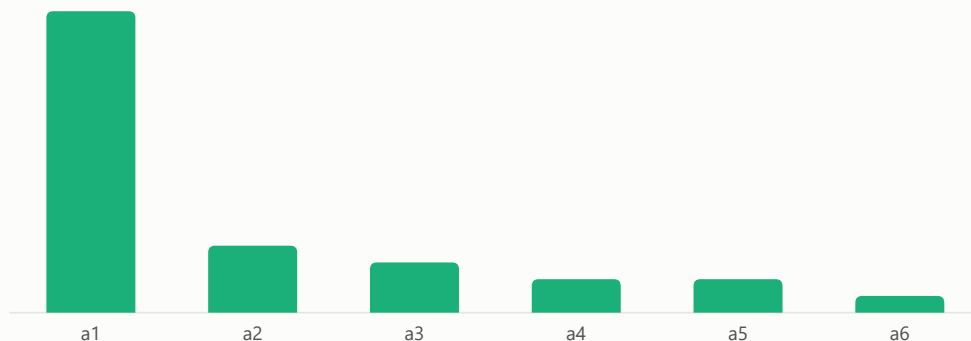


Figure 47: Several axes reduced to one verdict by a formula a person can read.

CONDITIONS

- A formula that reduces several axes to one verdict, here ten axes to a toxicity verdict. An affine contraction has a rank-one read operator; the outer product of its weights, and its attributions sum exactly to the change in verdict.
- The contraction found by chapter 6's tool is fairness minus the general valence component, moved accuracy from 0.779 to 0.863 on 1600 items, and is a consumer a person can read.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 14 section 14.5 of *Data Mining as Observation*, with the formula in `gtc-prototype/docs/SPECTRUM_FINDINGS.md:90-99`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 14 section 14.3	contraction 0.779 to 0.863, 77 of 1600, 4.8 percent, 0.872 to 0.863, weight 2.69	gtc-prototype/docs/SPECTRUM_FINDINGS.md:66-99
chapter 14 section 14.5	the contraction formula fairness minus the general component	gtc-prototype/docs/SPECTRUM_FINDINGS.md:90-99

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Attribution.lean](#), theorems `attr_sum_affine`, `attr_unread`, `sq_sensitivity_eq_readOp_diag`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 14.

RELATED

[attribution](#) (p. 29); [read operator](#) (p. 291); [residualization](#) (p. 310); [importance](#) (p. 181).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 14.5, 0.9.

Ledger rows that cite the entry's records without naming it: GO-EC-3.

contrastive objective

concept. Entry id `contrastive-objective`.

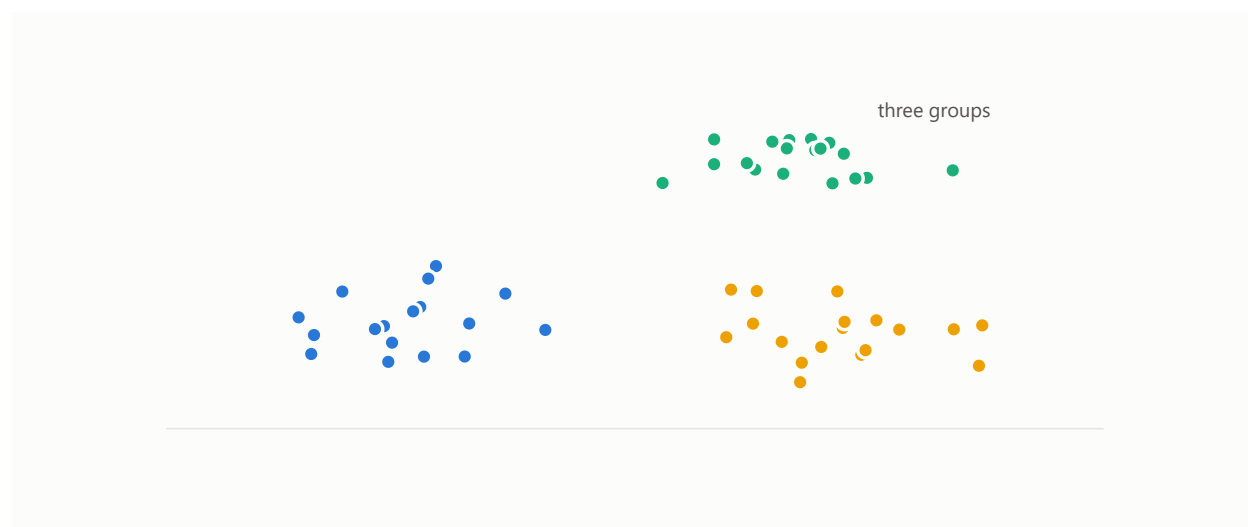


Figure 48: Pairs declared similar pulled together, others pushed apart.

DEFINITION

A training rule that pulls pairs declared similar together and pushes other pairs apart. The declared pairs are where an encoder learns what similar means. Chapter 12.

Example. Two paraphrases are pulled to cosine near one and an unrelated pair pushed toward zero, and the choice of pairs decides what the encoder can tell apart.

EQUATION

none

CONDITIONS

- A training rule that pulls pairs declared similar together and pushes other pairs apart. With a margin loss the loss is nonnegative, zero exactly when every declared pair scores above every other pair by the margin, and never lower for a larger margin.
- Swapping which pairs are declared similar reverses that condition, and with a positive margin no scorer satisfies both, so the declared pairs are where an encoder learns what similar means. Fine-tuning moves the reader, and the cross-corpus positives were the fix when within-corpus scores collapsed.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 12 section 12.4 of *Data Mining as Observation*, with the program's encoders in `xbse`, `xbse/README.md`: 104-130.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Contrastive.lean](#), theorems `loss_nonneg`, `loss_eq_zero_iff`, `not_both`, `loss_mono_margin`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 11, 12.

RELATED

[encoder](#) (p. 127); [embedding](#) (p. 126); [cross-corpus gate](#) (p. 94); [quotient](#) (p. 279).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 12.3, 11.1.

Ledger rows that cite the entry's records without naming it: NEG-2, GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 12 section 12.5.

control

instrument. Entry id `control`.

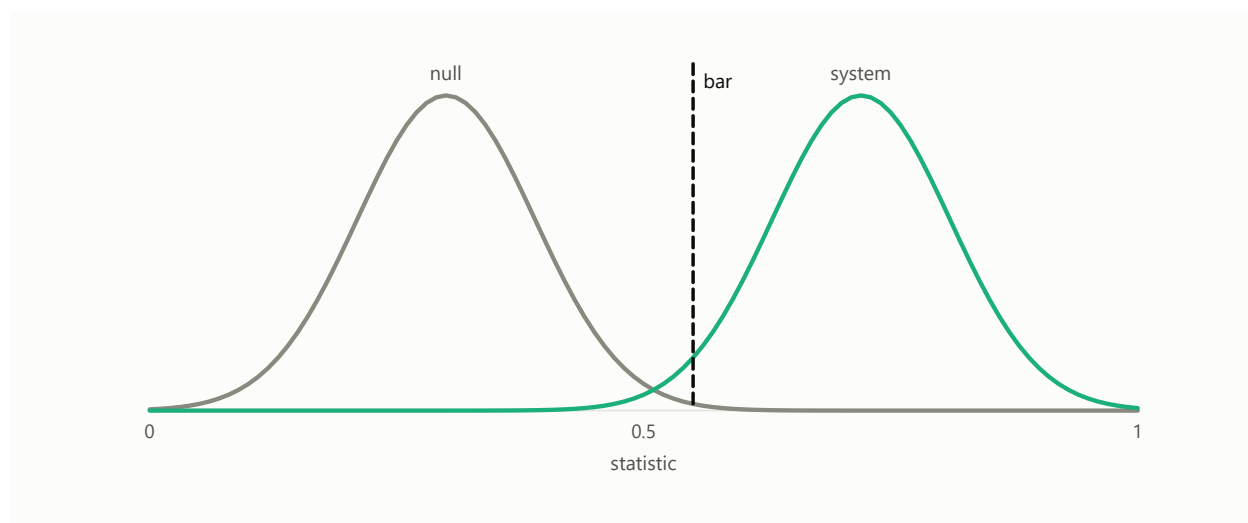


Figure 49: An arm that should show no effect, run beside the one that should.

DEFINITION

A comparison arm that should not show the effect, run beside the arm that should. A control that passes the bar makes the bar vacuous. Chapter 8 section 8.4.

Example. The first smooth-field control read 2.804 and was controlling for nothing; the second attempt read 0.184.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

CONDITIONS

- A comparison arm that should not show the effect, run beside the arm that should. A control that passes the bar makes the bar vacuous, so a claim names its control and reports both, and an isotropic control collapses the flip because there is no direction to read.
- The first smooth-field control read 2.804 and was controlling for nothing, because the gradient it was compared against was piecewise constant, and the second attempt read 0.184.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.4 of *Data Mining as Observation*, with the negative control in `constraint-gap-measurements/notes/negative_control.md:1-50`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	geometric-observation/claims/LEDGER.md row GO-6; geometric-observation/chapters/ch07_cost.md

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/NullModel.lean](#), theorems `card_pos_perm`, `scores_perm`, `stat_of_counts_const`, `excess_eq_zero_iff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Bar.lean](#), theorems `passes_anti`, `passes_mono`, `discriminates_iff`, `no_bar_of_null_ge`, `exists_bar_of_lt`, `vacuous_of_null_passes`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2, 3, 4, 6, 7, 8, 9, 12, 13.

RELATED

[null model](#) (p. 242); [bar](#) (p. 39); [chance level](#) (p. 62); [confound](#) (p. 77); [isotropic](#) (p. 187).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.17, 4.6.

Ledger rows that cite the entry's records without naming it: GO-6, NEG-2, NEG-4.

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.4.

correction

concept. Entry id correction.

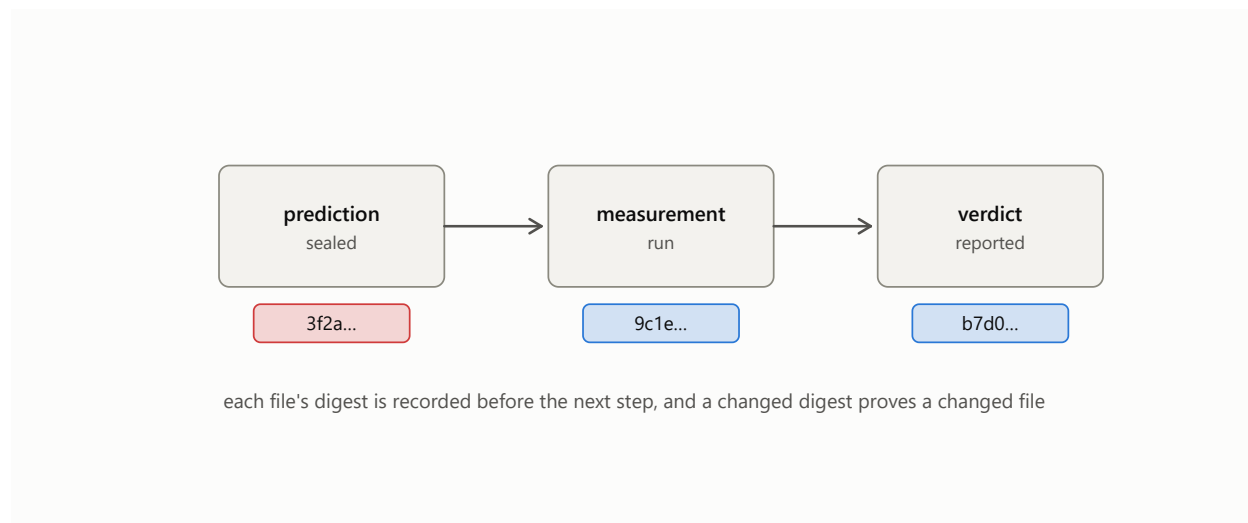


Figure 50: An erratum kept beside the original, itself sealed.

DEFINITION

An erratum or retraction, a change to a published number or claim that the record keeps beside the original, naming what was wrong and what replaced it. The Nadeau and Bengio adjustment is a statistical correction and has its own entry.

Example. The uncorrected t was stored as a sigma, and the corrected table changed nine wins to three, with seventeen ties.

EQUATION

none

CONDITIONS

- An erratum or retraction that the record keeps beside the original. A changed digest proves a changed file, so a correction is itself a sealed object, and the rows it corrects stay visible with the correction pointing at them.
- A correction can be the wrong shape. The uncorrected t was stored as a sigma, and the fitted correction that transferred with mean absolute error zero was the identity. The rescaling of a variance for a dependence the harness ignored is a different object, the Nadeau and Bengio correction.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 2 section 2.4 and chapter 8 section 8.5 of *Data Mining as Observation*, with the standing corrections in `geometric-observation/claims/LEDGER.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	constraint-gap/review/FINDINGS.md:1-35; constraint-gap/README.md:57-76

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Seal.lean](#), theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14.

RELATED

[Nadeau and Bengio correction](#) (p. 233); [ledger class](#) (p. 211); [sealed](#) (p. 324); [harness](#) (p. 168).

SEE ALSO

Ledger rows that cite the entry's records without naming it: NEG-9, NEG-13 → resolved.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.4, chapter 6 section 6.5, chapter 8 section 8.5.

correlation

concept. Entry id correlation.

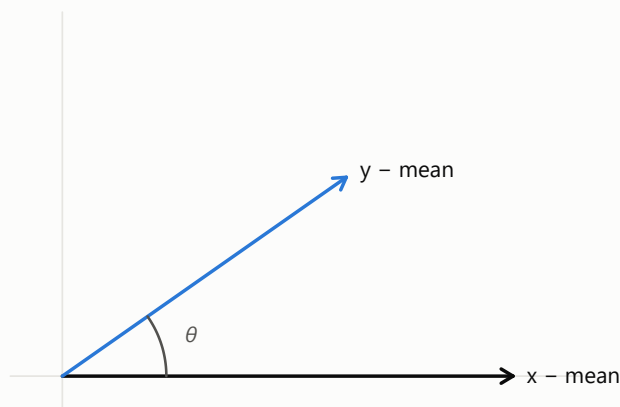


Figure 51: The cosine of two centred columns.

DEFINITION

The cosine of two centred columns, between minus one and one, unchanged by shifting or rescaling either. Its distance lives on the quotient that discards shift and scale. Chapter 0 section 0.6.

Example. Columns (1, 2, 3) and (2, 4, 6) have correlation 1, and (1, 2, 3) and (3, 2, 1) have correlation minus 1.

EQUATION

none

CONDITIONS

- The cosine of two centred columns. It lies between minus one and one, is one for a column against itself, is symmetric, and is unchanged by shifting or positively rescaling either column, so correlation distance lives on the quotient that discards shift and scale.
- Rank correlations read the order and discard the values, and a read distortion that correlates with a downstream ranking at 0.80 is a control and not a complete rank statistic, NEG-9.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Pearson, 1895, as chapter 0 section 0.6 and chapter 3 section 3.1 of *Data Mining as Observation* read it, with the radius-against-degree correlation in `the-angular-observer/README.md:135-139`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	<code>the-angular-observer/README.md:135-139</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Correlation.lean](#), theorems `abs_corr_le_one`, `corr_self`, `cosine_comm`, `corr_comm`, `center_shift`, `center_smul`, `corr_shift`, `corr_smul`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[cosine](#) (p. 87); [Spearman correlation](#) (p. 339); [Kendall correlation](#) (p. 194); [covariance matrix](#) (p. 91); [quotient](#) (p. 279).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.1, 3.1, 0.18.

Ledger rows that cite the entry's records without naming it: NEG-9.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.4, chapter 8 section 8.5.

cosine

concept. Entry id cosine.

DEFINITION

The dot product divided by the product of the two lengths, between minus one and one, unchanged by rescaling either vector. Its distance is the angle. Equation 0.1.

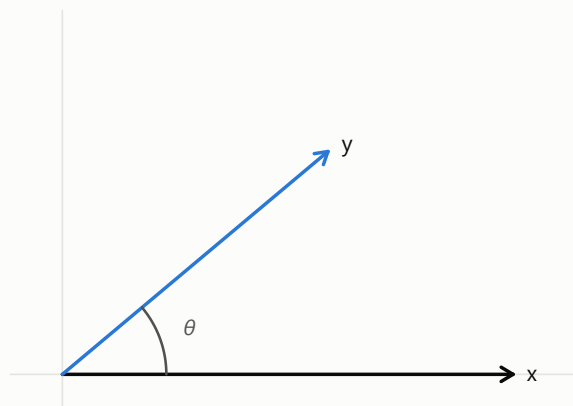


Figure 52: The dot product over the two lengths, the cosine of the angle between the vectors.

Example. Vectors $(1, 0)$ and $(1, 1)$ have cosine one over root 2, about 0.707, and doubling either vector leaves it unchanged.

EQUATION

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

CONDITIONS

- The dot product divided by the product of the two lengths. By Cauchy and Schwarz it lies between minus one and one, a vector has cosine one with itself, rescaling either vector by a positive factor does not change it, and it is the dot product of the two row-normalized vectors, so its distance is the angle on the quotient that discards length.
- A reconstruction cosine of 0.995 sat beside a consumer's perplexity of ten thousand, so the cosine is never an acceptance metric for a code, and on a spectral embedding the cosine is the right similarity because the length has converged to degree noise.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* GO-B-legal (035→036) [predicted]. Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [geometric-observation/claims/LEDGER.md:119](#).
- *refutes or corrects.* NEG-2 [refuted]. Reconstruction cosine as a proxy for key quality. [geometric-observation/claims/LEDGER.md:95](#).

FIRST STATED

Chapter 0 section 0.1 of *Data Mining as Observation*, equation 0.1, with the program's cosine-against-consumer negative in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation/chapters/ch02_failure_of_observer_free_measurement.md:40-60 ; geometric-observation/chapters/ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro/docs/KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49 ; geometric-observation/chapters/ch16_honest_negatives.md NEG-2

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality. [geometric-observation/claims/LEDGER.md:95](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/Cosine.lean](#), theorems `abs_dot_le`, `cosine_le_one`, `neg_one_le_cosine`, `cosine_self`, `cosine_smul`, `cosine_eq_dot_rowNormalize`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 6, 8, 10, 11, 12.

RELATED

[dot product](#) (p. 118); [Euclidean distance](#) (p. 133); [spectral embedding](#) (p. 342); [quotient](#) (p. 279); [identity reader](#) (p. 180).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 3.1, 3.3.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.3, chapter 9 section 9.1, chapter 11 section 11.2, chapter 12 section 12.3.

coupling null

concept. Entry id `coupling-null`.

DEFINITION

The case, alignment near one, in which the read subspace and the high-variance subspace coincide and the reconstruction-optimal code is also the read-preserving one. The flip's stated boundary. Chapter 4.

Example. Queries drawn from the corpus itself inflated the busiest count from about 61 to 339.9.

EQUATION

Book equation 4.6.

$$\kappa = \text{tr}(\bar{P}_C \bar{\Sigma}_x) \in [0, 1], \quad \kappa \rightarrow 1 \text{ is the coupling null (no flip).}$$

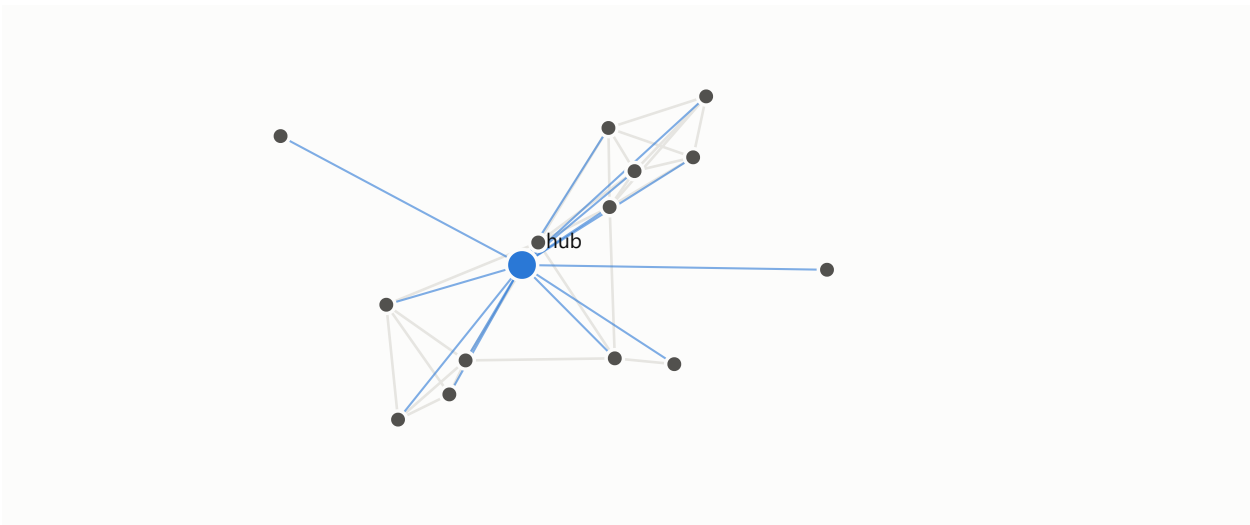


Figure 53: Hubs that appear because the queries were drawn from the corpus.

CONDITIONS

- The coupling null is the case in which the read subspace and the high-variance subspace coincide, alignment near one, so that the reconstruction-optimal code is also the read-preserving one and no flip is possible.
- It is the flip’s stated boundary, and a comparison that finds no flip there has confirmed the boundary rather than refuted the flip.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 8, [geometric-observation/chapters/ch08_value.md:108-116](#), the gradient-compression case, and chapter 12 for the alignment dial, DOI 10.5281/zenodo.21776291.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 4 section 4.4	gradient compression anti 300 of 300, flip 27 percent, coupling boundary	geometric-observation/chapters/ch08_value.md:108-116

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Alignment.lean](#), theorems `overlap_sq_le`, `alignment_le_one`, `alignment_nonneg`, `alignment_eq_one_of_proportional`, `coupling_null`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 4.

RELATED

[alignment](#) (p. 15); [the flip](#) (p. 147); [read distortion](#) (p. 290).

SEE ALSO

Ledger rows that cite the entry's records without naming it: GO-4, GO-B-optim-D4 (034 · D4).

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.4, chapter 7 section 7.2.

covariance matrix

concept. Entry id covariance-matrix.

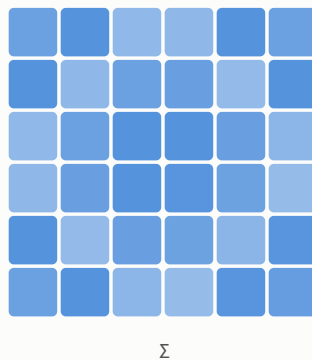


Figure 54: Variances on the diagonal and covariances off it, symmetric and positive semidefinite.

DEFINITION

The matrix of pairwise covariances of a set of rows, whose trace is the total variance. Equation 0.3.

Example. Two columns with variances 0.3 and 1.7 and no covariance have covariance matrix $\text{diag}(0.3, 1.7)$ and trace 2.0.

EQUATION

Book equation 0.3.

$$\Sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)], \quad \text{tr } \Sigma = \sum_i \Sigma_{ii}.$$

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

CONDITIONS

- The weighted average of the centred rows' outer products, the same construction as the read operator with the centred row in place of the sensitivity. Its quadratic form along a unit direction is the variance of the projection onto it, its trace is the total variance, and it is positive semidefinite.

- Chapter 4 pairs it with the read operator in one basis. Where the two are proportional there is no flip, and where they are not the water-filling allocation reads both.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. OT-7 [demonstrated]. The damage form, trace pairing, rank, and loading covariance are $GL(d)$ -invariant under $P' = A^{-T}PA^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and ... [geometric-observation/claims/LEDGER.md:30](#).
- *measures*. OT-2 [predicted]. Loading is a covariance, not a distance: reading error under a change of measure is priced by $\varepsilon \cdot \|E[h \cdot A]\|$ — predicted from the base measure alone — and a full-magnitude shift orthogonal to the operator's variation does nothing. [geometric-observation/claims/LEDGER.md:33](#).

FIRST STATED

Chapter 0 section 0.2 of *Data Mining as Observation*, paired with the read operator in chapter 4, and Volume 14 chapter 5, [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48](#).

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Covariance.lean](#), theorems `var_eq_quad`, `trace_eq_total`, `quad_nonneg`, `cov_symm`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 9, 10, 11, 12.

RELATED

[read operator](#) (p. 291); [whitening](#) (p. 381); [water-filling](#) (p. 378); [alignment](#) (p. 15); [explained variance](#) (p. 137).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.4.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 4 section 4.2.

coverage

concept. Entry id coverage.

DEFINITION

The fraction of decisions a certificate clears, reported beside its false-clear rate, since a certificate that clears nothing has a false-clear rate of zero and no use. Equation 0.26.

Example. A certificate that clears 500 of 1000 decisions has coverage one half.

EQUATION

Book equation 0.26.

$$FC = \Pr [W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

CONDITIONS

- A naive certificate clears every decision, so its coverage is one and its conditional and joint false-clear rates coincide.

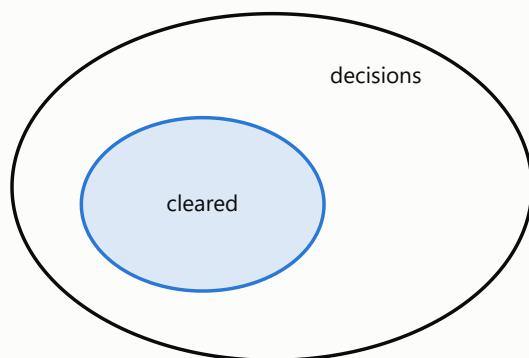


Figure 55: The fraction of decisions the certificate cleared.

- A witnessed certificate clears fewer decisions, and its false-clear rate is read beside its coverage, never alone.
- The word also names, in chapter 14, what a moderation instrument can read of the signal its inputs carry, which is a different quantity and is not this entry.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Data Mining as Observation, chapter 0 section 0.13 and chapter 13, introduced on 2026-09-03 when the false-clear rate was restated as a conditional rate, and Volume 14 chapter 19 at geometric-observation 7d91883.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Certificate.lean](#), theorems `falseClear_mul_coverage`, `coverage_empty`, `falseClear_mem_unit`, `minOverStrata_passes_iff`, `minOverStrata_le_weighted_mean`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 10, 12, 13, 14.

RELATED

[false-clear rate](#) (p. 141); [certificate](#) (p. 59); [witness](#) (p. 382).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 12 section 12.6, chapter 13 section 13.3.

cross-corpus gate

instrument. Entry id cross-corpus-gate.

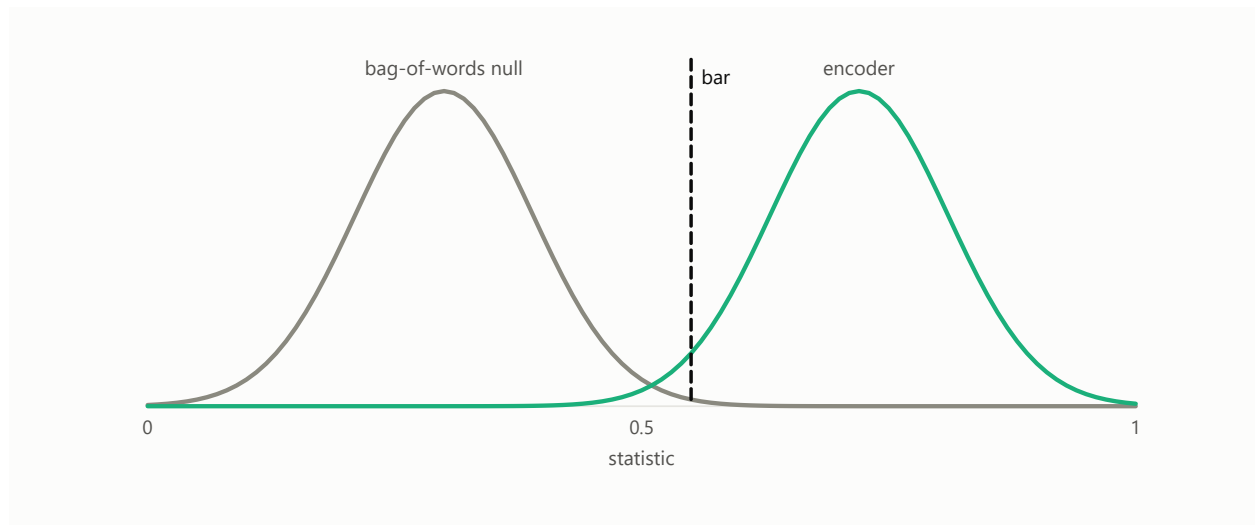


Figure 56: An encoder votes only if its AUROC on a corpus it never saw clears the margin over a bag-of-words null.

DEFINITION

The validation rule that an encoder votes only if its held-out AUROC on a corpus it was not trained on clears a preregistered margin over a bag-of-words null. Chapters 12 and 14.

Example. The rights encoder needed a margin of 0.10 over a bag-of-words null on a corpus it never saw; it scored 0.509 against the null's 0.512 and failed.

EQUATION

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

CONDITIONS

- An encoder votes only if its held-out AUROC on a corpus it was not trained on clears both nulls, the untrained encoder and the bag-of-words baseline, by the preregistered margin of one tenth. A validated encoder therefore beats each null by the margin.
- Since AUROC is at most one, a null above nine tenths cannot be cleared by any encoder, and the gate's verdict is unchanged by any strictly monotone recalibration of the score.
- The rights encoder fell below the untrained baseline and was retired rather than tuned, and the within-corpus scores that had collapsed on a second corpus were the reason the gate is cross-corpus.
- The gate compares the AUROC as the encoder orients its score, so a reversed encoder fails rather than passing at 1 minus A, and the bag-of-words null is scored the same way.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

The moral-embedding program, `xbse/README.md:104-130,160-172`, and chapter 12 section 12.5 and chapter 14 section 14.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.6	rights encoder 0.467 below untrained baseline, the AUROC lesson, cross-corpus same-sign fix	xbse/README.md:160-172; xbse/experiments/rights_r6_summary.json
chapter 12 section 12.5	within 0.75 to 0.955 collapsing to 0.47 to 0.55, cross-corpus positives as the fix, adversary not load-bearing	xbse\README.md:104-130,160-172

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/CrossCorpusGate.lean](#), theorems `clears_bow`, `clears_untrained`, `not_validated_of_saturated`, `margin_example`, `validated_comp`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 8, 12, 14.

RELATED

[reliability weight](#) (p. 306); [harness](#) (p. 168); [leakage](#) (p. 210); [preregistration](#) (p. 267).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.16.

cross-support ratio

concept. Entry id cross-support-ratio.

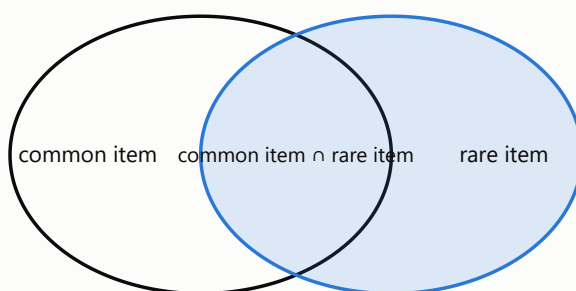


Figure 57: The support of the rarer item over the support of the commoner one.

DEFINITION

The smallest item support in an itemset over the largest, with a threshold below which the itemset is discarded. Chapter 5.

Example. An item in 2 percent of baskets paired with one in 40 percent has cross-support ratio 0.05.

EQUATION

none

CONDITIONS

- The smallest item support in an itemset over the largest. It lies in the unit interval, is one exactly when every item has the same support, and adding an item can only lower it, so a threshold on it prunes supersets safely in chapter 5's sense.
- It is a property of the item supports alone and says nothing about the itemset's own support, so it discards an itemset for pairing a common item with a rare one, not for being rare.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

TSK chapter 5 on cross-support patterns, as chapter 5 section 5.5 of *Data Mining as Observation* states it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/CrossSupport.lean](#), theorems `ratio_mem_unit`, `ratio_eq_one_iff`, `ratio_anti`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 5.

RELATED

[Apriori principle](#) (p. 25); [safe pruning](#) (p. 320); [lift](#) (p. 214); [multiple comparisons](#) (p. 229).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 5.3, 5.1.

cross-validation

instrument. Entry id `cross-validation`.

DEFINITION

Splitting the data into folds and evaluating on each fold a model trained on the others. Repeated cross-validation multiplies confidence it did not earn, chapter 8 section 8.5.

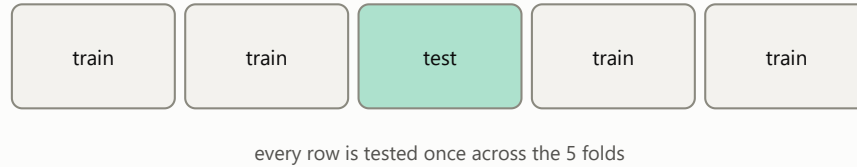


Figure 58: Folds tested once each, sharing training data.

Example. Ten folds of 100 rows each test every row once, and the overall accuracy is the mean of the ten fold accuracies.

EQUATION

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \quad \frac{\widehat{\text{Var}}_{\text{NB}}}{\hat{\sigma}^2/J} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

CONDITIONS

- Splitting the rows into folds and evaluating on each fold a model trained on the others, so every row is tested exactly once. The overall accuracy is the size-weighted mean of the fold accuracies, and with equal folds it is their plain mean.
- The folds share training data, so repeated cross-validation multiplies confidence it did not earn, which the Nadeau and Bengio correction prices at one plus the fold count times the test-to-train ratio.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Stone, cross-validatory choice, 1974, as chapter 8 section 8.5 of *Data Mining as Observation* reads it, with the program's fold accounting in `constraint-gap/review/FINDINGS.md:1-35`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/CrossValidation.lean](#), theorems `sizes_sum`, `accuracy_weighted`, `accuracy_mean_of_equal`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 5, 6, 7, 8, 14.

RELATED

[Nadeau and Bengio correction](#) (p. 233); [harness](#) (p. 168); [standard error](#) (p. 347); [leakage](#) (p. 210).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.4, chapter 8 section 8.5.

curvature

concept. Entry id curvature.

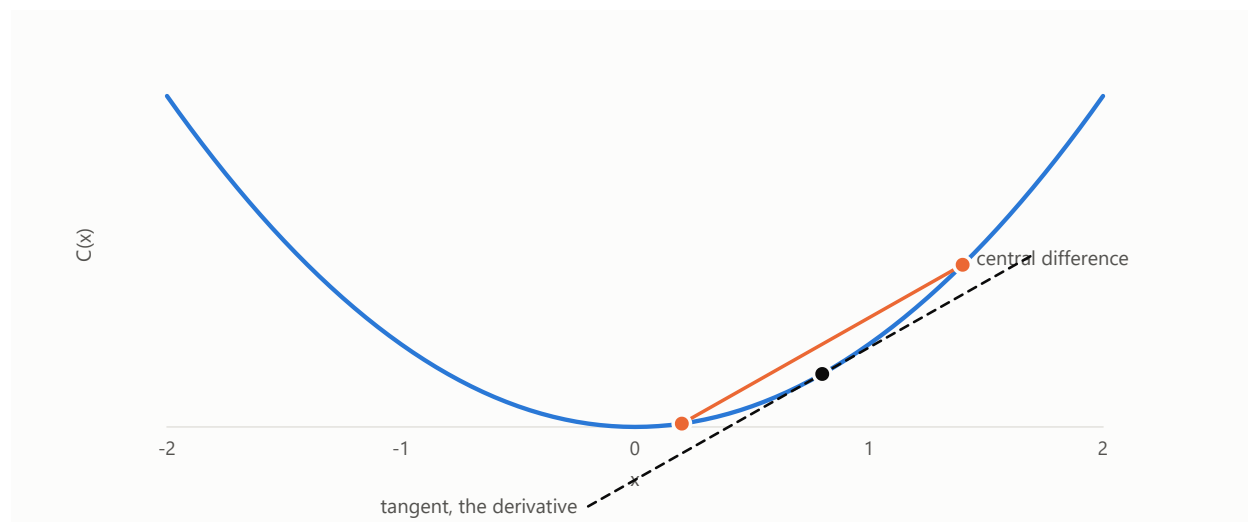


Figure 59: The second derivative, which the finite difference does not read and which sets the linear model's error.

DEFINITION

The second derivative of a consumer, which the finite difference of equation 0.8 does not read and which sets the error of the linear model at a step. Chapter 0 section 0.5 and chapter 7 section 7.2.

Example. For $C(x)$ equal to x squared, the central difference at 0.8 with step 0.6 gives exactly 1.6, and the one-sided difference gives 2.2.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

CONDITIONS

- The second derivative of a consumer, which the finite difference of equation 0.8 does not read. For a quadratic the second difference recovers it exactly at every step, the error of the linear model at a step is the curvature times the step squared, and the gradient at the two ends of a step differs by the curvature times the step. An affine consumer has none.
- The reader of a gradient is the optimizer, and its read operator is the loss curvature, which chapter 4’s flip was measured against on a real logistic model with its exact Hessian, anti three hundred of three hundred and the flip in 82 of 300.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-B-optim-D4 (034 · D4) [predicted]. Optimization — gradient compression, curvature (Hessian) read operator, on a REAL model (logistic regression); optional stretch [geometric-observation/claims/LEDGER.md:122](#).

FIRST STATED

Chapter 0 section 0.5 and chapter 7 section 7.2 of *Data Mining as Observation*, with the curvature reader’s flip in [geometric-observation/claims/LEDGER.md](#) row GO-B-optim-D4.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 7 section 7.2	second-order leaf values, logistic curvature	ESL 2e 10.9 to 10.13; XGBoost introduction to boosted trees

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Curvature.lean](#), theorems `second_quad`, `second_affine`, `linear_error`, `gradient_change`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 4, 7.

RELATED

[Hessian](#) (p. 172); [finite difference](#) (p. 146); [sensitivity](#) (p. 328); [boosting](#) (p. 48); [read operator](#) (p. 291).

SEE ALSO

Book equations stated beside the entry’s terms, not defining it: 0.9, 6.1.

Sources-table rows that share a record with the entry without naming it: chapter 7 section 7.2.

D

DBSCAN

instrument. Entry id dbscan.



Figure 60: Core points by neighbour count within a radius, clusters by reachability, the rest noise.

DEFINITION

A clustering that calls a point a core point when at least a minimum number of points lie within a radius, grows clusters by reachability from core points, and leaves the rest as noise. Chapter 9.

Example. With radius 0.5 and a minimum of 4 neighbours, a point with 6 neighbours within 0.5 is a core point and a point with none is noise.

EQUATION

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

CONDITIONS

- A clustering that calls a point a core point when at least a minimum number of points lie within a radius, grows a cluster by reachability from core points, and leaves the rest as noise. A wider radius or a smaller count makes more core points, every reachability chain starts at a core point, and a point that is not a core point reaches nothing.
- Its read subspace is the density coordinate, the one the geodesic reader of a spectral embedding discards, so two clustering consumers of one embedding read complementary coordinates.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Ester, Kriegel, Sander, and Xu, a density-based algorithm for discovering clusters, 1996, as chapter 9 section 9.1 of *Data Mining as Observation* reads it.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Dbscan.lean](#), theorems `ball_mono`, `core_mono`, `core_anti`, `reach_from_core`, `noise_unreachable`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 9, 10.

RELATED

[density](#) (p. 106); [k-means](#) (p. 198); [hierarchical clustering](#) (p. 173); [spectral clustering](#) (p. 340); [outlier](#) (p. 250).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.19, 10.3.

Ledger rows that cite the entry's records without naming it: NEG-11.

Sources-table rows that share a record with the entry without naming it: chapter 9 section 9.1, chapter 10 section 10.1.

decision boundary

concept. Entry id decision-boundary.

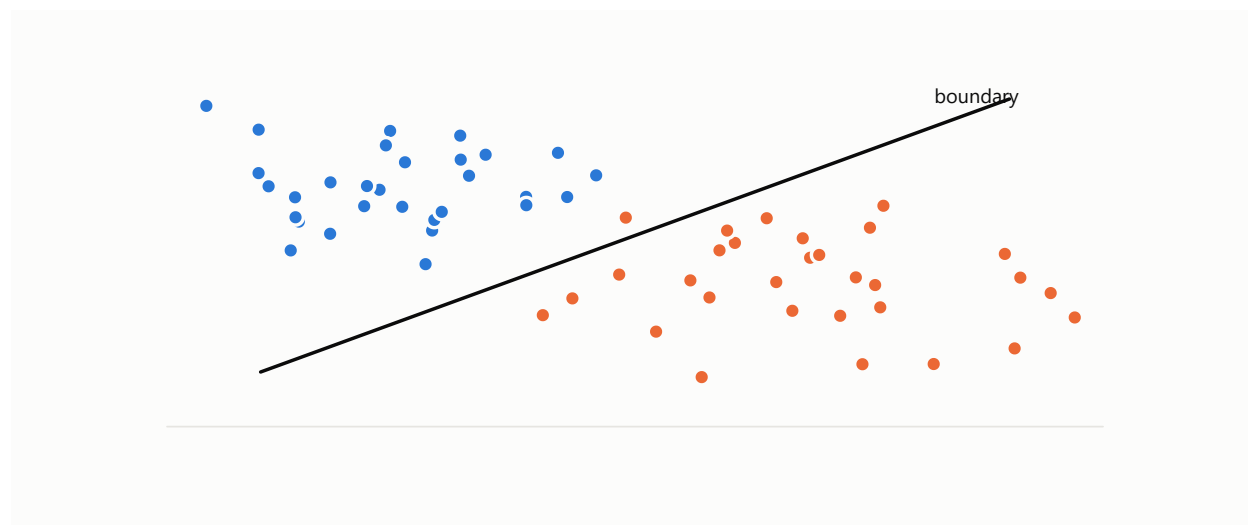


Figure 61: Where the score crosses the threshold.

DEFINITION

The set of inputs whose score equals the threshold. Chapter 6.

Example. For the score $2x_1 + x_2 - 3$ at threshold zero, the boundary is the line $2x_1 + x_2 = 3$.

EQUATION

none

CONDITIONS

- The set of inputs whose score equals the threshold. For a linear classifier it is an affine hyperplane. Moving orthogonally to the weights never crosses it, moving along the weights crosses it exactly once, and every boundary point has the same projection on the weights.
- The boundary is where the read direction and the threshold meet and nothing else about the input enters it, which is the classifier's nuisance drawn as a picture.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, where the classifier's read direction and its threshold meet.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/DecisionBoundary.lean](#), theorems `orth_stays`, `cross_once`, `same_projection`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0.

RELATED

[classifier](#) (p. 65); [threshold](#) (p. 362); [margin](#) (p. 221); [nuisance](#) (p. 240).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 6.1, 6.2.

Ledger rows that cite the entry's records without naming it: GO-1.

decision tree

instrument. Entry id `decision-tree`.

DEFINITION

A classifier that scores by a sequence of axis-aligned splits, flat within each leaf, so that its sensitivity is zero almost everywhere and its importances count splits. Chapter 6.

Example. Split on x_1 below 5, then on x_2 below 4 in the left branch; a row at x_1 equal to 5.1 lands in the right leaf whatever x_2 is.

EQUATION

none

CONDITIONS

- A classifier that scores by a sequence of axis-aligned splits and is flat within each leaf. Its finite difference is zero at every row that does not straddle a split, a split on one coordinate reads nothing of the others, and the Gini impurity it splits on is at most one half for two classes and zero exactly for a pure leaf.
- Its read subspace is spanned by the coordinates it splits on and a feature that appears in no split is in its nuisance exactly, which is why a finite-difference probe of a tree returns zero at most rows and importances that count splits are the right sensitivity for this reader.

Conditions are curated in `entries.toml` rather than read from a record.

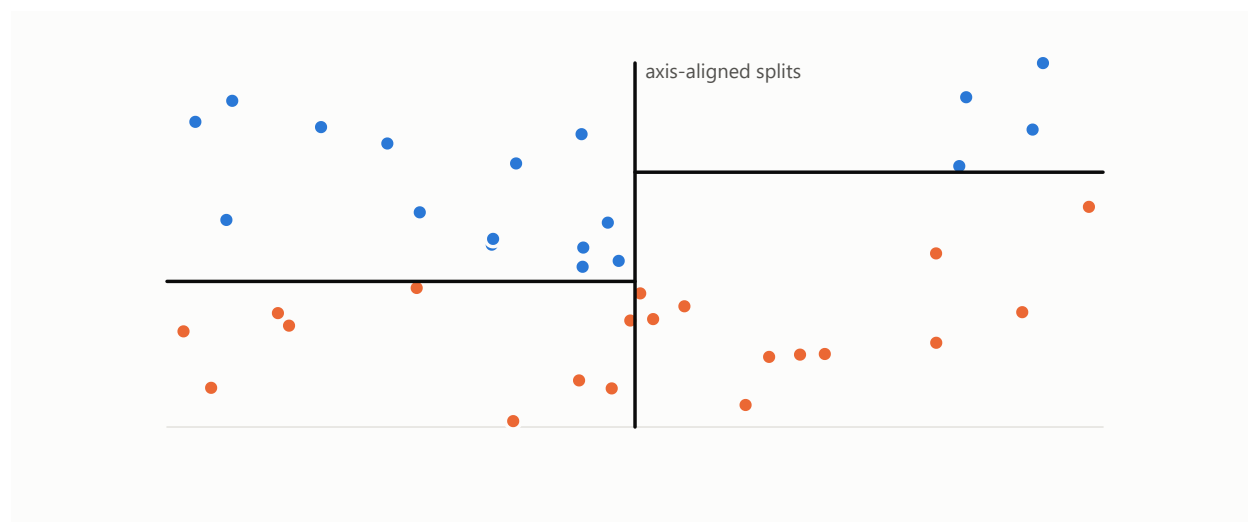


Figure 62: Axis-aligned splits, flat within each leaf.

LEDGER

none

FIRST STATED

Breiman, Friedman, Olshen, and Stone, classification and regression trees, 1984, as chapter 6 section 6.1 of *Data Mining as Observation* reads it, with the selection-consumer regime in `readscope/readscope/regimes.py:1-60`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/DecisionTree.lean](#), theorems `stump_flat_left`, `stump_flat_right`, `split_reads_one_axis`, `gini_le_half`, `gini_eq_zero_iff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 4, 5, 6, 7, 8, 9, 11, 14.

RELATED

[classifier](#) (p. 65); [importance](#) (p. 181); [boosting](#) (p. 48); [ensemble](#) (p. 129); [sensitivity](#) (p. 328).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 6.1, 0.8, 14.5.

Ledger rows that cite the entry's records without naming it: GO-1.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.3, chapter 6 section 6.1, chapter 6 section 6.3, chapter 7 section 7.3.

declaration

instrument. Entry id declaration.

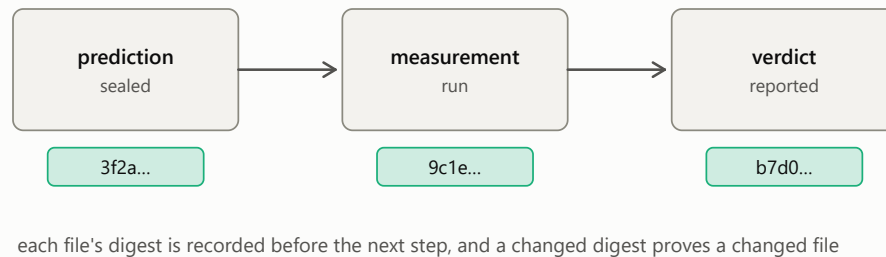


Figure 63: What the study will count, drop, or treat as a failure, sealed before the run.

DEFINITION

A sealed statement of what a study will count, drop, or treat as a failure, written before the run. Chapter 8 section 8.8.

Example. The declaration named timeouts and numerical failures as counted trajectories before the run, and one count was corrected the next day.

EQUATION

none

CONDITIONS

- A statement of what a study will count, drop, or treat as a failure, sealed before the run. A changed digest would prove a changed declaration, and a declaration drafted on one day and sealed the next is the record's normal shape.
- The declaration for the fourth crucible was drafted 2026-08-17, sealed 2026-08-18, and had one count corrected, and the missing-data rule is a required declared field.
- The declaration's chronology rests on the seal's public receipt, signed tag, or archival deposit, since a commit's dates are supplied fields.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.8 of *Data Mining as Observation*, with the sealed declaration in `geometric-observation/crucible/DECLARATION-V1.md:1-20`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.8	declaration drafted 2026-08-17, sealed 2026-08-18, one count corrected, G2	geometric-observation/crucible/DECLARATION-V1.md:1-20 ; geometric-observation/crucible/OT-CRUCIBLE-4.md:31-35

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Seal.lean](#), theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 7, 8, 10, 11, 12.

RELATED

[registered](#) (p. 305); [sealed](#) (p. 324); [preregistration](#) (p. 267); [commit hash](#) (p. 70).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.1, 8.3.

Ledger rows that cite the entry's records without naming it: OT-11.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.3.

degree

concept. Entry id degree.

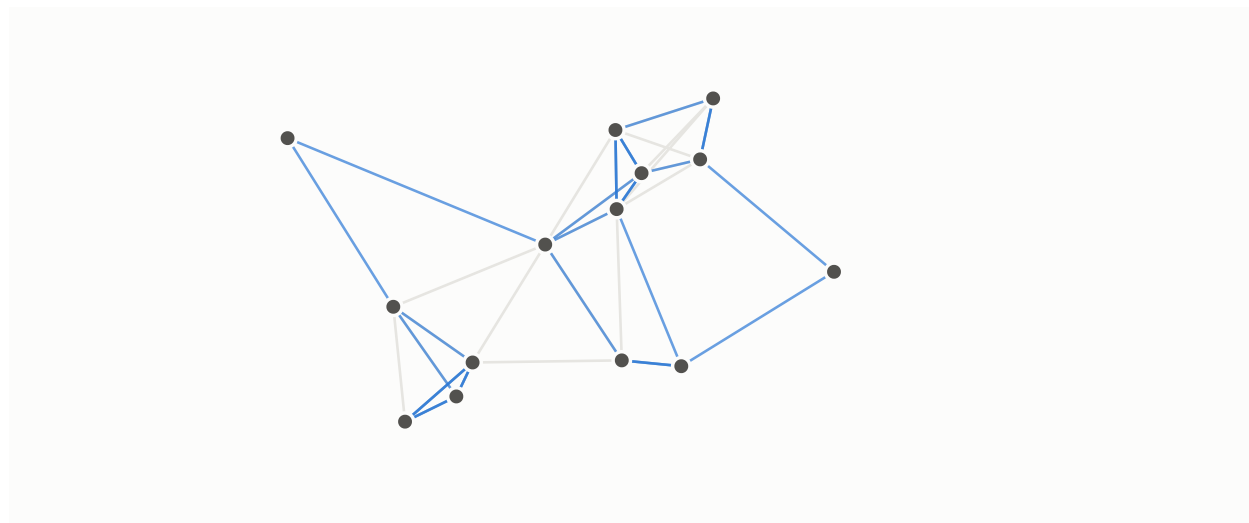


Figure 64: The number of edges at a node.

DEFINITION

Of a node in a graph, the number of edges it has. Chapter 0 section 0.10.

Example. A node with edges to three others has degree 3, and the degrees of a graph sum to twice its edge count.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

CONDITIONS

- The number of edges a node has. Summed over the graph the degrees count every edge twice, the sum is even, and no node has degree more than the number of other nodes.
- The radius of a spectral embedding tracks the degree at correlation 0.92 to 0.99 on the substrates tested, which is the measured half of chapter 3's claim that the magnitude is degree noise and the angle is what the geodesic-rank reader reads.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.10 and chapter 3 section 3.3 of *Data Mining as Observation*, with the radius-against-degree measurement in the-angular-observer, the-angular-observer/README.md:135-139.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular-observer/README.md:135-139

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Degree.lean](#), theorems `sum_degrees`, `degree_lt_card`, `sum_degrees_even`, `average_degree`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 3, 4, 8, 9.

RELATED

[Laplacian](#) (p. 207); [geodesic distance](#) (p. 162); [hubness](#) (p. 176); [recognizer](#) (p. 296).

SEE ALSO

Ledger rows that cite the entry's records without naming it: NEG-1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.2, chapter 3 section 3.3, chapter 9 section 9.1, chapter 9 section 9.2, chapter 11 section 11.1.

density

concept. Entry id density.

DEFINITION

The local crowding of rows around a point, the coordinate the geodesic reader of a spectral embedding discards and DBSCAN and the density detector read. Chapter 3 section 3.3 and chapter 9.

Example. A point with 20 neighbours within radius 1 sits in a denser region than one with 2.

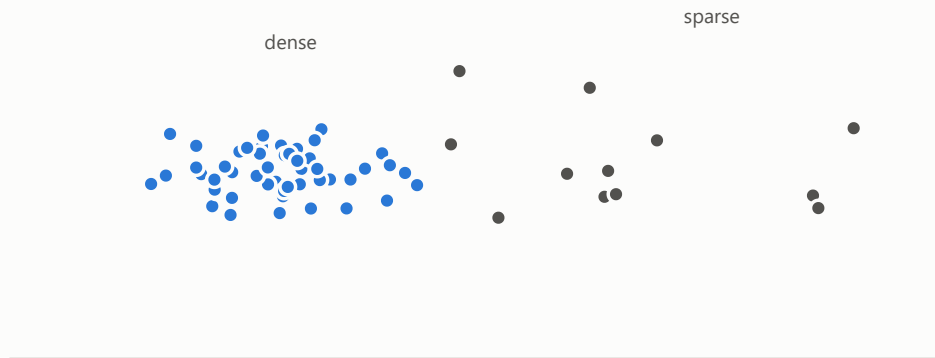


Figure 65: The local crowding of rows, the coordinate the geodesic reader discards.

EQUATION

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

CONDITIONS

- The local crowding of rows around a point, which on a neighbourhood graph is the degree. The radius of the spectral embedding converges to one over the square root of the degree, so the geodesic reader discards the density and DBSCAN and the density detector read it. A wider radius or a smaller count makes more core points, and the sum of degrees is twice the edge count.
- The radius-against-degree correlation was 0.92 to 0.99 across substrates, and the quotient that removed density to restore invariant fidelity was refuted four times, NEG-11.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects.* NEG-11 [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [geometric-observation/claims/LEDGER.md:104](#).

FIRST STATED

Chapter 3 section 3.3 of *Data Mining as Observation*, with the radius-against-degree measurement in the-`angular-observer/README.md:135-139`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [geometric-observation/claims/LEDGER.md:104](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/Dbscan.lean](#), theorems `ball_mono`, `core_mono`, `core_anti`, `reach_from_core`, `noise_`

unreachable, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).
[lean/DataMiningAsObservation/Degree.lean](#), theorems `sum_degrees`, `degree_lt_card`, `sum_degrees_even`, `average_degree`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 3, 8, 9, 10, 11.

RELATED

[degree](#) (p. 105); [spectral embedding](#) (p. 342); [DBSCAN](#) (p. 100); [geodesic distance](#) (p. 162); [hubness](#) (p. 176).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.19, 9.1.

Ledger rows that cite the entry's records without naming it: NEG-1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.3, chapter 9 section 9.1, chapter 10 section 10.1.

deployment mismatch

concept. Entry id `deployment-mismatch`.

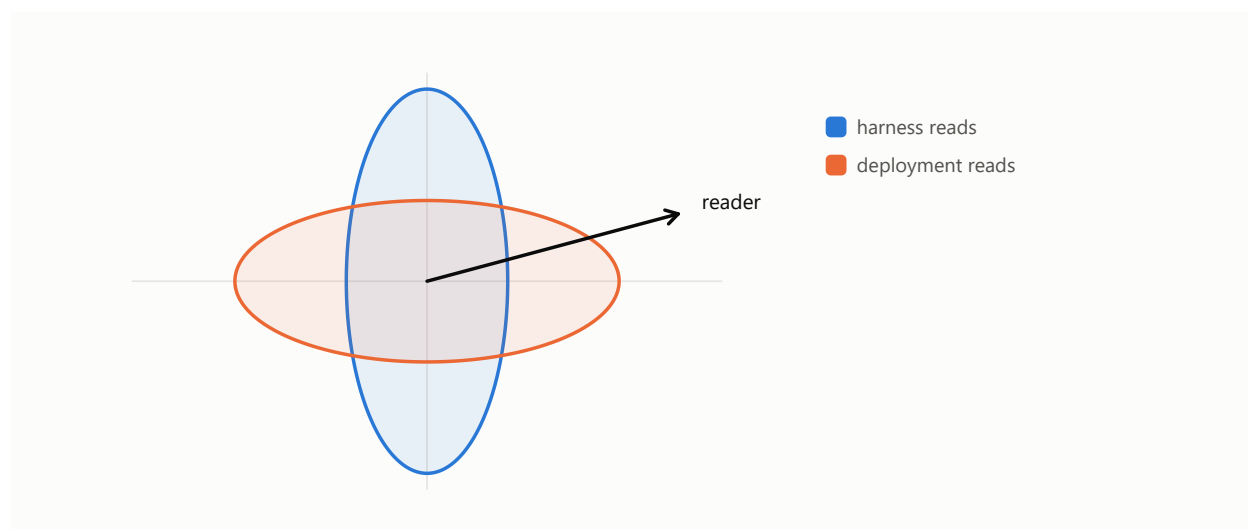


Figure 66: The harness reads a direction the deployed consumer never sees.

DEFINITION

The failure in which the consumer that was evaluated is not the consumer that was deployed, or time moved between the two. Chapters 1 and 13.

Example. The harness read a direction the deployed consumer never sees, and its 0.909 became 0.736 in deployment.

EQUATION

none

CONDITIONS

- The failure in which the consumer that was evaluated is not the consumer that was deployed, or time moved between the two. A deployment score is a weighted mean over the slices actually served, so it lies between the worst slice and the best, and a benchmark drawn from the best slice overstates it unless every weighted slice matches.

- The chapter's case scores 0.909 on the benchmark against a target of 0.8 and 0.736 on the deployment mean, and the six bars that separate the two are preregistered.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 1 section 1.7 and chapter 13 of *Data Mining as Observation*, with the program's case in the radio slice sweep, `observation-theory-campaigns/analysis/llm/PREREG-XPROTO-LLM.md`, and the serving-stack measurement of ledger row GO-2/GO-12/GO-13 operational.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.6	XPROTO-LLM, benchmark 0.909, 0.920, 0.909, thirty slices, target 0.8, naive 0.333, aware 0.033, spread 0.380, deployment mean 0.736, six bars on three seeds, sealed 2026-08-25 at b61f7f1	observation-theory-campaigns/experiments/LLM-EVAL-TRACK.md:1-50 ; observation-theory-campaigns/analysis/llm/XPROTO-LLM-graded.json ; observation-theory-campaigns/experiments/SEALS.md:85

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/DeploymentMismatch.lean](#), theorems `deployment_le_max`, `min_le_deployment`, `deployment_lt_max_of_gap`, `book_numbers`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 12.

RELATED

[observer](#) (p. 244); [certificate](#) (p. 59); [coherence time](#) (p. 67); [min-over-strata](#) (p. 226).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 13.1.

Ledger rows that cite the entry's records without naming it: OT-4, GO-2/GO-12/GO-13 operational (KV serving, 077).

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.6.

detector

instrument. Entry id detector.

DEFINITION

A scorer that calls a row an outlier, of four families reading distance from the mean, distance to neighbours, density relative to neighbours, or cluster membership. Chapter 10 section 10.1.

Example. A row 3.6 spreads from the mean is an outlier to the statistical detector and may be ordinary to the density detector.

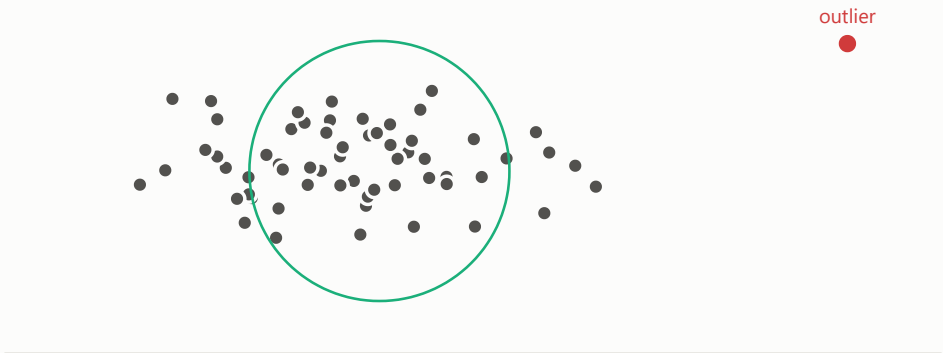


Figure 67: A scorer that calls a row an outlier, of four families.

EQUATION

none

CONDITIONS

- A scorer that calls a row an outlier, of four families. The statistical detector reads distance from the mean in units of spread, the proximity detector reads distance to neighbours, the density detector reads density relative to neighbours, and the clustering detector reads membership. Each reads a coordinate the others discard.
- No detector of the data alone sees the observer’s outliers, the anti-hubs, because they are a property of the queries, and a detector’s positives are few by Chebyshev’s arithmetic.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 10 section 10.1 of *Data Mining as Observation*, after TSK chapter 9, with the hierarchical typing in `turboquant-pro/turboquant_pro/anatomy.py:98-170`.

MEASUREMENTS

	Numbers, as the book’s sources table records them	Source
Where the book states it		
chapter 10 section 10.1	the four detector families	TSK 2e chapter 9

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Outlier.lean](#), theorems `zscore_mean`, `zscore_affine`, `outlier_iff`, `zscore_reflect`, `chebyshev`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).
[lean/DataMiningAsObservation/Mahalanobis.lean](#), theorems `dM2_nonneg`, `dM2_mean`, `dM2_scale`, `dM2_eq_whitened`, `dM2_antitone_in_variance`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

[lean/DataMiningAsObservation/Dbscan.lean](#), theorems `ball_mono`, `core_mono`, `core_anti`, `reach_from_core`, `noise_unreachable`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 3, 8, 10, 13.

RELATED

[outlier](#) (p. 250); [Mahalanobis distance](#) (p. 219); [density](#) (p. 106); [DBSCAN](#) (p. 100); [anti-hub](#) (p. 20).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.33, 10.3, 9.1.

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.3.

direction-only quantizer

instrument. Entry id `direction-only-quantizer`.

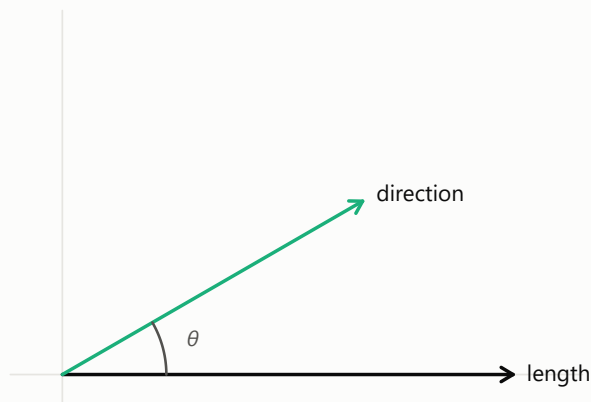


Figure 68: The length stored exactly and the direction rounded.

DEFINITION

A quantizer that stores each vector's length exactly and rounds its direction. Chapter 11.

Example. A vector of length 5 at 30 degrees is stored as the length 5 exactly and the direction rounded to the nearest of 16 angles.

EQUATION

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

CONDITIONS

- A quantizer that stores each vector's length exactly and rounds its direction to a unit codeword. The quantized vector keeps the original's length, its cosine with any query is the codeword's cosine, and its score against a query is the length times the query's dot product with the codeword, so the score error is the length times the query's dot product with the direction error.

- A key's cosine with its own quantized form can be near one while its score against a query moves by a multiple of its length, which is why keys at cosine 0.995 raised the perplexity by three orders of magnitude.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-B-LOCATA [predicted]. Real microphone-array recordings (LOCATA), DOA consumer — held-out confirmation with the PolarQuant compressor [geometric-observation/claims/LEDGER.md:117](#).

FIRST STATED

Chapter 11 section 11.2 of *Data Mining as Observation*, with PolarQuant in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49` and the LOCATA comparison of the ledger.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/DirectionQuantizer.lean](#), theorems `length_preserved`, `score_eq`, `score_error`, `co-sine_eq`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 4, 11.

RELATED

[quantization](#) (p. 275); [dot product](#) (p. 118); [KV cache](#) (p. 200); [attention](#) (p. 26); [the flip](#) (p. 147).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.1.

Ledger rows that cite the entry's records without naming it: NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 8 section 8.2, chapter 11 section 11.1.

discretization

instrument. Entry id discretization.

DEFINITION

Collapsing an ordered quantity to a few labels by binning it, so that values in one cell become indistinguishable. It forms a quotient. Chapter 2 section 2.5.

Example. Ages 23, 27, and 29 all become the label 20 to 29 in ten-year bins.

EQUATION

none

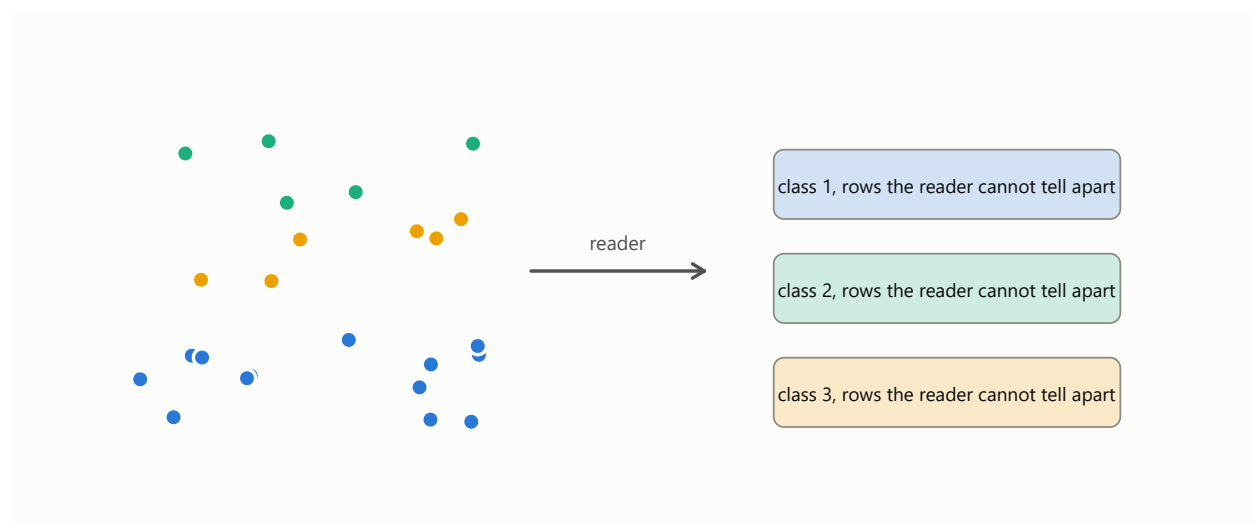


Figure 69: Values in one cell become one label.

CONDITIONS

- Collapsing an ordered quantity to a few labels by binning it. The bin index is nondecreasing in the value, every value in a cell gets the cell's index, two values in the same cell become indistinguishable, and a sample confined to m cells has at most m distinct labels.
- A discretization forms a quotient before anyone has said who reads the result, and a consumer that read the values it merged has lost them for every stage after.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 2 section 2.5 of *Data Mining as Observation*, after TSK section 2.3.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Discretization.lean](#), theorems `bin_mono`, `bin_of_mem`, `bin_eq_of_same_cell`, `card_bins_le`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2.

RELATED

[quotient](#) (p. 279); [standardization](#) (p. 348); [attribute type](#) (p. 28); [bit](#) (p. 43); [quantization](#) (p. 275).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 2.1, 0.12a.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.1, chapter 2 section 2.3.

disparate impact ratio

concept. Entry id disparate-impact-ratio.

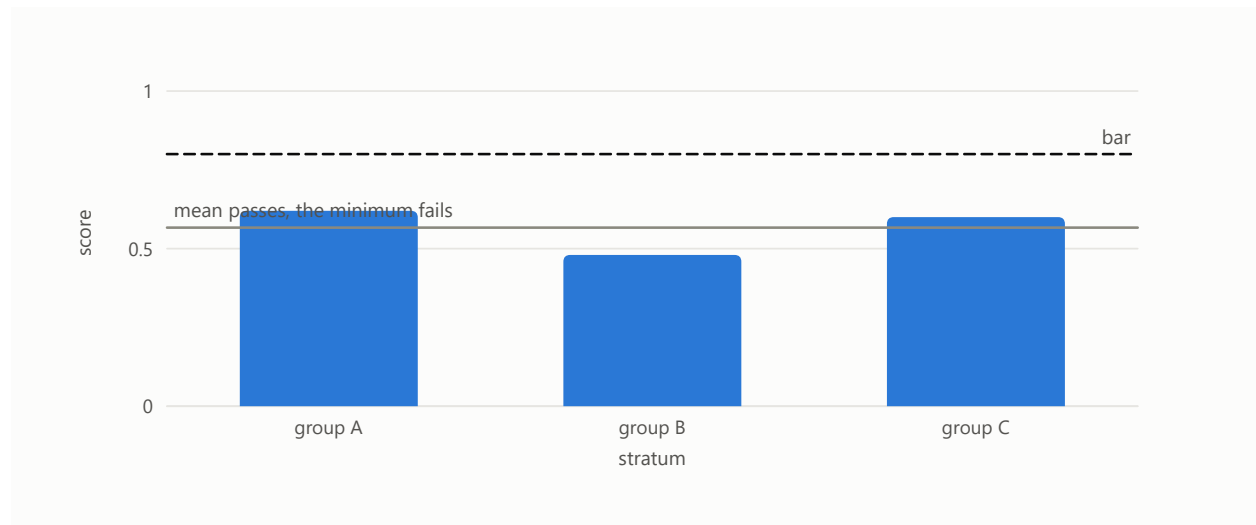


Figure 70: The positive rate of one group over another's.

DEFINITION

The rate of a favourable decision in one group divided by the rate in another. Equation 0.37.

Example. Positive rates of 0.48 for one group and 0.60 for another give a ratio of 0.8.

EQUATION

Book equation 0.37.

$$DI = \frac{\Pr[\hat{y} = 1 \mid g = a]}{\Pr[\hat{y} = 1 \mid g = b]}, \quad \text{equalized odds : } TPR_a = TPR_b, \text{ FPR}_a = \text{FPR}_b.$$

Book equation 14.5.

$$FC_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} FC_g, \quad \text{ABSTAIN otherwise.}$$

CONDITIONS

- The ratio of the favourable-decision rate in one group to the rate in another. It is one exactly when the rates agree, it inverts when the groups are swapped, and the four-fifths rule is a statement about the rate gap.
- The book's verdict is per group with a null, the worst group's false-clear rate over the groups large enough to score, which clears a bar exactly when every scored group does and is never below the pooled rate.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.18 and chapter 14 section 14.6 of *Data Mining as Observation*, with the per-group verdict of equation 14.5.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/DisparateImpact.lean](#), theorems `ratio_eq_one_iff`, `ratio_swap`, `four_fifths_iff`, `verdict_le_iff`, `weighted_le_verdict`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 14.

RELATED

[equalized odds](#) (p. 130); [min-over-strata](#) (p. 226); [false-clear rate](#) (p. 141); [abstention](#) (p. 12).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 14 section 14.1.

distance concentration

concept. Entry id distance-concentration.

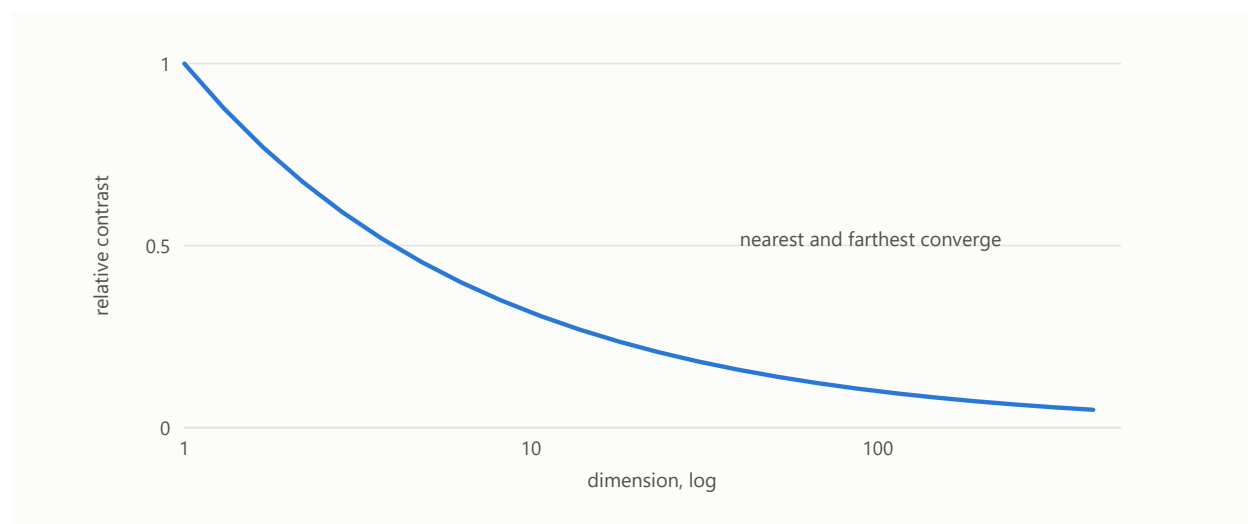


Figure 71: Nearest and farthest converge as the dimension grows.

DEFINITION

The narrowing of the spread of pairwise distances as dimension grows, read in the book as the reader running out of resolution. Chapter 3.

Example. In 500 dimensions the nearest and the farthest of 1000 random points differ in distance by a few percent.

EQUATION

none

CONDITIONS

- When a squared distance is a sum of independent coordinate contributions with a common mean and variance, its relative spread is the variance over the dimension times the squared mean, which falls with the dimension and tends to zero. The independence is the model's assumption.
- The book reads the narrowing as a property of the reader, the identity reader on all coordinates, and not of the data. A consumer that reads a low-dimensional subspace does not see it, which is why the effective rank and not the ambient dimension is the number that matters.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

The classical result on nearest neighbours in high dimension, as TSK chapter 2 presents it, and chapter 3 section 3.4 of *Data Mining as Observation*, where it is read as the reader running out of resolution.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/DistanceConcentration.lean](#), theorems `relSpread_eq`, `relSpread_antitone`, `relSpread_tendsto_zero`, `exists_dim_relSpread_lt`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 3.

RELATED

[hubness](#) (p. 176); [effective rank](#) (p. 123); [read subspace](#) (p. 293); [rank certificate](#) (p. 285).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.7.

Ledger rows that cite the entry's records without naming it: NEG-11.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.5, chapter 4 section 4.2.

distortion

concept. Entry id distortion.

DEFINITION

The error a code costs a reader, the trace of the read operator against the error covariance. For the identity reader it is the reconstruction error. Chapter 4 section 4.2.

Example. Two codes with the same total error 2.0 cost a reader at 15 degrees 0.394 and 1.606, four to one.

EQUATION

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

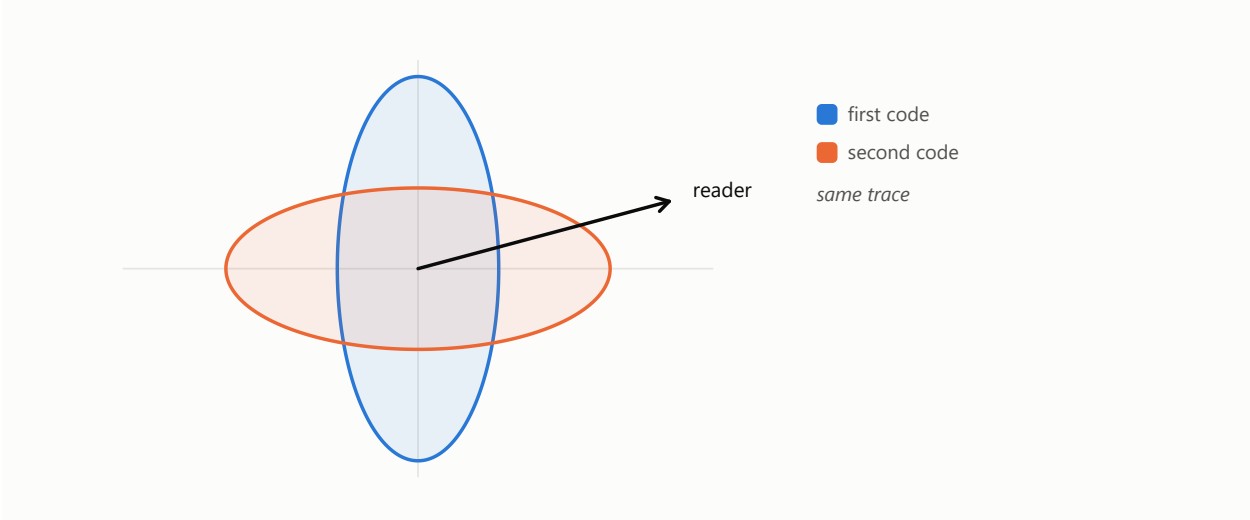


Figure 72: The error a code costs a reader, which two readers rank differently.

CONDITIONS

- The error a code costs a reader, the trace of the read operator against the error covariance. For the identity reader it is the reconstruction error, and for a reader with one direction it is the error along that direction alone, so two codes with equal reconstruction error can differ four to one for a reader at fifteen degrees.
- At matched bits the downstream preservation is not controlled by reconstruction and is controlled by the consumer-projected error, twelve of twelve, and read distortion controls but is not a complete rank statistic.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 1 section 1.4 and chapter 4 section 4.2 of *Data Mining as Observation*, with the read distortion in `geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:49`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:49-75
chapter 3 section 3.2	traces 2.0, diagonals 0.3 and 1.7, consumers at 15 and 75 degrees, distortion 0.39 vs 1.61, 4.1 to one, computed not drawn	<code>observation-theory/assets/make_flip_figure.py:4-24,107-119</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadDistortion.lean](#), theorems `read_distortion`, `identity_reader`, `quad_one`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Quantization.lean](#), theorems `error_le_half_step`, `quantize_level`, `half_step_bound`, `sq_error_le`, `finer_step`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 7, 8, 9, 11, 13.

RELATED

[read distortion](#) (p. 290); [reconstruction error](#) (p. 298); [identity reader](#) (p. 180); [quantization](#) (p. 275).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 2.1, 0.11.

Ledger rows that cite the entry's records without naming it: OT-7, GO-2 (neg. half: not reconstruction), GO-6.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.2.

dot product

concept. Entry id dot-product.

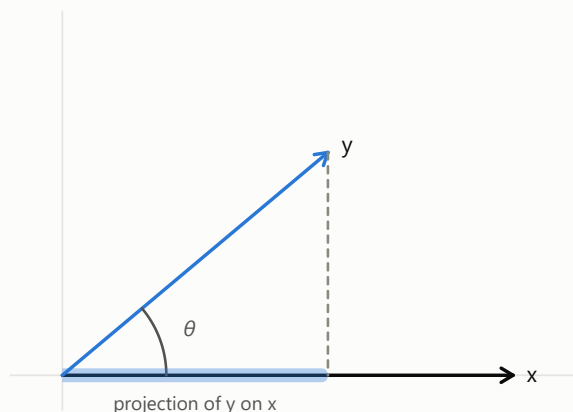


Figure 73: The sum of the products of matching coordinates.

DEFINITION

The sum of the coordinatewise products of two vectors. Equation 0.1.

Example. $(1, 2) \cdot (3, 4) = 3 + 8 = 11$.

EQUATION

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

CONDITIONS

- The sum of the coordinatewise products of two vectors, symmetric and bilinear. The cosine, the dot product over the two lengths, lies between minus one and one by Cauchy–Schwarz and is unchanged when either vector is scaled by a positive factor, while the dot product scales with the vector.
- A cosine reader has declared length a nuisance and a dot-product reader has not, which is the reader difference of chapter 3, and cosine 0.995 between keys and their reconstruction said nothing about the softmax that read them.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.1 and chapter 3 section 3.2 of *Data Mining as Observation*, with the program’s cosine-versus-consumer case in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/DotProduct.lean](#), `theorems self_nonneg, dot_comm, dot_sq_le, cosine_mem, dot_smul, cosine_smul`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 8, 11, 12.

RELATED

[projection](#) (p. 272); [Euclidean distance](#) (p. 133); [quotient](#) (p. 279); [nuisance](#) (p. 240).

SEE ALSO

Ledger rows that cite the entry’s records without naming it: NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 8 section 8.2, chapter 11 section 11.1, chapter 11 section 11.2.

drift

concept. Entry id `drift`.

DEFINITION

A change over time in what a consumer reads or in the data it reads. Chapter 8 tests a drift claim against its null and finds half of it was noise.

Example. A read operator measured at rank one moved to correlation 0.667 with its earlier self, against a null of 0.933.

EQUATION

Book equation 13.1.

$$\text{FC} = \Pr[W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr}(P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

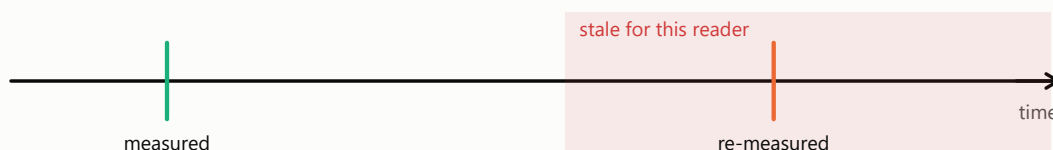


Figure 74: The read operator moving over time, so an old certificate no longer describes the reader.

CONDITIONS

- A drift is a change over time in what a consumer reads or in the data it reads, and its damage is the read distortion of the drift, so it is consumer-relative. The same drift is read as its full squared size by one consumer and as nothing by another, and a drift confined to the nuisance is read as zero however large.
- The operator-drift claim was tested against a paired null and half of it was noise. The rank-one drift held in fourteen of sixteen cells on one model, and the refresh intervention derived from it was refuted, which the ledger carries.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects*. OT-4 [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [geometric-observation/claims/LEDGER.md:36](#).
- *refutes or corrects*. OT-11 [void]. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [geometric-observation/claims/LEDGER.md:48](#).

FIRST STATED

Volume 14, chapter 19, [geometric-observation/chapters/ch19_the_certificate_that_ages.md](#), DOI 10.5281/zenodo.21776291, and the operator-drift record in readscope, [readscope/calibration/records/c11c-operator-drift.json](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.9	drift at rank one 0.667 vs null 0.933, sixteen cells	readscope/CALIBRATION.md:600-660 F-24 ; readscope/calibration/records/c11c-operator-drift.json

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [geometric-observation/claims/LEDGER.md:36](#).
- OT-11, [void]. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [geometric-observation/claims/LEDGER.md:48](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/Drift.lean](#), theorems `damage_rank_one`, `same_drift_two_consumers`, `damage_eq_zero_of_nuisance`, `damage_nonneg`, at [observation-data-mining 95af30c](#); what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2, 8, 11, 13, 14.

RELATED

[coherence time](#) (p. 67); [freshness](#) (p. 156); [certificate](#) (p. 59); [read distortion](#) (p. 290).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.10.

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.4, chapter 11 section 11.9, chapter 13 section 13.2.

E

early stopping

instrument. Entry id early-stopping.

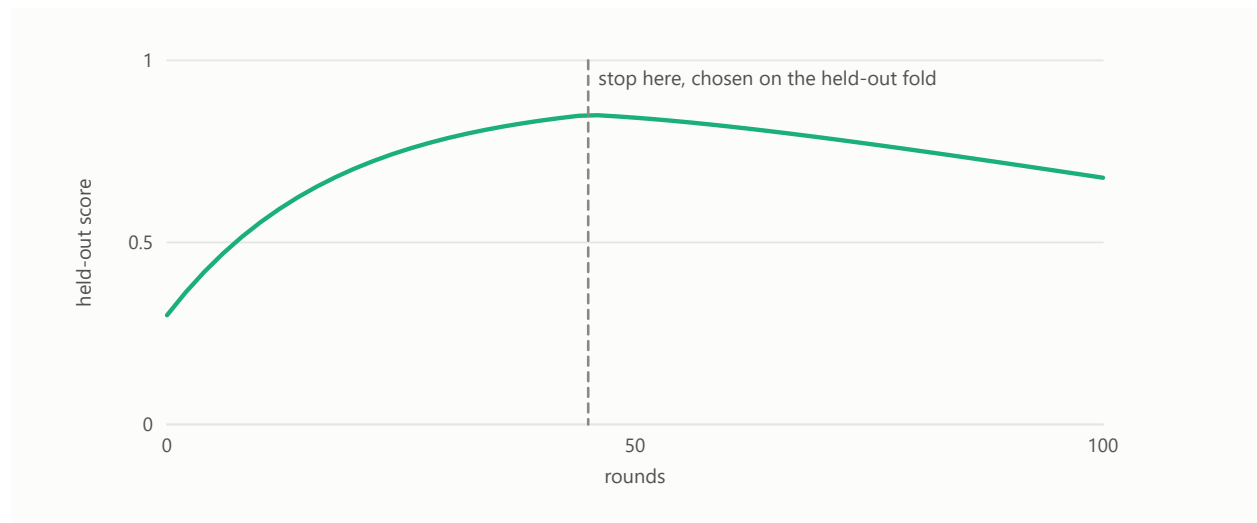


Figure 75: Halting a boosting run when a held-out score stops improving.

DEFINITION

Halting a boosting run when a held-out score stops improving. The rule reads the held-out fold, so it is a harness and the round count is a fitted parameter. Chapter 7 section 7.5.

Example. Boosting for 200 rounds and keeping round 45, where the held-out score peaked, makes 45 a fitted parameter.

EQUATION

none

CONDITIONS

- Halting a boosting or gradient run when a held-out score stops improving. The stopping rule reads the held-out fold, so the fold it reads is no longer held out for the reported score, and a scorer that has read the test fold scores it perfectly.
- Early stopping is a harness. The round count it picks is a fitted parameter, and the report states which fold picked it and which fold scored it.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 7 section 7.5 of *Data Mining as Observation*, after ESL 10.12.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Leakage.lean](#), theorems `errors_lookup_eq_zero`, `lookup_default`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/CrossValidation.lean](#), theorems `sizes_sum`, `accuracy_weighted`, `accuracy_mean_of_equal`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 7.

RELATED

[harness](#) (p. 168); [leakage](#) (p. 210); [cross-validation](#) (p. 96); [boosting](#) (p. 48); [split](#) (p. 346).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.1, 8.2.

Ledger rows that cite the entry's records without naming it: NEG-4.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.4, chapter 6 section 6.5, chapter 7 section 7.2.

effective rank

concept. Entry id effective-rank.

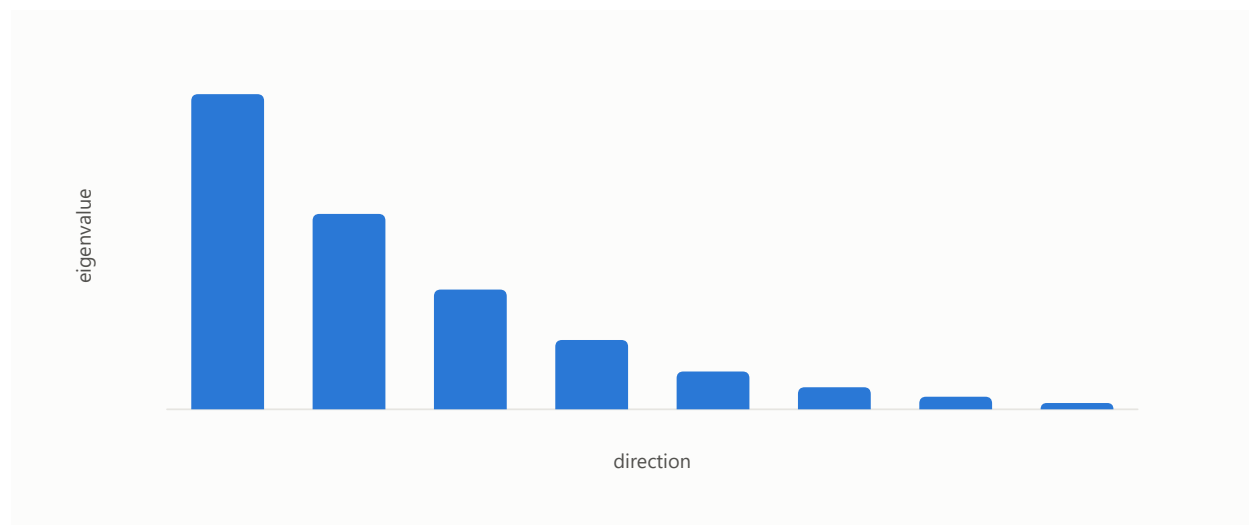


Figure 76: The number of directions a spectrum really uses, with no cutoff to choose.

DEFINITION

The number of directions a matrix really uses, the square of the sum of its eigenvalues over the sum of their squares, defined for a nonnegative spectrum that is not all zero, with no cutoff to choose. Equation 0.7.

Example. Eigenvalues (1, 1, 1, 1) have effective rank 4, and (1, 0.12, 0.06, 0.02) have effective rank about 1.4.

EQUATION

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

CONDITIONS

- The effective rank is the square of the sum of the eigenvalues over the sum of their squares, between one and the dimension for a nonnegative spectrum, with no cutoff to choose.
- It is not invariant under a change of basis of the input, unlike the trace pairing and the rank, and the ledger row that says so is the one that licenses comparing read operators across coordinate systems.
- An allocation over a spectrum with effective rank below two concentrates on one direction, and the allocation report says so before giving a gain.
- It is defined for a nonnegative spectrum that is not all zero, and it lies between one and the dimension.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. OT-7 [demonstrated]. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-T}PA^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and ... [geometric-observation/claims/LEDGER.md:30](#).

FIRST STATED

The participation ratio of a spectrum, applied to read operators in `readscope`, `readscope/readscope/spectrum.py:35-70`, and chapter 0 section 0.4 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	effective rank as participation ratio, energy rank	readscope/readscope/spectrum.py:35-70
chapter 4 section 4.2	allocation report, gain over uniform, concentration caution below effective rank 2	turboquant-pro/turboquant_pro/read_allocation.py:244-307
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	gtc-prototype/docs/SPECTRUM_FINDINGS.md:10-18

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/EffectiveRank.lean](#), theorems `sq_sum_le`, `effRank_le`, `sum_sq_le_sq_sum`, `one_le_effRank`, `effRank_const`, `effRank_single`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 4, 9, 11, 14.

RELATED

[read operator](#) (p. 291); [read subspace](#) (p. 293); [water-filling](#) (p. 378); [recognizer](#) (p. 296).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 4.2.

Sources-table rows that share a record with the entry without naming it: chapter 14 section 14.3, chapter 14 section 14.5, chapter 14 section 14.7.

eigenvalue, eigenvector

concept. Entry id eigenvalue-eigenvector.

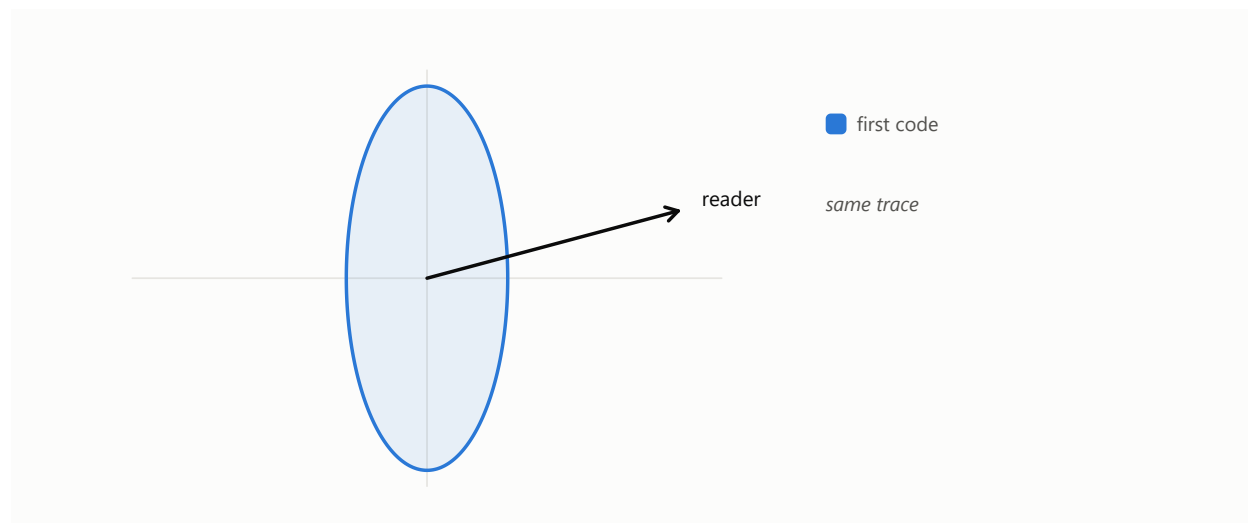


Figure 77: The directions a symmetric matrix stretches along, and by how much.

DEFINITION

A direction a symmetric matrix only stretches, and the factor by which it stretches it. The eigenvectors of a covariance are its principal directions. Equation 0.5.

Example. $\text{diag}(0.3, 1.7)$ has eigenvalues 0.3 and 1.7 with eigenvectors along the two axes.

EQUATION

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

CONDITIONS

- A direction a symmetric matrix only stretches, and the factor by which it stretches it. Eigenvectors with distinct eigenvalues are orthogonal, every eigenvalue of a semidefinite matrix is nonnegative, and the quadratic form along an eigenvector is the eigenvalue times the squared length.
- The eigenvectors of a covariance are its principal directions, the basis in which chapter 4 pairs the covariance with the read operator, and the eigenvalues of a Laplacian are what the recognizer reads.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.4 of *Data Mining as Observation*, with the program's spectrum records in readscope and the recognizer battery.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Eigen.lean](#), theorems `pairing_symm`, `orthogonal_of_ne`, `eigenvalue_nonneg`, `quad_eigen`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 4, 9, 11.

RELATED

[covariance matrix](#) (p. 91); [effective rank](#) (p. 123); [whitening](#) (p. 381); [recognizer](#) (p. 296); [Laplacian](#) (p. 207).

SEE ALSO

Ledger rows that cite the entry's records without naming it: OT-7.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.2, chapter 9 section 9.4, chapter 14 section 14.3, chapter 14 section 14.5, chapter 14 section 14.7.

embedding

concept. Entry id embedding.

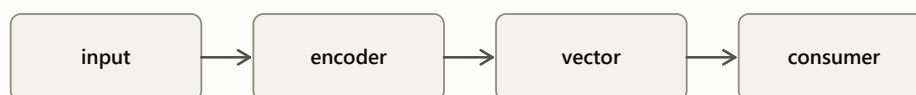


Figure 78: A row mapped to a vector by an encoder whose quotient is learned.

DEFINITION

A vector assigned to an object, a word, a sentence, a node, a document, so that nearness in the vector space stands for a relation between the objects. Chapter 11.

Example. A sentence becomes a vector of 768 numbers, and two paraphrases become vectors with cosine near one.

EQUATION

none

CONDITIONS

- A vector assigned to an object so that nearness in the vector space stands for a relation between the objects. Objects with the same embedding are indistinguishable to every consumer of the embedding, so the embedding is a quotient, and a cosine reader takes a further quotient in which the nearest neighbour is unchanged when the query is rescaled, where a dot-product reader does not.
- What an embedding is worth to a consumer is the consumer's number. The keys that reconstructed at cosine 0.995 and raised the perplexity by three orders of magnitude are the case, and the rank certificate is the instrument that says which neighbour rankings a compressed embedding preserved.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* GO-B-legal (035→036) [predicted]. Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [geometric-observation/claims/LEDGER.md:119](#).

FIRST STATED

Chapter 11 section 11.1 of *Data Mining as Observation*, with the program's embeddings in the legal-citation flip, the GloVe corpus of readscope, and the attention keys of the KV finding.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Embedding.lean](#), theorems quotient, cosine_scale_free, nearest_scale_free, dot_not_scale_free, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 3, 4, 8, 9, 10, 11, 12, 13, 14.

RELATED

[encoder](#) (p. 127); [dot product](#) (p. 118); [rank certificate](#) (p. 285); [hubness](#) (p. 176); [the flip](#) (p. 147).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.1, 11.2.

Ledger rows that cite the entry's records without naming it: NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 4 section 4.5, chapter 8 section 8.2, chapter 11 section 11.1, chapter 11 section 11.2.

encoder

concept. Entry id encoder.

DEFINITION

A model that maps an input to an embedding. Chapter 12.

Example. LaBSE maps a sentence in any of its languages to a 768-dimensional vector.

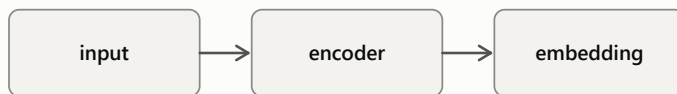


Figure 79: A model that maps an input to an embedding.

EQUATION

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

Book equation 14.1.

$$S = \frac{\sum_i w_i s_i}{\sum_i w_i}, \quad w_i = \max(0, 2 \cdot \text{AUROC}_i - 1).$$

CONDITIONS

- A model that maps an input to an embedding. An encoder votes only when its held-out AUROC on a corpus it was not trained on clears both nulls by the preregistered margin, its authority is its reliability weight, and a strictly monotone recalibration of its score changes neither the verdict nor the weight.
- The rights encoder fell below the untrained baseline and was retired rather than tuned, and the legal-citation encoder's flip is the ledger's demonstrated row.
- An encoder's held-out AUROC depends on the orientation of its score. A score that ranks the wrong way scores 1 minus A, and the gate reads the score as the encoder produces it and not the better of the two orientations.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 12 section 12.5 of *Data Mining as Observation*, with the program's encoders in `xbse` and the calibrated authority of `gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.6	rights encoder 0.467 below untrained baseline, the AUROC lesson, cross-corpus same-sign fix	xbse/README.md:160-172; xbse/experiments/rights_r6_summary.json

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/CrossCorpusGate.lean](#), theorems `clears_bow`, `clears_untrained`, `not_validated_of_saturated`, `margin_example`, `validated_comp`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 7, 8, 9, 10, 11, 12, 14.

RELATED

[embedding](#) (p. 126); [cross-corpus gate](#) (p. 94); [reliability weight](#) (p. 306); [retrieval-augmented pipeline](#) (p. 312).

SEE ALSO

Ledger rows that cite the entry's records without naming it: GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 12 section 12.5, chapter 14 section 14.1, chapter 14 section 14.2.

ensemble

instrument. Entry id ensemble.

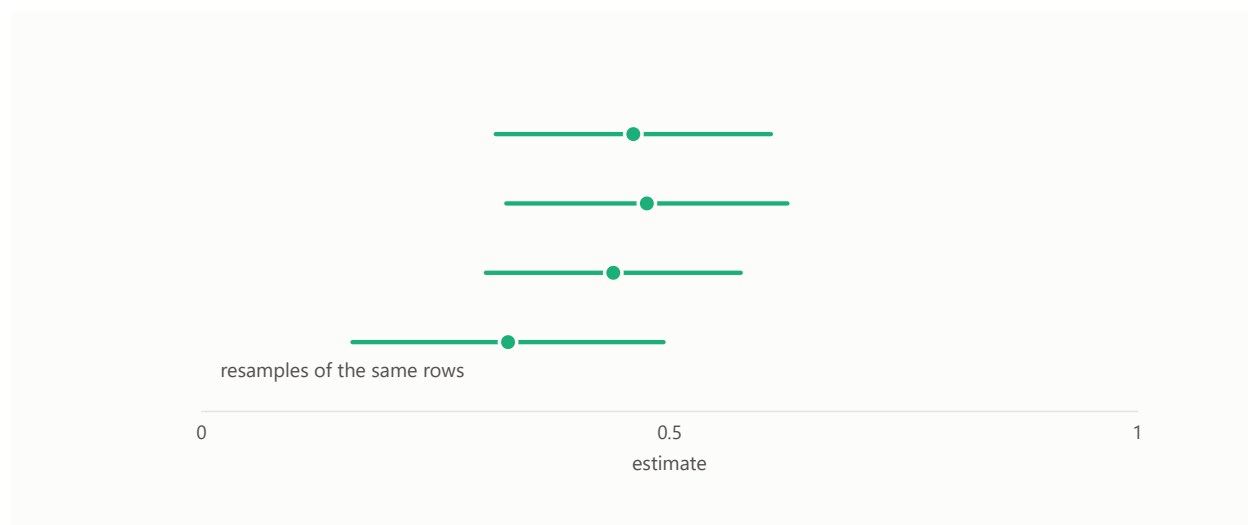


Figure 80: An average of several scorers, between its members.

DEFINITION

A scorer that averages or votes several scorers. Its average lies between its members and its squared error is at most their mean squared error. Chapter 7.

Example. Scores 0.6, 0.7, and 0.8 average 0.7, and the average's squared error is at most the mean of the three

squared errors.

EQUATION

none

CONDITIONS

- A scorer that averages or votes several scorers. Its average lies between the smallest and the largest member, its squared error is at most the mean of the members' squared errors, and a constant ensemble is its constant. That arithmetic is why bagging reduces variance.
- Whether an ensemble beats a formula is a measured question and not a principle, and the comparison is fair only inside one fold protocol with the Nadeau and Bengio correction applied, where nine wins to three became three wins, seventeen ties, and eleven losses.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Breiman, bagging predictors, 1996, as chapter 7 section 7.1 of *Data Mining as Observation* reads it, with the fair comparison in `constraint-gap/review/INDETERMINATES.md:1-40`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.1	bagging variance, out-of-bag estimation, random forests, importances, AdaBoost	Hastie, Tibshirani, Friedman, ESL 2e chapters 15 and 10.1; TSK 2e 4.10

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Ensemble.lean](#), theorems `average_between`, `sq_average_le`, `average_const`, `at_observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 6, 7.

RELATED

[boosting](#) (p. 48); [decision tree](#) (p. 102); [standard error](#) (p. 347); [Nadeau and Bengio correction](#) (p. 233).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 6.1, 0.18, 8.2.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.3, chapter 7 section 7.3, chapter 7 section 7.4.

equalized odds

concept. Entry id `equalized-odds`.

DEFINITION

The requirement that the true-positive rate and the false-positive rate be the same across groups. Equation 0.37.

Example. True positive rates of 0.90 and 0.88 with false positive rates of 0.10 and 0.11 across two groups are nearly equalized odds.

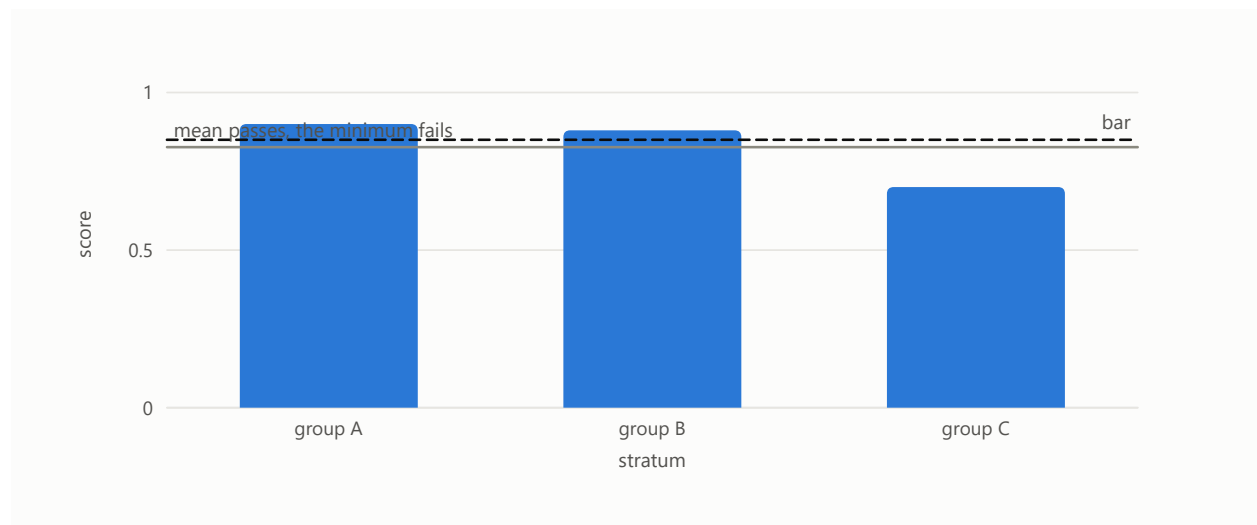


Figure 81: Equal true and false positive rates across groups.

EQUATION

Book equation 0.37.

$$DI = \frac{\Pr[\hat{y} = 1 \mid g = a]}{\Pr[\hat{y} = 1 \mid g = b]}, \quad \text{equalized odds : } TPR_a = TPR_b, FPR_a = FPR_b.$$

CONDITIONS

- Equal true-positive and false-positive rates across groups. A group's favourable rate is its true-positive rate times its base rate plus its false-positive rate times the rest, so under equalized odds the favourable rates agree exactly when the base rates do, given that the classifier separates at all.
- Equalized odds and parity of outcomes therefore conflict whenever the base rates differ. With rates 0.8 and 0.2 and base rates one half and one tenth the favourable rates are 0.5 and 0.26, a ratio below four fifths.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.18 and chapter 14 section 14.6 of *Data Mining as Observation*, where fairness cannot be had three ways.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/EqualizedOdds.lean](#), theorems `parity_of_equal_base`, `favourable_sub`, `parity_iff_equal_base`, `example_conflict`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 10, 12, 14.

RELATED

[disparate impact ratio](#) (p. 114); [calibration](#) (p. 56); [min-over-strata](#) (p. 226); [certificate](#) (p. 59).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 14 section 14.1.

escalation

concept. Entry id escalation.



Figure 82: A verdict handed up when the certificate cannot clear it.

DEFINITION

The decision to hand an item to a person rather than decide it, the single-decision form of abstention. Chapter 14.

Example. A decision the certificate cannot clear at its bar is handed to a person, and the count of escalations is reported.

EQUATION

none

CONDITIONS

- The decision to hand an item to a person rather than decide it. A rule that escalates every item whose margin is below a bar decides a set that can only shrink as the bar rises, so coverage can only fall, every decided item carries at least the bar's margin, and a bar of zero escalates nothing.
- Coverage times the decided false-clear rate is the fraction of all items cleared and wrong, so an escalation policy trades coverage for a false-clear rate and a deployment report states both.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 14 section 14.9 of *Data Mining as Observation*, the single-decision form of abstention, with the refusal counts of the re-description test.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Escalation.lean](#), theorems `decided_anti`, `coverage_anti`, `decided_margin`, `decided_zero`, `joint_rate`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 8, 9, 10, 11, 14.

RELATED

[abstention](#) (p. 12); [margin](#) (p. 221); [coverage](#) (p. 92); [false-clear rate](#) (p. 141); [certificate](#) (p. 59).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 14.5, 10.7, 0.26.

Euclidean distance

concept. Entry id euclidean-distance.

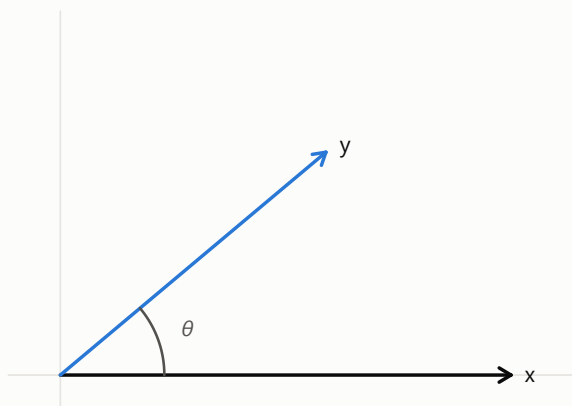


Figure 83: The length of the difference of two vectors.

DEFINITION

The straight-line distance between two vectors, which reads every coordinate at the scale it arrives in. Chapter 3.

Example. The distance between $(0, 0)$ and $(3, 4)$ is 5.

EQUATION

none

CONDITIONS

- The straight-line distance between two vectors, whose square is the identity reader's quadratic form on the difference, so it reads every coordinate at the scale it arrives in. It is symmetric and zero exactly between a row and itself.

- Rescaling one coordinate changes which of two rows is nearer. A choice of scale is a choice of reader, and reconstruction error, the Euclidean distance between a row and its code, is the identity reader's number and not the consumer's.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 3 section 3.2 of *Data Mining as Observation*, where a distance is a choice of what to ignore.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/EuclideanDistance.lean](#), theorems `distSq_eq_quad_one`, `distSq_comm`, `distSq_eq_zero_iff`, `ranking_flips`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 6, 10, 11, 12.

RELATED

[identity reader](#) (p. 180); [dot product](#) (p. 118); [Mahalanobis distance](#) (p. 219); [distance concentration](#) (p. 115).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.1, 4.1.

Ledger rows that cite the entry's records without naming it: NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 8 section 8.2, chapter 11 section 11.1, chapter 11 section 11.2.

eviction

concept. Entry id eviction.

DEFINITION

Choosing what to drop when a cache is full. Chapter 13.

Example. A cache of 4 entries receiving a fifth drops the least recently used one.

EQUATION

none

CONDITIONS

- Choosing what to drop when a cache is full. Evicting tokens replaces attention over all tokens with attention over the kept subset, so the output lies between the kept values, equals them when they agree, and the whole cache is the case of no eviction.
- Which tokens can go is consumer-relative, since a head reads a key only through its score, and the serving-stack measurement of ninety-seven percent eviction at a five percent keep against an oracle is the ledger's number, after one attempt whose control did not exist.

Conditions are curated in `entries.toml` rather than read from a record.

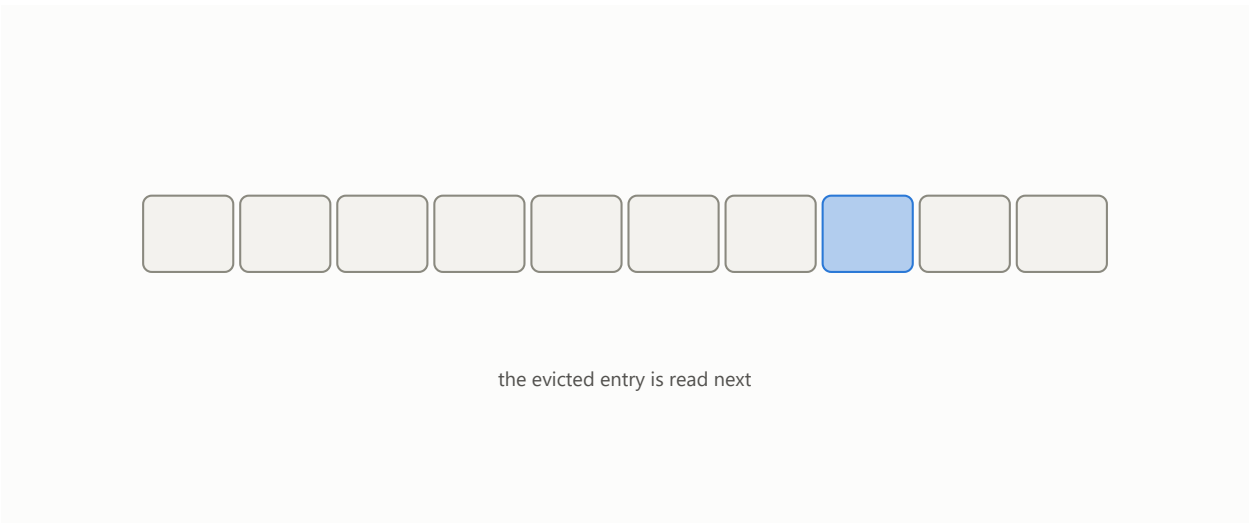


Figure 84: Choosing what to drop when the cache is full.

LEDGER

- *measures.* GO-2/GO-12/GO-13 operational (KV serving, 077) [demonstrated]. Consumer-relative access width measured on a production serving stack (Qwen2.5-7B KV-cache eviction, matched budget): task quality tracks measured predictive uncertainty u about the consumer’s future reads, not nominal scorer width — the ... [geometric-observation/claims/LEDGER.md:81](#).

FIRST STATED

Chapter 13 section 13.6 of *Data Mining as Observation*, with the program’s serving-stack case in [geometric-observation/prereg/GO-P-2026-077-kv-consumer-relative-eviction.md](#) and the ledger row that carries it.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 13 section 13.6	attempt three, windows 1024, 256, 32, uncertainty 0.982 to 0.892, 5 of 6, 0.4375 with SE 0.070 at 5 percent keep vs 0.30, 97 percent eviction, oracle-miss 0.370 vs 0.25, V4 0.078 vs 0.0625, contrast 0.359 with SE 0.068, n 64, seed 20260812, 89 duty cycles	geometric-observation/claims/LEDGER.md row GO-2/GO-12/GO-13 operational; geometric-observation/prereg/GO-P-2026-077-kv-consumer-relative.md ; geometric-observation/results/G013-kvaw2-governed.json

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Eviction.lean](#), theorems kept_weights_sum, keptOutput_le_max, min_le_keptOutput, keptOutput_const, keptOutput_univ, at observation-data-mining 95af30c; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 13.

RELATED

[KV cache](#) (p. 200); [attention](#) (p. 26); [coherence time](#) (p. 67); [budget](#) (p. 51).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 13.1, 0.23.

Ledger rows that cite the entry's records without naming it: GO-12.

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.6.

expected calibration error

instrument. Entry id expected-calibration-error.

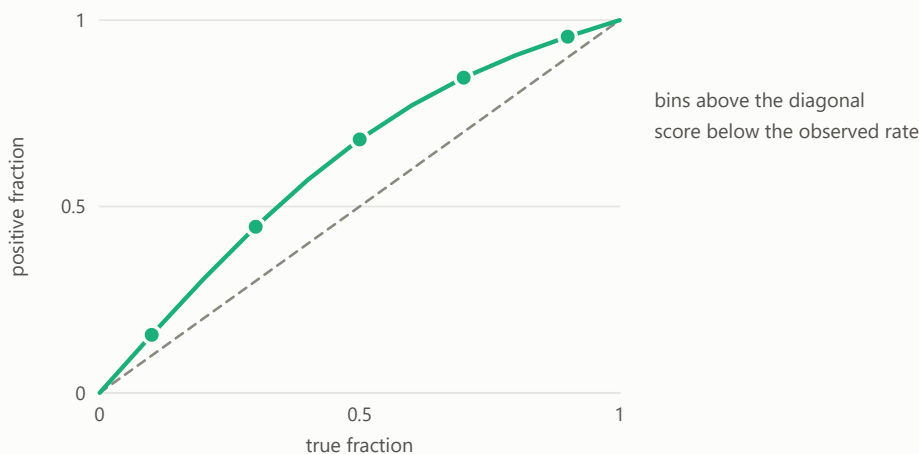


Figure 85: The bin-weighted gap between the mean score and the fraction of positives.

DEFINITION

The average over score bins of the absolute difference between the mean score and the fraction of positives in the bin. Equation 0.29.

Example. Bins with mean scores 0.2 and 0.8 and positive fractions 0.3 and 0.7, each holding half the rows, give an error of 0.1.

EQUATION

Book equation 0.29.

$$\text{ECE} = \sum_b \frac{n_b}{n} |\bar{s}_b - \bar{y}_b|.$$

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

CONDITIONS

- The average over score bins, weighted by bin size, of the absolute difference between the bin's mean score and its fraction of positives. It lies in the unit interval, it is zero exactly when every bin's mean score equals its positive fraction, and a calibrated scorer has error zero.

- The program’s validated encoders carry errors of 0.018 to 0.101 on held-out splits against raw values up to 0.223, and the reliability weight that gates an encoder’s authority is computed beside it.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.14 of *Data Mining as Observation*, equation 0.29, with the program’s per-encoder errors in `xbse/README.md:175-195` and `gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 12 section 12.5	ECE 0.018 to 0.101 vs raw up to 0.223, reliability weight, audit binding	<code>xbse/README.md:175-195</code> ; gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Calibration.lean](#), theorems `ece_nonneg`, `ece_le_one`, `ece_eq_zero_iff`, `ece_of_calibrated`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[calibration](#) (p. 56); [reliability weight](#) (p. 306); [AUROC](#) (p. 32); [threshold](#) (p. 362).

SEE ALSO

Book equations stated beside the entry’s terms, not defining it: 14.2.

Ledger rows that cite the entry’s records without naming it: NEG-4.

Sources-table rows that share a record with the entry without naming it: chapter 14 section 14.2.

explained variance

concept. Entry id explained-variance.

DEFINITION

The fraction of total variance retained by a set of principal components, the identity reader’s criterion for a reduction. Chapter 4.

Example. Keeping 2 of the eigenvalues 4, 3, 2, and 1 explains 7 of 10, or 70 percent.

EQUATION

Book equation 4.1.

$$d_O = \text{tr}(P_C M_\delta) \quad \text{against} \quad \text{tr } M_\delta = d_O|_{P_C=I}.$$

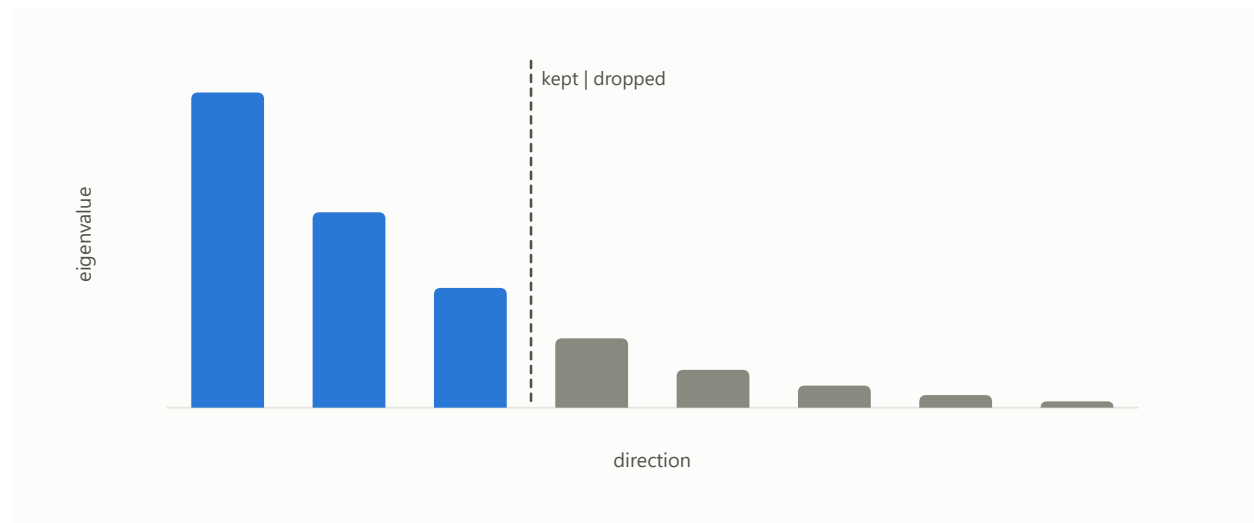


Figure 86: The fraction of the spectrum the kept components carry.

CONDITIONS

- The fraction of total variance the first k components carry. It lies in the unit interval, cannot fall as k grows, and reaches one at the full dimension. With unit sensitivities it is the fraction a consumer retains, which is why it is the identity reader's criterion.
- A consumer that reads other directions retains a different fraction. With variances 9 and 1 and a consumer reading only the second direction, keeping the first component explains nine tenths of the variance and none of what the consumer reads, which is the flip in its smallest form.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

TSK appendix B on principal component analysis, as chapter 4 section 4.1 of *Data Mining as Observation* reads it, the identity reader's criterion for a reduction, with the program's spectrum record in `gtc-prototype/docs/SPECTRUM_FINDINGS.md:10-18`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	gtc-prototype/docs/SPECTRUM_FINDINGS.md:10-18

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ExplainedVariance.lean](#), theorems `explained_mem_unit`, `explained_mono`, `explained_full`, `retained_identity`, `retained_example`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 4, 8.

RELATED

[identity reader](#) (p. 180); [the flip](#) (p. 147); [effective rank](#) (p. 123); [water-filling](#) (p. 378).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.7.

Ledger rows that cite the entry's records without naming it: GO-1.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.2, chapter 11 section 11.7, chapter 14 section 14.3, chapter 14 section 14.5, chapter 14 section 14.7.

F

F1

concept. Entry id f1.

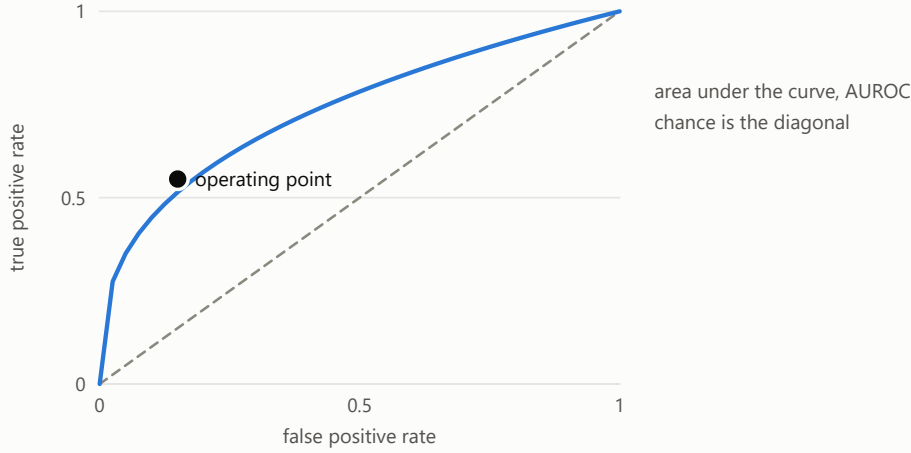


Figure 87: The harmonic mean of precision and recall at a threshold.

DEFINITION

Twice the product of precision and recall over their sum. The optimal thresholded F1 of a score is the best F1 over every threshold and depends only on the score's ranking. Equation 0.28.

Example. Precision 0.6 and recall 0.75 give F1 of 2 times 0.45 over 1.35, or 0.667.

EQUATION

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y) = \max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(f(X), \tau)], y)$$

for every strictly monotone g .

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (TPR - FPR), \quad \pi = \text{prevalence},$$

$$J \leq 2A - 1 \text{ when the ROC curve is concave, and not in general.}$$

CONDITIONS

- Twice the product of precision and recall over their sum. The optimal thresholded F1 of a score is the best F1 over every threshold in both directions, and it depends only on the score's ranking, so a strictly monotone transform of the score leaves it unchanged.

- Its ceiling from the Youden index holds for every score, and the AUROC form of that ceiling only for concave ROC curves, which is the theory-radar erratum.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.14 and chapter 6 section 6.3 of *Data Mining as Observation*, with the program's optimal thresholded F1 in theory-radar, DOI 10.5281/zenodo.20660206.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	Monotone Invariance and AUROC invariance theorems, the AUROC to F1 bound via the Youden index, stated in the source for every ROC curve and corrected in the book to concave curves, with the counterexample at AUROC 0.75 and F1 0.857	theory-radar/paper/astar_paper.tex:94-170 ; theory-radar/paper/theory_radar_paper.tex:290-304

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/MonotoneInvariance.lean](#), theorems `aurocNum_comp`, `auroc_comp`, `predicted_comp`, `predictedBelow_comp`, `sweptF1_comp`, `sweptF1Below_comp`, `optF1_comp`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 5, 6, 7.

RELATED

[precision, recall](#) (p. 266); [Monotone Invariance Theorem](#) (p. 227); [Youden F1 bound](#) (p. 385); [formula search](#) (p. 154); [threshold](#) (p. 362).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.3.

false-clear rate

concept. Entry id false-clear-rate.

DEFINITION

The fraction of the decisions a certificate cleared in which the witness said otherwise, conditional on clearing and reported beside the certificate's coverage. Equation 0.26.

Example. Of 1000 clearances the witness refuted 206, a false-clear rate of 0.206 at whatever coverage the certificate had.

EQUATION

Book equation 0.26.

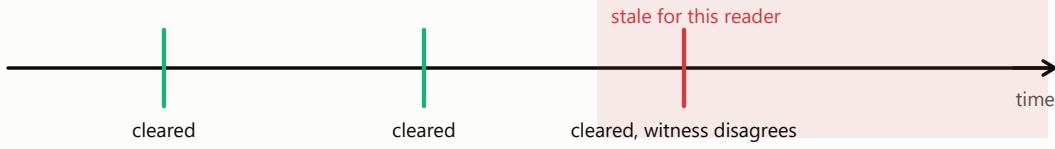


Figure 88: How often a clearance was wrong, conditional on clearing.

$$FC = \Pr [W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

Book equation 13.1.

$$FC = \Pr [W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr} (P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

Book equation 12.4.

$$FC_{\text{naive}} = \Pr [\text{acc}_s < \text{target} \mid \text{benchmark} \geq \text{target}] = \frac{1}{S} \sum_{s=1}^S \mathbf{1}[\text{acc}_s < \text{target}]$$

when the benchmark clears, so that the naive certificate's coverage is 1.

CONDITIONS

- A witness must exist. Where none exists the rate is not measurable and the certificate is unassessed, not safe.
- The rate is per consumer, given per read operator, slice, group, or footprint, never as one aggregate.
- Coverage is reported with it. A witnessed certificate clears fewer decisions than a naive one, and the two numbers are read together.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14 chapter 19, `geometric-observation/chapters/ch19_the_certificate_that_ages.md:61-76`, and the freshness program's umbrella document `observation-theory-campaigns/experiments/FRESHNESS-PROGRAM.md`, re-homed into the campaigns repository on 2026-08-24.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.6	XPROTO-LLM, benchmark 0.909, 0.920, 0.909, thirty slices, target 0.8, naive 0.333, aware 0.033, spread 0.380, deployment mean 0.736, six bars on three seeds, sealed 2026-08-25 at b61f7f1	observation-theory-campaigns/experiments/LLM-EVAL-TRACK.md:1-50 ; observation-theory-campaigns/analysis/llm/XPROTO-LLM-graded.json ; observation-theory-campaigns/experiments/SEALS.md:85
chapter 13 section 13.3	ZooKeeper hot 0.99 cold 0.01 witnessed 0.0, Postgres 0.50 to 0.06, MongoDB 0.47 to 0.03, production Postgres 0.47 to 0.02, real substrates, disjoint seeds	observation-theory-campaigns/experiments/DATABASE-FRESHNESS-TRACK.md:1-45
chapter 13 section 13.3	BGP 0.351, IS-IS 0.184, OSPF 0.083, second collector	observation-theory-campaigns/analysis/mongo/PREREG-XPROTO-MG.md:42 ; observation-theory-campaigns/experiments/ROUTING-TELEMETRY-TRACK.md:130-150 ; geometric-observation/BOOK-OUTLINE.md:75
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35
chapter 13 section 13.4	refuted fit 0.177 T_coh R ² 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	observation-theory-campaigns\analysis\csi\CSI-refreshfloor.json:2,111-114 ; observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-47
chapter 13 section 13.4	sealed correction, 0.5 dB calibration, floors 4/6/4, 2/2/3, 1, slopes 0.1008, 0.1398, 0.1086, R ² 0.7376, 0.9218, 0.6174, bars B1 to B4 and MC1 to MC4 all true, seeds 20260827 to 20260829, sealed 1d3de4a	observation-theory-campaigns/analysis/csi/PREREG-XPROTO-CSI-SWEEP2.md:1-60 ; observation-theory-campaigns/analysis/csi/XPROTO-CSI-SWEEP2-graded.json ; observation-theory-campaigns/experiments/SEALS.md:90

From [observation-theory-campaigns/experiments/DATABASE-FRESHNESS-TRACK.md](#) at f6d1fe2.

- line 23. XPROTO-PG (analysis/pgrep); Postgres, recovery_min_apply_delay; WAL LSN; ~0.50 → ~0.06
- line 24. XPROTO-MG (analysis/mongo); MongoDB delayed secondary; oplog ts; ~0.47 → ~0.03
- line 25. XPROTO-PGX (analysis/pgx); production PG, netem lag; WAL LSN, pg_stat_statements footprint; ~0.47 → ~0.02
- line 27. XPROTO-ZK (analysis/zk); ZooKeeper 3.9 ensemble; zxid, sync(); hot 0.99 / cold 0.01, witnessed 0.0

From [observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md](#) at f6d1fe2.

- line 25. XPROTO-CSI (analysis/csi); CQI → MCS; HARQ; 0.34–0.37 → 0.10 (OLLA); pass 08-23
- line 26. XPROTO-BEAM (analysis/beam); mmWave beam index; HARQ; 0.31 → 0.02 (BFR); pass 08-23
- line 27. XPROTO-AICSI (analysis/aicsi); neural-CSI recon (turboquant bridge); precoder/HARQ; recon wins yet 0.28 → 0.13; pass 08-23, scope-corrected 08-25 warning
- line 28. XPROTO-HO (analysis/ho); RSRP → serving cell; RLF; 0.31–0.44 → 0.09–0.12; pass 08-23
- line 30. XPROTO-PHY (analysis/phy); PMI / RI / TA; HARQ; 0.27–0.42 → 0.055–0.13; pass 08-24

FAILURES AND CORRECTIONS

- [observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:33-39](#) at f6d1fe2. warning **XPROTO-AICSI scope correction (2026-08-25)**. The v1 seal stands for what it tested, but the reconstruction-vs-consumer dissociation does **not** survive the community-standard substrate. On real 3GPP CDL-C with a CsiNet-class codec the NMSE-optimal codec reconstructs near-perfectly (NMSE ≈ 0.03) and false-clears 0.0 on all seeds; the pre-registered kill fired. See [analysis/aicsi/PREREG-XPROTO-AICSI-V2.md](#) (REFUTED AT SHAKEDOWN, NOT SEALED, kept negative). This is a scope correction, not a retraction. Do not headline the AICSI row; WCNC §III-C is dropped.
- [observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-50](#) at f6d1fe2. **The age horizon** ([analysis/csi/](#), XPROTO-CSI-SWEEP2 sealed 2026-08-27, graded PASS): at the calibrated 0.10 budget the largest compliant report period is a few TTI at 10 Hz and 1 TTI at 50 Hz and above, so there is no proportionality law to fit. The earlier CSI-refreshfloor.* claim of a floor $\approx 0.177 \cdot T_{\text{coh}}$ ($R^2=0.915$) was an **unsealed exploration** measured at a relaxed 0.15 threshold against the 0.10 target, with a fresh baseline that never met the budget. It is refuted; the record is kept, not cited. Separately, the optimal linear predictor cannot beat the one-coherence-time wall (Gaussian fading $\rightarrow$ Wiener optimal), and that wall sits well outside the budget horizon. The **RAN governor** ([analysis/ran](#)) is the reference O-RAN rApp core: observe $\rightarrow$ measure false-clear $\rightarrow$ refresh at the floor $\rightarrow$ escalate (diversity / re-route).
- [geometric-observation/chapters/ch19_the_certificate_that_ages.md:69-76](#) at 787a933. — the rate at which a clearance is wrong, conditional on the certificate having cleared, reported beside the certificate's coverage. (An earlier draft wrote the joint probability $\Pr[\text{clears} \wedge \text{refutes}]$; the joint rate is the product of the two and is what a naive certificate that clears everything reports, since its coverage is one. Corrected 2026-09-03.) A certificate whose false-clear rate exceeds the target it purports to guarantee is **vacuous** for that consumer, however confident it looks. FC is measurable wherever a witness exists, which — per the table — is nearly everywhere it matters.

MACHINE CHECKED

[lean/DataMiningAsObservation/Certificate.lean](#), theorems `falseClear_mul_coverage`, `coverage_empty`, `falseClear_mem_unit`, `minOverStrata_passes_iff`, `minOverStrata_le_weighted_mean`, at [observation-data-mining 95af30c](#); what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 12, 13, 14.

RELATED

[certificate](#) (p. 59); [witness](#) (p. 382); [coverage](#) (p. 92); [coherence time](#) (p. 67); [refresh floor](#) (p. 299); [min-over-strata](#) (p. 226).

SEE ALSO

none

fine-tuning

instrument. Entry id fine-tuning.

DEFINITION

Continuing to train an encoder on a new objective or corpus, so that its quotient changes. Chapter 11 section 11.8 and chapter 12 section 12.4.

Example. An encoder tuned on translation pairs put seven of thirteen backbone areas in one class that the general encoder kept apart.

EQUATION

none

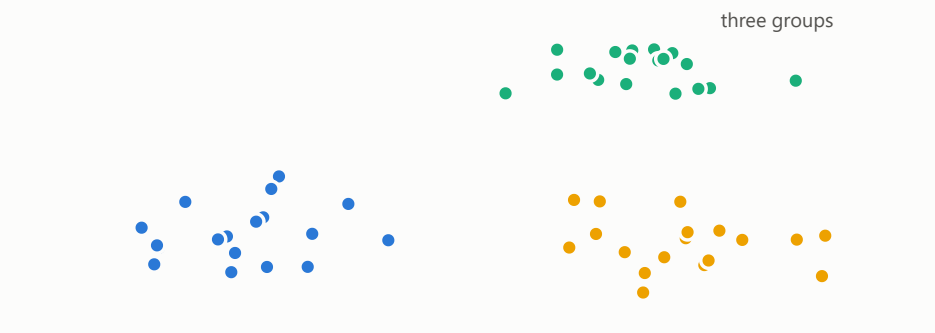


Figure 89: Continuing to train an encoder so that its quotient changes.

CONDITIONS

- Continuing to train an encoder on a new objective or corpus, so that its quotient changes. A contrastive objective pulls declared pairs together, and the cross-corpus gate asks whether the tuned encoder still separates on a corpus it never saw.
- An encoder fine-tuned on one relation is evaluated on that relation and on one it never saw, with a paired interval on each, and the within-corpus score of 0.75 to 0.955 collapsed to 0.47 to 0.55 across corpora until cross-corpus positives were added.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 11 section 11.8 and chapter 12 section 12.4 of *Data Mining as Observation*, with the translation-trained encoder in `turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Contrastive.lean](#), theorems `loss_nonneg`, `loss_eq_zero_iff`, `not_both`, `loss_mono_margin`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/CrossCorpusGate.lean](#), theorems `clears_bow`, `clears_untrained`, `not_validated_of_saturated`, `margin_example`, `validated_comp`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 11, 12.

RELATED

[contrastive objective](#) (p. 82); [encoder](#) (p. 127); [cross-corpus gate](#) (p. 94); [leakage](#) (p. 210); [paraphrase class](#) (p. 254).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 12.3, 11.1.

Ledger rows that cite the entry's records without naming it: GO-B-Llama, GO-B-Llama-rematch, GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.4, chapter 11 section 11.8, chapter 12 section 12.4, chapter 12 section 12.5.

finite difference

instrument. Entry id finite-difference.

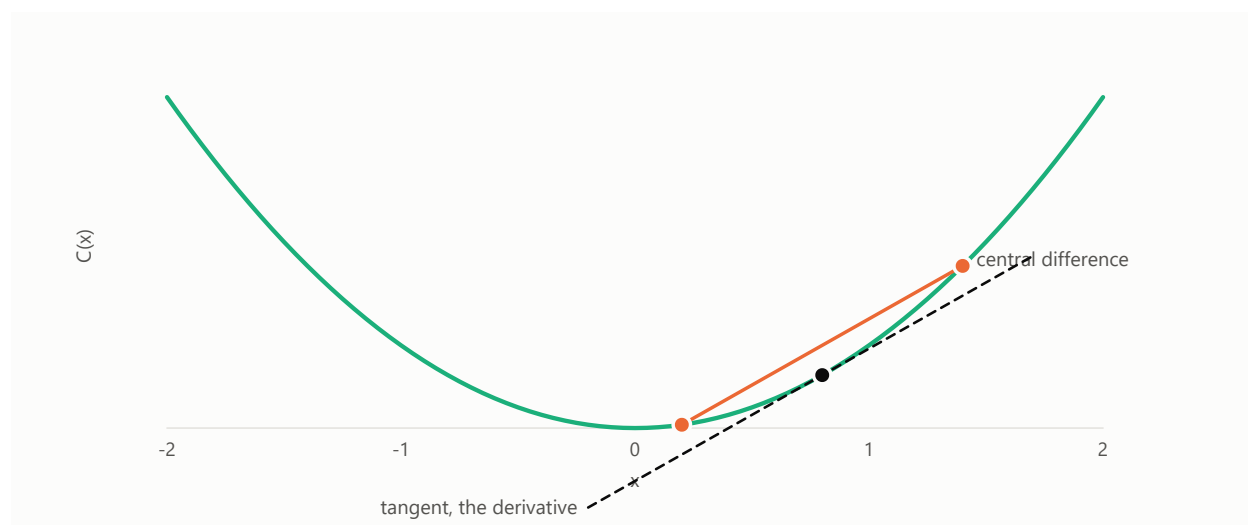


Figure 90: A derivative from two evaluations a step apart.

DEFINITION

An estimate of a derivative from two evaluations of the function at points a small step apart. Equation 0.8.

Example. For $C(x)$ equal to x squared, evaluating at 0.8 plus and minus 0.6 gives $(1.96 - 0.04)$ over 1.2, exactly 1.6.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

CONDITIONS

- An estimate of a derivative from two evaluations of the function at points a small step apart. The central difference recovers the derivative of a quadratic exactly at every step size, while the one-sided difference is off by the curvature times the step, which is why doing it in both directions is more accurate.
- Doing it once per coordinate costs d evaluations and in both directions costs $2d$, which chapter 11 shows is the price at one operating point for a probe confined to the directions it chooses. The measurement instrument for that is the blind probe.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.5 of *Data Mining as Observation*, equation 0.8, with the budget law measured in `readscope/README.md:118-141` and `readscope/CALIBRATION.md:419-434`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/FiniteDifference.lean](#), theorems `central_quad`, `forward_quad`, `forward_error`, `central_cost`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 7, 11, 12, 14.

RELATED

[sensitivity](#) (p. 328); [read operator](#) (p. 291); [blind probe](#) (p. 45); [budget](#) (p. 51); [budget cliff](#) (p. 52).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.9, 11.4.

Ledger rows that cite the entry's records without naming it: GO-1.

Sources-table rows that share a record with the entry without naming it: chapter 11 section 11.7.

the flip

result. Entry id `flip`.

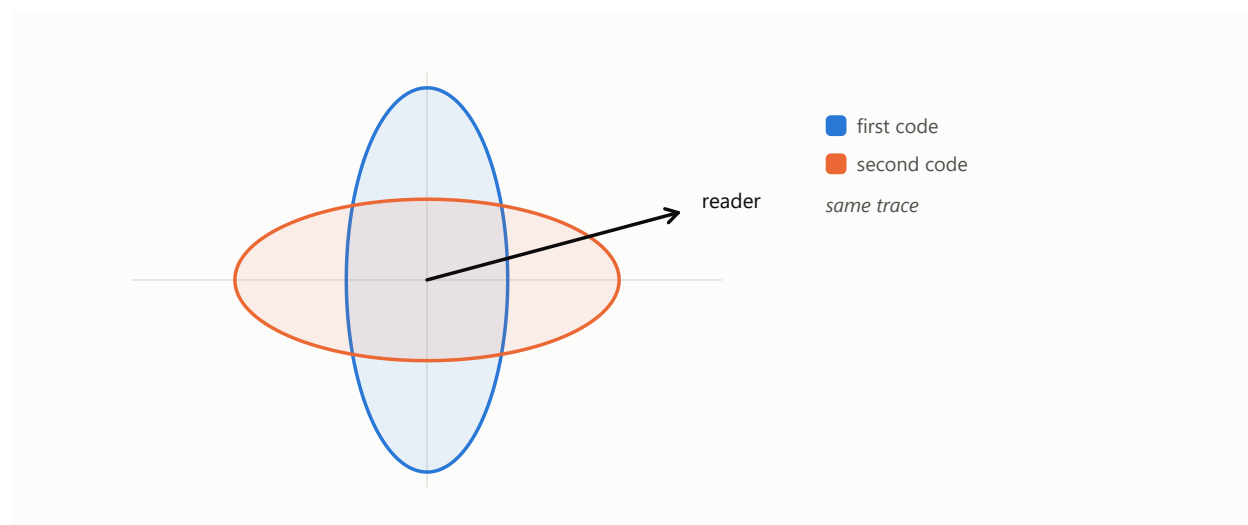


Figure 91: At matched bits, the code that reconstructs worse can serve the consumer better.

DEFINITION

At matched bits, the outcome in which the code with the worse reconstruction error scores better on the consumer's task, while a code built to destroy the read subspace scores worst. The book's central empirical claim. Chapter 4.

Example. Codes `diag(0.3, 1.7)` and `diag(1.7, 0.3)` have the same total error 2.0, and a reader at 15 degrees pays

0.394 for one and 1.606 for the other.

Known as, or related to prior art. Consumer-relative compression sits beside the information bottleneck of Tishby, Pereira, and Bialek and beside task-based quantization. The flip is the measured claim that at matched bits the code that reconstructs worse can serve the consumer better.

EQUATION

Book equation 4.5.

$$\text{flip} : \quad \text{task}(O) > \text{task}(R) \text{ and } \text{tr } M_\delta^O > \text{tr } M_\delta^R, \quad \text{task}(\text{anti}) < \text{task}(R), \quad \text{bits}(O) = \text{bits}(R).$$

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

Book equation 0.10.

$$d_O = \text{tr}(P_C M_\delta) = \mathbb{E}[\delta^\top P_C \delta], \quad M_\delta = \mathbb{E}[\delta \delta^\top], \quad P_C = I \Rightarrow d_O = \text{tr } M_\delta.$$

CONDITIONS

- Matched bits, and reconstruction matched or reported, for every comparison of two codes.
- A read operator misaligned with where the signal has its energy. The alignment of book equation 4.6 is the dial, and near one, the coupling null, no flip is possible.
- The read operator is the average of local linearizations, and the ordering it predicts is the ordering on average over the workload.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-2 (neg. half: not reconstruction) [demonstrated]. At matched bits, downstream preservation is not controlled by reconstruction error. [geometric-observation/claims/LEDGER.md:63](#).
- *measures*. GO-2 (pos. half: consumer-projected covariance *controls*) [replicated]. Downstream preservation is controlled by the error covariance projected on the consumer's read subspace, $\text{tr}(P_C \Sigma_\delta)$. [geometric-observation/claims/LEDGER.md:64](#).
- *measures*. GO-4 [replicated]. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [geometric-observation/claims/LEDGER.md:66](#).
- *refutes or corrects*. NEG-16 (KV serving, end-task) [refuted]. *At matched bits and matched reconstruction error, steering KV quantization error into the attention read subspace degrades LongBench task score by the registered floors on a deployed-class model.* Refuted at its registered effect sizes on ... [geometric-observation/claims/LEDGER.md:92](#).

FIRST STATED

Volume 14, chapter 8, [geometric-observation/chapters/ch08_value.md:1-30](#), DOI 10.5281/zenodo.21776291. The two-dimensional case is the cover and equation 3.1 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation/chapters/ch02_failure_of_observer_free_measurement.md:40-60 ; geometric-observation/chapters/ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro/docs/KV_KEYS_FINDING.md:1-49
chapter 1 section 1.4	the flip sealed in twelve domains and three physics, held in at least five domains and all three	geometric-observation/chapters/ch08_value.md:40-70 ; geometric-observation/claims/LEDGER.md rows GO-2, GO-B-AV163, D3, 038
chapter 2 section 2.5	whitened code wins at every budget	geometric-observation/chapters/ch08_value.md:84-97
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer/README.md:170-175 ; geometric-observation/chapters/ch16_honest_negatives.md NEG-1
chapter 4 section 4.3	the flip definition, flip versus (A2) verdict	geometric-observation\chapters\ch08_value.md:1-30,100-108
chapter 4 section 4.3	GO-2 0.0934 vs 0.0938, 2.53 times, 12 of 12, anti 21 times; retrieval 0.0964, negative 4.70 and positive 4.65, recon 0.40	geometric-observation/claims/LEDGER.md rows GO-2 negative and positive halves
chapter 4 section 4.3	acoustic 148 of 201, 201 of 201, 152 of 201; seismic 13 of 17, 17 of 17, 13 of 17; whale 0.934 vs 0.883, 2 times, 300 of 300; at least 5 domains and 3 physics; battery prediction met	geometric-observation/chapters/ch08_value.md:40-70 ; geometric-observation/claims/LEDGER.md rows GO-B-AV163, D3, 038
chapter 4 section 4.3	whitened code on whale 0.83, 0.85, 0.97 vs 0.41, 0.80, 0.89	geometric-observation/chapters/ch08_value.md:84-97
chapter 4 section 4.4	gradient compression anti 300 of 300, flip 27 percent, coupling boundary	geometric-observation/chapters/ch08_value.md:108-116
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation/claims/LEDGER.md row GO-4
chapter 4 section 4.6	NEG-4 through NEG-10, the 25 percent codebook confound, Spearman 0.80, the recon-matched precondition	geometric-observation/chapters/ch16_honest_negatives.md NEG-4 to NEG-10
chapter 6 section 6.2	real model, median 0.567 vs bar 0.60, about 4.5 times chance, heads at 0.815 and 0.958, sixteen of sixteen both, two of three triggers, NEG-12, later rematch four of four	geometric-observation/claims/LEDGER.md rows GO-B-Llama and NEG-12; geometric-observation/chapters/ch16_honest_negatives.md:73-81
chapter 6 section 6.2	whale clan classifier, 8718 codas, 0.934 vs 0.883, reconstructs twice as well, 300 of 300	geometric-observation/chapters/ch08_value.md:40-70 ; geometric-observation/claims/LEDGER.md row 038

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49 ; geometric-observation/chapters/ch16_honest_negatives.md NEG-2
chapter 8 section 8.2	recalibration improves reconstruction and worsens the consumer	geometric-observation/chapters/ch16_honest_negatives.md NEG-4

FAILURES AND CORRECTIONS

- NEG-16 (KV serving, end-task), [refuted]. *At matched bits and matched reconstruction error, steering KV quantization error into the attention read subspace degrades LongBench task score by the registered floors on a deployed-class model.* Refuted at its registered effect sizes on ... [geometric-observation/claims/LEDGER.md:92](#).
- [geometric-observation/chapters/ch16_honest_negatives.md:28-30](#) at 787a933. **NEG-4 — lightweight on-line (Lloyd) key calibration beats the default.** [refuted]. *It improves reconstruction and is worse on the consumer metric* — reconstruction-is-not-the- target, in its purest, most anti-correlated form.
- [geometric-observation/chapters/ch16_honest_negatives.md:32-34](#) at 787a933. **NEG-5 ... NEG-9 — the KV-key ladder that found the right statistic.** Five refuted proxies, and together the most instructive sequence in the ledger, because each refutation *located* the next hypothesis:
- [geometric-observation/chapters/ch16_honest_negatives.md:36-50](#) at 787a933. - **NEG-5:** the originally-registered test missed (2/4 bars); the apparent reversal was a matched-bits confound (an uncounted per-block codebook, $\sim +25\%$ bits). Refutes the *test*, not GO-2 — and forced a bit-matched redesign. - **NEG-6:** a demeaned-error-norm proxy fails to order the arms. The controlling quantity is the across-token *variance* of the query-projected error, not an error-magnitude scalar. *Magnitude $\neq$ downstream-relevant structure* (Chapter 6's lemma, learned here). - **NEG-7:** at *fixed reconstruction*, the downstream ranking **flips** with the consumer. Invariant-preservation is a property of the (compressor, consumer) *pair*, not the compressor — the sharpest form of GO-2's negative half. - **NEG-8:** the variance-ratio proxy reaches only Spearman 0.80; it is a noisy linear estimator of $\text{tr}(P_C \Sigma_\delta)$. Gate on the *direct* projection instead. - **NEG-9:** even the direct trace $\text{tr}(P_C \Sigma_\delta)$, while nailing the discriminating flip 12/12, is *not a complete rank statistic* — it misorders a middle pair whose per-token error has structure the trace misses. The mechanism is real; the scalar is not exhaustive.
- [geometric-observation/chapters/ch16_honest_negatives.md:56-61](#) at 787a933. **NEG-10 — the flip appears on any independent representation (prospective).** [refuted]. On embedding retrieval, arm b reconstructed strictly better *and* Pareto-dominated downstream — no flip — so reconstruction was not falsified there. **Precondition discovered: the flip is observable only for reconstruction-matched arms.** The anti-probe half still transferred. This negative is why every later flip row is built on recon-matched arms; the miss defined the protocol.
- [geometric-observation/chapters/ch16_honest_negatives.md:106-112](#) at 787a933. - **Identifiability** (fixable): NEG-5, NEG-10, legal-035, NEG-12 — the read operator was mis-estimated or the arms weren't recon-matched. Fixed by the blind probe and by protocol. - **Coupling** (a true boundary): D4 — read and signal energy intrinsically aligned; not rehabilitatable by any read-operator recovery. - **Precondition** (needs a working consumer): moral-on-frozen-embeddings — no read direction to protect until the consumer is competent. - **Mechanism-absent** (a genuine refutation): NEG-11 — the claimed effect does not exist.

MACHINE CHECKED

[lean/DataMiningAsObservation/Flip.lean](#), theorems `trace_eq`, `reading_difference`, `reader_swap_reverses`, `flip`, `cos_sq_pi_div_twelve`, `sin_sq_pi_div_twelve`, `reading_15_first`, `reading_15_second`, `sqrt_three_bounds`, `readings_15_to_three_decimals`, `reader_75_is_swap`, `four_to_one_flip`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2, 3, 4, 6, 7, 11, 12, 13, 14.

RELATED

[read distortion](#) (p. 290); [alignment](#) (p. 15); [anti arm](#) (p. 19); [rank certificate](#) (p. 285); [coupling null](#) (p. 89).

SEE ALSO

none

floor, ceiling

concept. Entry id floor-ceiling.

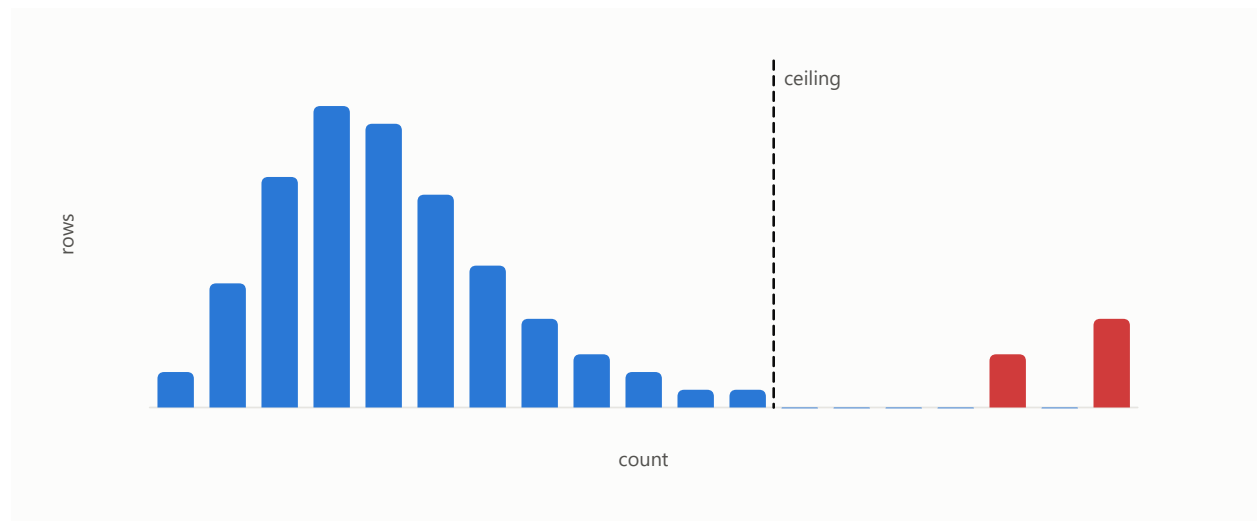


Figure 92: A lower bound a quantity cannot fall below, an upper bound it cannot exceed.

DEFINITION

A floor is a lower bound a quantity cannot fall below and a ceiling an upper bound it cannot exceed, each proved or measured and named as such. Chapter 0 section 0.9 and chapter 13.

Example. The refresh floor was 4 slots at 10 hertz, and the Poisson ceiling at 10 expected retrievals per row is about 20.

EQUATION

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$J \leq 2A - 1$ when the ROC curve is concave, and not in general.

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6$ TTI at 10 Hz, 1 TTI at ≥ 50 Hz.

CONDITIONS

- A floor is a lower bound a quantity cannot fall below, and a ceiling an upper bound it cannot exceed, each proved or measured and named as such. The refresh floor is the shortest renewal that keeps a certificate within its error, the Poisson ceiling is the count chance allows, and the Youden ceiling bounds the best F1.
- A floor that was fitted as a fraction of the coherence time was refuted, and the omission floor that was first read as unbounded was corrected to a floor, so each is a measured object with a record.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects*. NEG-13 $\rightarrow$ resolved [missed]. (GO-P-2026-026, prospective, real LLM) Appendix-E's omission floor is a rate-irreducible downstream wall on trained Llama read operators. [geometric-observation/claims/LEDGER.md:106](#).

FIRST STATED

Chapter 0 section 0.9 and chapter 13 section 13.4 of *Data Mining as Observation*, with the omission floor in `geometric-observation/chapters/ch07_cost.md` and the Youden ceiling in `theory-radar/paper/astar_paper.tex: 94-110`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	commission tax of order half $r \log M$ over m , omission floor, unbounded corrected to floor, naive projector 30 percent over, VI-6	geometric-observation/chapters/ch07_cost.md section on mismatch; geometric-observation/claims/LEDGER.md VI-6
chapter 4 section 4.2	floor measured on 16 of 16 heads, naive assumption off 20 to 60 percent, about 100000 times, 0.10 nats against about 6e-7, first gate missed 0 of 16, NEG-13	geometric-observation/chapters/ch07_cost.md ; geometric-observation/claims/LEDGER.md GO-P-2026-027 and NEG-13

FAILURES AND CORRECTIONS

- NEG-13 $\rightarrow$ resolved, [missed]. (GO-P-2026-026, prospective, real LLM) Appendix-E's omission floor is a rate-irreducible downstream wall on trained Llama read operators. [geometric-observation/claims/LEDGER.md:106](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/CoherenceTime.lean](#), theorems `clarke_to_three_decimals`, `examples`, `refuted_law_exceeds_sealed`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/PoissonCeiling.lean](#), theorems `mass_nonneg`, `tail_antitone`, `tail_zero`, `tail_le_one`, `expectedAtLeast_antitone`, `example_mean`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/YoudenF1.lean](#), theorems `f1_eq`, `f1_le_of_youden`, `auroc_eq`, `youden_eq`, `youden_exceeds_auroc_form`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14.

RELATED

[refresh floor](#) (p. 299); [Poisson ceiling](#) (p. 262); [Youden index](#) (p. 387); [budget cliff](#) (p. 52); [bar](#) (p. 39).

SEE ALSO

Ledger rows that cite the entry's records without naming it: OT-4.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.2, chapter 6 section 6.3, chapter 13 section 13.3, chapter 13 section 13.4.

footprint

concept. Entry id footprint.

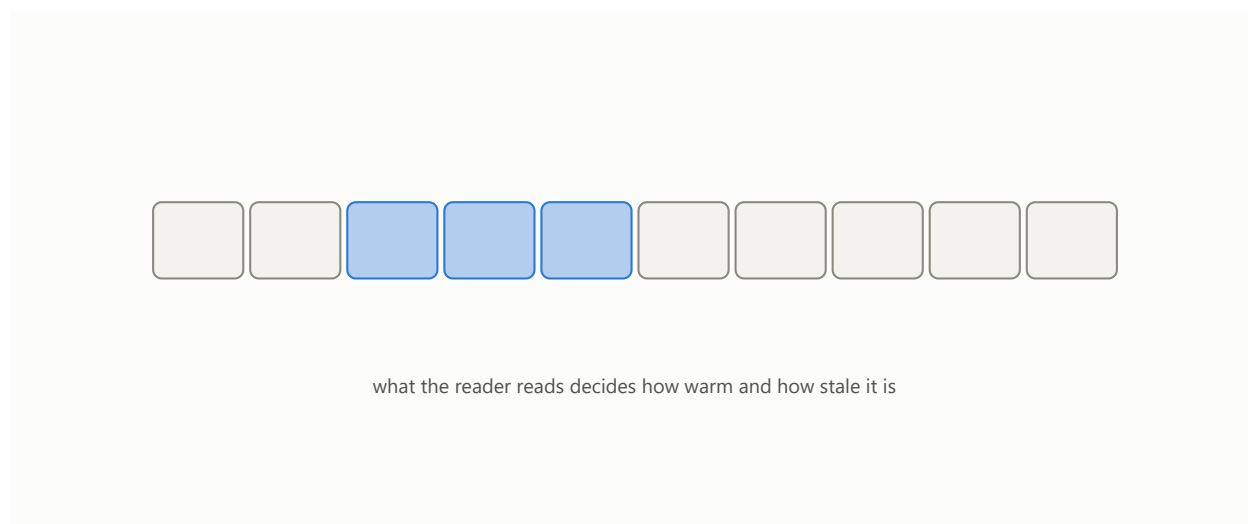


Figure 93: The set of entries a reader reads.

DEFINITION

The set of entries a reader reads. What the reader reads decides how warm and how stale the reader is. Chapter 13 section 13.3.

Example. A reader that touches 3 of a cache's 10 entries has a footprint of 3, and only those 3 decide its hit rate.

EQUATION

none

CONDITIONS

- The set of entries a reader reads. The hit rate is one when the cache holds the footprint and does not move when entries outside it are added, so what the reader reads decides how warm, and how stale, the reader is.
- A replica is stale because of what the reader asks for. The hot reader read a false-clear rate of 0.99 and the cold reader 0.01 on the same store, and the aggregate described neither.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 13 section 13.3 of *Data Mining as Observation*, with the reader's footprint in `geometric-observation/chapters/ch19_the_certificate_that_ages.md:1-95`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Cache.lean](#), theorems `hitRate_mem_unit`, `hitRate_cold`, `hitRate_warm`, `hitRate_mono`, `hitRate_footprint`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 11, 12, 13, 14.

RELATED

[cache](#) (p. 55); [cold, warm](#) (p. 69); [replica](#) (p. 307); [false-clear rate](#) (p. 141); [read subspace](#) (p. 293).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.26, 13.2, 13.1.

Ledger rows that cite the entry's records without naming it: OT-11, GO-12.

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.2, chapter 13 section 13.3.

formula search

instrument. Entry id `formula-search`.

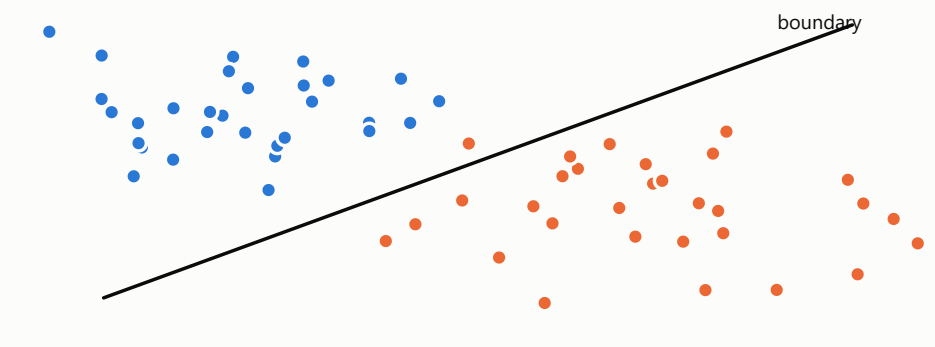


Figure 94: A search over short formulas scored by optimal thresholded F1.

DEFINITION

A score written as a short expression in the features, such as the smaller of two features plus a third, with a threshold. Chapter 6.

Example. Depth three over 18 operations found a feature minus a principal component plus a feature at F1 0.963 on breast cancer.

EQUATION

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$J \leq 2A - 1$ when the ROC curve is concave, and not in general.

CONDITIONS

- The search enumerates short expressions over the features and scores each by its optimal thresholded F1, pruning monotone unary nodes by the Monotone Invariance Theorem and, in the A* companion, leaves by an F1 ceiling.
- Its first comparison reported nine wins that became three under the Nadeau and Bengio correction, with seventeen datasets indeterminate, and the record carries the retraction at the size of the result.
- The main comparison used a fold-local meta-learned pruning rule and never rested on the AUROC form of the ceiling. The A* companion did, and the rerun shows what that cost, nothing on those datasets.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

theory-radar, DOI 10.5281/zenodo.20660206, theory-radar/README.md, with the corrected comparison in the constraint-gap review, constraint-gap/review/REDESIGN.md:1-110.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- [theory-radar/ERRATA.md:3-75](#) at 37c4e6c. 2026-09-03. The AUROC to F1 bound holds only for concave ROC curves **What was claimed.** Theorem “AUROC-F1 Bound” in `paper/astar_paper.tex`, the derivation in `src/symbolic_search/_auroc_proof.py`, and the pruning bounds in `run_astar_v2.py` and `src/symbolic_search/_heuristic_dag.py` all rested on the statement that any ROC curve with area A has maximum Youden index $J = \max(\text{TPR} - \text{FPR})$ at most $2A - 1$. **Why it is wrong.** The statement holds for concave ROC curves (equivalently, scores whose likelihood ratio is monotone), because a concave curve lies above the two chords through its Youden point and so has area at least $(1 + J)/2$. It fails for non-concave curves. Counterexample, found by an external reviewer of the companion textbook: rank three quarters of the positives above every negative and the remaining quarter below every negative. AUROC is 0.75, the maximum Youden index is 0.75, and at prevalence one half the best thresholded F1 is 0.857, above the 0.80 the AUROC form of the ceiling allows. The sound ceiling for that score, from $J = 0.75$, is 0.889, and the score sits under it. The `auroc_f1_bound` formula in `run_astar_v2.py`, $2A\pi/(A\pi + (1 - A)(1 - \pi))$, is not a valid upper bound either: at prevalence 0.1 the same score reaches F1 0.857 against a “bound” of 0.5. **Consequence for reported results.** The searches that used these bounds to prune could have discarded a candidate formula whose ROC curve was not concave and whose true best F1 exceeded the computed ceiling. The formulas and wins reported by those runs are therefore what a pruned search found, and are lower bounds on what an unpruned search would find. The Monotone Invariance Theorem and the results that do not depend on the AUROC bound are unaffected. **Correction.** The sound form of the ceiling substitutes the measured maximum Youden index of the candidate’s own scores for $2A - 1$. It holds for every score, costs the same sort as AUROC, and is what `max_f1_for_youden` and `youden_f1_bound` now compute. The AUROC form is retained under its original names with a docstring stating the concavity condition, and the pruning sites now use the measured index. A rerun of the published searches under the corrected bound was done on 2026-09-03 and 2026-09-04, `rerun_youden_2026_09_03.py`, results in `results/rerun_youden_2026_09_03.json`, run on the Atlas workstation, CPU only. **Rerun, part A, the pre-filter admissibility table (tab:auroc).** Depth-2 pairwise enumeration on the eight datasets, exhaustive against the AUROC pre-filter at the seven published thresholds, with the sound Youden ceiling beside it. The evaluation counts under the pre-filter reproduce

the published table (Breast Cancer 8700 pairs, 2576 evaluated at 0.75; Wine 1560, 413; Circles 20, 4; Moons 20, 2). The pre-filter lost no optimum on any dataset at any threshold, which is what the paper reported. That was luck of the data, not the theorem: the retracted ceiling sat below the F1 a pair formula actually reach

- [theory-radar/paper/REVISION_PLAN.md:39-45](#) at 37c4e6c. **Fix:** - Drop “ σ significance” everywhere - Report: mean $\Delta F1$, std, 95% CI, and corrected resampled t-statistic - Table columns: “Test F1”, “GB F1”, “ $\Delta F1$ ”, “95% CI”, “Direction” - Text: “the formula outperforms GB by $\Delta F1=0.031$, 95% CI [0.028, 0.034]” - Add Nadeau-Bengio corrected t-test (accounts for CV fold correlation) - Keep effect size as supplementary but don’t call it “ σ ”

MACHINE CHECKED

[lean/DataMiningAsObservation/MonotoneInvariance.lean](#), theorems `aurocNum_comp`, `auroc_comp`, `predicted_comp`, `predictedBelow_comp`, `sweptF1_comp`, `sweptF1Below_comp`, `optF1_comp`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

[lean/DataMiningAsObservation/YoudenF1.lean](#), theorems `f1_eq`, `f1_le_of_youden`, `auroc_eq`, `youden_eq`, `youden_exceeds_auroc_form`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 5, 6, 7, 14.

RELATED

[Monotone Invariance Theorem](#) (p. 227); [safe pruning](#) (p. 320); [Youden F1 bound](#) (p. 385); [read operator](#) (p. 291).

SEE ALSO

Book equations stated beside the entry’s terms, not defining it: 6.2, 7.4, 7.6.

Sources-table rows that share a record with the entry without naming it: chapter 5 section 5.4, chapter 6 section 6.3, chapter 6 section 6.4, chapter 6 section 6.5, chapter 7 section 7.3, chapter 7 section 7.4, chapter 8 section 8.4, chapter 8 section 8.5, chapter 8 section 8.9.

freshness

concept. Entry id freshness.

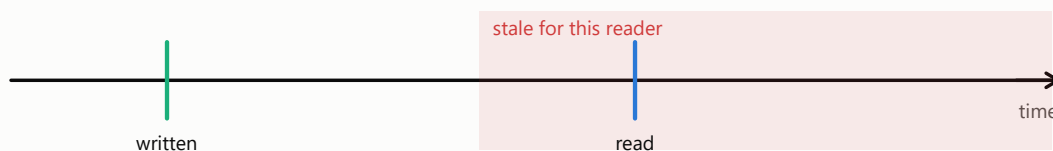


Figure 95: Whether a read is still what the quantity is.

DEFINITION

Whether a stored value is still within its coherence time when it is read. Consumer-relative, since two readers of the same replica can have different freshness. Chapter 13.

Example. A value written at 10:00 and read at 10:05 with a coherence time of 3 minutes is stale.

EQUATION

Book equation 0.26.

$$FC = \Pr[W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr[\mathcal{C}_t \text{ clears}].$$

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz, } 1 \text{ TTI at } \geq 50 \text{ Hz.}$

CONDITIONS

- A stored value is fresh for a reader when its age is within that reader's coherence time. Freshness is consumer-relative. Two readers of the same replica with different coherence times disagree about every age between the two, and the staleness rate of a mixed population is a weighted mean of the two readers' rates, strictly between them when they differ, so it describes neither.
- The measured false-clear rates on the named substrates, ZooKeeper, Postgres, MongoDB, and the radio link, are the freshness program's numbers, and the refuted refresh law is its standing negative.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects.* OT-4 [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [geometric-observation/claims/LEDGER.md:36](#).
- *refutes or corrects.* OT-11 [void]. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [geometric-observation/claims/LEDGER.md:48](#).

FIRST STATED

Volume 14, chapter 19, [geometric-observation/chapters/ch19_the_certificate_that_ages.md](#), DOI 10.5281/zenodo.21776291, and the freshness track, [observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md](#): 20-35.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.2	the grammar, certificate, witness, refresh floor, false-clear rate, vacuity, the witness table	geometric-observation/chapters/ch19_the_certificate_that_ages.md:1-95 ; observation-theory-campaigns/experiments/FRESHNESS-PROGRAM.md:1-40
chapter 13 section 13.4	refuted fit 0.177 T_{coh} R^2 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	observation-theory-campaigns/analysis/csi/CSI-refreshfloor.json:2,111-114 ; observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-47

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [geometric-observation/claims/LEDGER.md:36](#).
- OT-11, [void]. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [geometric-observation/claims/LEDGER.md:48](#).
- [observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-50](#) at f6d1fe2. **The age horizon** (analysis/csi/, XPROTO-CSI-SWEEP2 sealed 2026-08-27, graded PASS): at the calibrated 0.10 budget the largest compliant report period is a few TTI at 10 Hz and 1 TTI at 50 Hz and above, so there is no proportionality law to fit. The earlier CSI-refreshfloor.* claim of a floor $\approx 0.177 \cdot T_{\text{coh}}$ ($R^2=0.915$) was an **unsealed exploration** measured at a relaxed 0.15 threshold against the 0.10 target, with a fresh baseline that never met the budget. It is refuted; the record is kept, not cited. Separately, the optimal linear predictor cannot beat the one-coherence-time wall (Gaussian fading $\rightarrow$ Wiener optimal), and that wall sits well outside the budget horizon. The **RAN governor** (analysis/ran) is the reference O-RAN rApp core: observe $\rightarrow$ measure false-clear $\rightarrow$ refresh at the floor $\rightarrow$ escalate (diversity / re-route).

MACHINE CHECKED

[lean/DataMiningAsObservation/Freshness.lean](#), theorems disagree, mixedRate_between, mixedRate_eq_left_iff, stale_for_all, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 8, 11, 12, 13, 14.

RELATED

[coherence time](#) (p. 67); [certificate](#) (p. 59); [false-clear rate](#) (p. 141); [refresh floor](#) (p. 299); [drift](#) (p. 119).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.25.

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.3.

G

gate

instrument. Entry id gate.

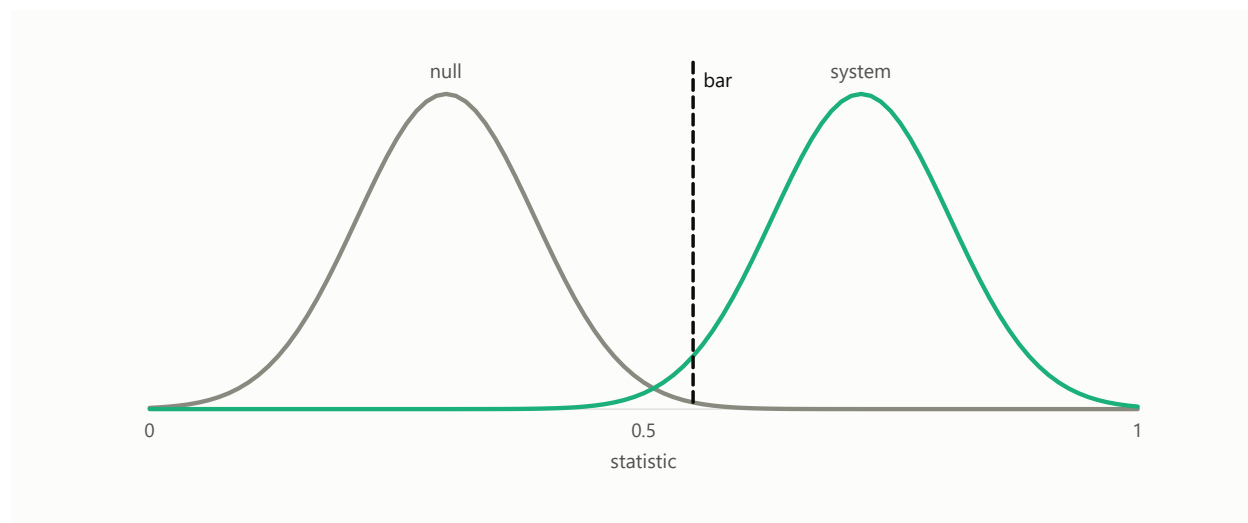


Figure 96: A bar a build must pass before it proceeds.

DEFINITION

A bar that a build or a claim must pass before it proceeds, with the bar and the null written down first. Chapter 8 section 8.3 and chapter 11 section 11.6.

Example. The build gate required anti-hub recall of at least 0.90 in every stratum; the compressed index read 0.663 and failed.

EQUATION

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

CONDITIONS

- A bar that a build or a claim must pass before it proceeds, with the bar and the null written down first. Passing a higher bar passes every lower one, a gate discriminates only when the null fails it and the real system passes it, and a gate the null passes is vacuous.
- The hubness gate was predicted to be blind to the category mix and was not, so the instrument claim was withdrawn, and the gate meta-rule says relative contrast discriminates nothing.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.3 of *Data Mining as Observation*, with the build gate in `turboquant-pro/docs/HUBNESS_PRIMER.md:86-131` and the cross-corpus gate in `xbse/README.md:130-160`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.5	Gate A design error, A prime skew 3.970 to 3.177, max 287 to 213, Robin Hood 0.372 to 0.261, fraction 0.117 to 0.079, compressed path 0.663 vs 0.90, seven strata, 0.62 to 0.69 vs 0.76 to 0.84	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:1-45
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131
chapter 12 section 12.5	the gate, both nulls, margin 0.10, scorecard 0.622 to 0.853, bag of words 0.46 to 0.54, rights 0.509 vs 0.512 and 0.467	xbse\README.md:130-160,195-219

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Bar.lean](#), theorems `passes_anti`, `passes_mono`, `discriminates_iff`, `no_bar_of_null_ge`, `exists_bar_of_lt`, `vacuous_of_null_passes`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/CrossCorpusGate.lean](#), theorems `clears_bow`, `clears_untrained`, `not_validated_of_saturated`, `margin_example`, `validated_comp`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 4, 8, 9, 10, 11, 12, 13, 14.

RELATED

[bar](#) (p. 39); [cross-corpus gate](#) (p. 94); [anti-hub recall](#) (p. 22); [preregistration](#) (p. 267); [null model](#) (p. 242).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 10.7, 8.3.

Ledger rows that cite the entry's records without naming it: NEG-14, GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.2, chapter 10 section 10.5.

Gaussian

concept. Entry id `gaussian`.

DEFINITION

The normal distribution, in many dimensions the one whose density falls with the Mahalanobis distance. An isotropic Gaussian has every direction the same. Chapter 9 and chapter 10.

Example. Values 1, 2, and 3 from a normal distribution with mean 2 and spread 1 have z-scores −1, 0, and 1.

EQUATION

none

CONDITIONS

- The normal distribution, in many dimensions the one with density falling with the Mahalanobis distance. An isotropic Gaussian has every direction the same, so every reader reads the same variance and no code

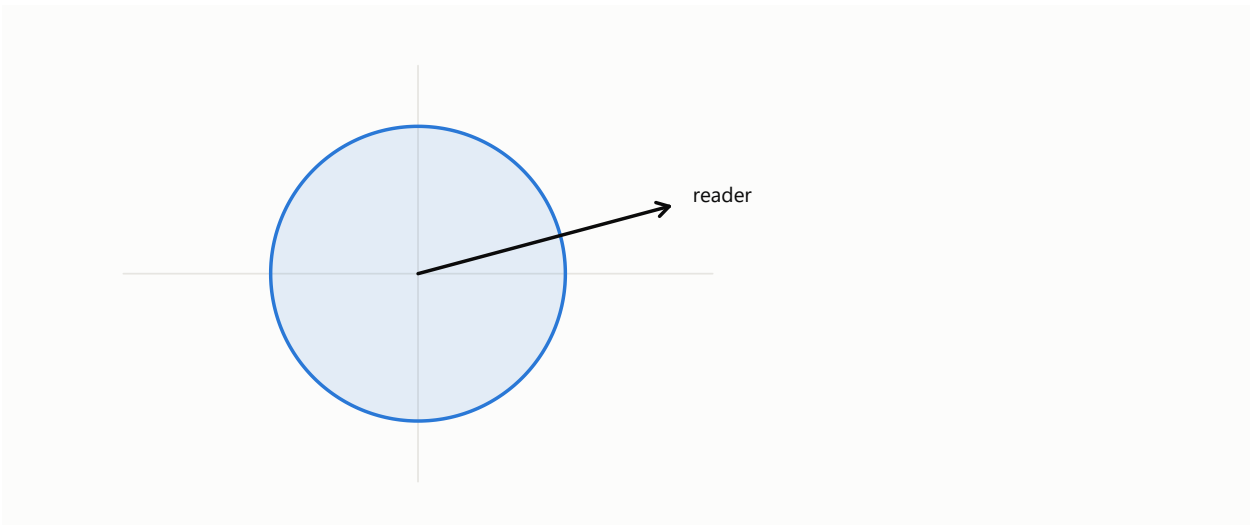


Figure 97: The normal distribution, isotropic when every direction is the same.

can flip, and whitening turns any Gaussian into an isotropic one.

- The isotropic Gaussian is the control on which hub typing correlated above 0.8 with nothing, and the Gaussian pass of GO-8 was five of five with the control-statistic caveat.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-8 [replicated]. On two independent source families (binary Markov; Gaussian AR(1)), a fixed stored record’s operational reset threshold rises with the age of the retained side information exactly as the staleness–work complement prices it: same record, ... [geometric-observation/claims/LEDGER.md:71](#).

FIRST STATED

Chapter 9 section 9.3 and chapter 10 section 10.3 of *Data Mining as Observation*, with the isotropic Gaussian control in `turboquant-pro/turboquant_pro/anatomy.py:98-170`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 10 section 10.3	hierarchical typing, tails 0.95 and 0.85, prescriptions, two designs that died, correlation above 0.8 on an isotropic Gaussian	turboquant-pro/turboquant_pro/anatomy.py:98-170
chapter 13 section 13.5	GO-8, 0.10 to 0.55 across ages 0 to 64, flip probability 0.05, 1 percent to 100 percent at age 32, Gaussian pass 5 of 5, the control-statistic caveat	geometric-observation/claims/LEDGER.md row GO-8 ; geometric-observation/experiments/GO-landauer-gaussian-secondsettings-NOTES.md

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Isotropy.lean](#), theorems `isotropic_reads_same`, `isotropic_no_flip`, `anisotropic_readers_differ`, `flip_iff_anisotropic`, at `observation-data-mining 95af30c`; what the check covers is stated in

the book's [appendix C](#).

[lean/DataMiningAsObservation/Mahalanobis.lean](#), theorems `dm2_nonneg`, `dm2_mean`, `dm2_scale`, `dm2_eq_whitened`, `dm2_antitone_in_variance`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 9, 10, 11, 13.

RELATED

[isotropic](#) (p. 187); [Mahalanobis distance](#) (p. 219); [null model](#) (p. 242); [whitening](#) (p. 381); [standard error](#) (p. 347).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.33, 0.6, 9.3.

Ledger rows that cite the entry's records without naming it: GO-3.

geodesic distance

concept. Entry id geodesic-distance.

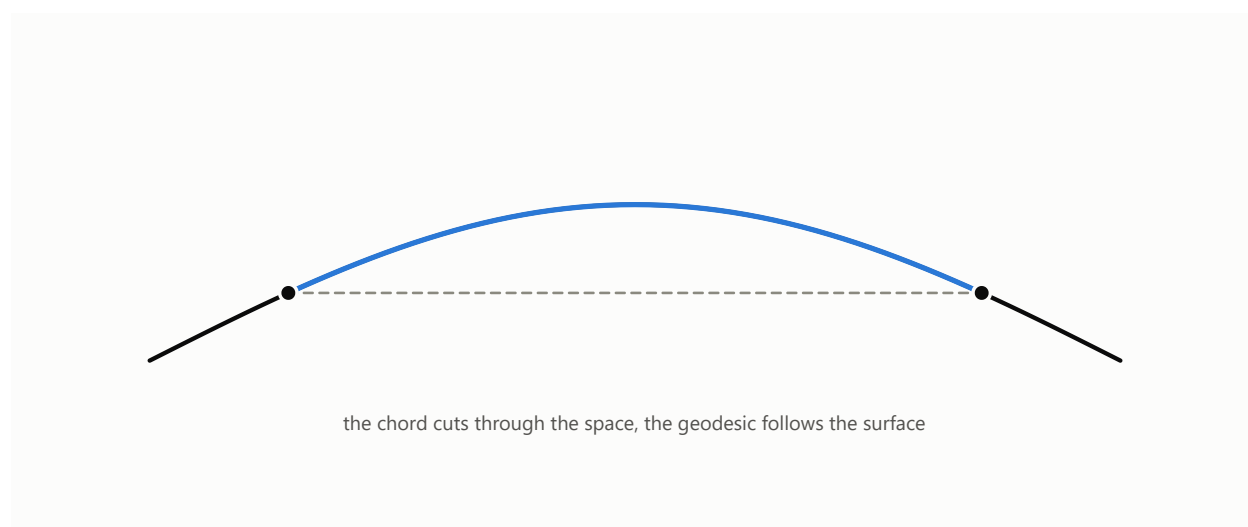


Figure 98: The shortest path along the surface.

DEFINITION

The length of the shortest path along edges between two nodes of a graph. Chapter 0 section 0.10.

Example. On a circle of radius 1, two points a quarter turn apart are 1.414 apart through the plane and 1.571 apart along the circle.

EQUATION

Book equation 3.2.

$$R(i, j) = \sum_{k \geq 2} \frac{1}{\lambda_k} \left(\frac{u_k(i)}{\sqrt{d_i}} - \frac{u_k(j)}{\sqrt{d_j}} \right)^2 \rightarrow \frac{1}{d_i} + \frac{1}{d_j} \quad (n \rightarrow \infty, \dim \geq 3).$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

CONDITIONS

- The length of the shortest path along edges between two nodes. It is zero from a node to itself, symmetric, one between adjacent nodes, and on a connected graph obeys the triangle inequality.
- The geodesic-rank reader reads only the ordering of these distances, so a strictly increasing transform of them leaves its nearest neighbour unchanged. That the angle of the spectral embedding carries the ordering and the radius carries degree is the measured claim of chapter 3, with the refuted commute-filter row beside it.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.10 and chapter 3 section 3.3 of *Data Mining as Observation*, with the geodesic-rank reader of Volume 14 chapter 9, `geometric-observation/chapters/ch09_legibility.md:10-41`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	row normalization is the projection onto the read subspace of the geodesic-rank consumer	geometric-observation/chapters/ch09_legibility.md:31-41 ; geometric-observation/chapters/ch03_historical_precursors.md:98-110
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation/chapters/ch09_legibility.md:10-41 ; the- angular-observer/theorem.md:110-146

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/GeodesicDistance.lean](#), theorems `dist_self`, `dist_comm`, `dist_triangle`, `dist_adj`, `geodesic_rank_invariant`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 9, 10.

RELATED

[rank-faithful](#) (p. 286); [Laplacian](#) (p. 207); [recognizer](#) (p. 296); [bi-Lipschitz](#) (p. 42).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.20.

Ledger rows that cite the entry's records without naming it: GO-3, NEG-1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.2, chapter 3 section 3.3, chapter 9 section 9.2, chapter 11 section 11.1.

geometric decision cost

concept. Entry id geometric-decision-cost.

DEFINITION

The cost of an alternative in a decision task is the Mahalanobis distance between the encoding of the alternative and the encoding of the task's reference point under the inverse covariance. It is the pullback distance of the observer whose consumer is the scenario encoding, whose output metric is the inverse covariance, and whose inactive dimensions are directions in the kernel of that metric. The read operator of the program, written for a displacement from a reference point instead of for a code's error.

Example. In the ultimatum encoding an offer of 48 percent has social, identity, and epistemic displacements 0.373, 0.278, and 0.003 from the reference, so its cost is the square root of 0.373 squared over 78.26 plus 0.278 squared over 32.28 plus 0.003 squared over 0.01262, about 0.071, and no offer on the integer grid costs less.

Known as, or related to prior art. Mahalanobis, 1936, for the distance. Luce, 1959, for the logit choice rule the lottery predictions use. Kahneman and Tversky, 1979, for the six lottery problems, and Fehr and Schmidt, 1999, for the inequality-aversion baseline. The specialization is Corollary 2 of the observer-representation draft, which identifies this cost with the pullback of Theorem 1(b) and identifies nothing about the map from cost gaps to lottery frequencies beyond the rule as written.

EQUATION

none

CONDITIONS

- For a discrete menu no Jacobian is needed and the cost is the exact pullback of the quadratic form through the encoding. On a continuous scenario manifold the local metric is the encoding's Jacobian transposed, times the inverse covariance, times the Jacobian, which is the read operator of read-operator with the inverse covariance as output metric, geometric-observation/paper/observer-representation.tex:303-330.
- The consumer is the hand-coded encoding of each task as a reference point and a menu of alternatives in nine dimensions, erism1-lib/docs/papers/foundations/submission/ieee-tcss/final_manuscript.tex:339-376, and the three active dimensions are a sparsity pattern of the metric chosen by a search, not a demonstrated budget of the reader.
- The three active variances are 78.26, 32.28, and 0.01262 on social impact, virtue and identity, and epistemic status. They were fitted on nine game targets by a search over every active set of size at most five, with a 20-point log grid from 0.01 to 100 for one and two dimensions and 5,000 seeded log-uniform draws on the same range for three or more, eris-econ/src/eris_econ/structural_fuzz.py:94-186. Candidate active sets were then ranked on all sixteen targets, eris-econ/src/eris_econ/structural_fuzz.py:693-740, so no lottery target is out of sample and the lottery matches are parameter reuse within a jointly selected architecture.
- The reference implementation, eris-econ 0.1.0, reproduces every reported prediction, 16 of 16 targets within tolerance at an unweighted mean absolute error of 2.70 percentage points and 3.95 on the six Ruggeri lottery items, eris-econ/src/eris_econ/targets.py:338-461 and erism1-lib/docs/papers/foundations/submission/ieee-tcss/final_manuscript.tex:692-722. The calibration objective is a weighted mean absolute error, with weights 1 on the ultimatum mean, the dictator mean, and the responder threshold and 0.5 on the other game targets.
- Game predictions are cost minima on an integer percentage grid and involve no temperature. Lottery predictions are a binary logit whose temperature is the larger of 0.5 and 0.24 times the cost gap to the power 2.13, eris-econ/src/eris_econ/targets.py:40-42. That rule is not monotone above a cost gap of 1.41, four of the six lottery targets lie above it with gaps 3.48, 2.21, 2.24, and 2.21, and the ordering of the Allais problem against the strong-certainty problem is carried by the temperature and not by the geometry, since the encoding gives the Allais problem the larger gap while the data give it the higher risky-choice rate and no constant temperature reproduces both, erism1-lib/docs/papers/foundations/submission/ieee-tcss/final_manuscript.tex:234-266. The two temperature constants were set from those two targets.
- An inactive dimension is a direction in the kernel of the output metric. With the monetary dimension

inactive, multiplying every stake by a constant leaves every prediction unchanged, and conversely a model whose predictions are unchanged for two distinct multipliers has no monetary weight, `geometric-observation/paper/observer-representation.tex:333-345`. The reference implementation also normalizes the monetary coordinate to the stake, so the encoding is stake-blind for every metric and a rejection of invariance refutes encoding and metric together.

- On the high-stakes ultimatum data of Andersen, Ertac, Gneezy, Hoffman, and List, collected in eight villages of Meghalaya in northeast India with 458 responders and stakes from 20 to 20,000 rupees against an average yearly income of about 17,000, three pre-specified tests reject invariance, Kruskal-Wallis H 60.7 at $p = 4.17 \times 10^{-13}$, chi-squared 16.18 at $p = 0.0010$, and a likelihood-ratio test of a logit with a log-stake term at $p = 8.77 \times 10^{-5}$ with AIC falling from 584.9 to 571.5, `erisml-lib/docs/papers/foundations/submission/ieee-tcss/final_manuscript.tex:1005-1019`. The calibrated model predicts invariance and that prediction fails. The result motivates and does not test a monetary coordinate in absolute or income units with finite sensitivity, and no stake at which invariance fails is located, `erisml-lib/docs/papers/foundations/submission/ieee-tcss/final_manuscript.tex:1031-1043`.
- The fitted model is not a strong-utility model, because the Allais problem has both a larger cost gap and a higher risky-choice rate than the strong-certainty problem. On binary menus it predicts nothing outside the random-utility class. The non-reducibility theorem of the observer-representation draft concerns the budgeted observer whose read subspace is selected from the menu, and not this model.
- The cumulative-prospect-theory baseline passes all six lottery problems at a mean absolute error of 5.7 percentage points with its canonical parameters, `erisml-lib/docs/papers/foundations/submission/ieee-tcss/final_manuscript.tex:1194-1221`, and the baseline used does not specify strategic behavior. The two models differ in flexibility and fitting protocol, so the difference in error is not a ranking of the theories.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

The IEEE Transactions on Computational Social Systems paper, accepted 2026-09-06, `erisml-lib/docs/papers/foundations/submission/ieee-tcss/final_manuscript.tex:1-60`, and its reading as a pullback distance in `geometric-observation/paper/observer-representation.tex:303-330`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- [erisml-lib/docs/papers/foundations/submission/ieee-tcss/FINAL-FILES-NOTES.md:60-71](#) at c368921. Round 2 (2026-09-07, after the owner's reviewer-style feedback on the final build) All three blockers and the temperature issue are addressed. Every number below was re-derived from `eris-econ` this session. 19. Holdout language removed everywhere. Because the reference implementation ranks candidate active sets on all sixteen targets, no lottery target is fully out of sample. The abstract, introduction, Section V.A, Section V.D (retitled "What Was Fitted or Selected on Which Targets"), Table VII (role column now "ranking only"), Section VI.F, the discussion, Limitation 5, and the conclusion now say: variances fitted on games, structure selected on the full benchmark, lottery results demonstrate cross-domain parameter reuse within a jointly selected architecture, not out-of-sample prediction. P11 is stated to test the reduction assumption rather than predict from distinct inputs. 20. CPT appendix corrected. The canonical absolute errors (4.7, 7.6, 5.3, 4.3, 3.4, 8.8 pp) are all within 10 pp, so canonical CPT passes 6/6 at MAE 5.7%, not 4/6. The accepted text's arithmetic was wrong. The "optimized" run (MAE 11.2%, claimed as the best MAE-minimizing solution while a 5.7% point was known) is withdrawn entirely, together with "pass-rate ceiling", "CPT does not exceed 4/6", and "both parameterizations fail P3 and P11". Table IX now reads CPT 6/6 at 5.7%, geometric 6/6 at 3.95% on the lottery subset, with a sentence that the two differ in flexibility and fitting protocol. "CPT cannot be applied to games" is now "the CPT baseline used here does not specify strategic behavior". The CPT fitting script was not found in any repo, so the canonical per-target numbers remain as reported in the accepted text and the pass count is corrected from those numbers. 21. Andersen interpretation made consistent with the stake-normalization caveat.

Because the encoding normalizes money to the stake, no value of the monetary variance would predict a stake effect. The calibrated model predicts invariance and that prediction fails. Removed throughout: “as the model predicts”, “becomes active at consequential stakes”, “in the direction the theory anticipates”, “locates the boundary”. Abstract, introduction, contribution 4, Section VII.A (retitled “Money-Zero as a Falsifiable Prediction”), Fig. 5 caption (“prediction of the calibrated share-normalized model”), VII.B, VII.D, Limitation 8 (“Failure of invariance demonstrated, boundary not located”), and conclusion now say the result motivates but does not test an absolute- or income-scaled monetary coordinate with finite sensitivity. 22. Temperature nonmonotonicity reported. Turnover at cost gap 1.41. Cost gaps computed from eris-econ: P1 3.48, P3 2.21, P7 2.24, P11 2.21, P16 0.014, P17 0.013, so four of six targets sit on the nonmonotone branch. The encoding gives P1 a larger gap than P3 while the data show more risky choice on P1, and no constant temperature reproduces both ($T = 0.5$ gives P1 0.1%, $T = 3.42$ gives P3 34.4%). T

MACHINE CHECKED

none

USED IN

none

RELATED

[Mahalanobis distance](#) (p. 219); [read operator](#) (p. 291); [observer](#) (p. 244); [consumer](#) (p. 79); [output metric](#) (p. 251); [quotient](#) (p. 279); [nuisance](#) (p. 240); [whitening](#) (p. 381); [softmax](#) (p. 336); [read direction](#) (p. 288).

SEE ALSO

none

graph

concept. Entry id graph.

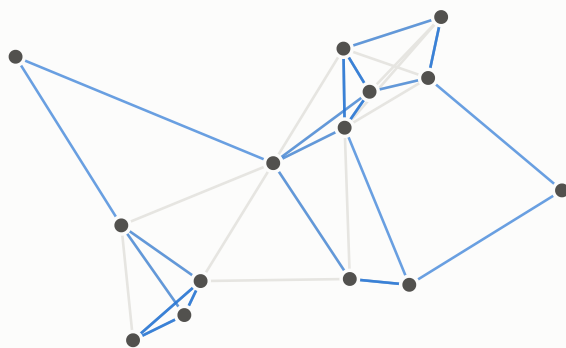


Figure 99: Nodes and the edges that join pairs of them.

DEFINITION

A set of nodes and a set of edges joining pairs of them. A neighbourhood graph joins each row of a dataset to its nearest rows. Chapter 0 section 0.10.

Example. Four nodes joined in a square have four edges, and every node has degree 2.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

CONDITIONS

- A set of nodes and a set of edges joining pairs of them. A neighbourhood graph joins each row to its k nearest rows, so every node has exactly k out-neighbours. The mutual form keeps an edge only when each node is among the other's neighbours, is symmetric, is a subgraph of the directed form, and gives no node more than k mutual neighbours.
- The Laplacian of the graph carries the dimension in its eigenvalue count and the geodesic ranking in its low eigenvectors' angles, which is what chapters 3 and 9 read.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.10 of *Data Mining as Observation*, with the neighbourhood graphs of the-angular-observer and the recognizer battery.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Graph.lean](#), theorems `outDegree`, `mutual_symm`, `mutual_sub`, `mutualNbrs_card_le`, `mem_mutualNbrs`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 9, 10, 11.

RELATED

[degree](#) (p. 105); [Laplacian](#) (p. 207); [geodesic distance](#) (p. 162); [spectral embedding](#) (p. 342); [manifold](#) (p. 220).

SEE ALSO

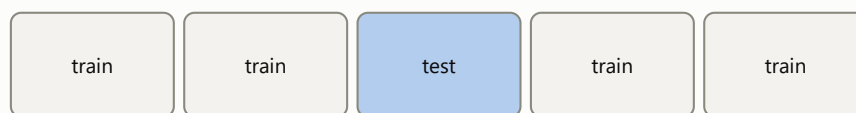
Ledger rows that cite the entry's records without naming it: GO-3, NEG-1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.2, chapter 3 section 3.3, chapter 9 section 9.1, chapter 9 section 9.2, chapter 11 section 11.1.

H

harness

concept. Entry id harness.



every row is tested once across the 5 folds

Figure 100: The evaluation reads folds, samples, and seeds, and what it reads it can leak.

DEFINITION

The code that evaluates a model. It is itself a consumer with a read subspace, and chapter 8 shows how it reads the test fold when allowed to.

Example. Ten folds, five seeds, and 100 samples per fold is a budget of 5000 reads of the data.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

CONDITIONS

- The harness is the code that evaluates a model, and it is a consumer with its own read subspace, its output metric, and a budget of folds, samples, and seeds. What it reads decides what the reported score measures.
- A harness that is allowed to see the test fold, through feature selection before the split, through repeated selection on one split, or through a metric that discriminates nothing, reports a score of itself.
- Repeated cross-validation folds are not independent, so the variance of a fold mean is understated unless corrected, and the program's own retracted comparison is the case.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.1 of *Data Mining as Observation*, with the program's own harness failures in `openvector-bench/results/QUERY_COUPLING_ARTIFACT.md:1-20` and `constraint-gap/review/FINDINGS.md:1-35`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- [theory-radar/paper/REVISION_PLAN.md:39-45](#) at 37c4e6c. **Fix:** - Drop “ σ significance” everywhere - Report: mean $\Delta F1$, std, 95% CI, and corrected resampled t-statistic - Table columns: “Test F1”, “GB F1”, “ $\Delta F1$ ”, “95% CI”, “Direction” - Text: “the formula outperforms GB by $\Delta F1=0.031$, 95% CI [0.028, 0.034]” - Add Nadeau-Bengio corrected t-test (accounts for CV fold correlation) - Keep effect size as supplementary but don’t call it “ σ ”

MACHINE CHECKED

[lean/DataMiningAsObservation/Bonferroni.lean](#), theorems `family_error_le`, `bonferroni`, `one_look`, at observation-data-mining 95af30c; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 2, 4, 6, 7, 8, 10, 12, 14.

RELATED

[leakage](#) (p. 210); [observer](#) (p. 244); [preregistration](#) (p. 267); [ledger class](#) (p. 211).

SEE ALSO

Book equations stated beside the entry’s terms, not defining it: 8.2, 8.3.

Ledger rows that cite the entry’s records without naming it: OT-11, NEG-2, NEG-4.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 6 section 6.4, chapter 8 section 8.1, chapter 8 section 8.2, chapter 8 section 8.4, chapter 8 section 8.5, chapter 11 section 11.1, chapter 11 section 11.2.

hash

instrument. Entry id hash.

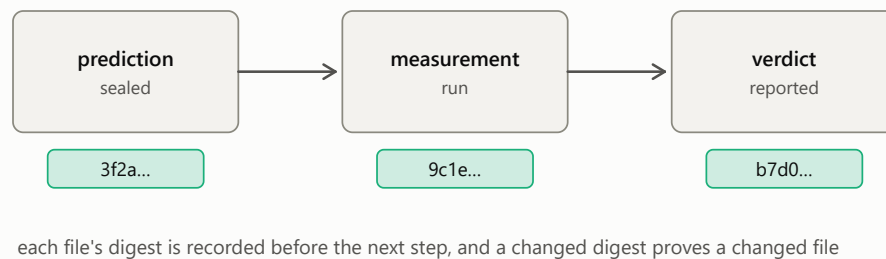


Figure 101: A fixed-length fingerprint that changes whenever the file does.

DEFINITION

A short fixed-length fingerprint of a file, computed so that any change to the file changes the fingerprint. Chapter 0 section 0.12.

Example. Changing one byte of a file changes its SHA-256 digest, and two files with different digests are different files.

EQUATION

none

CONDITIONS

- A short fixed-length fingerprint of a file, computed so that any change to the file changes the fingerprint. A changed hash proves a changed file, an unchanged file has an unchanged hash, and a digest with fewer values than there are files sends two distinct files to the same digest, so an equal hash is evidence and not proof.
- The seal ledger records the hash beside the commit, and the recognizer's templates and the probe's records carry theirs, so that a reader with the repository can check what was fixed before the measurement.
- A hash proves content and not time. When a file existed in the hashed state is established by a public push, a signed tag, a transparency log, or an archival deposit.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.12 of *Data Mining as Observation*, with the program's seal ledger, `observation-theory-campaigns/experiments/SEALS.md:1-10`, where every seal records a SHA-256.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Seal.lean](#), theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[sealed](#) (p. 324); [preregistration](#) (p. 267); [ledger class](#) (p. 211); [certificate](#) (p. 59).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.3.

Ledger rows that cite the entry's records without naming it: OT-7.

head

concept. Entry id head.

DEFINITION

One attention operation. A model has many, each with its own queries, keys, and values. Chapter 0 section 0.11.

Example. One head reads a query against 512 keys and returns one weighted mix of the values.

EQUATION

Book equation 0.23.

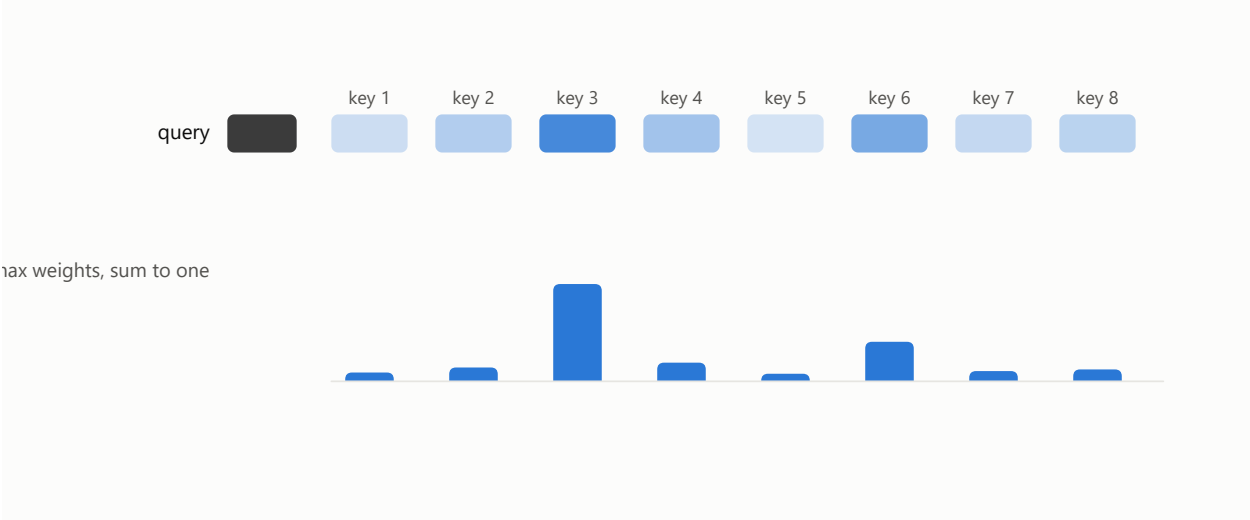


Figure 102: One attention head, a query against every key.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

CONDITIONS

- One attention operation, with its own queries, keys, and values. A head’s output lies between the smallest and largest value, it reads the keys only through their scores against its query, and a key change the query does not read leaves its output unchanged however large the change.
- Each head has its own read subspace, a few query-weighted directions of each key, so a compression that serves one head can fail another, and the serving-stack measurement of chapter 13 is per head.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program’s head-level measurements in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49` and the planted probe of chapter 6.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Attention.lean](#), theorems `output_le_max`, `min_le_output`, `output_congr`, `output_nuisance`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 7, 8, 11, 12, 13.

RELATED

[attention](#) (p. 26); [KV cache](#) (p. 200); [read subspace](#) (p. 293); [softmax](#) (p. 336).

SEE ALSO

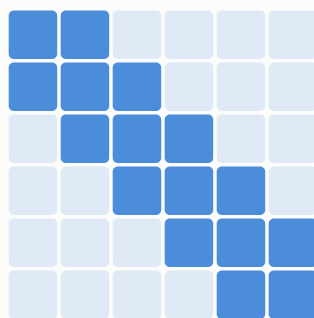
Book equations stated beside the entry's terms, not defining it: 11.1.

Ledger rows that cite the entry's records without naming it: GO-2/GO-12/GO-13 operational (KV serving, 077), NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 8 section 8.2, chapter 11 section 11.1, chapter 11 section 11.2.

Hessian

concept. Entry id `hessian`.



diagonal for naive Bayes

Figure 103: Second derivatives, with the interactions off the diagonal.

DEFINITION

The matrix of second derivatives of a score, whose off-diagonal entries are the interactions. Naive Bayes has a diagonal one. Chapter 6.

Example. For the score $3x_1 + 2x_2 + x_1 x_2$ the Hessian has 1 off the diagonal; for naive Bayes it has 0.

EQUATION

none

CONDITIONS

- The matrix of second derivatives of a score, whose off-diagonal entries are the interactions. The second difference recovers a quadratic's curvature exactly, and a score that is additive across features, naive Bayes, has a diagonal Hessian because one feature's contribution is the same whatever the others are.
- The read operator is not the Hessian. Two features' sensitivities can rise and fall together across rows even

when neither changes the other's effect, so a diagonal Hessian does not give a diagonal read operator. Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-B-optim-D4 (034 · D4) [predicted]. Optimization — gradient compression, curvature (Hessian) read operator, on a REAL model (logistic regression); optional stretch [geometric-observation/claims/LEDGER.md:122](#).

FIRST STATED

Chapter 6 section 6.1 and chapter 7 section 7.2 of *Data Mining as Observation*, with the exact Hessian of a real logistic model in [geometric-observation/claims/LEDGER.md](#) row GO-B-optim-D4.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.2	real logistic model, exact Hessian, anti 300 of 300, flip 82 of 300, coupling diagnosis, bound not refutation	geometric-observation/claims/LEDGER.md row GO-B-optim-D4

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Curvature.lean](#), theorems `second_quad`, `second_affine`, `linear_error`, `gradient_change`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/NaiveBayes.lean](#), theorems `log_likelihood_sum`, `log_odds_sum`, `contribution_independent`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 6, 7, 14.

RELATED

[curvature](#) (p. 98); [naive Bayes](#) (p. 235); [boosting](#) (p. 48); [read operator](#) (p. 291); [Jacobian](#) (p. 192).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 6.1, 0.9.

Sources-table rows that share a record with the entry without naming it: chapter 7 section 7.2.

hierarchical clustering

instrument. Entry id `hierarchical-clustering`.

DEFINITION

A clustering that merges the closest pair of clusters at each step, under a linkage that reads the nearest pair, the farthest, or the average, so that merge heights never decrease. Chapter 9.

Example. Points at 0, 1, 5, and 6 merge 0 with 1 and 5 with 6 at height 1, then the two pairs at height 4 under single linkage.

EQUATION

Book equation 9.1.

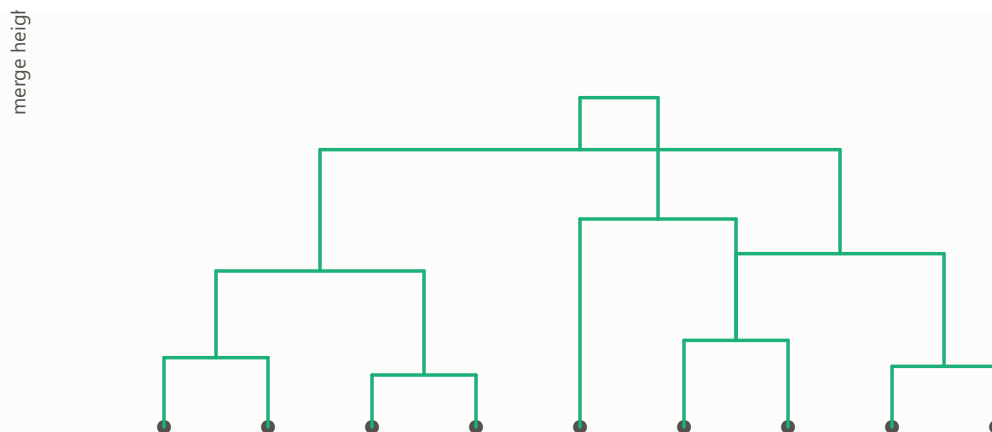


Figure 104: The closest pair merged at each step, with heights that never decrease.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

CONDITIONS

- A clustering that merges the closest pair of clusters at each step under a linkage that reads the nearest pair, the farthest pair, or the average. When the merged pair was the closest at some height, no updated distance falls below it, so the merge heights never decrease, single linkage is never above complete, and average linkage lies between them.
- The dendrogram inherits the distance's quotient at every level, so the tree is a certificate about the chosen quotient and not about the data.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 9 section 9.1 of *Data Mining as Observation*, after TSK chapter 7.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Hierarchical.lean](#), theorems `single_ge`, `complete_ge`, `average_ge`, `single_le_complete`, `average_between`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 9.

RELATED

[k-means](#) (p. 198); [DBSCAN](#) (p. 100); [Euclidean distance](#) (p. 133); [validity index](#) (p. 372); [quotient](#) (p. 279).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 3.1.

Sources-table rows that share a record with the entry without naming it: chapter 9 section 9.1, chapter 10 section 10.1.

hub

concept. Entry id hub.

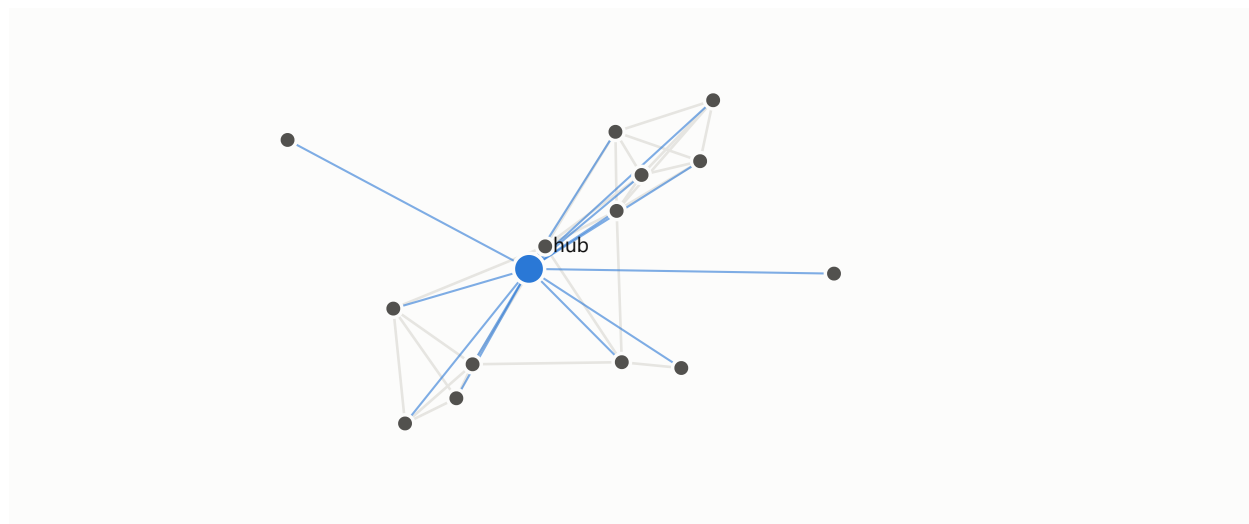


Figure 105: A row retrieved far more often than chance allows.

DEFINITION

A row retrieved as a nearest neighbour far more often than chance allows. The book shows hubness is almost entirely a property of the queries and the reader, not of the corpus. Chapters 3 and 10.

Example. One row appeared in 287 of the queries' neighbour lists, far above what chance allows.

EQUATION

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

Book equation 10.5.

$$\begin{aligned} \text{central} &: \text{pct}_{\text{centrality}} \geq 0.95; \\ \text{dense} &: \neg \text{central} \wedge \text{pct}_{\text{density}} \geq 0.85; \quad \text{prescribe by } f_{\text{central}}, f_{\text{dense}} \text{ at } 0.75. \end{aligned}$$

CONDITIONS

- A row retrieved as a nearest neighbour far more often than chance allows, a count above the Poisson ceiling. Since the counts sum to the slots handed out, the number of rows with count above a ceiling is at most the slots over the ceiling plus one, so hubs are few by arithmetic, and a higher ceiling names fewer of them.
- Whether a row is a hub depends on the queries and the retrieval rule alone, which is the sense in which hubness is a query property, and the program's first hub counts fell by a factor of several when query coupling was removed.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects*. NEG-11 [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [geometric-observation/claims/LEDGER.md:104](#).

FIRST STATED

Radovanović, Nanopoulos, and Ivanović, hubs in space, 2010, as chapter 3 section 3.5 of *Data Mining as Observation* reads it, with the program's finding in openvector-bench that hubness is a query property, openvector-bench/results/QUERY_COUPLING_ARTIFACT.md:1-20.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	Poisson null, hub excess, budget parameter	openvector-bench/openvector_bench/hubness.py:41-100
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131
chapter 11 section 11.6	count of ten, hubs and anti-hubs, max 78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, abstain below 2.5k, centering vs mutual-proximity rescaling	turboquant-pro/docs/HUBNESS_PRIMER.md:1-60, 60-170
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [geometric-observation/claims/LEDGER.md:104](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/Hub.lean](#), theorems `card_hubs_le`, `hubs_congr`, `hubs_anti`, `hubs_empty`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 3, 5, 10, 11, 12, 13.

RELATED

[hubness](#) (p. 176); [Poisson ceiling](#) (p. 262); [anti-hub](#) (p. 20); [Robin Hood index](#) (p. 313); [null model](#) (p. 242).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.17.

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.5, chapter 8 section 8.1, chapter 10 section 10.3, chapter 11 section 11.6.

hubness

result. Entry id hubness.

DEFINITION

The excess of a few rows in the nearest-neighbour lists of many queries over what the Poisson ceiling allows, measured by the busiest count and the Robin Hood index, a property of the queries. Chapter 3 section 3.5 and

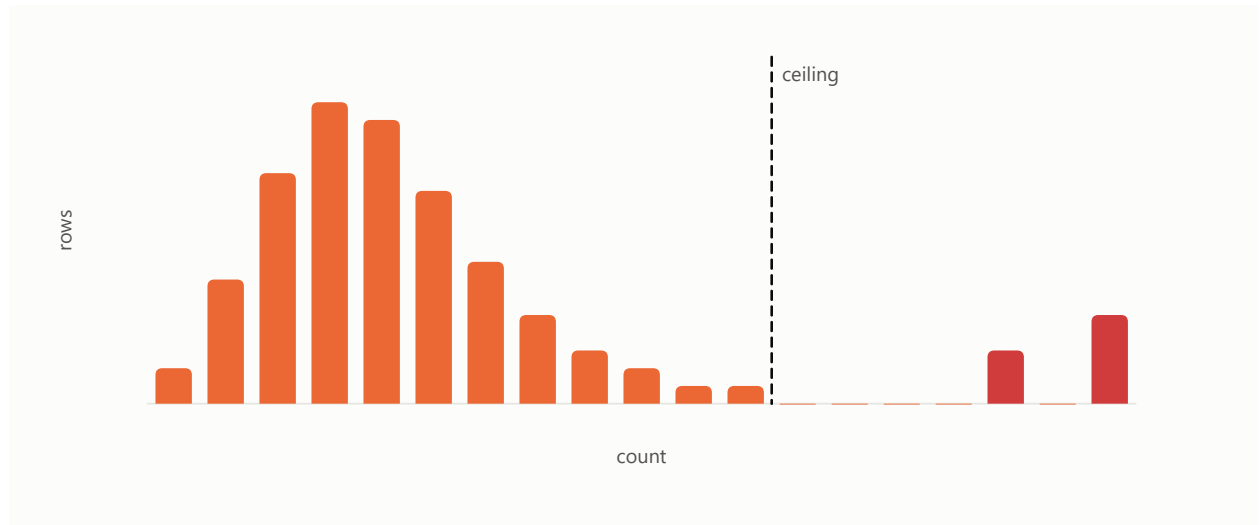


Figure 106: The count distribution's tail above the Poisson ceiling.

chapter 10.

Example. The busiest row's count fell from 287 to 213 and the skew from 3.970 to 3.177 when the query coupling was removed.

EQUATION

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

Book equation 0.35.

$$\text{RH} = \sum_i \max\left(0, \frac{N_i}{\sum_j N_j} - \frac{1}{n}\right).$$

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

CONDITIONS

- A hub is a row retrieved as a nearest neighbour far more often than the Poisson null allows, and the ceiling of that null is derived from the query count, the neighbourhood size, and the corpus size.
- Hubness is measured against a query workload and belongs to it. The two corpus-side mechanisms that were registered to explain it were refuted under seal.
- Anti-hubs come in kinds, and one kind is manufactured by the query budget through the pigeonhole floor, so a count of never-retrieved rows changes with the number of slots per row.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. NEG-15 (Bell boundary) [demonstrated]. *Query-conditioned hubness supplies a mechanism for Bell-inequality violation without action at a distance*. Refuted as a mechanism; the settings-as-queries re-framing survives only as vocabulary. [geometric-observation/claims/LEDGER.md:93](#).

- *refutes or corrects*. GO-5 [refuted]. An $\alpha=1$ density/hubness quotient restores invariant fidelity in ≥ 1 non-spectral domain. [geometric-observation/claims/LEDGER.md:67](#).
- *refutes or corrects*. NEG-11 [refuted]. (GO-5, prospective $\times 4$) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [geometric-observation/claims/LEDGER.md:104](#).

FIRST STATED

Radovanović, Nanopoulos, and Ivanović for the phenomenon. The observation program's reading, that hubness is almost entirely a property of the queries and the reader rather than the corpus, in openvector-bench and turboquant-pro, and in chapter 3 section 3.5 and chapter 10 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.6	ball-growth heat a missing-mean artifact, eleven campaigns, 3 of 12 to 11 of 12 cells	openvector-bench/README.md:160-170; openvector-bench/results/RC13_VERDICT.md:1-20
chapter 3 section 3.5	Poisson null, hub excess, budget parameter	openvector-bench/openvector_bench/hubness.py:41-100
chapter 3 section 3.5	query mass best single feature at every K for all four responses, seven features add at most 20 percent, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	openvector-bench/results/R13_STAGE0_RESULT.md:40-50; openvector-bench/results/R13_STAGE1_RESULT.md:1-40
chapter 3 section 3.6	Bell audit, max S 2.00000 across 72 configurations, d 3 to 128, correlation negative 0.036, post-selected 2.7308, controls 2.748, 2.386, 3.174, seed 20260817, sealed at 6e825d8	geometric-observation/articles/2026-08-03-hubness-does-not-weaken-bell.md:1-60 ; geometric-observation/claims/LEDGER.md row NEG-15; geometric-observation/results/GO-bell-geometry-audit.json
chapter 8 section 8.1	G1 339.9 vs 71.9 same corpus, real about 61, rounds 1 to 4 at 300 to 420, three families falsified	openvector-bench/results/QUERY_COUPLING_ARTIFACT.md:1-20
chapter 8 section 8.4	gate meta-rule, relative contrast discriminates nothing	openvector-bench/openvector_bench/score_rc1.py:1-14; turboquant-pro/docs/RESEARCH_ROADMAP.md:133-160
chapter 8 section 8.7	RC-7 five of ten, band 0.076 vs 0.19, gate missed by 0.007, 2.4 times, eight-block rule	openvector-bench/paper/profile/PROFILE_PAPER.md:576-598; openvector-bench/results/RC7_VERDICT.md
chapter 10 section 10.2	five kinds, balanced accuracy 0.611, 0.718, 0.684, the category table, per-category 0.57 to 0.82, the sweep, pigeonhole floor	openvector-bench/results/R13_STAGE1_RESULT.md:40-75
chapter 10 section 10.2	half 2 failed, at most 1.28 times against 2, withdrawal and its scope	openvector-bench/results/R13_STAGE1_RESULT.md:75-110
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro/docs/STRATA_RFC.md:24-98

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90 ; turboquant-pro/docs/STRATA_RFC.md:98-130
chapter 11 section 11.4	0.999 against own ranking vs 0.592 against fp32 truth, three truth layers, difficulty strata	openvector-bench/README.md:60-96
chapter 11 section 11.4	R80 table, 25x probe depth, 92 percent own cell, one third cross-article, 0.53 matches same-article share, relative contrast discriminates nothing	openvector-bench/results/R80_ANN.md:1-40 ; turboquant-pro/docs/RESEARCH_ROADMAP.md:133-160
chapter 11 section 11.6	query mass best single feature at every K for all four responses, seven features add at most 20 percent at 12 leaves, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	openvector-bench/results/R13_STAGE0_RESULT.md:40-50 ; openvector-bench/results/R13_STAGE1_RESULT.md:1-40
chapter 12 section 12.2	recall 0.999 vs 0.592, the derived death point within 6 percent across fourteen corpora	openvector-bench/README.md:60-96 ; geometric-observation/claims/LEDGER.md row GO-3
chapter 13 section 13.1	one hundred billion vectors on a preemptible fleet, systems result not a tier, corpus rejected by the admission battery	openvector-bench/README.md:250-263
chapter 13 section 13.1	twelve-figure corpus about 128 TB, kilobyte manifest	openvector-bench/README.md:60-80

FAILURES AND CORRECTIONS

- GO-5, [refuted]. An $\alpha=1$ density/hubness quotient restores invariant fidelity in ≥ 1 non-spectral domain. [geometric-observation/claims/LEDGER.md:67](#).
- NEG-11, [refuted]. (GO-5, prospective $\times 4$) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [geometric-observation/claims/LEDGER.md:104](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/Hubness.lean](#), theorems `sum_count`, `sum_count_eq`, `count_congr`, `antiHub_iff`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Hub.lean](#), theorems `card_hubs_le`, `hubs_congr`, `hubs_anti`, `hubs_empty`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 3, 5, 10, 11, 12, 13.

RELATED

[hub](#) (p. 175); [anti-hub](#) (p. 20); [Poisson ceiling](#) (p. 262); [quotient](#) (p. 279); [certificate](#) (p. 59); [rank certificate](#) (p. 285); [min-over-strata](#) (p. 226).

SEE ALSO

none

I

identity reader

concept. Entry id identity-reader.

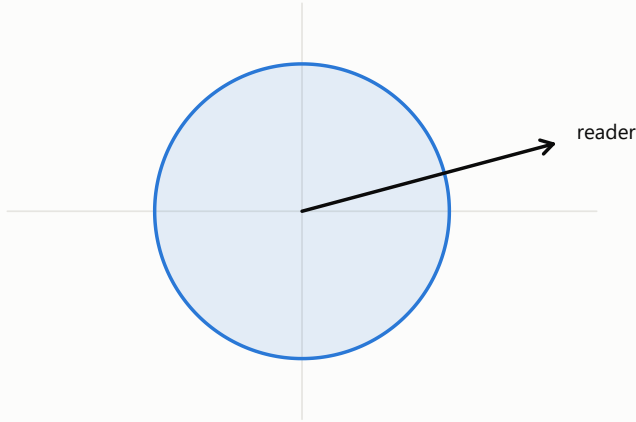


Figure 107: The reader that weighs every direction equally.

DEFINITION

The consumer whose read operator is the identity, for which read distortion is mean squared error. Every consumer-blind metric assumes it. Chapter 0 section 0.5.

Example. The identity reader charges $0.3 + 1.7 = 2.0$ for either of the two codes and cannot tell them apart.

EQUATION

Book equation 0.10.

$$d_O = \text{tr}(P_C M_\delta) = \mathbb{E}[\delta^\top P_C \delta], \quad M_\delta = \mathbb{E}[\delta \delta^\top], \quad P_C = I \Rightarrow d_O = \text{tr} M_\delta.$$

Book equation 1.2.

$$d_O = \text{tr}(P_C M_\delta), \quad (d_O = \text{tr} M_\delta \text{ for every admissible } M_\delta) \iff P_C = I.$$

CONDITIONS

- Read distortion equals the trace of the error's second moment for every admissible error exactly when the read operator is the identity. A particular error can make the two agree by accident.
- Explained variance and the rule of picking rank at 95 percent are the identity reader's criteria, and the book records them as the benchmark's rules rather than its own.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 2, `geometric-observation/chapters/ch02_failure_of_observer_free_measurement.md`, DOI 10.5281/zenodo.21776291, where reconstruction error is the distortion of the reader that reads every

direction equally.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadDistortion.lean](#), theorems `read_distortion`, `identity_reader`, `quad_one`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 8, 9, 10, 12, 13.

RELATED

[read operator](#) (p. 291); [read distortion](#) (p. 290); [the flip](#) (p. 147); [quotient](#) (p. 279).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 4.1.

Ledger rows that cite the entry's records without naming it: GO-2 (neg. half: not reconstruction).

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 4 section 4.3, chapter 4 section 4.5.

importance

instrument. Entry id importance.

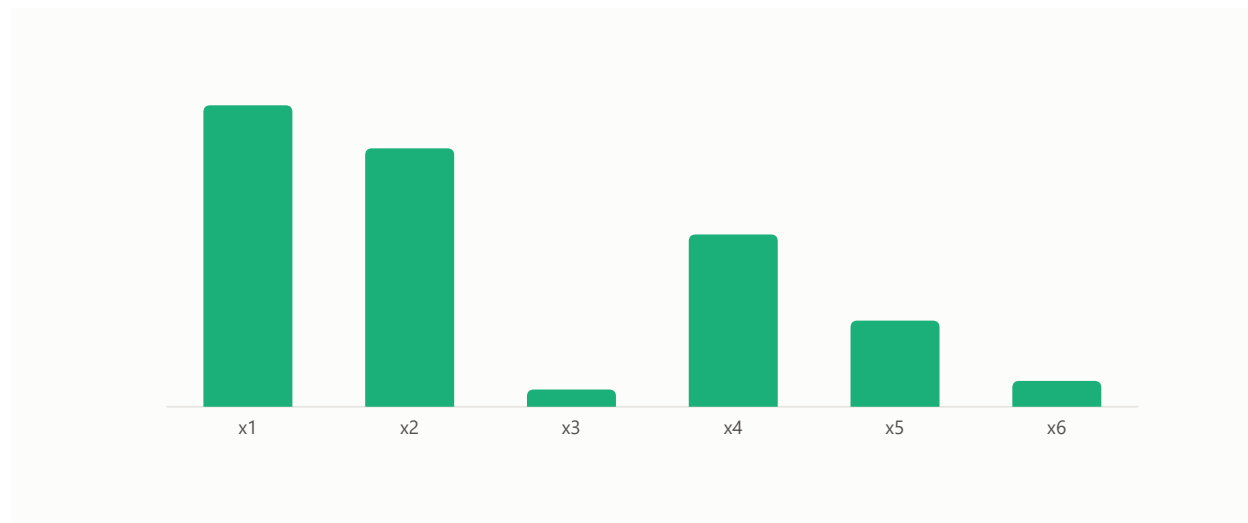


Figure 108: How much a classifier reads each feature, counting splits for a tree.

DEFINITION

A per-feature score of how much a classifier reads it. For a tree it counts splits, which is the right sensitivity for a reader that is flat within leaves. Permutation importance and partial dependence answer related questions. Chapter 7 and chapter 14.

Example. A tree that splits on x_1 twice and on x_3 once and never on x_2 gives importances 2, 0, and 1.

EQUATION

none

CONDITIONS

- A per-feature score of how much a classifier reads it. For a gradient reader the averaged squared sensitivity is the read operator's diagonal, and for a tree, whose finite difference is zero at every row that does not straddle a split, counting splits is the right sensitivity.
- Permutation importance averages the consumer over rows that may never occur together, and partial dependence reads it at inputs it was never trained on, so the three answer different questions and the report names which.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Breiman, random forests, 2001, as chapter 7 section 7.1 and chapter 14 section 14.5 of *Data Mining as Observation* read it, with the selection-consumer regime in `readscope/readscope/regimes.py:1-60`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.1	bagging variance, out-of-bag estimation, random forests, importances, AdaBoost	Hastie, Tibshirani, Friedman, ESL 2e chapters 15 and 10.1; TSK 2e 4.10

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Attribution.lean](#), theorems `attr_sum_affine`, `attr_unread`, `sq_sensitivity_eq_readOp_diag`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/DecisionTree.lean](#), theorems `stump_flat_left`, `stump_flat_right`, `split_reads_one_axis`, `gini_le_half`, `gini_eq_zero_iff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 4, 6, 7, 14.

RELATED

[attribution](#) (p. 29); [decision tree](#) (p. 102); [sensitivity](#) (p. 328); [read operator](#) (p. 291); [ensemble](#) (p. 129).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 14.5, 0.9, 6.1.

Ledger rows that cite the entry's records without naming it: GO-EC-3.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.3, chapter 6 section 6.1, chapter 14 section 14.5.

imputation

instrument. Entry id `imputation`.

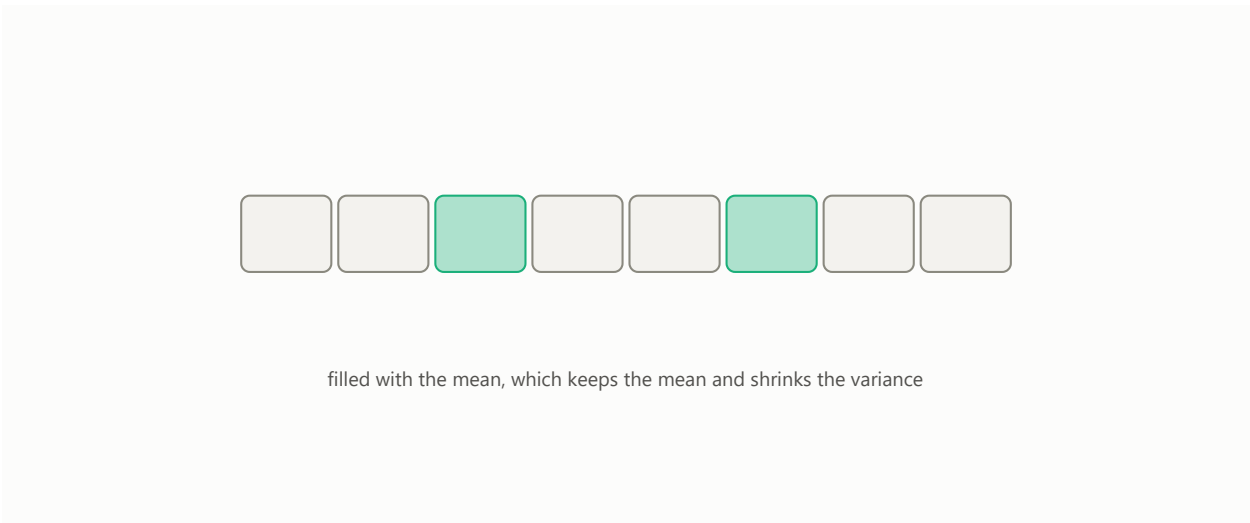


Figure 109: A value written into a missing cell that the consumer will read.

DEFINITION

Writing a value into a missing cell that the consumer will read. Mean imputation leaves the column mean unchanged and shrinks its variance. Chapter 2.

Example. Filling three missing cells of a column with mean 5 leaves the mean at 5 and shrinks the variance.

EQUATION

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

CONDITIONS

- Writing a value into a missing cell that the consumer will read. Mean imputation leaves the column mean unchanged, the filled cells add nothing to the sum of squared deviations, and the variance shrinks because the same sum of squares is spread over more rows.
- An imputation is a claim about the mechanism that produced the gap, and it is fit inside the fold and stated in the preregistration, because an imputation fit on all rows leaks the test fold into the training fold.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 2 section 2.3 of *Data Mining as Observation*, with the missing-data rule as a required preregistration field in `observation-theory-campaigns/experiments/PREREG-TEMPLATE.md:47-52`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.3	MCAR, MAR, MNAR and the remedies, imputation before splitting	TSK 2e section 2.2; instructor working documents, not public [@bond2026course], <code>ECE_514-01_FA26_session-outlines.md:64-72</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Imputation.lean](#), theorems `mean_imputed`, `sum_sq_imputed`, `variance_imputed_le`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 2, 8.

RELATED

[standardization](#) (p. 348); [read operator](#) (p. 291); [leakage](#) (p. 210); [preregistration](#) (p. 267).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.9.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.3.

intrinsic dimension

concept. Entry id intrinsic-dimension.

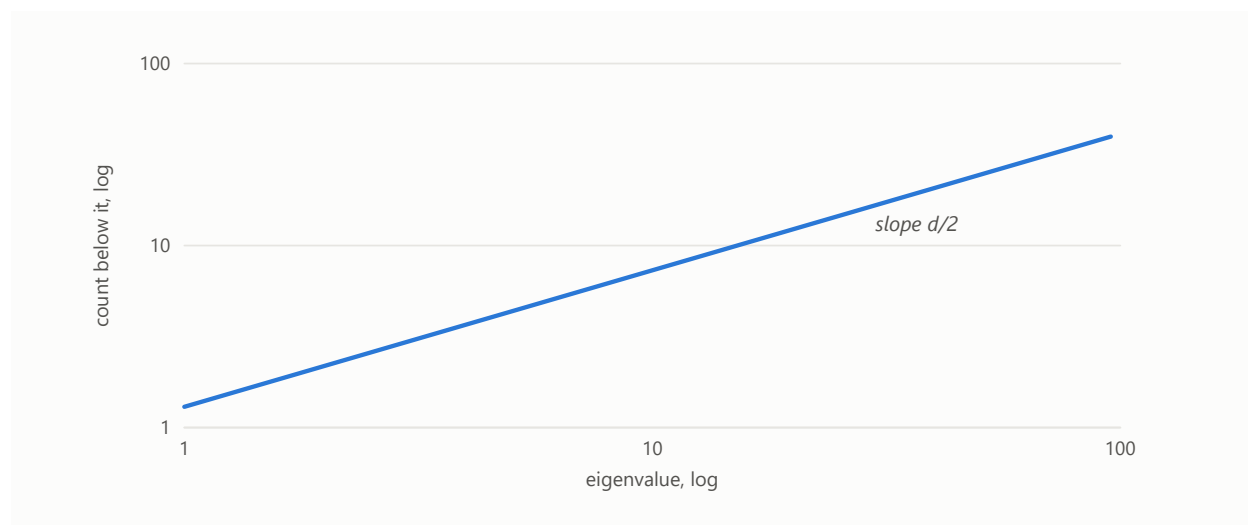


Figure 110: The dimension read from the slope of the eigenvalue count.

DEFINITION

The number of directions a dataset varies along locally, whatever the number of its coordinates and whatever the number of its covariance directions. Chapter 0 section 0.10.

Example. A circle drawn in three coordinates has intrinsic dimension 1; the recognizer read 1.81, 2.80, and 1.74 on its three templates.

EQUATION

Book equation 0.31.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

CONDITIONS

- The number of directions a dataset varies along locally, whatever its number of coordinates. Read from the slope of the Laplacian eigenvalue count on log axes by Weyl's law, it is an asymptotic estimate that depends on the graph's construction and the range of eigenvalues fitted. It is not the number of covariance directions, since a circle has intrinsic dimension one and two covariance directions.
- The recognizer reads dimension before shape, and the battery's dimensions of 1.81, 2.80, and 1.74 were read against sealed bars on templates built for the purpose.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Weyl's law, 1911, as chapter 0 section 0.15 states it, applied in the recognizer of Volume 14 chapter 11, [geometric-observation/chapters/ch11_the_recognizer.md:1-95](#), and chapter 9 section 9.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.2	the recognizer's mechanism, low multiplets and angular distances, dimension before shape by Weyl's law, refusal, the growth gotcha	geometric-observation/chapters/ch11_the_recognizer.md:1-95

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/IntrinsicDimension.lean](#), theorems `log_weyl`, `dimension_from_slope`, `weyl_double`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 8, 9, 11.

RELATED

[recognizer](#) (p. 296); [effective rank](#) (p. 123); [distance concentration](#) (p. 115); [vacuity threshold](#) (p. 371).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.7.

Ledger rows that cite the entry's records without naming it: GO-3.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.3, chapter 9 section 9.1, chapter 9 section 9.2, chapter 9 section 9.4, chapter 11 section 11.1, chapter 14 section 14.3, chapter 14 section 14.5, chapter 14 section 14.7.

inverted file

instrument. Entry id `inverted-file`.

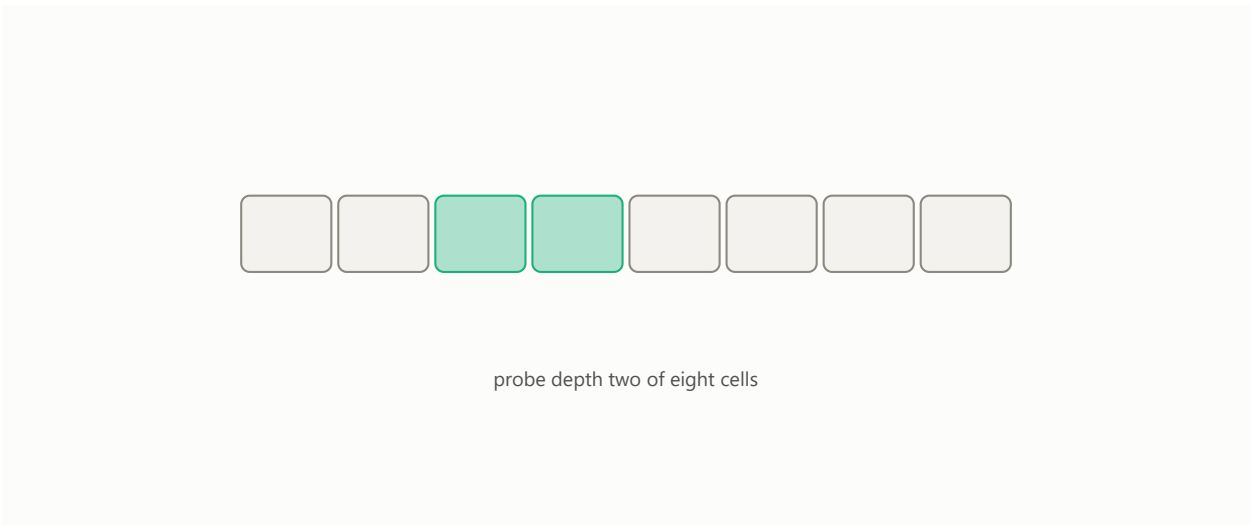


Figure 111: The vectors partitioned into cells, and a query that probes a few.

DEFINITION

An index that partitions vectors into cells by k-means and answers a query by searching only the cells nearest to it. The number of cells searched is the probe count. Chapter 11.

Example. A million vectors in 1024 cells, probing 8 cells, scores about 8000 candidates per query.

EQUATION

none

CONDITIONS

- An index that partitions vectors into cells by k-means and searches only the cells nearest the query. Probing more cells never loses a candidate, a neighbour is found exactly when its cell is probed, the scanned fraction is the probed cells’ size over the corpus, and with equal cells it is the probe count over the cell count.
- The recall at one probe against eight in the chapter’s table is a measurement, and the vacuity threshold of the certificate predicts where single-stage retrieval dies, which is the ledger’s demonstrated row.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Sivic and Zisserman’s video Google and the IVF index of Jégou, Douze, and Schmid, as chapter 11 section 11.3 of *Data Mining as Observation* presents them, with the program’s probe sweeps in `openvector-bench`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 14 section 14.8	quantum falsified, 320 probes, contextuality zero, DOI 10.5281/zenodo.20660110	non-abelian-sqnd/README.md:5-22; sqnd-probe/README.md:12-20; sqnd-probe/CITATION.cff:16

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/InvertedFile.lean](#), theorems `scanned_mono`, `found_iff`, `scannedFraction_mem_unit`, `equal_cells`, `scanned_univ`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 11, 13, 14.

RELATED

[recall at k](#) (p. 295); [product quantization](#) (p. 271); [vacuity threshold](#) (p. 371); [anti-hub](#) (p. 20).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.3, 11.4.

Ledger rows that cite the entry's records without naming it: GO-3, NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 11 section 11.2, chapter 11 section 11.7, chapter 12 section 12.3.

isotropic

concept. Entry id isotropic.

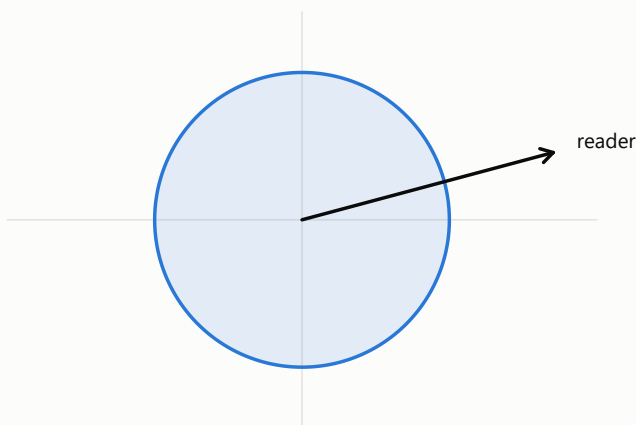


Figure 112: Every direction has the same variance.

DEFINITION

Of a covariance, having the same variance in every direction. Chapter 0.

Example. Covariance `diag(1, 1)` is isotropic, and a reader at any angle sees variance 1.

EQUATION

none

CONDITIONS

- An isotropic covariance has the same variance in every direction, so every unit reader reads it the same and no flip is possible. An anisotropic one has different variances in different directions, and two unit readers can read it differently.
- The flip needs anisotropy. Two codes with the variances exchanged are read the same by every reader exactly when the two variances agree, which is the boundary the coupling null names from the other side.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.3 and chapter 3 section 3.2 of *Data Mining as Observation*, with the isotropic-Gaussian control of chapter 10.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Isotropy.lean](#), theorems `isotropic_reads_same`, `isotropic_no_flip`, `anisotropic_readers_differ`, `flip_iff_anisotropic`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 7, 9, 10, 11, 12.

RELATED

[the flip](#) (p. 147); [coupling null](#) (p. 89); [covariance matrix](#) (p. 91); [alignment](#) (p. 15).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 3.1, 4.6.

Ledger rows that cite the entry's records without naming it: GO-2 (neg. half: not reconstruction).

itemset

concept. Entry id `itemset`.

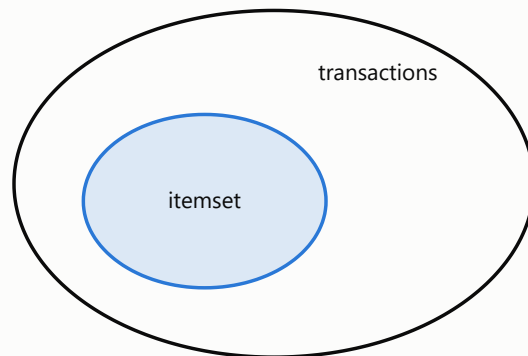


Figure 113: A set of items, with the transactions that contain it as its support.

DEFINITION

A set of items that appear together in a transaction. A closed itemset has no superset with the same support and a maximal one has no frequent superset. Chapter 5.

Example. Bread and butter together is an itemset, and its support is the number of baskets holding both.

EQUATION

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

CONDITIONS

- A set of items that appear together in a transaction. Its support lies in the unit interval, the empty itemset has support one, a superset has support at most that of a subset, and the support of a union is at most the smaller of the two supports.
- A closed itemset has no strict superset with the same support and a maximal frequent itemset has no frequent strict superset, and every maximal frequent itemset is closed, since a superset with the same support would be frequent too.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Agrawal, Imieliński, and Swami, mining association rules, 1993, as TSK chapter 5 presents it and chapter 5 section 5.1 of *Data Mining as Observation* states it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ItemSet.lean](#), theorems `supportFrac_mem_unit`, `supportFrac_empty`, `supportFrac_anti`, `supportFrac_union_le`, `closed_of_maximal`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 5.

RELATED

[support](#) (p. 353); [Apriori principle](#) (p. 25); [safe pruning](#) (p. 320); [confidence](#) (p. 75).

SEE ALSO

none

J

Jaccard

concept. Entry id jaccard.

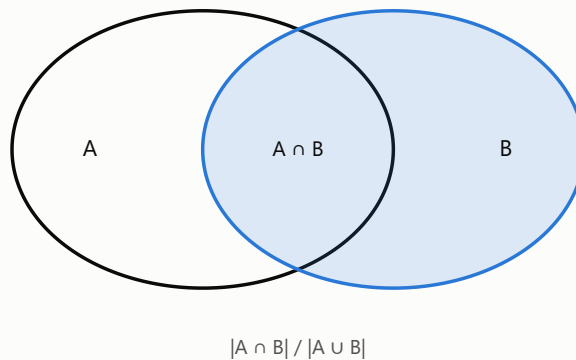


Figure 114: Intersection over union.

DEFINITION

The size of the intersection of two sets over the size of their union, between zero and one, zero exactly when the sets are disjoint, which discards joint absence. Chapter 0 section 0.6 and chapter 3.

Example. Sets {a, b, c} and {b, c, d} share 2 of 4 elements, a Jaccard coefficient of 0.5.

EQUATION

none

CONDITIONS

- The size of the intersection of two sets over the size of their union, between zero and one, one for a nonempty set against itself, symmetric, and zero exactly when the sets are disjoint. A subset's coefficient is its share of the larger set.
- It reads nothing outside the union, so its distance lives on the quotient that discards joint absence, which is the right quotient for a consumer that reads presence and the wrong one for a consumer that reads counts.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Jaccard, the distribution of the flora in the alpine zone, 1912, as chapter 0 section 0.6 and chapter 3 section 3.1 of *Data Mining as Observation* read it.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Jaccard.lean](#), theorems `jaccard_nonneg`, `jaccard_le_one`, `jaccard_self`, `jaccard_comm`, `jaccard_eq_zero_iff`, `jaccard_subset`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3.

RELATED

[cosine](#) (p. 87); [quotient](#) (p. 279); [bag of words](#) (p. 36); [Euclidean distance](#) (p. 133); [itemset](#) (p. 188).

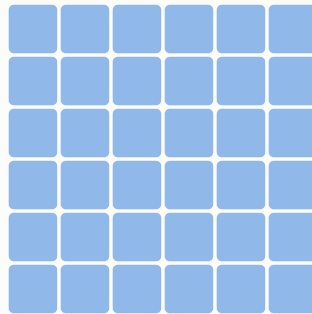
SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.1, 3.1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.2.

Jacobian

concept. Entry id `jacobian`.



rows are outputs, columns inputs

Figure 115: The partial derivatives of a vector-valued consumer.

DEFINITION

The matrix of partial derivatives of a vector-valued consumer, whose transpose times itself averages to the read operator. A discrete stage has none. Chapter 0 section 0.5 and chapter 12.

Example. A consumer mapping (x_1, x_2) to $(x_1 + x_2, x_1 - x_2)$ has Jacobian with rows $(1, 1)$ and $(1, -1)$.

EQUATION

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

CONDITIONS

- The matrix of partial derivatives of a vector-valued consumer, whose transpose times itself averages to the read operator. Along a chain of stages the rank of the composed operator is at most the rank of either stage, and a stage that identifies two inputs identifies them for every stage after it.
- Chunking and indexing have no Jacobian, since a document goes to a set of pieces and a query to a candidate list, and what survives without a derivative is the statement about distinctions.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.5 and chapter 2 section 2.2 of *Data Mining as Observation*, with the pullback composition and the rank bound in `geometric-observation/ch06_mathematical_preliminaries.md:10-27`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Pipeline.lean](#), theorems `quotient_inherited`, `quotient_inherited_chain`, `rank_comp_le_first`, `rank_comp_le_second`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 12.

RELATED

[read operator](#) (p. 291); [sensitivity](#) (p. 328); [Hessian](#) (p. 172); [retrieval-augmented pipeline](#) (p. 312).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.9.

Ledger rows that cite the entry's records without naming it: OT-7.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 12 section 12.2.

K

Kendall correlation

concept. Entry id `kendall-correlation`.

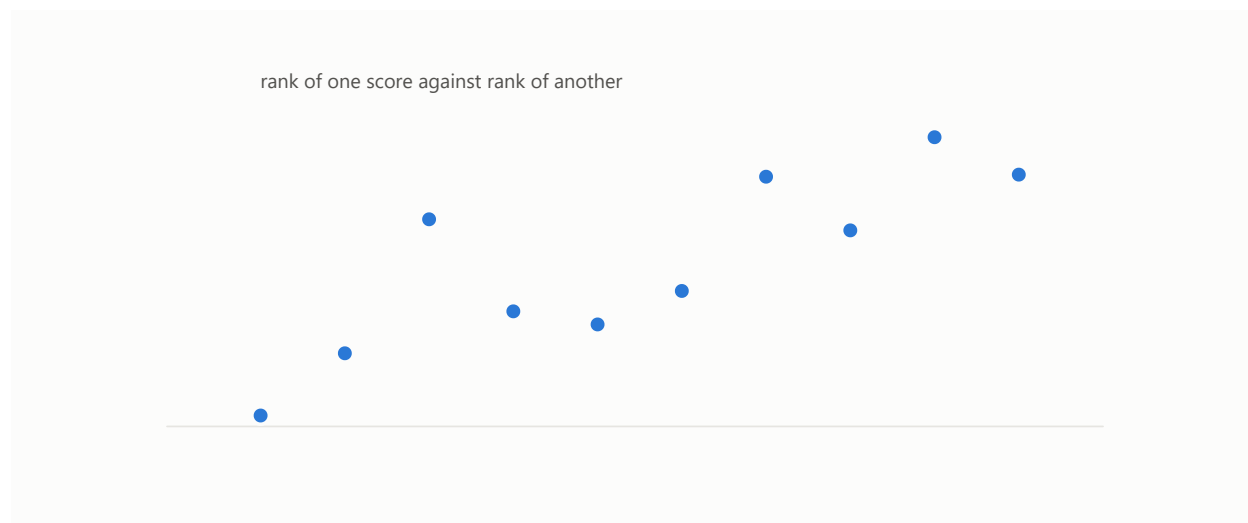


Figure 116: Concordant pairs less discordant pairs over all pairs.

DEFINITION

The fraction of item pairs two rankings order the same way, minus the fraction they order differently. Equation 0.15.

Example. Rankings (1, 2, 3) and (1, 3, 2) agree on two pairs and disagree on one, Kendall tau one third.

EQUATION

Book equation 0.15.

$$\tau = \frac{\#\{\text{concordant pairs}\} - \#\{\text{discordant pairs}\}}{n(n-1)/2}.$$

CONDITIONS

- The fraction of item pairs two rankings order the same way minus the fraction they order differently. It lies between minus one and one, it is one when no pair disagrees, and since it depends only on the orderings a strictly increasing transform of either score leaves it unchanged.
- It is the rank-agreement statistic the rank certificate bounds from below, and a floor on it is a floor on the consumer's rankings, not on any distance.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Kendall, a new measure of rank correlation, 1938, as chapter 0 section 0.9 states it, and the rank certificate's statistic in `readscope` and `openvector-bench`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Kendall.lean](#), theorems `not_both`, `card_add_le`, `tau_mem`, `concordant_comp`, `discordant_comp`, `tau_comp`, `discordant_self`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 8, 11.

RELATED

[rank certificate](#) (p. 285); [rank-faithful](#) (p. 286); [Monotone Invariance Theorem](#) (p. 227); [recall at k](#) (p. 295).

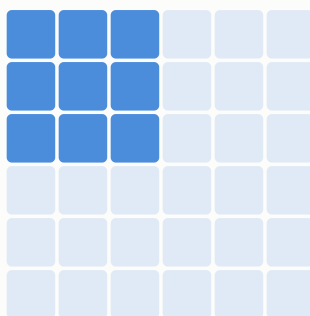
SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.2.

Ledger rows that cite the entry's records without naming it: GO-B-Llama, GO-B-Llama-rematch.

kernel

concept. Entry id `kernel`.



the unread block

Figure 117: The directions a matrix sends to zero, the nuisance for the read operator.

DEFINITION

The directions a matrix sends to zero. The kernel of the read operator is the nuisance. Chapter 0 section 0.5.

Example. The read operator of the consumer $x_1 + x_2$ sends $(1, -1)$ to zero, so $(1, -1)$ is in its kernel.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 0.12b.

$$x \sim_{\bar{P}_{C,\mu}} x' \iff x - x' \in \ker \bar{P}_{C,\mu}.$$

CONDITIONS

- The directions a matrix sends to zero. The kernel of the read operator is the nuisance, the directions the consumer never reads, and a direction is in it exactly when every row's sensitivity is orthogonal to it. The kernel's dimension is the dimension less the rank.
- The reconstruction coder spends bits on the kernel and the output coder does not, and the gap between them is governed by the share of the error's entropy that lies in the kernel.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.5 and section 0.6 of *Data Mining as Observation*, with the kernel entropy share in `geometric-observation/claims/LEDGER.md` row GO-6.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Rank.lean](#), theorems `rank_le_width`, `rank_le_height`, `rank_outer_le_one`, `rank_mul_le`, `rank_zero`, `rank_transpose`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 11.

RELATED

[nuisance](#) (p. 240); [read operator](#) (p. 291); [read subspace](#) (p. 293); [quotient](#) (p. 279); [rank](#) (p. 283).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.12a.

Ledger rows that cite the entry's records without naming it: OT-7, GO-6.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 4 section 4.2.

KL divergence

concept. Entry id `k1-divergence`.

DEFINITION

A measure of how far one probability distribution is from another, zero when they are identical. Equation 0.24.

Example. A fair coin coded as if heads had probability 0.9 costs about 0.74 bits of divergence per toss.

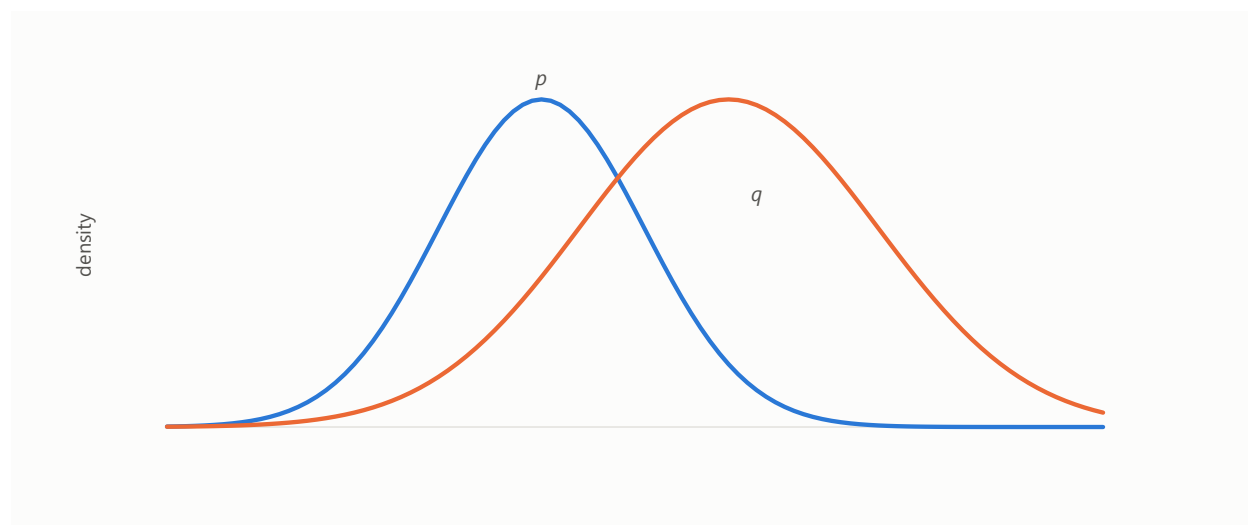


Figure 118: How many extra bits coding with q costs when the truth is p .

EQUATION

Book equation 0.24.

$$\text{KL}(p \parallel q) = \sum_i p_i \ln \frac{p_i}{q_i} \geq 0.$$

Book equation 0.22.

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(w_t \mid w_{<t}).$$

CONDITIONS

- The expected log ratio of two distributions' masses under the first. It is nonnegative by Gibbs' inequality, zero when the distributions agree, and not symmetric, so the direction of the comparison is part of the claim.
- The book's use is through perplexity. A reconstruction at cosine 0.995 raised the perplexity by three orders of magnitude, which is the case that reconstruction error is not the consumer's error.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Kullback and Leibler, on information and sufficiency, 1951, as chapter 0 section 0.11 states it beside perplexity, with the program's case in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation/chapters/ch02_failure_of_observer_free_measurement.md:40-60 ; geometric-observation/chapters/ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro/docs/KV_KEYS_FINDING.md:1-49
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49 ; geometric-observation/chapters/ch16_honest_negatives.md NEG-2

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/KL.lean](#), theorems `k1_nonneg`, `k1_self`, `k1_not_symm`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 8, 11.

RELATED

[identity reader](#) (p. 180); [read distortion](#) (p. 290); [calibration](#) (p. 56); [the flip](#) (p. 147).

SEE ALSO

Ledger rows that cite the entry's records without naming it: NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.5, chapter 3 section 3.2, chapter 11 section 11.1, chapter 11 section 11.2.

k-means

instrument. Entry id `k-means`.

DEFINITION

A clustering that minimizes the sum of squared errors to the cluster centres. The mean minimizes each cluster's error and assigning each row to its nearest centre never raises the total, so it is the identity reader's clustering. Chapter 9.

Example. Points 1, 2, and 6 with two centres settle at 1.5 and 6, with sum of squared errors 0.5.

EQUATION

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

CONDITIONS

- A clustering that minimizes the sum of squared errors to the cluster centres. The error about any centre is the error about the mean plus the row count times the squared distance between the two, so the mean minimizes each cluster's error, and assigning each row to its nearest centre never raises the total.

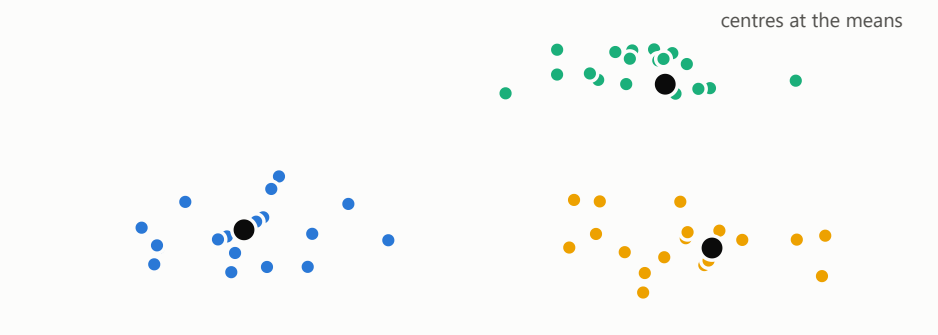


Figure 119: Centres at the means, rows assigned to the nearest.

- It is the identity reader's clustering. It weighs every direction equally, and a direction with large variance and no group structure pulls its centres exactly as variance-based reduction pulls its components.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Lloyd, least squares quantization in PCM, 1957, as chapter 9 section 9.1 of *Data Mining as Observation* reads it.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/KMeans.lean](#), theorems `sse_decomposition`, `mean_minimizes`, `assign_nearest`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 9, 10, 11.

RELATED

[identity reader](#) (p. 180); [validity index](#) (p. 372); [silhouette](#) (p. 332); [inverted file](#) (p. 185); [DBSCAN](#) (p. 100).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.30, 4.2.

Ledger rows that cite the entry's records without naming it: GO-3.

Sources-table rows that share a record with the entry without naming it: chapter 9 section 9.4, chapter 10 section 10.1.

KV cache

concept. Entry id kv-cache.



a window keeps the last tokens

Figure 120: One key and one value per token per head, kept for the next token.

DEFINITION

The stored keys and values of earlier tokens, kept so they need not be recomputed. Compressing it is where several of the book's examples go wrong. Chapters 0 and 11.

Example. Serving 14000 tokens with 32 heads keeps 14000 keys and values per head for the next token to read.

EQUATION

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

CONDITIONS

- The stored keys and values of every earlier token for every layer, so its size is the product of layers, tokens, two, the head width, and the bits per number. Compressing it changes the keys the heads read.
- A compression that preserves every query-key score preserves every head output, and one that preserves the keys' cosine need not, since two keys of the same direction and different lengths have cosine one and different scores. That is the attention-key finding in one sentence.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* GO-2/GO-12/GO-13 operational (KV serving, 077) [demonstrated]. Consumer-relative access width measured on a production serving stack (Qwen2.5-7B KV-cache eviction, matched budget): task quality tracks measured predictive uncertainty u about the consumer's future reads, not nominal scorer width — the ... [geometric-observation/claims/LEDGER.md:81](#).

FIRST STATED

Chapter 0 section 0.11 and chapter 11 section 11.1 of *Data Mining as Observation*, with the program's measurements in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49` and the serving-stack ledger row.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/KVCache.lean](#), theorems `cacheBits_tokens`, `cacheBits_half`, `ratios`, `output_of_scores_preserved`, `cosine_not_score`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 11.

RELATED

[attention](#) (p. 26); [eviction](#) (p. 134); [the flip](#) (p. 147); [budget](#) (p. 51).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.1.

Ledger rows that cite the entry's records without naming it: NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 8 section 8.2, chapter 8 section 8.9, chapter 11 section 11.1, chapter 11 section 11.2.

L

lag

concept. Entry id lag.



Figure 121: How far a replica trails the primary.

DEFINITION

How far a replica trails the primary. The same lag gives two readers two staleness rates. Chapter 0 section 0.13 and chapter 13.

Example. A replica 50 writes behind the primary is fresh for a reader that touches none of those 50 rows and stale for one that does.

EQUATION

none

CONDITIONS

- How far a replica trails the primary, in writes or in time. A read of the replica is stale for a consumer whose footprint the lag has touched and fresh for one it has not, so the same lag gives two readers two staleness rates, and a mixed workload's rate lies between them.
- Postgres read 0.50 to 0.06, MongoDB 0.47 to 0.03, and production Postgres 0.47 to 0.02 across readers on the same lag, with disjoint seeds.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.13 of *Data Mining as Observation*, with the replica measurements in `observation-theory-campaigns/experiments/DATABASE-FRESHNESS-TRACK.md`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Replica.lean](#), theorems `stale_mono`, `stale_zero`, `naive_certificate`, `witnessed_certificate`, `witnessed_coverage`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Freshness.lean](#), theorems `disagree`, `mixedRate_between`, `mixedRate_eq_left_iff`, `stale_for_all`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 13.

RELATED

[replica](#) (p. 307); [freshness](#) (p. 156); [coherence time](#) (p. 67); [false-clear rate](#) (p. 141); [footprint](#) (p. 153).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.26, 13.2, 0.25.

Ledger rows that cite the entry's records without naming it: OT-11.

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.2, chapter 13 section 13.3.

Landauer's principle

result. Entry id `landauers-principle`.

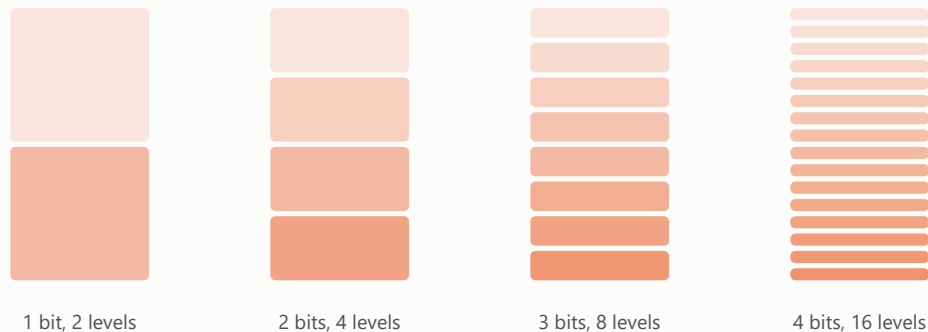


Figure 122: Erasing a bit costs at least $kT \ln 2$ of energy.

DEFINITION

Erasing one bit costs at least Boltzmann's constant times the absolute temperature times the natural log of two. Equation 0.27.

Example. Erasing one bit at 300 kelvin costs at least 2.87 times 10^{-21} joules.

EQUATION

Book equation 0.27.

$$E_{\min} = k_B T \ln 2 \approx 2.87 \times 10^{-21} \text{ J at 300 K.}$$

Book equation 13.3.

$$W_{\text{reset}} \geq k_B T \ln 2 \cdot H(M | S), \quad H(M | S) \leq H(M).$$

CONDITIONS

- Erasing one bit costs at least Boltzmann's constant times the absolute temperature times the natural logarithm of two, 2.87 times ten to the minus twenty-one joules at 300 kelvin. The reset bound of chapter 13 charges that price per bit of the record's conditional entropy given what the consumer keeps.
- Conditioning cannot raise entropy, so the consumer-relative bound is never above the consumer-free one, and a record that keeps nothing about the erased part costs nothing to reset. The entropies themselves are the ledger's measurements, two source families and six codebooks.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-7 [replicated]. A stored description's description rate and its conditional Landauer reset content are operationally separate resources: the same finite- n code index needing $\hat{R} \approx 0.67$ bits/symbol to describe is fully recoverable from retained ... [geometric-observation/claims/LEDGER.md:69](#).
- *measures*. GO-8 [replicated]. On two independent source families (binary Markov; Gaussian AR(1)), a fixed stored record's operational reset threshold rises with the age of the retained side information exactly as the staleness-work complement prices it: same record, ... [geometric-observation/claims/LEDGER.md:71](#).
- *measures*. GO-9 [replicated]. Coordinated reset is operationally cheaper than independent reset by the records' shared-structure information: with two consumer records sharing a component, recovering either record's bin residual with the *other record intact* lowers ... [geometric-observation/claims/LEDGER.md:73](#).

FIRST STATED

Landauer, irreversibility and heat generation in the computing process, 1961, as chapter 0 section 0.13 states it, and its consumer-relative form in Volume 14 and the rate-work paper of ledger rows GO-7 to GO-9.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	geometric-observation/claims/LEDGER.md row GO-6; geometric-observation/chapters/ch07_cost.md
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation/claims/LEDGER.md row GO-4
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	geometric-observation/claims/LEDGER.md row GO-1
chapter 7 section 7.2	curvature reader, identical reconstruction 0.298, flip twelve of twelve both, projected variance twelve of twelve, reconstruction Spearman negative 0.20, identity curvature tie	geometric-observation/experiments/G02-gradient-curvature-NOTES.md:1-40 ; geometric-observation/claims/LEDGER.md row GO-B

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.2	real logistic model, exact Hessian, anti 300 of 300, flip 82 of 300, coupling diagnosis, bound not refutation	geometric-observation/claims/LEDGER.md row GO-B-optim-D4
chapter 9 section 9.3	the margin certificate, mu crit as the expected maximum of N minus 1 standard normals, rho, death at 0.948 within 6 percent, Spearman 0.991 vs 0.873, fourteen corpora, six gates, the v1 to v3 path, the standing correction	geometric-observation/experiments/G03-certificate-vacuity-v3-NOTES.md:1-60 ; geometric-observation/claims/LEDGER.md row GO-3
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	geometric-observation/claims/LEDGER.md row GO-1; geometric-observation/chapters/ch10_the_blind_probe.md:34-52
chapter 12 section 12.2	recall 0.999 vs 0.592, the derived death point within 6 percent across fourteen corpora	openvector-bench/README.md:60-96 ; geometric-observation/claims/LEDGER.md row GO-3
chapter 12 section 12.3	uncompressed 0.79, centred 0.84	geometric-observation/claims/LEDGER.md row GO-B-legal
chapter 12 section 12.3	041 non-oracle, frozen LSA TF-IDF to SVD 100 train-only, AUROC 0.975 vs 0.910, flip tied, magnitude overshoot, partial	geometric-observation/chapters/ch10_the_blind_probe.md:100-118 ; geometric-observation/claims/LEDGER.md row GO-B-blind 041
chapter 13 section 13.5	GO-7, 0.67 bits per symbol, bin rate 0.26 equals 0.39, error 1.00 without side information, two families, six codebooks	geometric-observation/claims/LEDGER.md row GO-7
chapter 13 section 13.5	GO-8, 0.10 to 0.55 across ages 0 to 64, flip probability 0.05, 1 percent to 100 percent at age 32, Gaussian pass 5 of 5, the control-statistic caveat	geometric-observation/claims/LEDGER.md row GO-8; geometric-observation/experiments/G0-landauer-gaussian-secondsettings-NOTES.md
chapter 13 section 13.6	attempt three, windows 1024, 256, 32, uncertainty 0.982 to 0.892, 5 of 6, 0.4375 with SE 0.070 at 5 percent keep vs 0.30, 97 percent eviction, oracle-miss 0.370 vs 0.25, V4 0.078 vs 0.0625, contrast 0.359 with SE 0.068, n 64, seed 20260812, 89 duty cycles	geometric-observation/claims/LEDGER.md row GO-2/GO-12/GO-13 operational; geometric-observation/prereg/G0-P-2026-077-kv-consumer-relative.md ; geometric-observation/results/G013-kvaw2-governed.json

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Landauer.lean](#), theorems `eMin_300`, `resetBound_1e`, `resetBound_zero`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 13.

RELATED

[budget](#) (p. 51); [certificate](#) (p. 59); [read operator](#) (p. 291); [coherence time](#) (p. 67).

SEE ALSO

none

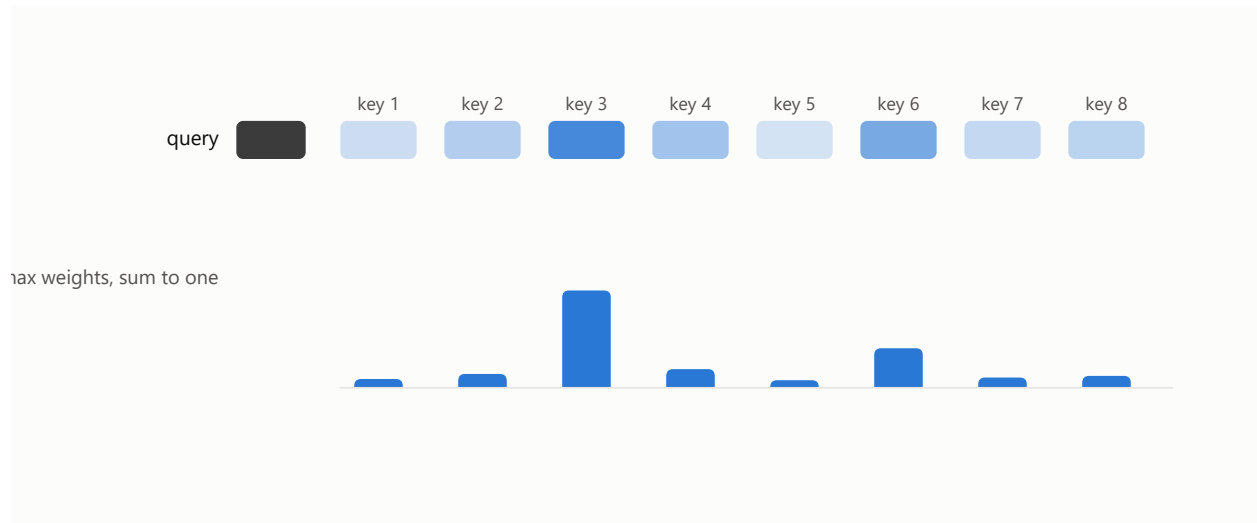
language model*concept.* Entry id language-model.

Figure 123: A model that reads tokens and outputs a probability for the next one.

DEFINITION

A model that reads a sequence of tokens and outputs a probability for the next one, scored by perplexity. Chapter 0 section 0.11.

Example. A model with perplexity 12 on a text is as uncertain, per token, as a fair choice among 12 tokens.

EQUATION

Book equation 0.22.

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(w_t \mid w_{<t}).$$

CONDITIONS

- A model that reads a sequence of tokens and outputs a probability for the next one. Its perplexity is two to the power of the average bits per token, at least one and equal to the vocabulary size for a uniform guess, and its attention weights are a softmax, positive and summing to one.
- A seven-billion-parameter model served passage retrieval over contexts of fourteen thousand tokens across three registered attempts, and a key quantizer with reconstruction cosine 0.995 broke it to a perplexity near ten thousand.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the serving measurements in `geometric-observation/experiments/GO-kv-serving-flip-NOTES.md`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Perplexity.lean](#), theorems `bits_nonneg`, `one_le_perplexity`, `perplexity_uniform`, `perplexity_mono`, `bits_of_finding`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Softmax.lean](#), theorems `denom_pos`, `softmax_pos`, `softmax_sum`, `softmax_shift`, `softmax_lt_iff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 4, 6, 8, 11, 12, 13.

RELATED

[token](#) (p. 363); [perplexity](#) (p. 258); [attention](#) (p. 26); [KV cache](#) (p. 200); [softmax](#) (p. 336).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.23, 11.1.

Ledger rows that cite the entry's records without naming it: NEG-16 (KV serving, end-task), NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 8 section 8.2, chapter 11 section 11.2, chapter 13 section 13.6.

Laplacian

concept. Entry id `laplacian`.

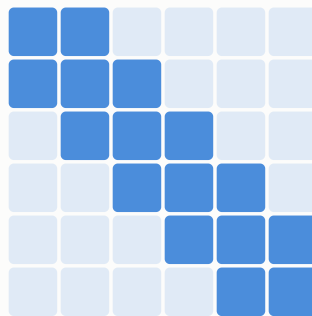


Figure 124: Degrees on the diagonal, minus the adjacency off it.

DEFINITION

A matrix built from a graph's edges whose eigenvectors, from the smallest eigenvalue up, are the smoothest functions on the graph. The unnormalized form is zero on the constant vector, and the normalized form of equation 0.19 is zero on the square root of the degree. Equation 0.19.

Example. A path of three nodes has unnormalized Laplacian with rows $(1, -1, 0)$, $(-1, 2, -1)$, and $(0, -1, 1)$,

which is zero on $(1, 1, 1)$.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2}AD^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad Lu_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

CONDITIONS

- The unnormalized Laplacian, the degree matrix less the adjacency, has quadratic form the sum over edges of the squared difference, so it is positive semidefinite, zero on the constant vector, and positive across any edge. The normalized form of equation 0.19 has the same ordering of eigenvalues, but its zero-eigenvalue vector is the square root of the degree times the constant vector, not the constant vector.
- The Lean file checks the unnormalized form. The embedding of equation 0.20 divides each eigenvector entry by the square root of the degree, which is the scaling that carries the normalized eigenvectors back to the unnormalized ones.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.10 of *Data Mining as Observation*, with the program's spectral work in Volume 14 chapter 9 and the-angular-observer.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Laplacian.lean](#), theorems `quad_eq`, `quad_nonneg`, `quad_const`, `quad_pos_of_edge`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 4, 9, 11.

RELATED

[intrinsic dimension](#) (p. 184); [recognizer](#) (p. 296); [rank-faithful](#) (p. 286); [bi-Lipschitz](#) (p. 42).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 9.2.

Ledger rows that cite the entry's records without naming it: NEG-1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.2, chapter 3 section 3.3, chapter 9 section 9.1, chapter 9 section 9.2, chapter 9 section 9.4, chapter 11 section 11.1, chapter 14 section 14.3, chapter 14 section 14.5, chapter 14 section 14.7.

latent semantic analysis

instrument. Entry id latent-semantic-analysis.

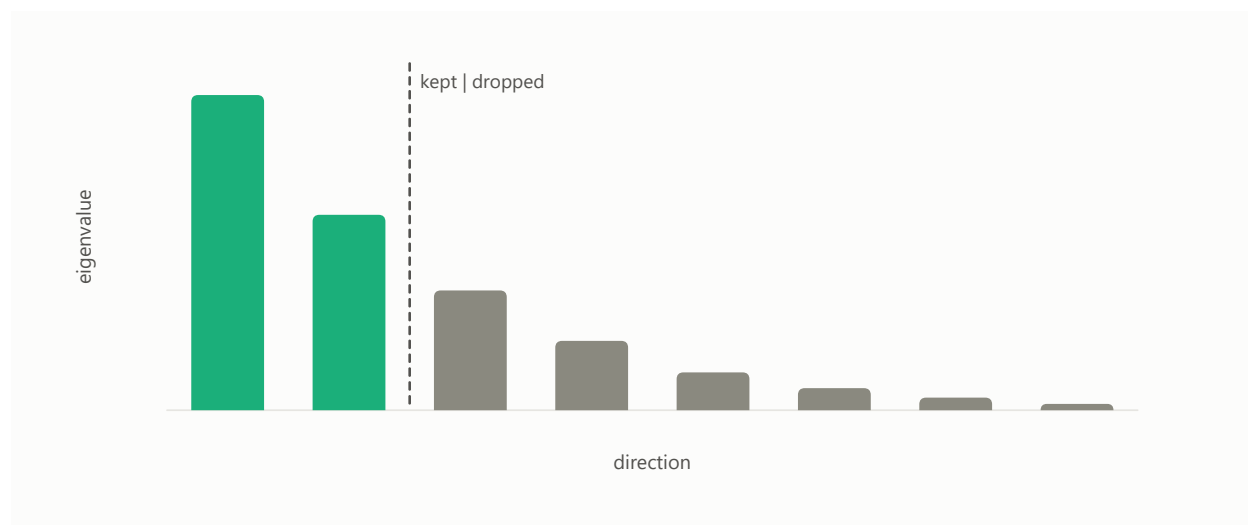


Figure 125: The top singular components of the term-document matrix.

DEFINITION

The truncated singular value decomposition of the term-document matrix, ordinarily uncentred, which coincides with principal components on documents only after centring. Chapter 12.

Example. A term-document matrix of 20000 terms by 5000 documents reduced to 100 singular components gives each document 100 coordinates.

EQUATION

none

CONDITIONS

- The truncated singular value decomposition of the term-document matrix, ordinarily uncentred, with the top components kept. It coincides with principal components on documents only after the matrix is centred, or under an explicitly uncentred convention, and the fraction of squared singular values kept grows with the components kept and reaches one at full rank.
- It is the identity reader on term-document variance, and the components it keeps are the directions of largest variance whether or not the consumer reads them. The non-oracle test fit a frozen embedding of 100 components on the training split only and recovered the reader blind.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Deerwester, Dumais, Furnas, Landauer, and Harshman, indexing by latent semantic analysis, 1990, as chapter 12 section 12.1 of *Data Mining as Observation* reads it, with the program's frozen embedding in `geometric-observation/chapters/ch10_the_blind_probe.md:100-118`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.3	041 non-oracle, frozen LSA TF-IDF to SVD 100 train-only, AUROC 0.975 vs 0.910, flip tied, magnitude overshoot, partial	geometric-observation/chapters/ch10_the_blind_probe.md:100-118 ; geometric-observation/claims/LEDGER.md row GO-B-blind 041

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/PCA.lean](#), theorems `varAlong_basis`, `varAlong_le`, `varAlong_ge`, `dropped_eq`, `dropped_nonneg`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/ExplainedVariance.lean](#), theorems `explained_mem_unit`, `explained_mono`, `explained_full`, `retained_identity`, `retained_example`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/TFIDF.lean](#), theorems `weight_everywhere`, `weight_nonneg`, `weight_antitone`, `tf_mem_unit`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 12.

RELATED

[TF-IDF](#) (p. 360); [principal component analysis](#) (p. 269); [explained variance](#) (p. 137); [bag of words](#) (p. 36); [blind probe](#) (p. 45).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.36, 0.5, 4.2.

Ledger rows that cite the entry's records without naming it: GO-B-legal (035→036).

leakage

concept. Entry id leakage.

DEFINITION

Any information in the training data that could only be known after the decision, or any identifier that lets the model recognize a row it will be tested on. Chapter 8.

Example. An imputation fit on all 1000 rows before the split has read the 100 test rows.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

CONDITIONS

- Leakage is any information in the training data that could only be known after the decision, or any identifier that lets the model recognize a row it will be tested on. It is a property of the harness's read subspace, not of the model.
- Selecting features on the whole data and then cross-validating leaks the test fold into the selection, and the reported error can fall from the true rate to a fraction of it, as the ESL example shows.
- Coupling between the queries and the corpus of a retrieval benchmark is leakage of the same kind, and the program's hubness numbers moved by a factor of several when the coupling was removed.

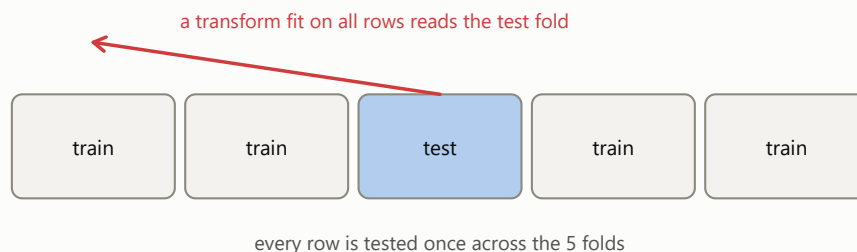


Figure 126: A transform fit on all rows reads the test fold.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.1 of *Data Mining as Observation*, with the ESL example of selection before the split, Hastie, Tibshirani, and Friedman, section 7.10.2.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Leakage.lean](#), theorems `errors_lookup_eq_zero`, `lookup_default`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 2, 4, 8.

RELATED

[harness](#) (p. 168); [preregistration](#) (p. 267); [sealed](#) (p. 324); [hubness](#) (p. 176).

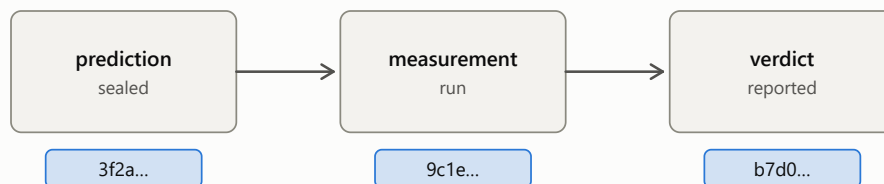
SEE ALSO

Ledger rows that cite the entry's records without naming it: NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.7, chapter 8 section 8.1.

ledger class

concept. Entry id `ledger-class`.



each file's digest is recorded before the next step, and a changed digest proves a changed file

Figure 127: Proved, demonstrated, replicated, predicted, exploratory, refuted, missed, or void.

DEFINITION

One of six labels every headline claim carries, proved, demonstrated, replicated, predicted, exploratory, or refuted, defined in the evidence ledger of Volume 14. Chapter 1.

Example. GO-1 is demonstrated, NEG-2 is refuted, and a row whose bar the null passed is void.

EQUATION

none

CONDITIONS

- Every headline claim carries one of six labels, proved, demonstrated, replicated, predicted, exploratory, or refuted. A claim's class can only rise by a sealed pass on held-out data and only fall by a sealed miss, and a refuted claim is reported at the prominence of a pass.
- A number produced without a sealed bar is exploratory whatever its size. It may motivate a claim and may not carry one.
- The authority for a claim's class is the ledger, and a chapter or an encyclopedia entry that disagrees with it is rebuilt.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, `geometric-observation/PROTOCOL.md:58-75` and `geometric-observation/OBSERVATION.md:43-45`, DOI `10.5281/zenodo.21776291`, with the ledger itself at `geometric-observation/claims/LEDGER.md`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Ledger.lean](#), theorems `step_none`, `step_miss_le`, `step_pass_ge`, `step_miss_refuted`, `exploratory_stays`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 8, 11.

RELATED

[preregistration](#) (p. 267); [sealed](#) (p. 324); [certificate](#) (p. 59); [false-clear rate](#) (p. 141).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 1.1.

Ledger rows that cite the entry's records without naming it: OT-11, GO-2 (neg. half: not reconstruction), NEG-15 (Bell boundary).

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.5, chapter 8 section 8.3, chapter 8 section 8.8, chapter 8 section 8.10, chapter 11 section 11.5.

licensed, not licensed

concept. Entry id licensed.

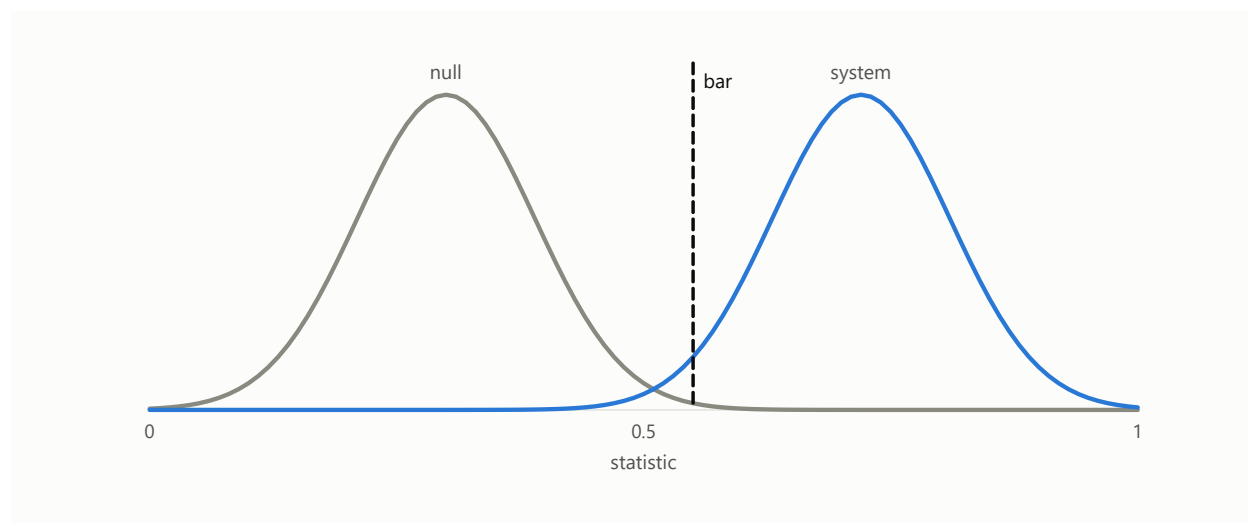


Figure 128: A claim is licensed when it was sealed before the run and its bar separated it from the null.

DEFINITION

A claim is licensed when it is a theorem checked in Lean in appendix C, or a measurement sealed with its commit hash before the run that survived its registered null and bar, and a chapter may rest on it. A claim is not licensed when it is exploratory, a retrospective fit, a single seed, or posited, and a chapter may mention it and not lean on it. How to use this book, and chapter 8 section 8.10.

Example. The flip across twelve domains is licensed, sealed and survived its bars; the alignment law's retrospective fit is not licensed, since it is an exploratory row.

EQUATION

none

CONDITIONS

- A claim is licensed when it is a theorem checked in Lean, or a measurement sealed with its commit hash before the run that survived its registered null and bar, so that a changed digest would prove a changed prediction and the bar discriminated the system from the null. A chapter may rest its thesis on a licensed claim.
- A claim is not licensed when it is exploratory, a retrospective fit, a single seed, or posited, and a chapter

may mention it and may not lean on it. The classes between, demonstrated, replicated, and predicted, are defined in chapter 8, and a row of the table names which one each claim carries.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

The table in how to use this book, and chapter 8 section 8.10 of *Data Mining as Observation*, with the six classes in `geometric-observation/PROTOCOL.md:58-75`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Bar.lean](#), theorems `passes_anti`, `passes_mono`, `discriminates_iff`, `no_bar_of_null_ge`, `exists_bar_of_lt`, `vacuous_of_null_passes`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Seal.lean](#), theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

none

RELATED

[ledger class](#) (p. 211); [sealed](#) (p. 324); [bar](#) (p. 39); [posited versus measured](#) (p. 263); [registered](#) (p. 305).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.3.

Ledger rows that cite the entry's records without naming it: OT-11, GO-2 (neg. half: not reconstruction).

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.5, chapter 8 section 8.10.

lift

concept. Entry id `lift`.

DEFINITION

Confidence divided by the consequent's support, so that lift one is the independence baseline. It fails at low support. Chapter 5.

Example. Confidence 0.8 for bread implies butter against a butter base rate of 0.4 is lift 2.

EQUATION

Book equation 5.2.

$$\text{lift}(X \rightarrow Y) = \frac{c(X \rightarrow Y)}{s(Y)} = \frac{s(X \cup Y)}{s(X) s(Y)}.$$

CONDITIONS

- Confidence over the consequent's support, the joint support over the product of the two supports. It is one under independence, symmetric in the two itemsets, and bounded by the reciprocal of the consequent's

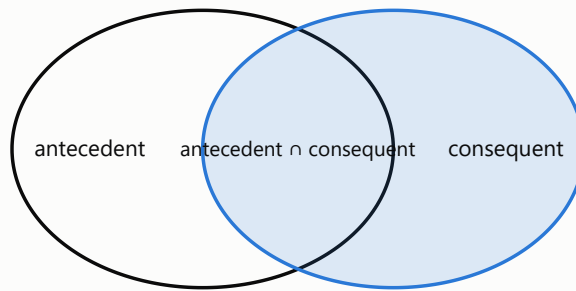


Figure 129: Confidence over the base rate of the consequent.

support.

- It fails at low support. One transaction in N containing both items, and neither elsewhere, gives lift N , which is why the rules with the highest lift are the ones chapter 8's multiple-comparison rule applies to.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

TSK chapter 5 on objective measures, as chapter 5 section 5.1 of *Data Mining as Observation* states it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Lift.lean](#), theorems `lift_indep`, `lift_symm`, `lift_le_inv`, `lift_single`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 5.

RELATED

[Apriori principle](#) (p. 25); [multiple comparisons](#) (p. 229); [safe pruning](#) (p. 320); [harness](#) (p. 168).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 5.1.

linear classifier

instrument. Entry id linear-classifier.

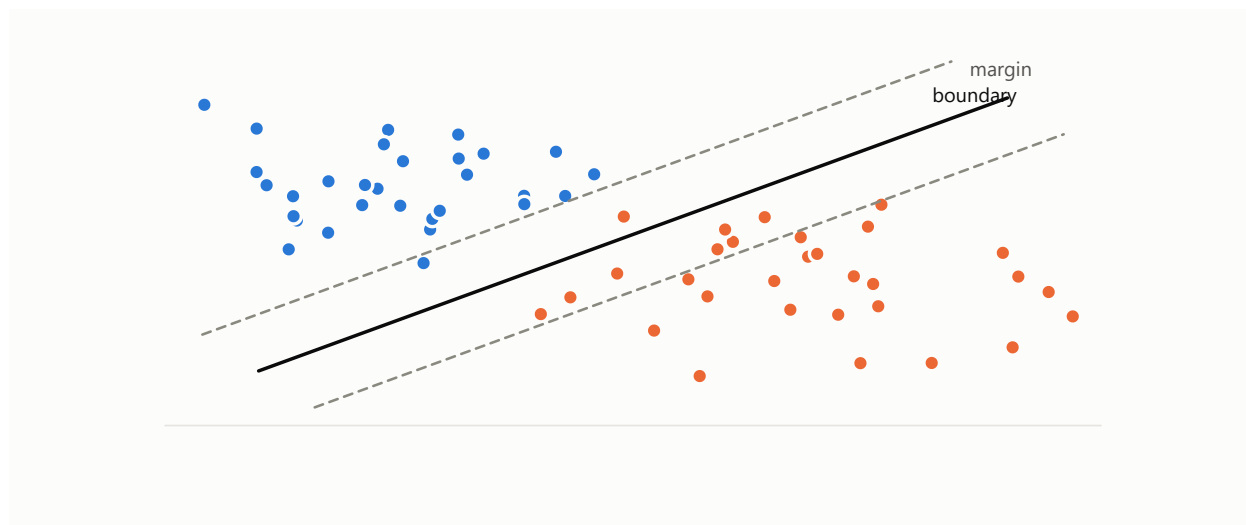


Figure 130: A weighted sum, whose boundary is a hyperplane.

DEFINITION

A classifier that scores by a weighted sum, logistic regression or a linear support vector machine. Its read subspace is the one direction of its weight vector. Chapter 6.

Example. The score $2x_1 + x_2 - 3$ reads only the direction $(2, 1)$, and $(1, -2)$ is in its nuisance.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

CONDITIONS

- A classifier that scores by a weighted sum, logistic regression or a linear support vector machine. Its decision boundary is the hyperplane where the weighted sum is zero, its read subspace is the one direction of its weight vector, and its read operator is the outer product of that vector with itself, scaled by the score's slope.
- Its nuisance is everything orthogonal to the weight vector, and a planted linear consumer is the case a probe is tested against before it is trusted on a real model.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the planted affine consumer in `geometric-observation/claims/LEDGER.md` row GO-1.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Logistic.lean](#), theorems `sigmoid_pos`, `sigmoid_lt_one`, `sigmoid_zero`, `sigmoid_neg`, `sigmoid_strictMono`, `decision_iff`, `decision_linear`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 6, 7, 11.

RELATED

[logistic regression](#) (p. 217); [decision boundary](#) (p. 101); [read subspace](#) (p. 293); [margin](#) (p. 221); [planted](#) (p. 260).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.9, 0.28.

Ledger rows that cite the entry's records without naming it: GO-1.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.1, chapter 6 section 6.2, chapter 11 section 11.7.

logistic regression

instrument. Entry id `logistic-regression`.

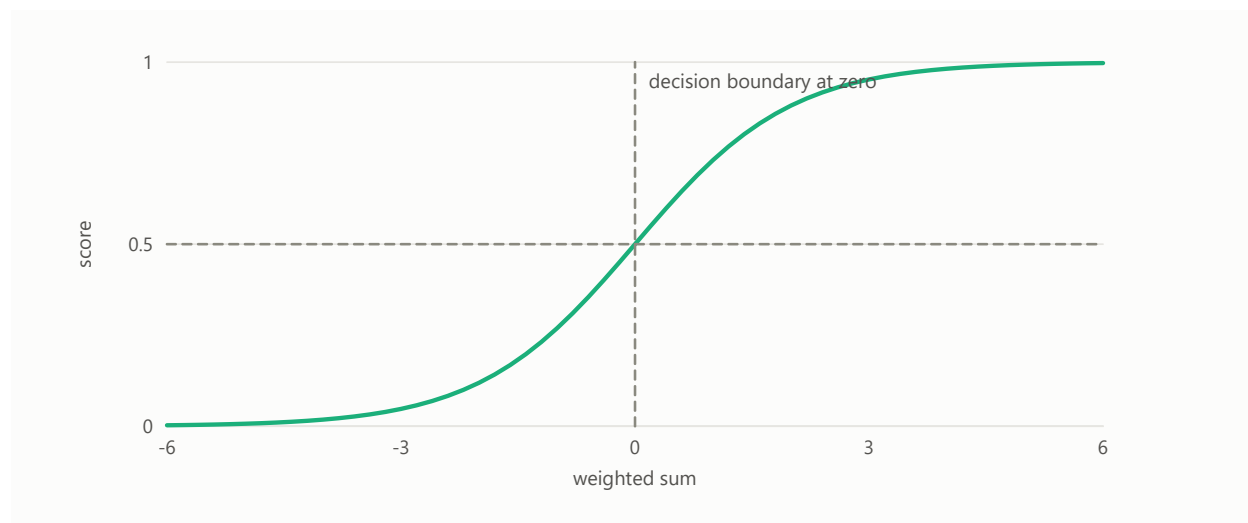


Figure 131: The sigmoid of a weighted sum, above one half on one side of the hyperplane.

DEFINITION

A linear classifier whose score is the sigmoid of a weighted sum, strictly between zero and one, above one half exactly on one side of the hyperplane. Chapter 6.

Example. A weighted sum of 0 gives score 0.5, of 2 gives 0.881, and of -2 gives 0.119.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

CONDITIONS

- A linear classifier whose score is the sigmoid of a weighted sum. The sigmoid lies strictly between zero and one, is one half at zero, is symmetric about one half, and is strictly increasing, so the decision at threshold one half is the sign of the weighted sum and the decision boundary is the hyperplane.
- Its score moves only along its weight vector, so its read subspace is one direction and its nuisance is everything orthogonal, and its read operator is the outer product of the weight vector with itself scaled by how steep the sigmoid is at each row.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-B-optim-D4 (034 · D4) [predicted]. Optimization — gradient compression, curvature (Hessian) read operator, on a REAL model (logistic regression); optional stretch [geometric-observation/claims/LEDGER.md:122](#).

FIRST STATED

Cox, the regression analysis of binary sequences, 1958, as chapter 6 section 6.1 of *Data Mining as Observation* reads it, with the real logistic model's Hessian in [geometric-observation/claims/LEDGER.md](#) row GO-B-optim-D4.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.2	real logistic model, exact Hessian, anti 300 of 300, flip 82 of 300, coupling diagnosis, bound not refutation	geometric-observation/claims/LEDGER.md row GO-B-optim-D4

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Logistic.lean](#), theorems `sigmoid_pos`, `sigmoid_lt_one`, `sigmoid_zero`, `sigmoid_neg`, `sigmoid_strictMono`, `decision_iff`, `decision_linear`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 6, 7, 11, 12.

RELATED

[classifier](#) (p. 65); [decision boundary](#) (p. 101); [threshold](#) (p. 362); [read subspace](#) (p. 293); [Hessian](#) (p. 172).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.9, 0.28.

Ledger rows that cite the entry's records without naming it: GO-1.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.1.

M

Mahalanobis distance

concept. Entry id mahalanobis-distance.

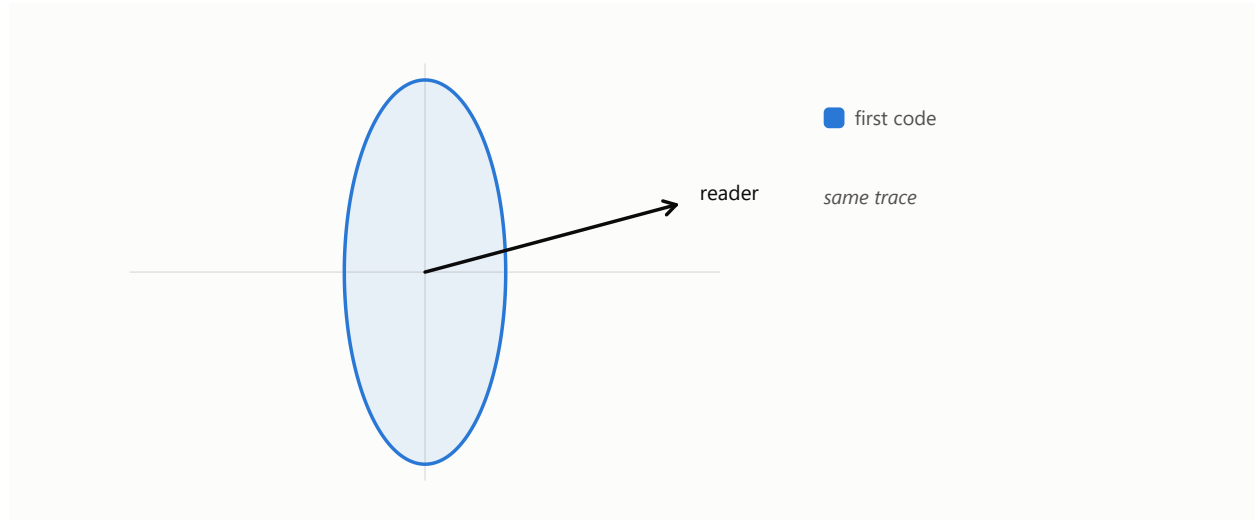


Figure 132: Euclidean distance after whitening.

DEFINITION

The Euclidean distance of a row from the mean after whitening, which needs a positive-definite covariance or a pseudoinverse on its support. Equation 0.33.

Example. With covariance $\text{diag}(1, 4)$, the row $(3, 4)$ has Euclidean distance 5 from the mean and Mahalanobis distance the square root of 13, or 3.61.

EQUATION

Book equation 0.33.

$$d_M(x) = \sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)}.$$

Book equation 10.1.

$$z(x) = \frac{x - \mu}{\sigma}, \quad d_M(x)^2 = (x - \mu)^\top \Sigma^{-1} (x - \mu) = \sum_i \frac{(v_i \cdot (x - \mu))^2}{\lambda_i}.$$

CONDITIONS

- The Euclidean distance of a row from the mean after whitening. In the eigenbasis its square is the sum of the squared centred coordinates over the eigenvalues, nonnegative, zero at the mean, unchanged when data and covariance are rescaled together, and a deviation along a direction of larger variance counts for less.
- The detector that scores by it treats every whitened direction equally, so it is the identity reader on whitened data, and a consumer that reads a subspace calls different rows outliers.
- It needs a positive-definite covariance, or a pseudoinverse restricted to the covariance's support, and it equals the Euclidean distance after whitening.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Mahalanobis, on the generalised distance in statistics, 1936, as chapter 0 section 0.16 states it, and chapter 10 section 10.1 of *Data Mining as Observation*, where the statistical detector is the identity reader on whitened data.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Mahalanobis.lean](#), theorems `dm2_nonneg`, `dm2_mean`, `dm2_scale`, `dm2_eq_whitened`, `dm2_antitone_in_variance`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 10.

RELATED

[whitening](#) (p. 381); [identity reader](#) (p. 180); [hubness](#) (p. 176); [abstention](#) (p. 12).

SEE ALSO

none

manifold

concept. Entry id manifold.

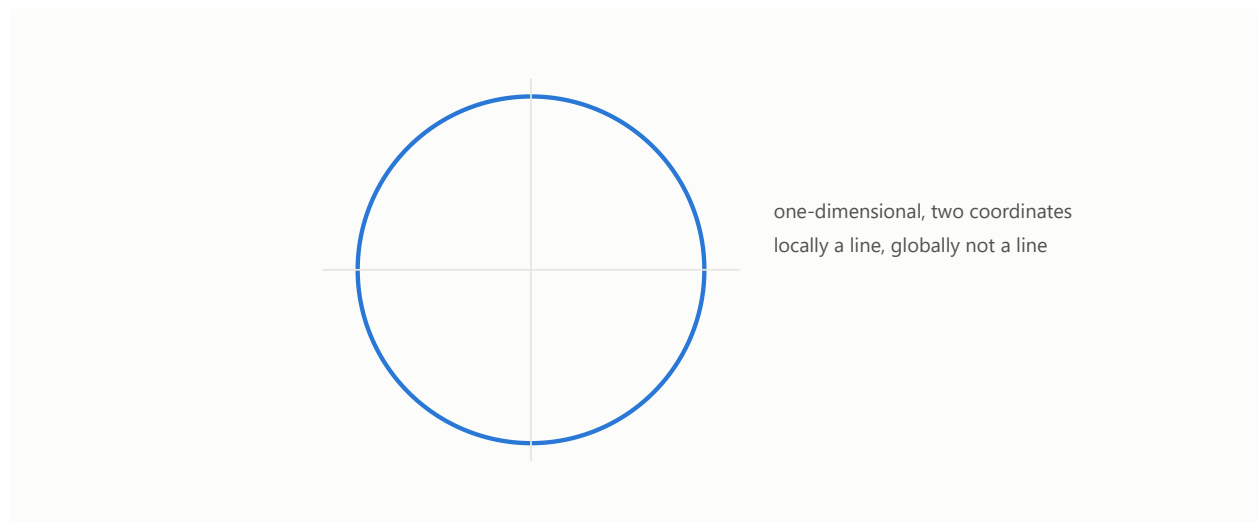


Figure 133: Locally like flat space, and not globally low-dimensional in the coordinates.

DEFINITION

A curved surface of some dimension that looks flat when viewed closely. A dataset lies on one when it is locally low-dimensional, which is not the same as varying along fewer covariance directions, since a circle is one-dimensional and uses two coordinates. Chapter 0 section 0.10.

Example. A circle in the plane is a one-dimensional manifold that uses both coordinates.

EQUATION

none

CONDITIONS

- A space that is locally like flat space of some dimension. A dataset lies on a manifold when it is locally low-dimensional, which does not mean it varies globally along fewer covariance directions. A circle in the plane is one-dimensional and uses both coordinates, so no linear projection recovers it.
- Convergence of the graph Laplacian to the manifold's holds under stated conditions on sampling density, graph construction, kernel bandwidth, normalization, and scaling, after Belkin and Niyogi, and a finite graph spectrum names a shape only against a finite list of candidates, which is the recognizer's limitation and applies here.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.10 and chapter 9 section 9.2 of *Data Mining as Observation*, with the recognizer of Volume 14 chapter 11 and its battery in the-angular-observer.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	hyperbolic rejected, curvature negative 0.98 to negative 0.14, trend 1.09 minus 0.157 d across 20 manifolds and 5 families	the-angular-observer\README.md: 111-147, 183-185

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/IntrinsicDimension.lean](#), theorems `log_weyl`, `dimension_from_slope`, `weyl_double`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 9.

RELATED

[intrinsic dimension](#) (p. 184); [recognizer](#) (p. 296); [graph](#) (p. 166); [Laplacian](#) (p. 207); [template match](#) (p. 359).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.31, 9.2, 0.32.

Ledger rows that cite the entry's records without naming it: GO-3.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.3, chapter 9 section 9.1, chapter 9 section 9.2, chapter 11 section 11.1.

margin

concept. Entry id margin.

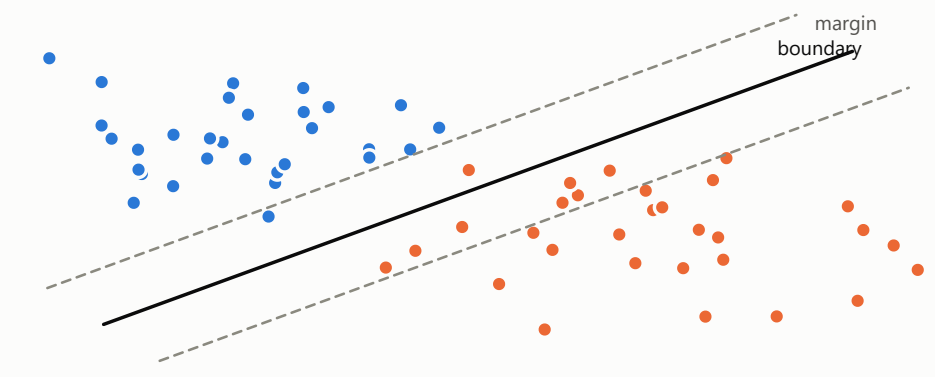


Figure 134: The distance from the boundary to the nearest rows.

DEFINITION

How far a row's score sits from the threshold. Chapter 6.

Example. Points at distance 0.5 and 0.8 from the boundary on either side give a margin of 0.5.

EQUATION

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E} \left[\max_{N-1} \mathcal{N}(0, 1) \right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous.}$$

CONDITIONS

- How far a row's score sits from the threshold. A perturbation whose length times the weight length is smaller than the margin cannot move a linear classifier's score across the threshold, by Cauchy-Schwarz, and a step along the weights of exactly that size reaches the boundary.
- The margin is therefore a certificate on a decision, and chapter 9's margin certificate is the same idea with a noise model in place of a perturbation bound and a vacuity threshold derived from it.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the margin certificate of chapter 9 section 9.3 and the neighbour margins of chapter 11.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Margin.lean](#), theorems `score_add`, `decision_stable`, `tight_along_weights`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 6, 7, 9, 11, 12, 14.

RELATED

[classifier](#) (p. 65); [decision boundary](#) (p. 101); [vacuity threshold](#) (p. 371); [rank certificate](#) (p. 285); [escalation](#) (p. 132).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.2.

Ledger rows that cite the entry's records without naming it: GO-3.

matched bits

instrument. Entry id matched-bits.



Figure 135: Two codes are compared only at the same bit count.

DEFINITION

The rule that two codes are compared only at the same bit count. At matched bits two codes with equal reconstruction error can still differ for a reader, which is the flip. Chapter 4 section 4.3.

Example. Two 4-bit codes of the same vector are compared; a 4-bit code against a 6-bit code is not a comparison.

EQUATION

none

CONDITIONS

- The rule that two codes are compared only at the same bit count, since each added bit halves the step and quarters the squared error. At matched bits two codes with equal reconstruction error can still differ for a reader, and the flip is that difference.
- A comparison that was not bit-matched was refuted by the twenty-five percent codebook confound, and the redesign matched bits before any verdict was read.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* GO-2 (neg. half: not reconstruction) [demonstrated]. At matched bits, downstream preservation is not controlled by reconstruction error. [geometric-observation/claims/LEDGER.md:63](#).

FIRST STATED

Chapter 4 section 4.3 of *Data Mining as Observation*, with the bit-matched redesign in `geometric-observation/chapters/ch08_value.md:1-30`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Bit.lean](#), theorems `levels_succ`, `levels_add`, `step_succ`, `sqError_succ`, `sqError_antitone`, `levels_zero`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Isotropy.lean](#), theorems `isotropic_reads_same`, `isotropic_no_flip`, `anisotropic_readers_differ`, `flip_iff_anisotropic`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 4, 6, 12.

RELATED

[bit](#) (p. 43); [budget](#) (p. 51); [quantization](#) (p. 275); [control](#) (p. 83).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 4.2, 4.6, 0.15.

Ledger rows that cite the entry's records without naming it: GO-2 (pos. half: consumer-projected covariance *controls*), NEG-4.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 3 section 3.3, chapter 4 section 4.3, chapter 4 section 4.6, chapter 6 section 6.2, chapter 8 section 8.2.

metric

concept. Entry id `metric`.

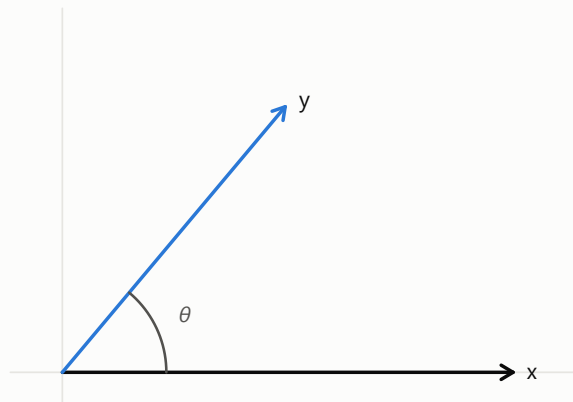


Figure 136: A rule for the distance between two rows or two outputs.

DEFINITION

A rule for the distance between two rows or two outputs. A distance is a choice of what to ignore, and a local metric on an output is a positive semidefinite matrix, a pseudometric until its kernel is quotiented out, which the book calls a metric by convention. Chapter 3 section 3.1 and chapter 0 section 0.5.

Example. Euclidean distance between (0, 0) and (3, 4) is 5, and a metric that ignores the second coordinate reads 3.

EQUATION

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

CONDITIONS

- A rule for the distance between two rows, or between two outputs. A distance function is a choice of what to ignore, and the local metric on a consumer's output is the positive semidefinite matrix that says how a small change is scored. The squared Euclidean distance is symmetric and zero exactly between equal rows.
- A dataset-level loss, accuracy, F1, or a rank correlation is not a local metric and cannot be inserted into the read-operator formula, and the same two codes get opposite verdicts from two metrics.
- A positive semidefinite read form is a pseudometric until the directions in its kernel are quotiented out. The book calls it a metric by convention, on the quotient.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 3 section 3.1 and chapter 0 section 0.5 of *Data Mining as Observation*, with the read metric in `geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/EuclideanDistance.lean](#), theorems `distSq_eq_quad_one`, `distSq_comm`, `distSq_eq_zero_iff`, `ranking_flips`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/OutputMetric.lean](#), theorems `neg_reverses`, `readOp_of_neg`, `cost_flips`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[output metric](#) (p. 251); [Euclidean distance](#) (p. 133); [read operator](#) (p. 291); [quotient](#) (p. 279); [geodesic distance](#) (p. 162).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 1.1.

Ledger rows that cite the entry's records without naming it: GO-EC-3.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 3 section 3.2.

min-over-strata

concept. Entry id min-over-strata.

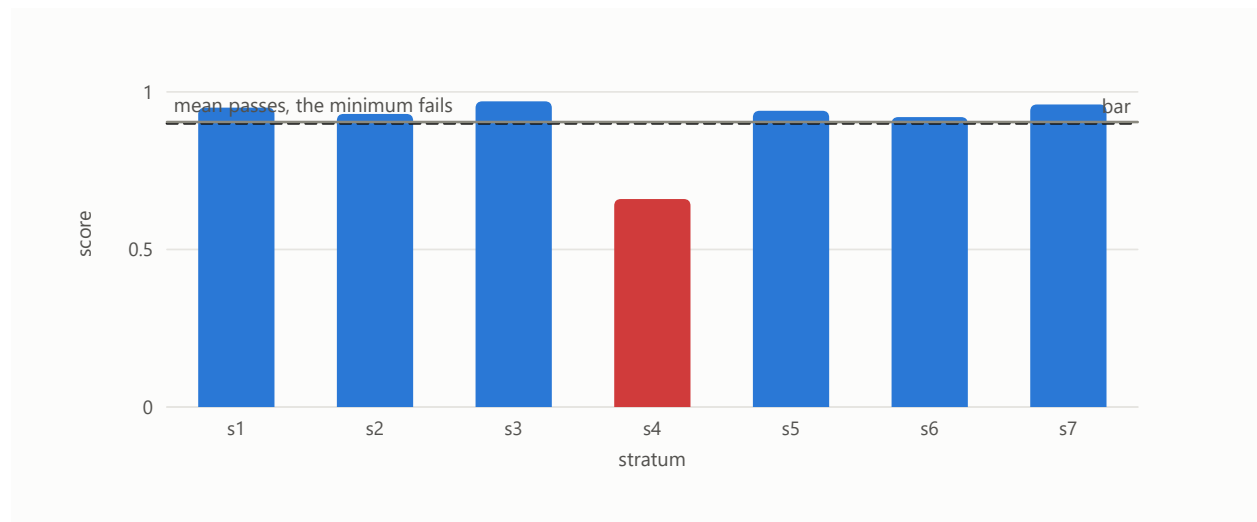


Figure 137: The worst stratum is the verdict.

DEFINITION

The rule that a verdict over several groups is the worst group's verdict, with abstention counted as a verdict, never the average. Chapter 10.

Example. Strata at 0.95, 0.93, 0.97, and 0.66 have a mean of 0.878 and a minimum of 0.66, and the minimum is the verdict.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 14.5.

$$\text{FC}_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} \text{FC}_g, \quad \text{ABSTAIN otherwise.}$$

CONDITIONS

- A verdict over several groups is the worst group's verdict, with a group too small to score counted as an abstention and reported, never averaged away.
- The first stratified run made two wrong predictions, and the record carries them at the size of the result.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

The compression program's stratified evaluation, STRATA, in turboquant-pro, DOI 10.5281/zenodo.20660087, and chapter 10 section 10.4 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro/docs/STRATA_RFC.md:24-98
chapter 10 section 10.4	first run, 350000 of 2391361 rows, 7 of 14 eligible, abstentions with 2 and 1 rows, Robin Hood 0.3447 to 0.4469 ratio 1.30 against 1.5, skew 2.68 to 4.40 ratio 1.64 against 3, 63 times sample range	turboquant-pro/docs/RESULTS_multilingual_strata.md:1-55

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Certificate.lean](#), theorems `falseClear_mul_coverage`, `coverage_empty`, `falseClear_mem_unit`, `minOverStrata_passes_iff`, `minOverStrata_le_weighted_mean`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 10, 11, 14.

RELATED

[false-clear rate](#) (p. 141); [coverage](#) (p. 92); [hubness](#) (p. 176); [certificate](#) (p. 59).

SEE ALSO

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.6, chapter 3 section 3.5, chapter 8 section 8.1, chapter 8 section 8.4, chapter 8 section 8.7, chapter 8 section 8.10, chapter 10 section 10.2, chapter 10 section 10.4, chapter 10 section 10.5, chapter 11 section 11.4, chapter 11 section 11.6, chapter 12 section 12.2, chapter 12 section 12.4, chapter 13 section 13.1.

Monotone Invariance Theorem

result. Entry id monotone-invariance.

DEFINITION

A strictly monotone transform of a score in either direction cannot change its optimal thresholded F1, because the sweep tries thresholds both ways. A strictly increasing transform leaves the AUROC unchanged, and a strictly decreasing one sends it to one minus itself. Equation 6.2.

Example. Squaring a positive score keeps every ranking, so its AUROC and its optimal F1 are unchanged, and negating it sends AUROC 0.75 to 0.25.

EQUATION

Book equation 6.2.

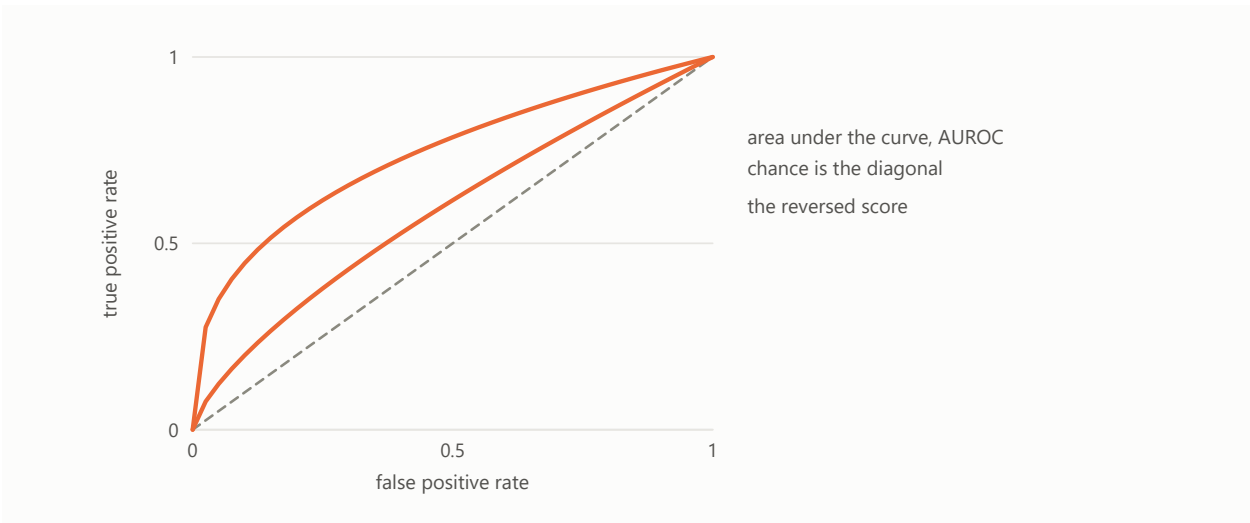


Figure 138: Optimal thresholded F1 survives any strictly monotone transform, AUROC only an increasing one.

$$\max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y) = \max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(f(X), \tau)], y)$$

for every strictly monotone g .

CONDITIONS

- Two statements with different scope. A strictly monotone transform of a score in either direction cannot change its optimal thresholded F1, because the sweep tries thresholds in both directions and so reaches every partition the transform can. A strictly increasing transform leaves the AUROC unchanged, and a strictly decreasing one sends it to 1 minus A.
- The F1 half licenses the pruning of monotone unary nodes in the formula search. The AUROC half licenses nothing about decreasing transforms. The Lean file states the increasing case, and the AUROC file states the reversal.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

theory-radar, theory-radar/paper/theory_radar_paper.tex:290-304 and theory-radar/paper/astar_paper.tex:94-110, DOI 10.5281/zenodo.20660206, and chapter 6 section 6.3 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 5 section 5.4	Monotone Invariance and AUROC invariance theorems, the AUROC to F1 bound via the Youden index, stated in the source for every ROC curve and corrected in the book to concave curves, with the counterexample at AUROC 0.75 and F1 0.857	theory-radar/paper/astar_paper.tex:94-170 ; theory-radar/paper/theory_radar_paper.tex:290-304

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory-radar/paper/theory_radar_paper.tex:290-304 ; theory-radar/paper/astar_paper.tex:94-110

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/MonotoneInvariance.lean](#), theorems `aurocNum_comp`, `auroc_comp`, `predicted_comp`, `predictedBelow_comp`, `sweptF1_comp`, `sweptF1Below_comp`, `optF1_comp`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 5, 6, 12.

RELATED

[formula search](#) (p. 154); [safe pruning](#) (p. 320); [Youden F1 bound](#) (p. 385).

SEE ALSO

none

multiple comparisons

concept. Entry id `multiple-comparisons`.

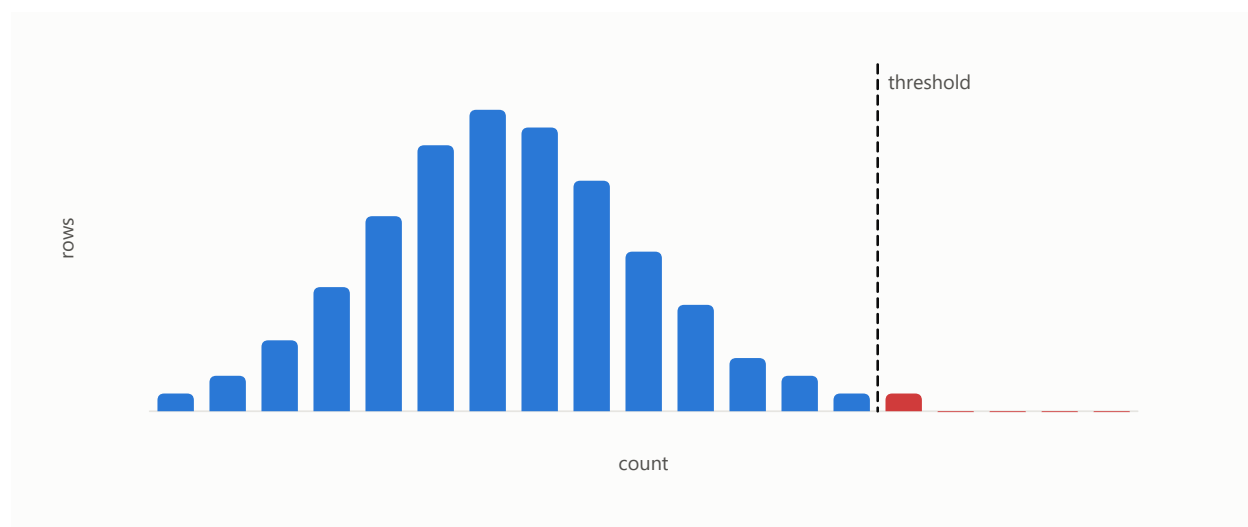


Figure 139: Many tests, and the best reported.

DEFINITION

The inflation of false positives when many hypotheses are tested and the best is reported. The rate one minus one minus alpha to the m assumes independent tests, and the Bonferroni bound, which divides the threshold by the number of tests, holds under any dependence. Chapter 0 section 0.9.

Example. Forty independent tests at 5 percent pass at least one by chance 87 percent of the time, and Bonferroni tests each at 0.125 percent.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

CONDITIONS

- The inflation of false positives when many hypotheses are tested and the best is reported. The chance that at least one of m independent tests at level α is a false positive is one minus one minus α to the m , which needs the tests to be independent. The Bonferroni bound, m times α , is a union bound and holds under any dependence, and dividing the level by m keeps the family-wise rate at α .
- Five hundred of ten thousand null hypotheses pass at five percent, and a harness that searches many formulas or thresholds and reports the best has run that many tests.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.9 and chapter 8 section 8.5 of *Data Mining as Observation*, with the program's own case in `constraint-gap/review/FINDINGS.md:1-35`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/MultipleComparisons.lean](#), theorems `max_ge_each`, `book_numbers`, `corrected_family_rate`, `expectedPasses_unbounded`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 5, 8.

RELATED

[harness](#) (p. 168); [Nadeau and Bengio correction](#) (p. 233); [preregistration](#) (p. 267); [formula search](#) (p. 154).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.4, chapter 7 section 7.3, chapter 7 section 7.4, chapter 8 section 8.5.

multiplet

concept. Entry `id multiplet`.

DEFINITION

A group of eigenvalues that are equal or nearly so, whose pattern names the shape a dataset lies on against a finite list of templates. On a finite graph the groups are approximate. Chapter 9.

Example. The cycle on 8 nodes has eigenvalues 0, 0.586, 0.586, 2, 2, 3.414, 3.414, and 4.

EQUATION

none

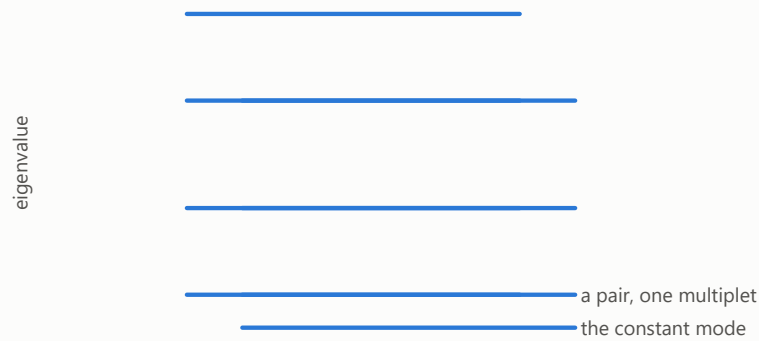


Figure 140: Eigenvalues that come in equal groups.

CONDITIONS

- A group of eigenvalues that are equal or nearly so. The cycle's Laplacian eigenvalue at k equals the one at n minus k , so every eigenvalue of the discrete circle other than the constant mode and, for even n , the alternating mode appears twice.
- Symmetric shapes produce multiplets in the continuum, the sphere in groups of three, five, seven. On a finite neighbourhood graph the multiplets are approximate, their spacing depends on sampling and graph construction, and the recognizer reads them against a finite list of templates and refuses when none matches.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.15 and chapter 9 section 9.2 of *Data Mining as Observation*, with the program's recognizer in `geometric-observation/chapters/ch11_the_recognizer.md:1-95`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.2	the recognizer's mechanism, low multiplets and angular distances, dimension before shape by Weyl's law, refusal, the growth gotcha	geometric-observation/chapters/ch11_the_recognizer.md:1-95

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Multiplet.lean](#), theorems `cycleEig_zero`, `cycleEig_nonneg`, `cycleEig_le_four`, `cycleEig_pair`, `pair_distinct`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 9.

RELATED

[Laplacian](#) (p. 207); [recognizer](#) (p. 296); [manifold](#) (p. 220); [Weyl's law](#) (p. 379); [eigenvalue, eigenvector](#) (p. 125).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.19, 0.31, 9.2.

Sources-table rows that share a record with the entry without naming it: chapter 9 section 9.2.

N

Nadeau and Bengio correction

correction. Entry id nadeau-and-bengio-correction.

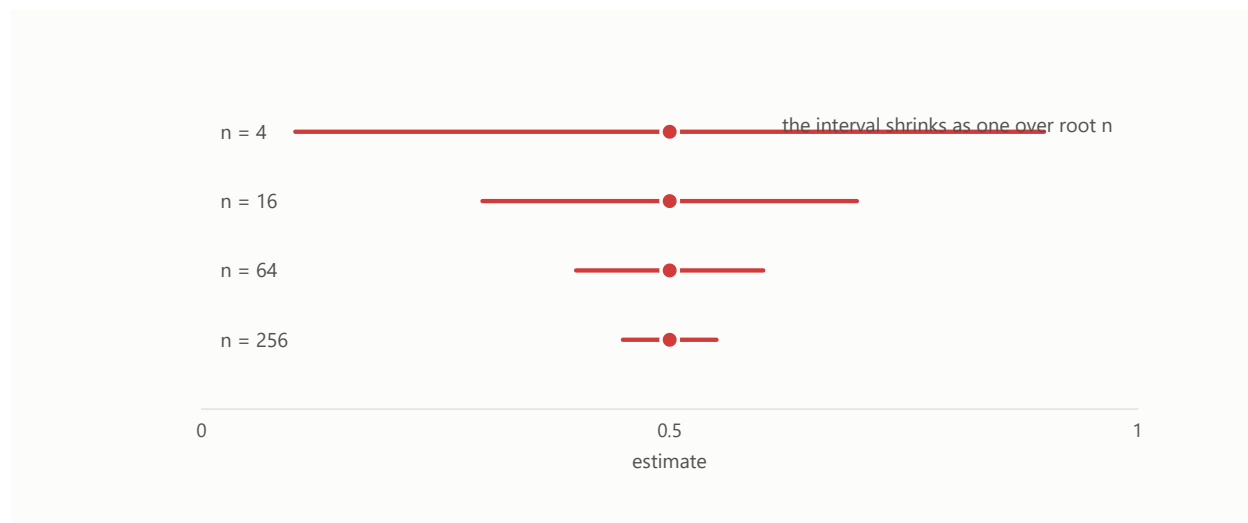


Figure 141: The variance of a cross-validated difference inflated for the folds' shared training data.

DEFINITION

The inflation of the variance of a repeated cross-validation estimate by one plus the ratio of test to training size times the number of repeats, because the folds are not independent. Equation 0.18.

Example. With 1000 folds and a test-to-train ratio of one quarter, the variance is inflated 251 times and the standard error 15.84 times.

EQUATION

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \quad \frac{\widehat{\text{Var}}_{\text{NB}}}{\hat{\sigma}^2/J} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

CONDITIONS

- Repeated cross-validation folds share training data, so the naive variance of a mean difference is multiplied by one plus the fold count times the test-to-train ratio. The corrected variance is never below the naive one and the inflation grows without bound in the fold count.
- With a thousand folds at a ratio of one quarter the inflation is 251, the standard error grows by 15.84, and about four folds' worth of independent evidence remain. The program's nine wins became three under it, with seventeen indeterminate.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Nadeau and Bengio, inference for the generalization error, 2003, as chapter 8 cites it, applied to the program's own comparison in `constraint-gap/review/FINDINGS.md:1-35` and chapter 8 section 8.5 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	<code>constraint-gap/review/FINDINGS.md:1-35</code> ; <code>constraint-gap/README.md:57-76</code>
chapter 6 section 6.4	the reviewer's revision plan, drop sigma, bounded claim, 19 at 200 by 5 vs 11 at 20 by 5, rerun in progress	theory-radar/paper/REVISION_PLAN.md:1-60
chapter 6 section 6.4	five analyses negative, 0.2 percent of floor, twelve times the band, learning-curve redesign	<code>constraint-gap/README.md:57-76</code> ; <code>constraint-gap/review/REDESIGN.md:1-45</code>
chapter 6 section 6.5	loading weights and stability across folds requested	theory-radar/paper/REVISION_PLAN.md issue 6
chapter 7 section 7.3	correlation negative 0.59 at p 4.8e-4 with size, negative 0.562 at p 0.001 with the ensemble's score, predictive 0.345 and 0.589, interval negative 0.165 to 0.125, the one-line summary	<code>constraint-gap/review/INDETERMINATES.md:60-80</code> ; <code>constraint-gap/review/REDESIGN.md:45-70</code>
chapter 7 section 7.4	the redesign, grid 50 to N, ten folds, twenty repetitions, crossing size, the 800-row boundary declared in advance, seventeen at median 0.006 and p 0.964, store the folds	<code>constraint-gap/review/REDESIGN.md:1-110</code>
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_{eff} 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	<code>constraint-gap/review/FINDINGS.md:1-35</code>
chapter 8 section 8.5	9/3/19 to 3/17/11, five negative analyses, 0.2 percent of floor, twelve times the band, learning-curve redesign	<code>constraint-gap/README.md:57-76</code> ; <code>constraint-gap/review/REDESIGN.md</code>
chapter 8 section 8.5	three-way inconsistency, lines 544, 548 to 556, 755	<code>constraint-gap/review/FINDINGS.md:7-31</code> against theory-radar/paper/theory_radar_v6_submission.tex

FAILURES AND CORRECTIONS

- [theory-radar/paper/REVISION_PLAN.md:39-45](#) at 37c4e6c. **Fix:** - Drop " σ significance" everywhere - Report: mean $\Delta F1$, std, 95% CI, and corrected resampled t-statistic - Table columns: "Test F1", "GB F1", " $\Delta F1$ ", "95% CI", "Direction" - Text: "the formula outperforms GB by $\Delta F1=0.031$, 95% CI [0.028, 0.034]" - Add Nadeau-Bengio corrected t-test (accounts for CV fold correlation) - Keep effect size as supplementary but don't call it " σ "

MACHINE CHECKED

[lean/DataMiningAsObservation/NadeauBengio.lean](#), theorems `correctedVar_eq`, `naiveVar_le_correctedVar`, `inflation_unbounded`, `book_numbers`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 6, 7, 8.

RELATED

[harness](#) (p. 168); [formula search](#) (p. 154); [preregistration](#) (p. 267); [multiple comparisons](#) (p. 229).

SEE ALSO

none

naive Bayes

instrument. Entry id `naive-bayes`.

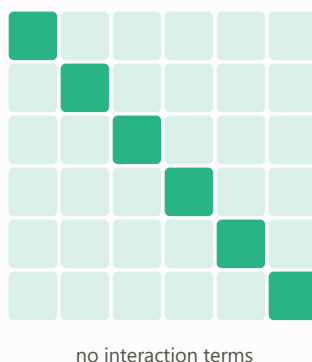


Figure 142: One-dimensional likelihoods multiplied, so the score is additive and the Hessian diagonal.

DEFINITION

A classifier that multiplies one-dimensional likelihoods and adds their logs, so that each feature's contribution is the same whatever the others are and it cannot read an interaction. Chapter 6.

Example. Two features with likelihood ratios 3 and 2 give a combined ratio of 6, whatever the other features are.

EQUATION

none

CONDITIONS

- A classifier that multiplies one-dimensional likelihoods and adds their logs. The log of the product is the sum of the logs, the log odds add across features, and one feature's contribution is the same whatever the others are, so the score has no interaction terms and its Hessian is diagonal.
- What naive Bayes cannot read is the interaction, which is a limit of the model class rather than a direction in its kernel, and its read operator is still not diagonal in general because two sensitivities can rise and fall together across rows.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, after TSK chapter 4.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/NaiveBayes.lean](#), theorems `log_likelihood_sum`, `log_odds_sum`, `contribution_independent`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 6.

RELATED

[classifier](#) (p. 65); [Hessian](#) (p. 172); [confidence](#) (p. 75); [logistic regression](#) (p. 217).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 6.1, 0.9.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.3, chapter 6 section 6.1.

nat

concept. Entry id nat.

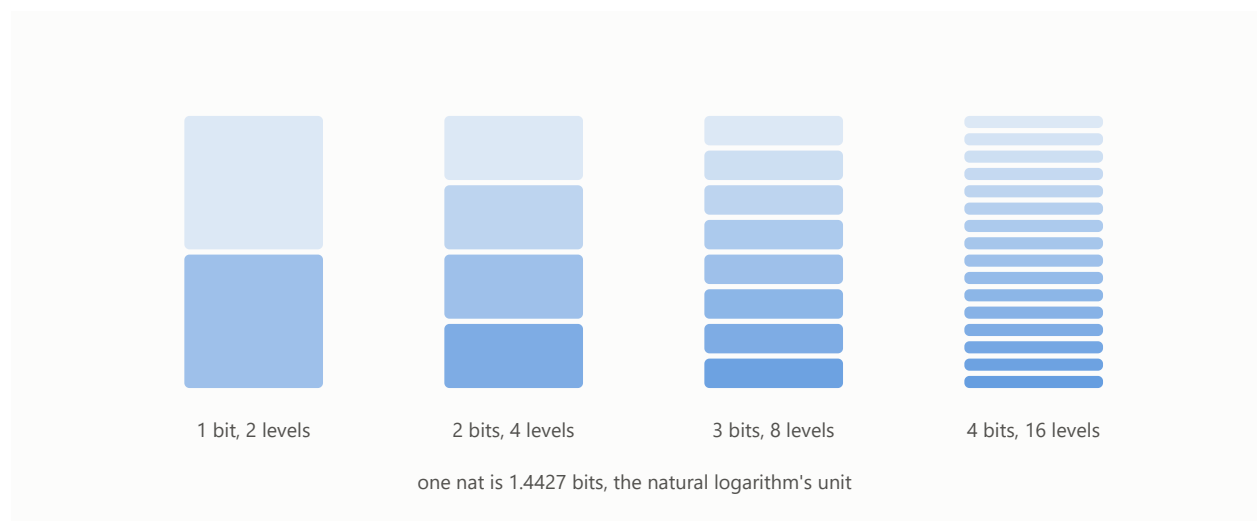


Figure 143: Information in the natural logarithm's unit.

DEFINITION

The unit of information measured with the natural logarithm, between 1.4426 and 1.4427 bits. Chapter 0 section 0.7.

Example. 0.10 nats is 0.144 bits, and 1 bit is 0.693 nats.

EQUATION

none

CONDITIONS

- The unit of information measured with the natural logarithm, as a bit is measured with the logarithm to base two. The two conversions undo each other; the conversion is monotone, and one nat is between 1.4426 and 1.4427 bits.
- The omission floor was measured at 0.10 nats against about six in ten million on sixteen of sixteen heads, and Landauer's bound is stated per nat as kT and per bit as kT times the log of two.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.7 of *Data Mining as Observation*, with the omission floor in nats in `geometric-observation/chapters/ch07_cost.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.4	C-7, sixteen points in 128 dimensions, rank fifteen, near ten to the eleventh nats, refuses when samples do not exceed dimension, warns below five per dimension, loading is a property of two distributions	readscope/SPEC.md:653-680
chapter 4 section 4.2	floor measured on 16 of 16 heads, naive assumption off 20 to 60 percent, about 100000 times, 0.10 nats against about 6e-7, first gate missed 0 of 16, NEG-13	geometric-observation/chapters/ch07_cost.md ; geometric-observation/claims/LEDGER.md GO-P-2026-027 and NEG-13

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/NatUnit.lean](#), theorems `log_two_pos`, `nats_bits`, `bits_nats`, `bitsOfNats_mono`, `one_nat_in_bits`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Bit.lean](#), theorems `levels_succ`, `levels_add`, `step_succ`, `sqError_succ`, `sqError_antitone`, `levels_zero`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 4.

RELATED

[bit](#) (p. 43); [budget](#) (p. 51); [Landauer's principle](#) (p. 203); [perplexity](#) (p. 258); [KL divergence](#) (p. 196).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.15, 4.2, 0.16.

Ledger rows that cite the entry's records without naming it: GO-7, NEG-13 → resolved.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.4, chapter 2 section 2.6, chapter 4 section 4.2, chapter 8 section 8.4, chapter 11 section 11.3, chapter 11 section 11.7, chapter 11 section 11.9.

nearest neighbour

concept. Entry id nearest-neighbour.

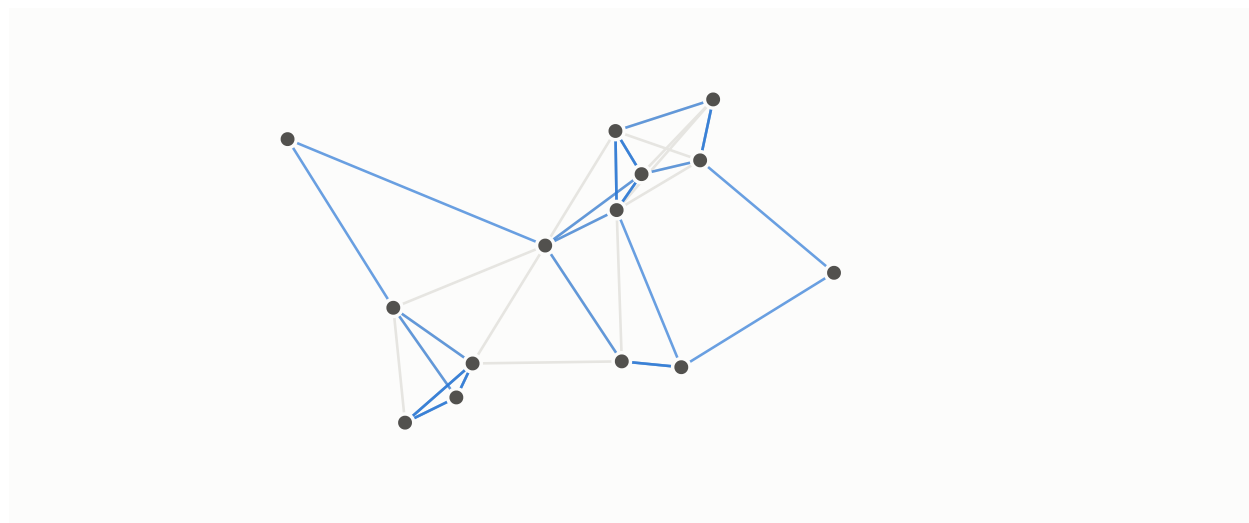


Figure 144: The row at minimal distance from a query.

DEFINITION

A row at minimal distance from a query. A strictly increasing transform of the distance names the same nearest neighbours, so the classifier reads the distance's order and nothing else. Chapter 6.

Example. Among rows at distances 0.9, 0.4, and 1.2 from a query, the second is the nearest, under any increasing rescaling of the distances.

EQUATION

none

CONDITIONS

- A row at minimal distance from a query under a chosen distance. A strictly increasing transform of the distance names the same nearest neighbours, so a nearest-neighbour classifier reads the distance's order and nothing else, a row is its own nearest neighbour, and two nearest neighbours sit at the same distance.
- Under Euclidean distance the classifier reads every standardized coordinate equally and under cosine it reads the angle, so it inherits the distance's quotient, and the rows that appear in many neighbour lists are the hubs.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Cover and Hart, nearest neighbor pattern classification, 1967, as chapter 6 section 6.1 of *Data Mining as Observation* reads it, with the neighbour lists of `openvector-bench/openvector_bench/hubness.py`: 41-100.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/NearestNeighbour.lean](#), theorems `nearest_comp`, `nearest_smul`, `self_nearest`, `nearest_same_distance`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Graph.lean](#), theorems `outDegree`, `mutual_symm`, `mutual_sub`, `mutualNbrs_card_le`, `mem_mutualNbrs`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 6, 8, 10, 11, 12.

RELATED

[neighbourhood graph](#) (p. 239); [hub](#) (p. 175); [recall at k](#) (p. 295); [Euclidean distance](#) (p. 133); [cosine](#) (p. 87).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 3.1, 0.19, 10.3.

Ledger rows that cite the entry's records without naming it: NEG-11.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.5.

neighbourhood graph

concept. Entry id neighbourhood-graph.

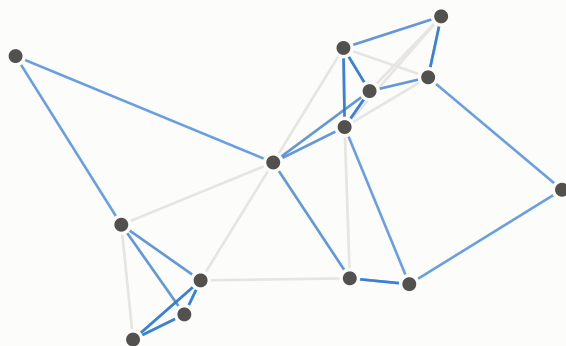


Figure 145: Each row joined to its k nearest rows.

DEFINITION

A graph on a dataset that joins each row to its k nearest rows, or to every row within a radius. Chapter 0 section 0.10.

Example. A 5-nearest-neighbour graph on 1000 rows has 5000 out-edges, five from each row.

EQUATION

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

CONDITIONS

- A graph on a dataset that joins each row to its k nearest rows, or to every row within a radius. Every node of a k-nearest graph has out-degree k, the mutual neighbours of a node are at most k, and being mutual neighbours is symmetric.
- As the graph grows on a manifold its low Laplacian eigenvectors converge to the smoothest functions on the manifold, so the embedding becomes a picture of the manifold, and the hubs of the graph are a property of the queries that built it.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.10 of *Data Mining as Observation*, with the substrate table in the-angular-observer/README.md:99-109.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Graph.lean](#), theorems `outDegree`, `mutual_symm`, `mutual_sub`, `mutualNbrs_card_le`, `mem_mutualNbrs`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 9, 11.

RELATED

[graph](#) (p. 166); [degree](#) (p. 105); [Laplacian](#) (p. 207); [spectral embedding](#) (p. 342); [geodesic distance](#) (p. 162); [hub](#) (p. 175).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.19, 3.3.

Ledger rows that cite the entry's records without naming it: NEG-1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.3, chapter 9 section 9.1, chapter 9 section 9.2.

nuisance

concept. Entry id nuisance.

DEFINITION

The directions of the data a consumer cannot distinguish, the kernel of its read operator. Chapter 1.

Example. For the consumer $x_1 + x_2$, the direction $(1, -1)$ is nuisance: moving along it changes nothing.

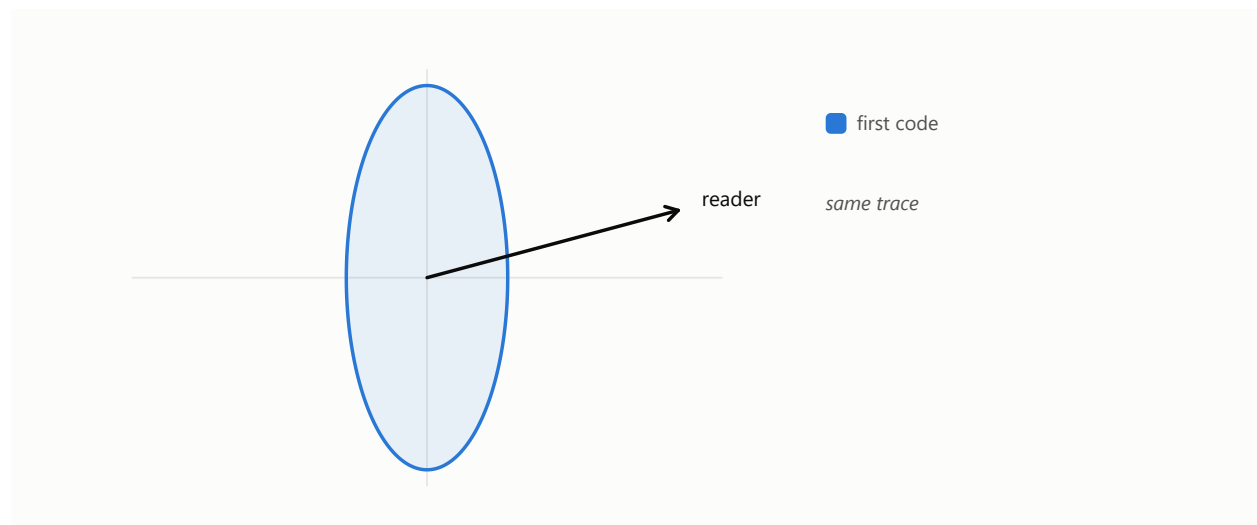


Figure 146: The directions the consumer never reads, the kernel of its read operator.

EQUATION

none

CONDITIONS

- The nuisance is the kernel of the workload-averaged read operator, the set of directions unread at almost every row of that workload, since the kernel of the average is the intersection of the local kernels.
- For a nonlinear consumer a null direction can turn with the row, so the nuisance of the average is not a global equivalence of inputs, and cosine similarity's radial direction is the case.
- A transform is safe when the reader it produces still resolves the read subspace, and discretization is safe only when the values it merges differ in nuisance directions.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-1 [predicted]. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [geometric-observation/claims/LEDGER.md:62](#).

FIRST STATED

Volume 14, chapter 5, [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48](#), DOI 10.5281/zenodo.21776291, where the nuisance is the kernel of the read operator.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at [observation-data-mining 95af30c](#); what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 2, 3, 6, 9, 11, 12, 13.

RELATED

[read subspace](#) (p. 293); [read operator](#) (p. 291); [quotient](#) (p. 279); [identity reader](#) (p. 180).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.12a, 0.12b.

Ledger rows that cite the entry's records without naming it: NEG-6, NEG-7.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 6 section 6.2, chapter 11 section 11.7.

null model

instrument. Entry id null-model.

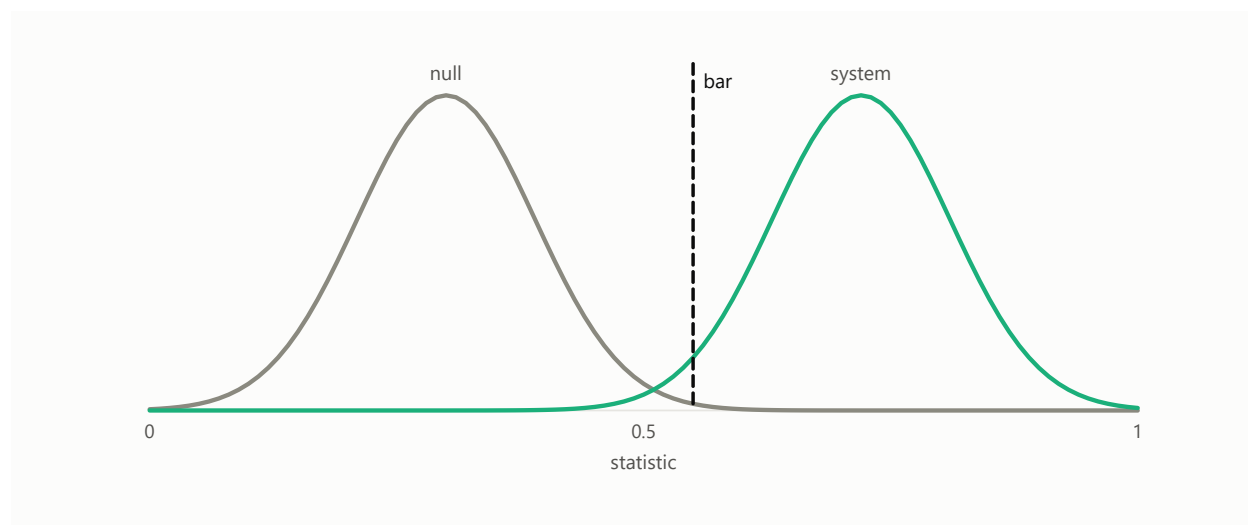


Figure 147: What the statistic looks like when the claimed effect is absent.

DEFINITION

A version of the data with the structure under test removed and everything else kept, the right comparison for any measured statistic. Chapter 0 section 0.9.

Example. Shuffling the labels 1000 times and rescoreing gives the null distribution the measured score is compared against.

EQUATION

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

CONDITIONS

- A version of the data with the structure under test removed and everything else kept. A label permutation keeps every class size and the bag of scores, so any statistic that reads only class sizes or only scores is unchanged by it, and only a statistic that reads the pairing between score and label can move.

- That is what makes the permuted statistic the right comparison, and the excess over the null mean is the number a chapter reports. A number without its null is not yet a finding, and the drift claim that lost half its size to a paired null is the ledger's case.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.9 and chapter 8 section 8.4 of *Data Mining as Observation*, with the program's Poisson, paired, and permutation nulls in `openvector-bench`, `readscope`, and the encoder gate.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/NullModel.lean](#), theorems `card_pos_perm`, `scores_perm`, `stat_of_counts_const`, `excess_eq_zero_iff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 5, 8, 9, 10, 11, 12, 14.

RELATED

[Poisson ceiling](#) (p. 262); [chance level](#) (p. 62); [cross-corpus gate](#) (p. 94); [p-value](#) (p. 273); [harness](#) (p. 168).

SEE ALSO

Ledger rows that cite the entry's records without naming it: OT-4, NEG-15 (Bell boundary).

O

observer

concept. Entry id observer.



the read operator is what the triple induces, and its kernel is the nuisance

Figure 148: A consumer, an output metric, and a budget.

DEFINITION

A consumer, its output metric, and its budget, written as the triple in equation 1.1. Naming all three is what every later chapter checks.

Example. A cosine ranker with rank order as its metric and 1000 candidates as its budget is one observer.

Known as, or related to prior art. The observer triple extends the active-subspace matrix with an output metric, a budget, and an audit discipline.

EQUATION

Book equation 1.1.

$$O = (C, G, B).$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

CONDITIONS

- An observer is a consumer, its output metric, and its budget. The consumer and the local geometry of its output metric determine the read operator. The budget bounds what of it can be measured and used and changes it only by changing the consumer.
- Two consumers with the same read operator on the same workload share a read geometry and are not thereby the same observer, since the consumer stays part of the triple and a sign change reverses every ranking.
- The output metric is a loss on the consumer's output. Where it has a local quadratic representation, that local geometry enters the read operator. A dataset-level loss has none, and the read operator is then taken on the score with the identity geometry.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 4, [geometric-observation/chapters/ch04_the_observer_triple.md:9-60](#), and [geometric-observation/OBSERVATION.md:1-10](#), DOI 10.5281/zenodo.21776291. Version 1.0 of the theory was declared on 2026-08-18 in [geometric-observation/crucible/DECLARATION-V1.md](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.2	the observer triple, read subspace, nuisance, same read operator means same read geometry	geometric-observation/chapters/ch04_the_observer_triple.md:9-60 ; geometric-observation/OBSERVATION.md:1-10
chapter 6 section 6.1	the classifier row of the consumer table, the output metric makes a different observer	geometric-observation/chapters/ch04_the_observer_triple.md:60-135

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at [observation-data-mining 95af30c](#); what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13.

RELATED

[read operator](#) (p. 291); [quotient](#) (p. 279); [read distortion](#) (p. 290); [certificate](#) (p. 59).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.9.

Ledger rows that cite the entry's records without naming it: OT-7, GO-1.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.2, chapter 1 section 1.3, chapter 1 section 1.5, chapter 8 section 8.8, chapter 8 section 8.10, chapter 11 section 11.7, chapter 11 section 11.9.

operating point

concept. Entry id `operating-point`.

DEFINITION

The row at which a consumer is read, or the threshold at which a classifier is scored. A probe pays 2d evaluations for one. Chapter 0 section 0.5 and chapter 11 section 11.7.

Example. A probe at one row with 16 directions spends 32 calls at that one operating point.

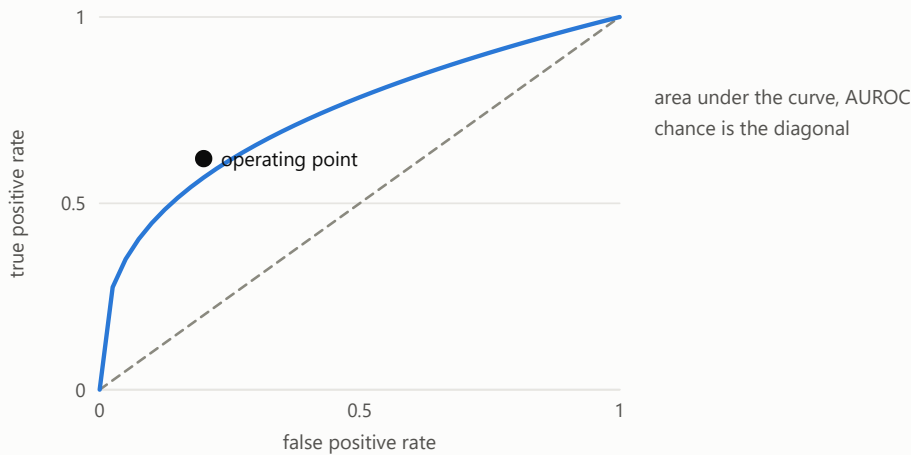


Figure 149: The row at which a consumer is read, or the threshold at which a classifier is scored.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

CONDITIONS

- The row at which a consumer is read, or the threshold at which a classifier is scored. A central-difference probe pays two evaluations per coordinate for one operating point, and raising a classifier's threshold can only shrink the set it calls positive.
- Whether many cheaper operating points can average their way back to the population operator was measured within a budget range and not proved, and the cliff at k equal to d does not soften for a probe confined to the directions it chooses.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.5 and chapter 11 section 11.7 of *Data Mining as Observation*, with the budget law in `read-scope/README.md:118-141`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/FiniteDifference.lean](#), theorems `central_quad`, `forward_quad`, `forward_error`, `central_cost`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Threshold.lean](#), theorems `predicted_anti`, `tp_anti`, `fp_anti`, `decision_comp`, `predicted_extremes`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 5, 6, 11, 14.

RELATED

[finite difference](#) (p. 146); [budget](#) (p. 51); [budget cliff](#) (p. 52); [threshold](#) (p. 362); [blind probe](#) (p. 45).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.28.

Ledger rows that cite the entry's records without naming it: OT-3, GO-4.

Sources-table rows that share a record with the entry without naming it: chapter 11 section 11.7.

orthogonal

concept. Entry id `orthogonal`.

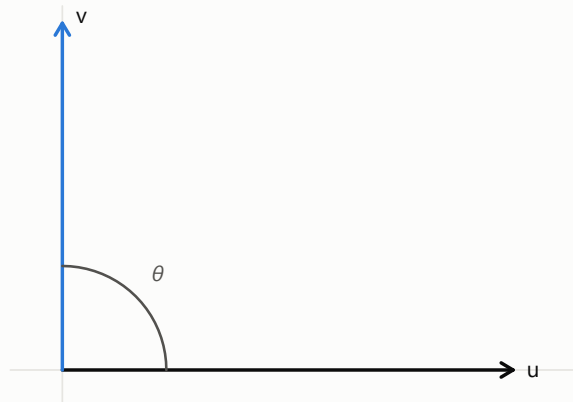


Figure 150: Dot product zero.

DEFINITION

Of two vectors, having dot product zero. The squared length of a sum of orthogonal vectors is the sum of the squared lengths. Chapter 0 section 0.1.

Example. $(1, 0)$ and $(0, 1)$ are orthogonal, and the squared length of their sum is $1 + 1 = 2$.

EQUATION

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

CONDITIONS

- Of two vectors, having dot product zero. The relation is symmetric, survives rescaling, holds between any vector and zero, the squared length of a sum of orthogonal vectors is the sum of the squared lengths, and the residual of a projection is orthogonal to the direction projected on.
- A linear classifier's nuisance is everything orthogonal to its weight vector, and the eigenvectors of a symmetric matrix with distinct eigenvalues are orthogonal, so the read subspace and the nuisance split every row.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.1 and section 0.2 of *Data Mining as Observation*.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Orthogonal.lean](#), theorems `orth_symm`, `pythagoras`, `orth_smul`, `orth_zero`, `residual_orth`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 4, 6, 9, 11, 12.

RELATED

[dot product](#) (p. 118); [projection](#) (p. 272); [eigenvalue, eigenvector](#) (p. 125); [nuisance](#) (p. 240); [read subspace](#) (p. 293).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.3, 0.9.

Ledger rows that cite the entry's records without naming it: OT-7.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2.

outer product

concept. Entry id `outer-product`.

DEFINITION

The matrix whose entry in row i and column j is the product of a vector's i -th and j -th coordinates, of rank at most one and positive semidefinite for a vector with itself. Chapter 0 section 0.5.

Example. $(1, 2)$ with itself gives the matrix with rows $(1, 2)$ and $(2, 4)$, which has rank one.

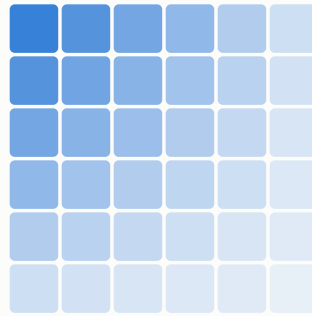
EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$



rank one

Figure 151: The entry in row i and column j is the product of the i -th and j -th coordinates.

Book equation 0.3.

$$\Sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)], \quad \text{tr } \Sigma = \sum_i \Sigma_{ii}.$$

CONDITIONS

- The matrix whose entry in row i and column j is the product of the i -th and j -th coordinates of a vector. Its action on any vector is a multiple of the vector it was built from, it has rank at most one, and the outer product of a vector with itself is positive semidefinite.
- The read operator is the weighted mean of the outer product of the sensitivity with itself, and a linear classifier's read operator is one outer product scaled by how steep the score is.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.5 of *Data Mining as Observation*.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/PositiveSemidefinite.lean](#), theorems `psd_add`, `psd_smul`, `psd_outer`, `psd_readOp`, `psd_diag_nonneg`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Rank.lean](#), theorems `rank_le_width`, `rank_le_height`, `rank_outer_le_one`, `rank_mul_le`, `rank_zero`, `rank_transpose`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's

appendix C.

USED IN

Data Mining as Observation chapters 0, 2, 6.

RELATED

[read operator](#) (p. 291); [rank](#) (p. 283); [positive semidefinite](#) (p. 265); [covariance matrix](#) (p. 91); [sensitivity](#) (p. 328).

SEE ALSO

Ledger rows that cite the entry's records without naming it: OT-7.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 6 section 6.1.

outlier

concept. Entry id outlier.



Figure 152: A row the reader cannot place.

DEFINITION

A row the reader cannot place. The statistical detector calls a row an outlier when its distance from the mean in units of spread passes a threshold, and no more than one over the threshold squared of the weight can. Chapter 10.

Example. A row 3.6 spreads from the mean is an outlier at threshold 3, and by Chebyshev at most one ninth of the weight lies past 3 spreads.

EQUATION

none

CONDITIONS

- A row the reader cannot place. The statistical detector scores a row by its distance from the mean in units of spread, which is zero at the mean, unchanged when data, mean, and spread are rescaled and shifted together, and past a threshold t exactly when the row lies more than t spreads from the mean. By Chebyshev the weight of rows beyond t is at most the mean squared score over t squared.
- The four detector families read different coordinates, distance from the mean, distance to neighbours,

density relative to neighbours, and cluster membership, and the observer's outliers, the anti-hubs, are the rows no detector of the data alone can see.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.16 and chapter 10 section 10.1 of *Data Mining as Observation*, after TSK chapter 9, with the hierarchical typing in `turboquant-pro/turboquant_pro/anatomy.py:98-170`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Outlier.lean](#), theorems `zscore_mean`, `zscore_affine`, `outlier_iff`, `zscore_reflect`, `chebyshev`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Mahalanobis.lean](#), theorems `dm2_nonneg`, `dm2_mean`, `dm2_scale`, `dm2_eq_whitened`, `dm2_antitone_in_variance`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 3, 10.

RELATED

[Mahalanobis distance](#) (p. 219); [anti-hub](#) (p. 20); [abstention](#) (p. 12); [whitening](#) (p. 381); [density](#) (p. 106).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.33, 10.3, 0.6.

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.1, chapter 10 section 10.3.

output metric

concept. Entry id `output-metric`.

DEFINITION

The rule by which a consumer's mistakes are scored, the second element of an observer. Chapter 1.

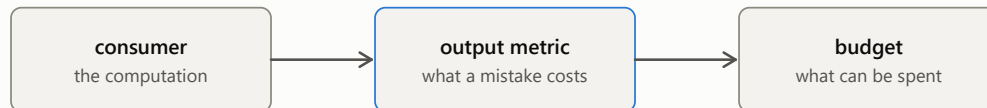
Example. Classifiers with one and three false positives against three and one false negatives tie on accuracy and split on any cost matrix.

EQUATION

none

CONDITIONS

- The rule by which a consumer's mistakes are scored, the second element of an observer. A consumer and its negation have the same read operator and reverse every comparison, so the read operator alone does not fix the observer, and two cost matrices score the same pair of classifiers in opposite orders, so naming the output metric is naming what a downstream mistake is.
- A dataset-level loss, accuracy, F1, a rank correlation, or dollars lost is not a local geometry on the score and



the read operator is what the triple induces, and its kernel is the nuisance

Figure 153: What a mistake costs, the second element of the observer.

cannot be inserted into the read-operator formula. Each interestingness measure of chapter 5 and each fairness metric of chapter 14 is an output metric, right for some consumer and wrong for others.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 1 section 1.2 of *Data Mining as Observation*, with the flip's verdict inversion in `geometric-observation/ch08_value.md:1-30`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/OutputMetric.lean](#), theorems `neg_reverses`, `readOp_of_neg`, `cost_flips`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 3, 5, 6, 8, 9, 11, 14.

RELATED

[observer](#) (p. 244); [consumer](#) (p. 79); [read operator](#) (p. 291); [budget](#) (p. 51).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 1.1, 0.9, 6.1.

Ledger rows that cite the entry's records without naming it: GO-2 (neg. half: not reconstruction), GO-2 (pos. half: consumer-projected covariance *controls*).

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.2, chapter 4 section 4.3.

P

paired

instrument. Entry id paired.

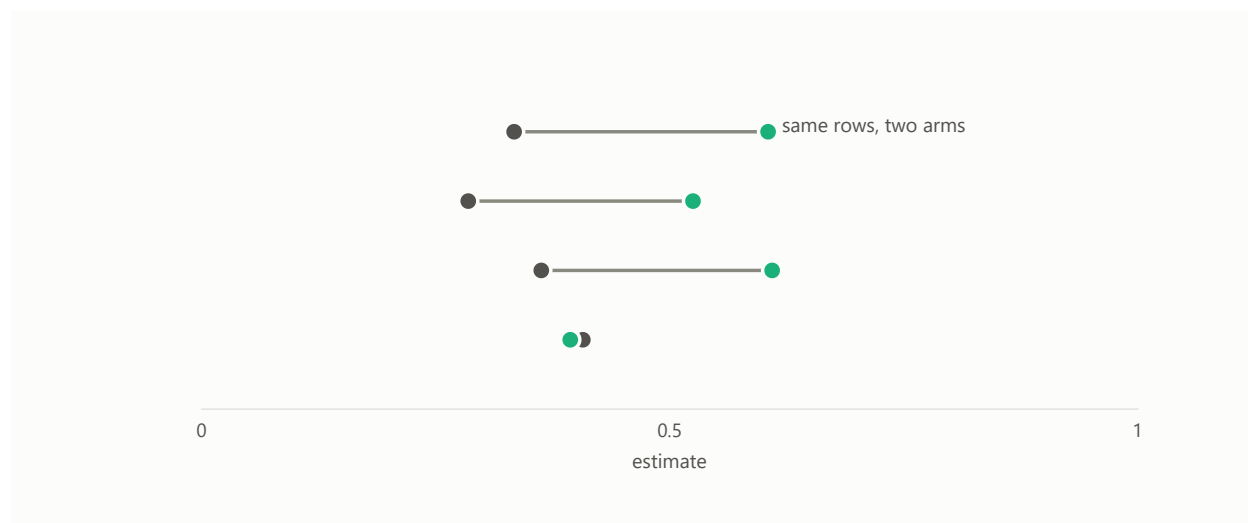


Figure 154: The same rows or seeds in both arms.

DEFINITION

Of a comparison, made on the same rows or seeds for both arms. Pairing helps exactly when the two arms covary. Chapter 0 section 0.9.

Example. Scoring both arms on the same 40 documents pairs the comparison, and the difference's variance drops when the arms covary.

EQUATION

none

CONDITIONS

- Of a comparison, made on the same rows or the same seeds for both arms, so that the difference is scored row by row. The variance of the difference is the sum of the variances less twice the covariance, so pairing helps exactly when the two arms covary.
- The paired null at rank two read 0.385 against 0.572 on fourteen of sixteen cells, and the encoder comparisons carry a paired bootstrap interval on each relation.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.9 and chapter 8 section 8.4 of *Data Mining as Observation*, with the paired null in `readscope/SPEC.md:806-857`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 8 section 8.4	C-11c paired null, rank 2 positional 0.385 vs null 0.572, 0.615 vs 0.224, 14 of 16 cells, scope 16 cells one 3B model 192 positions	readscope/SPEC.md:806-857 ; readscope/calibration/records/c11c-operator-drift.json

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Bootstrap.lean](#), theorems `mean_sub`, `var_sub`, `paired_lt_iff`, `cov_comm`, at observation-data-mining 95af30c; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 4, 6, 8, 11, 12.

RELATED

[bootstrap](#) (p. 49); [confidence interval](#) (p. 76); [seed](#) (p. 326); [standard error](#) (p. 347); [control](#) (p. 83).

SEE ALSO

Book equations stated beside the entry’s terms, not defining it: 0.18, 8.2, 0.29.

Ledger rows that cite the entry’s records without naming it: NEG-14, GO-B-Llama, GO-B-Llama-rematch.

Sources-table rows that share a record with the entry without naming it: chapter 11 section 11.8, chapter 12 section 12.4.

paraphrase class

concept. Entry id `paraphrase-class`.

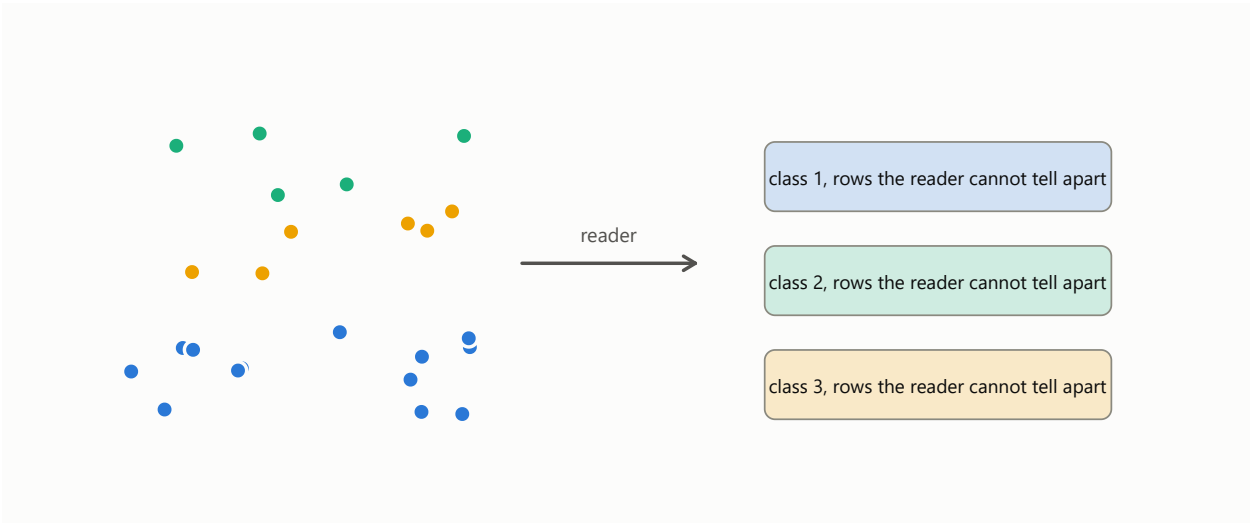


Figure 155: Texts the encoder cannot tell apart.

DEFINITION

A set of rewordings of one input that preserve its meaning. A decision is invariant to re-description when it does not move across the class. Chapter 14.

Example. The sentences ‘the row is stale’ and ‘the row is out of date’ land in one paraphrase class for the encoder.

EQUATION

Book equation 14.3.

$$\theta_d = \frac{\text{mean}_i |S_A(i) - S_B(i)|}{\text{mean}_{i \neq j} |S_A(i) - S_A(j)|}, \quad \text{bar } \theta_d \leq 0.5.$$

CONDITIONS

- A set of rewordings of one input that preserve its meaning. A verdict is invariant to re-description when it does not move across the class, and the ratio of equation 14.3 measures the movement under rewording over the spread between inputs. It is nonnegative, zero exactly when no item’s score moves, and at most one half exactly when the movement is at most half the spread.
- The measured ratios, 0.407 falling to 0.219 on sixty items and 0.482 to 0.301 under back-translation, with three of six gold items still flipping, are the program’s numbers, and the items that flip are routed to escalation.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 14 section 14.4 of *Data Mining as Observation*, with the program’s re-description test in `gtc-prototype/docs/REGATE.md:1-70` and the back-translation run.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ParaphraseClass.lean](#), theorems `movement_nonneg`, `ratio_nonneg`, `movement_eq_zero_iff`, `ratio_invariant`, `bar_iff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 12, 14.

RELATED

[quotient](#) (p. 279); [escalation](#) (p. 132); [certificate](#) (p. 59); [deployment mismatch](#) (p. 108).

SEE ALSO

none

percentile

concept. Entry id percentile.

DEFINITION

The value below which a given fraction of the rows fall. The fraction at or below a value is nondecreasing in the value. Chapter 11 section 11.3.

Example. Among 100 sorted ratios, the 97.5th over the 2.5th percentile is the rank certificate’s percentile setting.

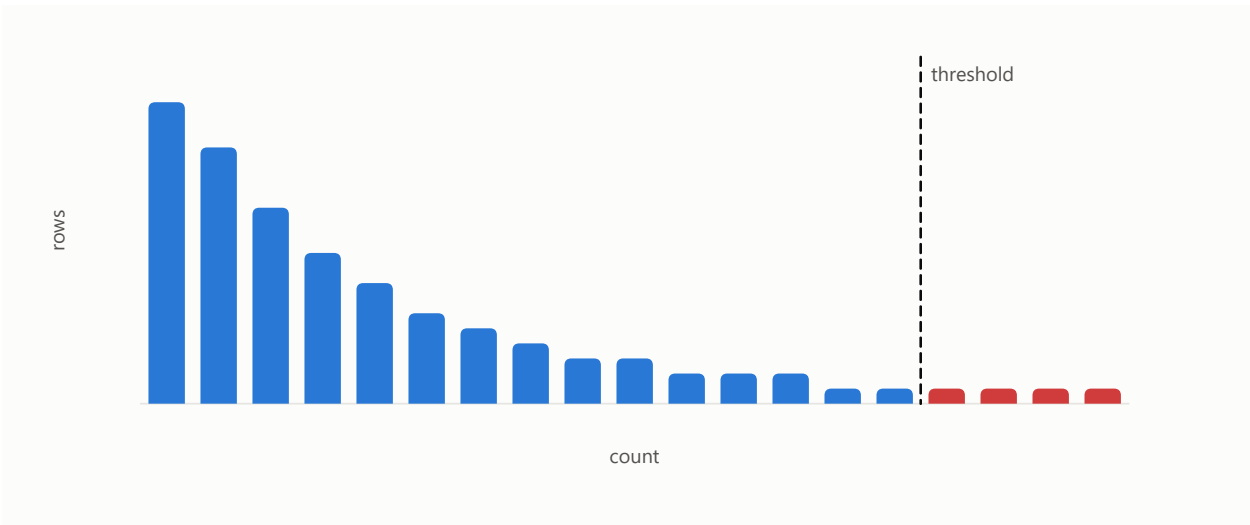


Figure 156: The value below which a given fraction of the rows fall.

EQUATION

Book equation 11.2.

$$r = \frac{d_{\text{compressed}}}{d_{\text{exact}}}, \quad \kappa_{\text{strict}} = \frac{\max r}{\min r}, \quad \tau \geq 1 - 2\hat{\mu}(\kappa_{\text{strict}}), \quad \rho_S \geq 1 - 3\hat{\mu}(\kappa_{\text{strict}}),$$
$$\kappa_{97.5/2.5} = \frac{q_{97.5}(r)}{q_{2.5}(r)} \text{ gives the same two expressions as estimates, not floors.}$$

CONDITIONS

- The value below which a given fraction of the rows fall. The fraction at or below a value is nonnegative, at most one, nondecreasing in the value, one at or above the largest row and zero below the smallest, and an upper percentile over a lower one is at least one.
- The rank certificate’s percentile setting reads the 97.5 over 2.5 percentile ratio as a robust estimate where the strict setting reads the max over min, and the anti-hub gate reads the fifth percentile of recall.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 11 section 11.3 of *Data Mining as Observation*, with the percentile setting of the rank certificate in `turboquant-pro/turboquant_pro`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Percentile.lean](#), theorems `cdf_nonneg`, `cdf_le_one`, `cdf_mono`, `cdf_of_all`, `cdf_of_none`, `ratio_ge_one`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 3, 10, 11.

RELATED

[rank certificate](#) (p. 285); [anti-hub recall](#) (p. 22); [skewness](#) (p. 335); [threshold](#) (p. 362); [Robin Hood index](#) (p. 313).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.35, 10.3.

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.2, chapter 10 section 10.3, chapter 11 section 11.2.

per-channel quantizer

instrument. Entry id `per-channel-quantizer`.



each channel has its own step

Figure 157: A step per coordinate.

DEFINITION

A quantizer that chooses a scale per coordinate. Chapter 11.

Example. With per-coordinate steps of 0.5 and 0.1, the vector (1.3, 0.27) is stored as (1.5, 0.3).

EQUATION

none

CONDITIONS

- A quantizer that chooses a step per coordinate. The squared error of a vector is at most the sum of the squared half steps, and a finer step on one coordinate lowers that coordinate's bound alone, which is the allocation water-filling makes by sensitivity.
- At four bits the per-channel uniform quantizer held the perplexity at 14.91 against 12.24 uncompressed,

where the direction-only quantizer at the same bits did not.
Conditions are curated in `entries.toml` rather than read from a record.

LEDGER
none

FIRST STATED
Chapter 11 section 11.2 of *Data Mining as Observation*, with the per-channel rows of the quantizer table in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS
none

MACHINE CHECKED
[lean/DataMiningAsObservation/Quantization.lean](#), theorems `error_le_half_step`, `quantize_level`, `half_step_bound`, `sq_error_le`, `finer_step`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN
Data Mining as Observation chapters 0, 3, 4, 11.

RELATED
[quantization](#) (p. 275); [direction-only quantizer](#) (p. 111); [water-filling](#) (p. 378); [KV cache](#) (p. 200).

SEE ALSO
Book equations stated beside the entry’s terms, not defining it: 11.1, 0.13.
Ledger rows that cite the entry’s records without naming it: NEG-2.
Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 8 section 8.2, chapter 11 section 11.1.

perplexity

concept. Entry id perplexity.

DEFINITION
Two to the power of the average number of bits a language model needs per token of a test text. Lower is better. Equation 0.22.

Example. An average of 3.585 bits per token gives perplexity 2 to the 3.585, which is 12.

EQUATION
Book equation 0.22.

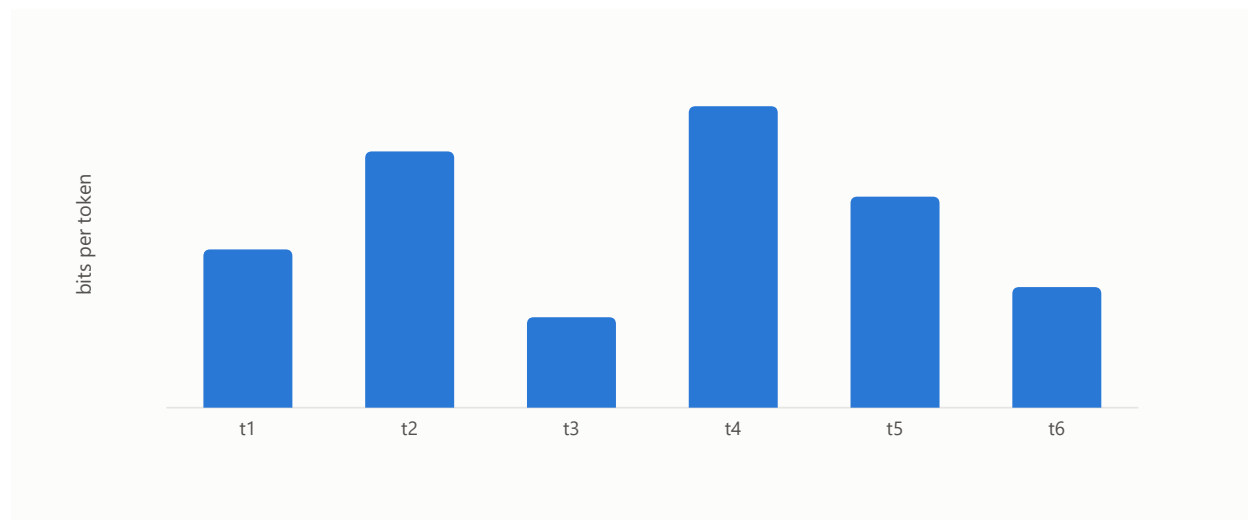


Figure 158: Two to the power of the average bits per token.

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(w_t | w_{<t}).$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

CONDITIONS

- Two to the power of the average number of bits a language model needs per token of a test text. It is at least one, equals the vocabulary size for a model that spreads its mass evenly, and is monotone in the bits.
- The attention-key finding raised it from 12.24 to 10643 at cosine 0.995, a rise of more than nine bits per token, and the recalibration that improved the reconstruction worsened it, the two standing negatives that keep reconstruction error off the list of acceptance metrics.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's case in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation/chapters/ch02_failure_of_observer_free_measurement.md:40-60 ; geometric-observation/chapters/ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro/docs/KV_KEYS_FINDING.md:1-49
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49 ; geometric-observation/chapters/ch16_honest_negatives.md NEG-2

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Perplexity.lean](#), theorems `bits_nonneg`, `one_le_perplexity`, `perplexity_uniform`, `perplexity_mono`, `bits_of_finding`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 8, 11.

RELATED

[KL divergence](#) (p. 196); [the flip](#) (p. 147); [identity reader](#) (p. 180); [read distortion](#) (p. 290).

SEE ALSO

Ledger rows that cite the entry's records without naming it: NEG-2, NEG-4.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.5, chapter 3 section 3.2, chapter 11 section 11.1, chapter 11 section 11.2.

planted

instrument. Entry id `planted`.

DEFINITION

Of a case, built so that the answer is known before the instrument reads it, such as an affine consumer whose read operator is its weight vector's outer product. Chapter 6 section 6.2.

Example. An affine consumer with weight vector (3, 4) has read operator proportional to the matrix with rows (9, 12) and (12, 16), which the probe must recover.

EQUATION

none

CONDITIONS

- Of a case, built so that the answer is known before the instrument reads it. An affine consumer has a known read operator, the outer product of its weight vector, so a probe that recovers it has been tested against a truth.
- The planted probe recovered the read subspace at overlap 0.936 against chance 0.059, twelve of twelve, and the real model then missed its bar at 0.567 against 0.60, which is the order the book keeps.

Conditions are curated in `entries.toml` rather than read from a record.

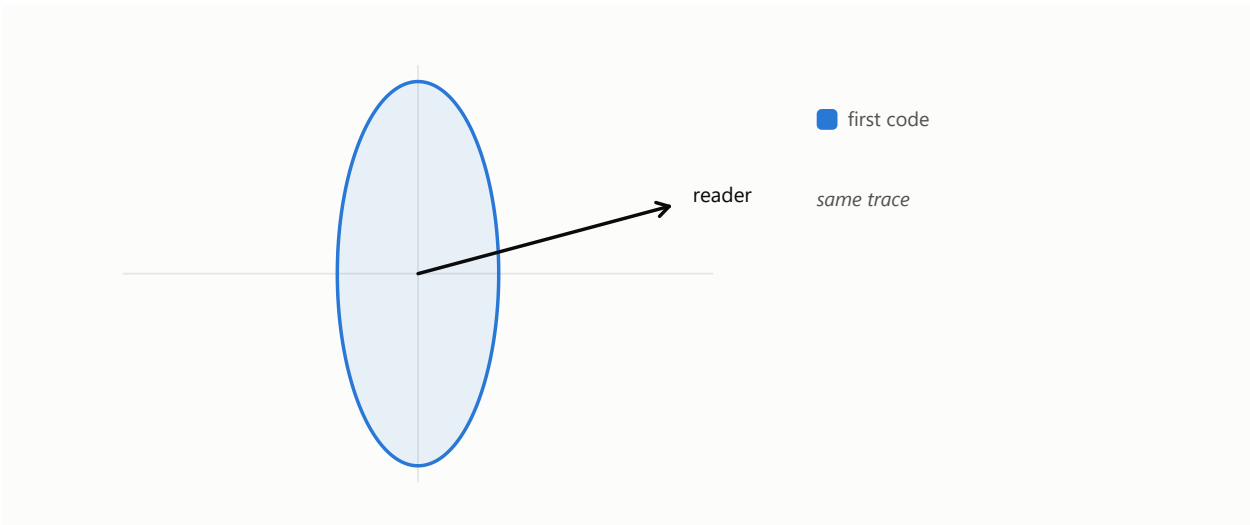


Figure 159: A case whose answer is known before the instrument reads it.

LEDGER

none

FIRST STATED

Chapter 6 section 6.2 of *Data Mining as Observation*, with the planted probe in `geometric-observation/claims/LEDGER.md` row GO-1.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	geometric-observation/claims/LEDGER.md row GO-1

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).
[lean/DataMiningAsObservation/ProbeCliff.lean](#), theorems `centralDiff_affine`, `centralDiff_basis`, `exists_blind_direction`, `indistinguishable`, `budget_cliff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 3, 6, 8, 11.

RELATED

[blind probe](#) (p. 45); [read operator](#) (p. 291); [control](#) (p. 83); [budget cliff](#) (p. 52).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.9, 6.1, 11.4.

Ledger rows that cite the entry's records without naming it: GO-1, NEG-12.

Sources-table rows that share a record with the entry without naming it: chapter 11 section 11.7.

Poisson ceiling

instrument. Entry id poisson-ceiling.

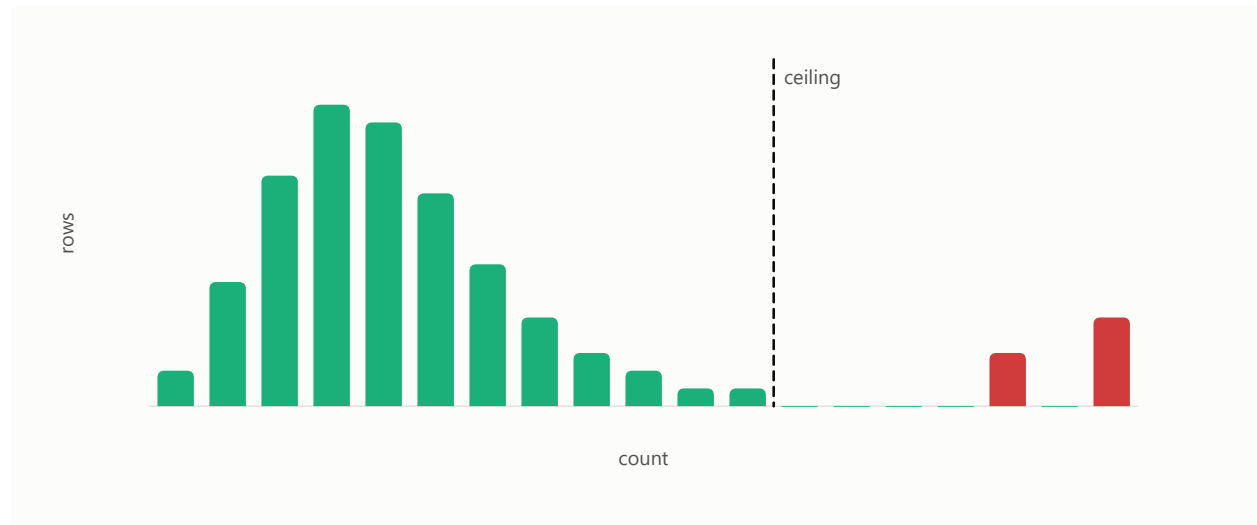


Figure 160: The count chance allows.

DEFINITION

The largest neighbour count that at least one point in a dataset would reach by chance under a Poisson null, above which a count is evidence of a hub. Equation 0.17.

Example. With 1000 queries at k equal to 10 over 1000 rows, each row expects 10 retrievals, and a count above about 20 is beyond chance.

EQUATION

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

CONDITIONS

- Under the null that the slots are handed out at random, a row's count is Poisson with mean the slots per row, and the ceiling is the largest count at least one row would reach by chance. A count above it is evidence of a hub, and a count below it is not evidence of anything.
- The ceiling depends on the query set, through the mean and through the rows the queries reach, so it is recomputed when the queries change, and the program's first hubness numbers moved by a factor of several when query coupling was removed.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

openvector-bench, openvector-bench/openvector_bench/hubness.py:41-100, and chapter 0 section 0.8 and chapter 3 section 3.5 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	Poisson null, hub excess, budget parameter	openvector-bench/openvector_bench/hubness.py:41-100

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/PoissonCeiling.lean](#), theorems `mass_nonneg`, `tail_antitone`, `tail_zero`, `tail_le_one`, `expectedAtLeast_antitone`, `example_mean`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3.

RELATED

[hubness](#) (p. 176); [anti-hub](#) (p. 20); [harness](#) (p. 168); [min-over-strata](#) (p. 226).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 10.3.

Ledger rows that cite the entry's records without naming it: NEG-11.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.5, chapter 8 section 8.1, chapter 10 section 10.2, chapter 10 section 10.3, chapter 11 section 11.6.

posited versus measured

concept. Entry id posited-versus-measured.

DEFINITION

The division of every claim in a report into those asserted by design and those backed by an artifact. Chapter 14.

Example. The four geometric volumes are posited; the twelve-domain flip is measured.

EQUATION

none

CONDITIONS

- Every claim in a report is either posited, asserted by design with no artifact behind it, or measured, backed by a committed artifact the reader can open. The division is stated at the top of the report and not inferred from the prose.
- A posited claim carries no ledger class above exploratory, and it stays there until a sealed measurement moves it. The ledger rule that says so is the one machine-checked in the book.
- Four volumes of the program are posited in their entirety and say so on their first page. Their claims are cited only as proposals.

Conditions are curated in `entries.toml` rather than read from a record.

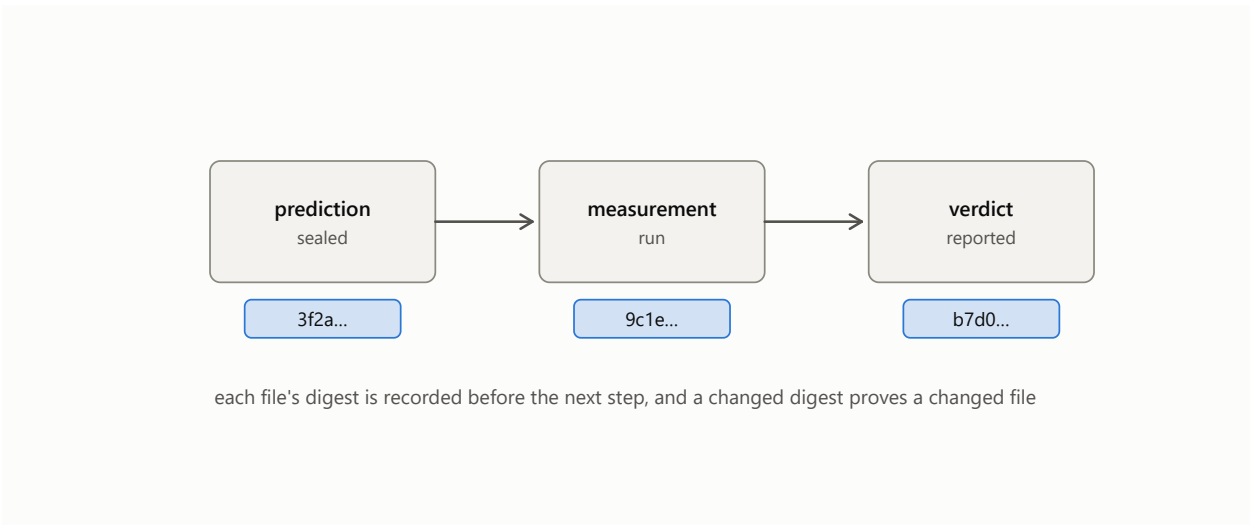


Figure 161: A claim written down before, and a number read after.

LEDGER
none

FIRST STATED
The epistemic banners of the four posited volumes, `geometric-ai/README.md`, `geometric-cognition/README.md`, `geometric-reasoning/README.md`, `geometric-education/README.md`, and chapter 14 section 14.8 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 14 section 14.8	registry rules, V4 proved in Lean and verified 2026-08-17 at commit ea7ee82, D4 posited with the drift list, reliability weight as one formula one source, 8 of 9	erism1-lib/docs/CONCEPT_REGISTRY.md:1-160
chapter 14 section 14.8	the four posited volumes	<code>geometric-ai/README.md</code> ; <code>geometric-cognition/README.md</code> ; <code>geometric-reasoning/README.md</code> ; <code>geometric-education/README.md</code> epistemic banners

FAILURES AND CORRECTIONS
none

MACHINE CHECKED
[lean/DataMiningAsObservation/Ledger.lean](#), theorems `step_none`, `step_miss_le`, `step_pass_ge`, `step_miss_refuted`, `exploratory_stays`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN
Data Mining as Observation chapters 14.

RELATED

[ledger class](#) (p. 211); [preregistration](#) (p. 267); [sealed](#) (p. 324); [reliability weight](#) (p. 306).

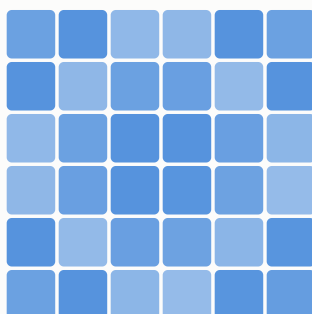
SEE ALSO

Book equations stated beside the entry's terms, not defining it: 1.1.

Ledger rows that cite the entry's records without naming it: OT-11, NEG-15 (Bell boundary).

positive semidefinite

concept. Entry id positive-semidefinite.



$$x^T A x \geq 0$$

Figure 162: Every quadratic form nonnegative.

DEFINITION

Of a symmetric matrix, having every quadratic form nonnegative, as the read operator and the covariance are. Chapter 0 section 0.3.

Example. The matrix with rows (1, 2) and (2, 4) is positive semidefinite, and the one with rows (1, 3) and (3, 1) is not, since (1, -1) gives -4.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^T], \quad g = \nabla C(x).$$

Book equation 0.11.

$$P_C(x) = J(x)^T G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

CONDITIONS

- Of a symmetric matrix, having every quadratic form nonnegative. A sum of such matrices is one, a nonnegative multiple is one, an outer product of a vector with itself is one, the read operator is one whenever the weights are nonnegative, and the diagonal entries are nonnegative.
- The local metric on a consumer's output is positive semidefinite and exists only where the output metric has a local quadratic representation, and every eigenvalue of a positive semidefinite matrix is nonnegative.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.3 and section 0.5 of *Data Mining as Observation*.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/PositiveSemidefinite.lean](#), theorems `psd_add`, `psd_smul`, `psd_outer`, `psd_readOp`, `psd_diag_nonneg`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2.

RELATED

[read operator](#) (p. 291); [covariance matrix](#) (p. 91); [outer product](#) (p. 248); [eigenvalue, eigenvector](#) (p. 125); [metric](#) (p. 224).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.3.

Ledger rows that cite the entry's records without naming it: OT-7.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2.

precision, recall

concept. Entry id `precision-recall`.

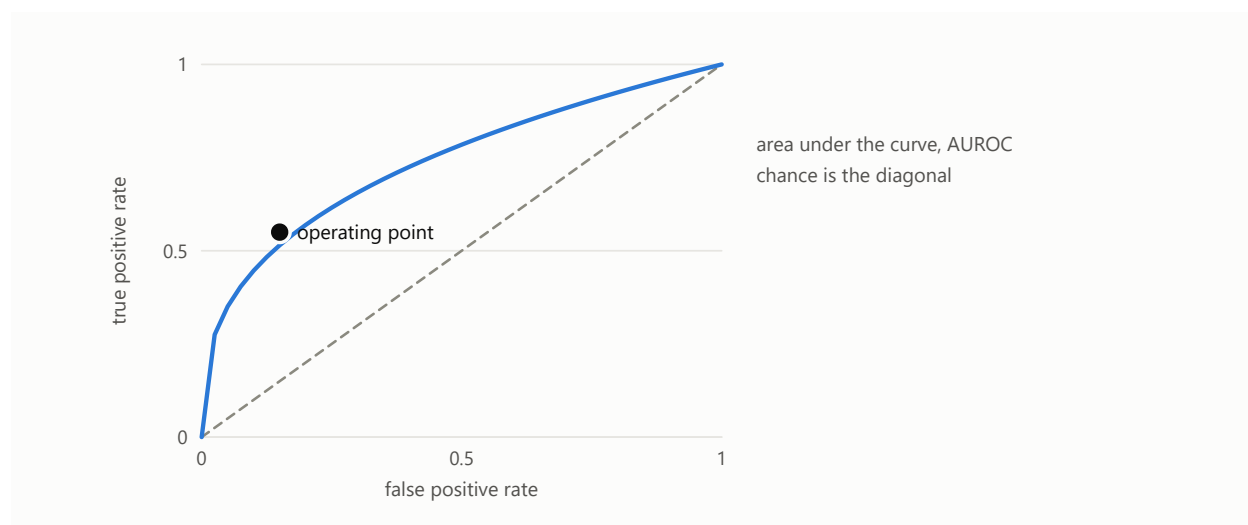


Figure 163: Of the rows called positive, how many are, and of the positives, how many are called.

DEFINITION

The fraction of predicted positives that are truly positive, and the fraction of true positives that were predicted. Equation 0.28.

Example. Ten rows called positive, six of them correct, out of eight true positives, is precision 0.6 and recall 0.75.

EQUATION

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

CONDITIONS

- The fraction of predicted positives that are truly positive, and the fraction of true positives that were predicted, both in the unit interval. F_1 is their harmonic mean, between the smaller of the two and their arithmetic mean, equal to both when they agree, and zero when either is zero.
- In counts F_1 is twice the true positives over twice the true positives plus the false positives and false negatives, the form the Youden ceiling bounds.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.14 of *Data Mining as Observation*, with the program's precision targets in `gtc-prototype/docs/SPECTRUM_FINDINGS.md:84-88`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 14 section 14.7	51 percent moderated at 80 and 95 percent precision on the balanced set	gtc-prototype/docs/SPECTRUM_FINDINGS.md:84-88

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/PrecisionRecall.lean](#), theorems `precision_mem_unit`, `recall_mem_unit`, `f1_le_mean`, `min_le_f1`, `f1_self`, `f1_zero`, `f1_counts`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 4, 6, 8, 9, 10, 11, 12, 14.

RELATED

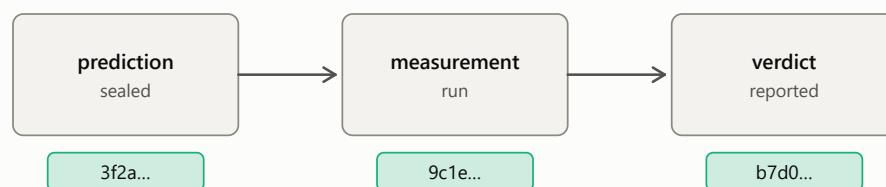
[F1](#) (p. 140); [threshold](#) (p. 362); [recall at k](#) (p. 295); [Youden F1 bound](#) (p. 385).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 6.2.

preregistration

instrument. Entry id `preregistration`.



each file's digest is recorded before the next step, and a changed digest proves a changed file

Figure 164: The claim, bar, null, and budget written before the measurement.

DEFINITION

Committing the hypothesis, the bar, and the analysis before the measurement is run, so that the record shows what was predicted. Chapter 8.

Example. The prediction, the bar, the null, and the seeds were committed before the run, and the commit hash is in the report.

EQUATION

none

CONDITIONS

- A preregistration commits the hypothesis, the bar, the analysis, and the missing-data rule before the measurement is run, so that the record shows what was predicted rather than what was found.
- Its bar must be anti-vacuous. A pass that any instrument would have produced, a threshold met by every reading, proves nothing, and the program's own first PF5 pass is the case.
- Multiple looks at the data need a corrected threshold, and a preregistration that names one look needs none.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.8 of *Data Mining as Observation*, with the program's template `observation-theory-campaigns/experiments/PREREG-TEMPLATE.md:1-71` and Volume 14's protocol `geometric-observation/PROTOCOL.md:58-75`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.3	the missing-data rule as a required preregistration field	observation-theory-campaigns/experiments/PREREG-TEMPLATE.md:47-52

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.6	the preregistration fields	instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md : 157-176; chapter 8 section 8.8 of this book

FAILURES AND CORRECTIONS

- [observation-theory-campaigns/ERRATA.md:90-110](#) at [f6d1fe2](#). E3. Two wrong bar values in the campaign notes **Found** 2026-08-07, same pass. Not a published error. [experiments/CAMPAIGN.md](#) twice described the PF4-006 entry-point and timestep clause as having a bar of 1e-10. The sealed declaration and the committed record both set that bar at 1e-8. The 1e-10 figure is the separate translation-covariance bar. The narrative was wrong and the verdict was not, since the clause failed at 1.3e-2 against either number. The same document described PREREG-PF4-009 as passing all six bars. The sealed document groups its requirements into six bars and the governed runner reports eight items, and the mapping between them was never declared. Both statements are now made explicitly. Corrected in place, since these are working notes rather than sealed documents or published records. —

MACHINE CHECKED

[lean/DataMiningAsObservation/Bonferroni.lean](#), [theorems family_error_le](#), [bonferroni](#), [one_look](#), at [observation-data-mining 95af30c](#); what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[sealed](#) (p. 324); [ledger class](#) (p. 211); [harness](#) (p. 168); [certificate](#) (p. 59).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.3.

Ledger rows that cite the entry's records without naming it: OT-7, GO-2 (neg. half: not reconstruction).

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.1, chapter 1 section 1.3, chapter 1 section 1.6, chapter 1 section 1.7, chapter 2 section 2.3, chapter 5 section 5.3, chapter 8 section 8.3, chapter 8 section 8.8.

principal component analysis

instrument. Entry id [principal-component-analysis](#).

DEFINITION

Projection of the centred data onto the eigenvectors of its covariance with the largest eigenvalues, the linear orthogonal projection of rank k that minimizes reconstruction error for the identity reader. Chapter 4.

Example. With eigenvalues 4, 3, 2, and 1, keeping the top two components drops an error of 3.

EQUATION

none

CONDITIONS

- Projection of the centred data onto the eigenvectors of its covariance with the largest eigenvalues. Among linear orthogonal projections of rank k it minimizes the reconstruction error. In the eigenbasis the variance along a unit direction lies between the smallest and the largest eigenvalue, a basis direction attains its own eigenvalue, and the error of keeping k components is the sum of the dropped eigenvalues.

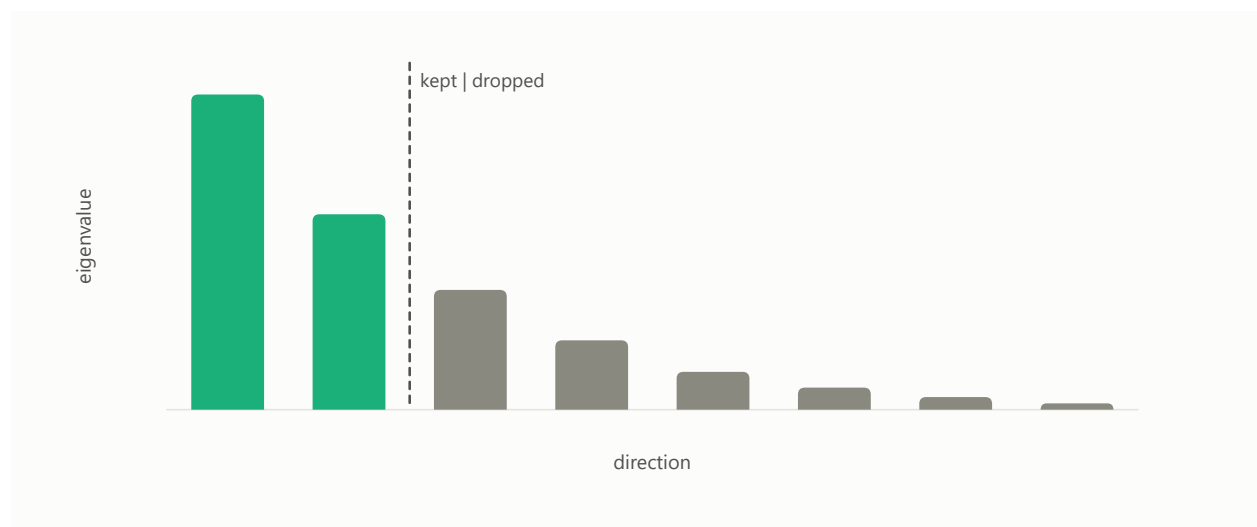


Figure 165: Projection onto the top eigenvectors, the identity reader's code.

- It is the reduction that minimizes reconstruction error for the identity reader, which weighs all directions equally, which is to say for no particular reader at all. At matched reconstruction the consumer-aware code beat it in twelve of twelve domains, and truncation loses on compact sets.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Pearson, on lines and planes of closest fit, 1901, and Hotelling, 1933, as chapter 4 section 4.1 of *Data Mining as Observation* reads them, with the program's truncation record in `turboquant-pro/benchmarks/RESULTS_glove.md:1-40`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.5	9.6x at recall 0.999 CI-gated on GloVe 1.18M; 32x at 0.9993 on private 199k LaBSE, ties OPQ, beats RaBitQ, 20x build; 27.7x and 114x reported; PCA truncation loses on compact sets; 20x at 199k and 4x at 1M over OPQ; RaBitQ builds in under a second	turboquant-pro/CLAIMS.md:28-49

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/PCA.lean](#), theorems `varAlong_basis`, `varAlong_le`, `varAlong_ge`, `dropped_eq`, `dropped_nonneg`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 4, 6, 12.

RELATED

[covariance matrix](#) (p. 91); [explained variance](#) (p. 137); [effective rank](#) (p. 123); [projection](#) (p. 272); [identity reader](#) (p. 180).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.5, 0.7, 4.2.

Ledger rows that cite the entry's records without naming it: GO-1, GO-6.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.2, chapter 4 section 4.5.

product quantization

instrument. Entry id product-quantization.



four sub-vectors, four codebooks

Figure 166: A vector split into pieces, each with its own codebook.

DEFINITION

Splitting a vector into pieces and quantizing each piece with its own codebook. Chapter 11.

Example. A 768-dimensional vector split into 8 pieces of 96, each coded with 256 centroids, costs 8 bytes.

EQUATION

none

CONDITIONS

- Splitting a vector into pieces and quantizing each with its own codebook. The squared error of the whole is the sum of the pieces' errors, so the best code for the whole is the best code for each piece separately, and K entries per piece over M pieces address K to the $M \log_2 K$ bits.
- It is evaluated by recall at k per stratum and by the rank certificate, never by reconstruction error, and the anti-hub stratum is where it fails first.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Jégou, Douze, and Schmid, product quantization for nearest neighbor search, 2011, as chapter 11 section 11.3 of *Data Mining as Observation* presents it, with the program's comparisons in openvector-bench and turboquant-pro.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ProductQuantization.lean](#), theorems `sq_error_add`, `inf_add`, `bits_of_codebooks`, `table_row`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 4, 8, 11, 13.

RELATED

[recall at k](#) (p. 295); [anti-hub](#) (p. 20); [water-filling](#) (p. 378); [rank certificate](#) (p. 285).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.3, 4.2.

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.4.

projection

concept. Entry id projection.

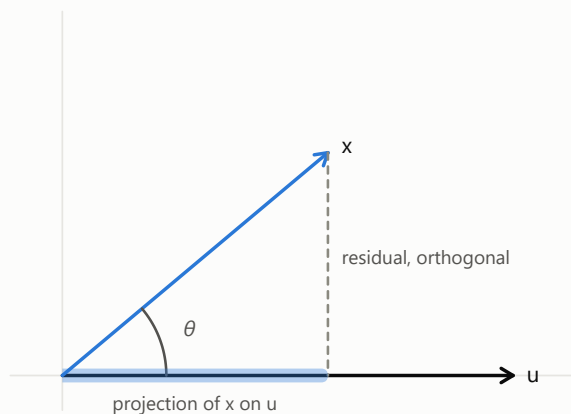


Figure 167: The component of one vector along another.

DEFINITION

The component of a vector along a unit direction. Equation 0.2.

Example. The projection of $(3, 4)$ onto the unit direction $(1, 0)$ is $(3, 0)$, and the residual $(0, 4)$ is orthogonal to it.

EQUATION

Book equation 0.2.

$$\text{proj}_u(x) = (u \cdot x) u, \quad \|u\| = 1.$$

CONDITIONS

- The component of a vector along a unit direction, the direction scaled by their dot product. Projecting twice is projecting once, the remainder is orthogonal to the direction, the projection is no longer than the vector, and adding anything orthogonal to the direction leaves it unchanged.
- The last is the quotient a one-direction reader takes. Row normalization is the projection onto the read subspace of the geodesic-rank consumer, and the invariant-nuisance split of the ledger is the same statement for a general consumer.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.2 of *Data Mining as Observation*, with the program's angular projection in Volume 14 chapter 9, `geometric-observation/chapters/ch09_legibility.md:31-41`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Projection.lean](#), theorems `proj_proj`, `dot_sub_proj`, `proj_sq_le`, `proj_add_orth`, `proj_add`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 3, 4, 6, 7, 9, 11.

RELATED

[dot product](#) (p. 118); [read subspace](#) (p. 293); [quotient](#) (p. 279); [nuisance](#) (p. 240).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.12a, 0.12b.

Ledger rows that cite the entry's records without naming it: GO-1.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.2, chapter 11 section 11.7.

p-value

concept. Entry id p-value.

DEFINITION

The probability of seeing a difference at least as large as the one observed if the true difference were zero. Chapter 0 section 0.9.

Example. A score more extreme than 970 of 1000 null scores has p-value 0.03.

EQUATION

none

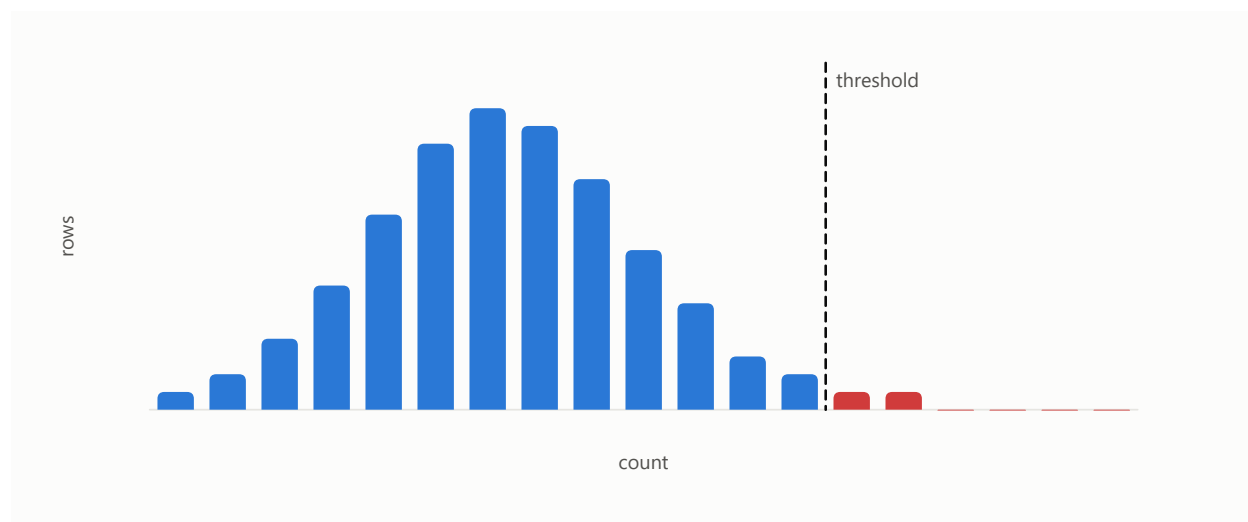


Figure 168: The fraction of the null distribution at least as extreme as the observation.

CONDITIONS

- Against a finite null sample, the fraction of null draws at least as large as the observation. It lies in the unit interval, never rises as the observation grows, and under the null the fraction of draws with p-value at or below a level is at most that level, which is what makes a threshold on it a false-positive rate.
- A p-value counts one look. Forty looks at five percent pass two by chance, and the book's rule is to count every comparison made and correct for it.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.9 of *Data Mining as Observation*, with the program's own p-values in the ensemble comparison of chapter 7 and the C-12 record in readscape.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/PValue.lean](#), theorems `pvalue_mem_unit`, `pvalue_antitone`, `uniform_bound`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 11.

RELATED

[multiple comparisons](#) (p. 229); [confidence interval](#) (p. 76); [harness](#) (p. 168); [preregistration](#) (p. 267).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.3.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.4, chapter 8 section 8.5.

Q

quantization

concept. Entry id quantization.



Figure 169: Replacing each number with one of a small set of values.

DEFINITION

Replacing each number with one of a small set of allowed values so that it can be stored in a few bits. Chapter 0 section 0.11.

Example. Rounding 1.37 to a step of 0.5 gives 1.5, an error of 0.13.

EQUATION

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

CONDITIONS

- Replacing each number with one of a small set of allowed values so that it can be stored in a few bits. A uniform quantizer with a given step has error at most half the step and zero error on a level, and halving the step halves the bound.
- Those are the identity reader's bounds. What a quantizer does to a consumer is the flip's question, and the key quantizer table of chapter 11 is the case in which the smaller reconstruction error was the worse code.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.11 and chapter 11 section 11.2 of *Data Mining as Observation*, with the program's quantizer table in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Quantization.lean](#), theorems `error_le_half_step`, `quantize_level`, `half_step_bound`, `sq_error_le`, `finer_step`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 7, 8, 10, 11.

RELATED

[bit](#) (p. 43); [codebook](#) (p. 66); [per-channel quantizer](#) (p. 257); [direction-only quantizer](#) (p. 111); [the flip](#) (p. 147).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.13.

Ledger rows that cite the entry's records without naming it: GO-2 (neg. half: not reconstruction), NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 8 section 8.2, chapter 8 section 8.9, chapter 11 section 11.1, chapter 11 section 11.2.

query, key, value

concept. Entry id query-key-value.

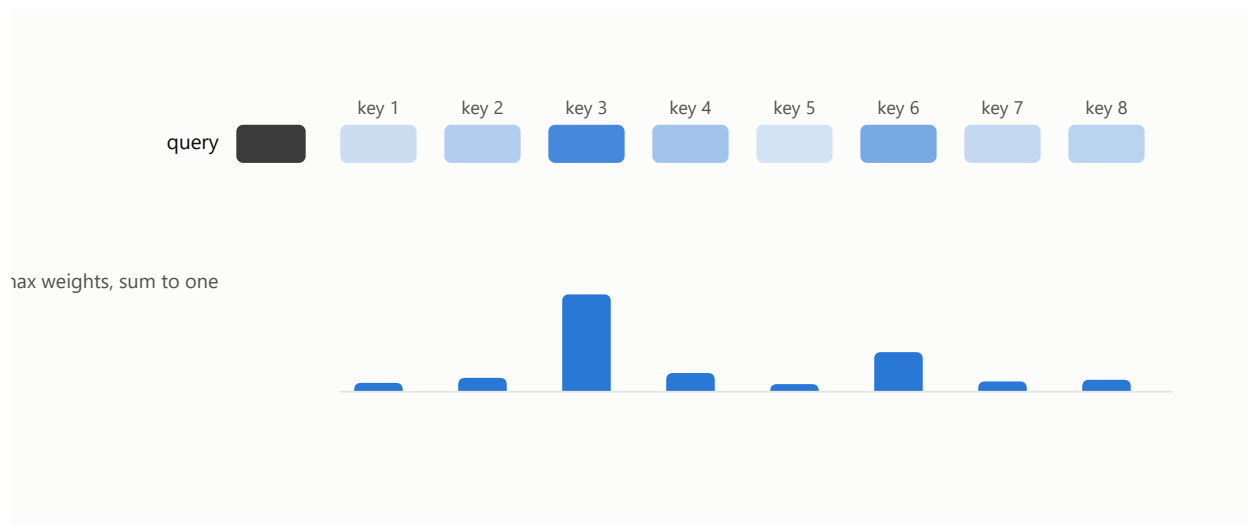


Figure 170: The query scores each key, and the weights mix the values.

DEFINITION

The three vectors each token produces inside attention. The query is compared to earlier keys and the result weights the values. Chapter 0 section 0.11.

Example. A query scores each key, the scores become softmax weights, and the weights mix the values into one output.

EQUATION

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

CONDITIONS

- The three vectors each token produces inside attention. The query is compared to earlier keys and the result weights the values, so a head reads the keys only through their scores against the query and its output lies between the smallest and largest value.
- Keys are read along the query and values are averaged, so a compression that preserves query-key scores preserves every output, and one that preserves the keys' cosine need not.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects.* NEG-2 [refuted]. Reconstruction cosine as a proxy for key quality. [geometric-observation/claims/LEDGER.md:95](#).

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's key-side measurements in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro/docs/KV_KEYS_FINDING.md:61-86 ; the-angular-observer/README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality. [geometric-observation/claims/LEDGER.md:95](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/Attention.lean](#), theorems `output_le_max`, `min_le_output`, `output_congr`, `output_nuisance`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[attention](#) (p. 26); [head](#) (p. 170); [KV cache](#) (p. 200); [softmax](#) (p. 336).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 8 section 8.2.

query set

concept. Entry id query-set.

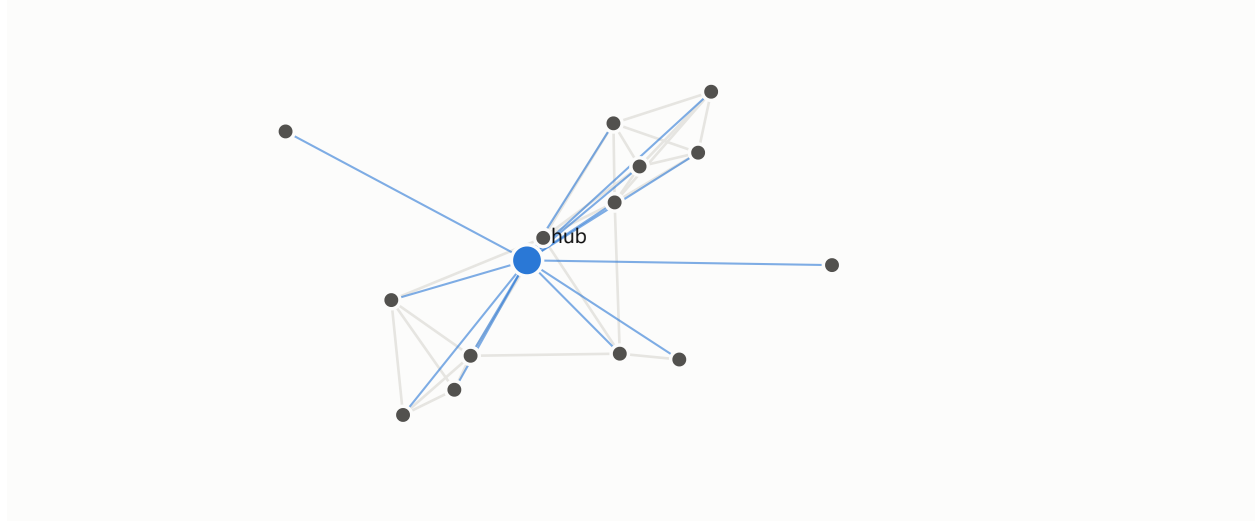


Figure 171: The rows a benchmark asks about, which decide the hubs.

DEFINITION

The rows a retrieval benchmark asks about. Hubness and the never-retrieved floor are properties of it. Chapter 3 section 3.5 and chapter 10 section 10.2.

Example. The 1000 queries of a benchmark decide which of the corpus's rows become hubs.

EQUATION

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

Book equation 10.4.

$$\text{never retrieved} \geq 1 - \frac{|Q|k}{n} \quad \text{whenever} \quad |Q|k < n.$$

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

CONDITIONS

- The rows a retrieval benchmark asks about. Two query sets that retrieve the same lists have the same hubs, no queries give no hubs, and the rows never retrieved number at least the row count less the queries times k, so hubness and the never-retrieved floor are properties of the query set.

- Queries drawn from the corpus inflated the first hub counts by a factor of several, and a generator matched on the query budget would be matched on an artifact.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. NEG-15 (Bell boundary) [demonstrated]. *Query-conditioned hubness supplies a mechanism for Bell-inequality violation without action at a distance*. Refuted as a mechanism; the settings-as-queries re-framing survives only as vocabulary. [geometric-observation/claims/LEDGER.md:93](#).

FIRST STATED

Chapter 3 section 3.5 and chapter 10 section 10.2 of *Data Mining as Observation*, with the query-coupling artifact in `openvector-bench/results/QUERY_COUPLING_ARTIFACT.md:1-20`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	query mass best single feature at every K for all four responses, seven features add at most 20 percent, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	<code>openvector-bench/results/R13_STAGE0_RESULT.md:40-50</code> ; <code>openvector-bench/results/R13_STAGE1_RESULT.md:1-40</code>
chapter 11 section 11.6	query mass best single feature at every K for all four responses, seven features add at most 20 percent at 12 leaves, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	<code>openvector-bench/results/R13_STAGE0_RESULT.md:40-50</code> ; <code>openvector-bench/results/R13_STAGE1_RESULT.md:1-40</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Hub.lean](#), theorems `card_hubs_le`, `hubs_congr`, `hubs_anti`, `hubs_empty`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Hubness.lean](#), theorems `sum_count`, `sum_count_eq`, `count_congr`, `antiHub_iff`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 8, 10, 11, 12, 13.

RELATED

[hub](#) (p. 175); [hubness](#) (p. 176); [Poisson ceiling](#) (p. 262); [recall at k](#) (p. 295); [harness](#) (p. 168).

SEE ALSO

Ledger rows that cite the entry's records without naming it: NEG-11.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.5, chapter 8 section 8.1, chapter 10 section 10.2.

quotient

concept. Entry id quotient.

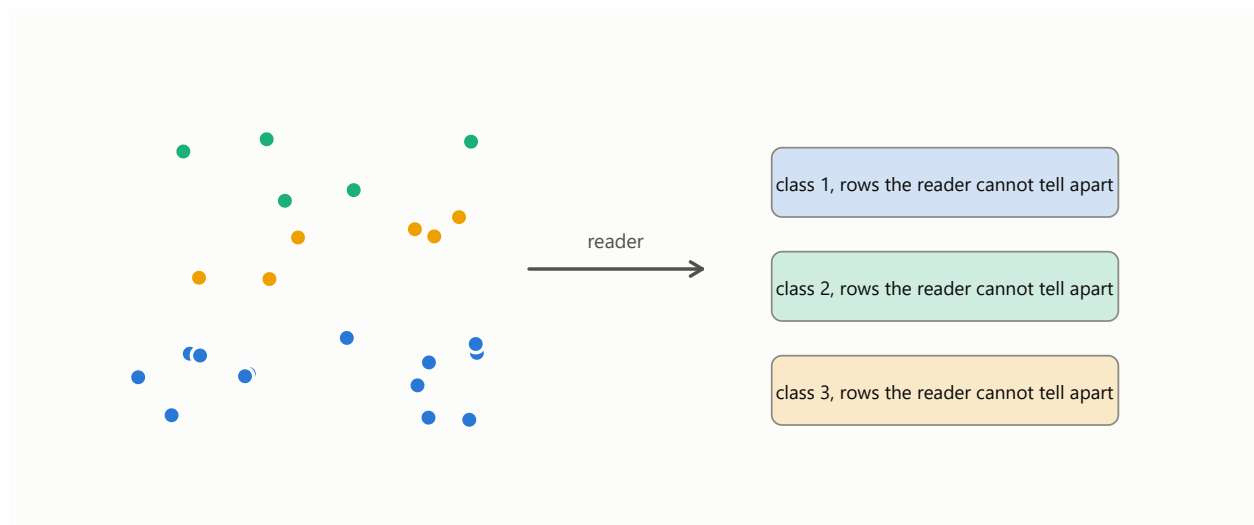


Figure 172: What the reader cannot tell apart.

DEFINITION

The space of data with a set of transformations declared not to matter, so that two rows differing only by such a transformation are the same point. Every distance is a distance on some quotient. Equation 0.12.

Example. The cosine reader cannot tell $(1, 1)$ from $(2, 2)$, so the two are one row on its quotient.

Known as, or related to prior art. The information bottleneck's view of relevant information, what the consumer does not read may be discarded, stated for a fixed consumer without a variational objective.

EQUATION

Book equation 0.12a.

$$x \sim_C x' \iff d_G(C(x), C(x')) = 0.$$

CONDITIONS

- The equivalence x related to x' when their difference lies in the kernel of the read operator is exact for an affine consumer with a fixed output geometry.
- Three things are kept apart. Exact equivalence, that the consumer's outputs agree under its output metric. A local null direction, in the kernel of the operator at one row. A workload-common null direction, in the kernel of the workload average, unread at almost every row. For a nonlinear consumer the third does not deliver the first, and cosine similarity is the case, whose radial direction turns with the row.
- An invertible transform identifies no two rows and forms no quotient. It changes the geometry a reader sees. Discretization changes the quotient.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects.* GO-5 [refuted]. An $\alpha=1$ density/hubness quotient restores invariant fidelity in ≥ 1 non-spectral domain. [geometric-observation/claims/LEDGER.md:67](#).
- *refutes or corrects.* NEG-6 [refuted]. Relative per-channel-demeaned error norm is the quotient-tangential quantity that controls softmax-KL. [geometric-observation/claims/LEDGER.md:99](#).

FIRST STATED

Volume 14, chapter 5, [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:78-90](#), DOI 10.5281/zenodo.21776291, and chapter 3 section 3.1 of *Data Mining as Observation*, where every similarity measure induces a distance on some quotient.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- GO-5, [refuted]. An $\alpha=1$ density/hubness quotient restores invariant fidelity in ≥ 1 non-spectral domain. [geometric-observation/claims/LEDGER.md:67](#).
- NEG-6, [refuted]. Relative per-channel-demeaned error norm is the quotient-tangential quantity that controls softmax-KL. [geometric-observation/claims/LEDGER.md:99](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 5, 6, 8, 9, 11, 12.

RELATED

[read operator](#) (p. 291); [identity reader](#) (p. 180); [the flip](#) (p. 147); [hubness](#) (p. 176).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.12b.

Ledger rows that cite the entry's records without naming it: GO-1.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 3 section 3.6, chapter 6 section 6.2, chapter 11 section 11.7.

R

random forest

instrument. Entry id random-forest.

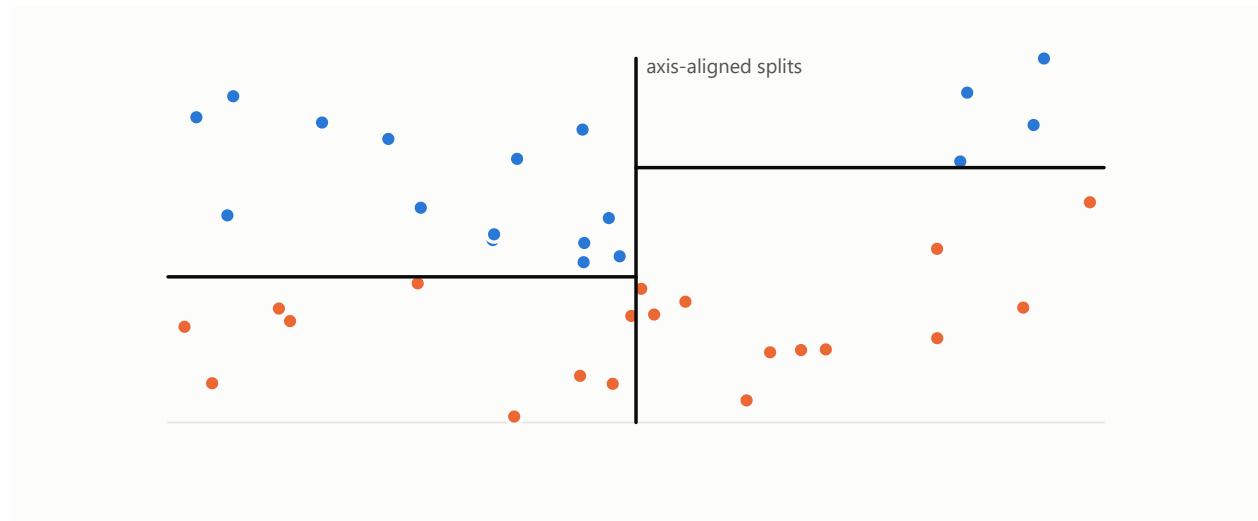


Figure 173: Trees on bootstrap samples with random features at each split, averaged.

DEFINITION

An ensemble of decision trees, each fit on a bootstrap sample with a random subset of features at each split. Chapter 7.

Example. Five hundred trees, each on a bootstrap sample with 5 of 30 features tried per split, vote.

EQUATION

none

CONDITIONS

- An ensemble of decision trees, each fit on a bootstrap sample with a random subset of features at each split, that averages or votes. The average lies between its members and its squared error is at most their mean squared error, and each tree is flat within a leaf so the forest's importances count splits.
- On the five-dataset benchmark the forest and the boosted trees win when the boundary needs many features and lose to a formula when it needs few.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Breiman, random forests, 2001, as chapter 7 section 7.1 of *Data Mining as Observation* reads it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.1	bagging variance, out-of-bag estimation, random forests, importances, AdaBoost	Hastie, Tibshirani, Friedman, ESL 2e chapters 15 and 10.1; TSK 2e 4.10

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Ensemble.lean](#), theorems `average_between`, `sq_average_le`, `average_const`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/DecisionTree.lean](#), theorems `stump_flat_left`, `stump_flat_right`, `split_reads_one_axis`, `gini_le_half`, `gini_eq_zero_iff`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 6, 7.

RELATED

[ensemble](#) (p. 129); [decision tree](#) (p. 102); [bagging](#) (p. 35); [importance](#) (p. 181); [boosting](#) (p. 48).

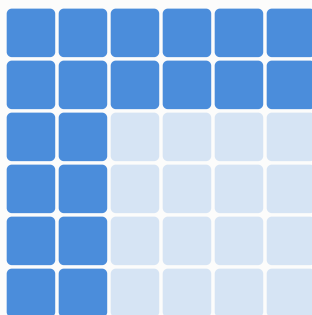
SEE ALSO

Book equations stated beside the entry's terms, not defining it: 6.1, 14.5.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.3, chapter 7 section 7.3.

rank

concept. Entry id rank.



rank two

Figure 174: The dimension of the column space.

DEFINITION

The dimension of a matrix's column space, at most the number of rows and of columns. An outer product has rank at most one, and a product has rank at most either factor's. Chapter 0 section 0.2.

Example. The matrix with rows (1, 2) and (2, 4) has rank 1, and an outer product always does.

EQUATION

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

CONDITIONS

- The dimension of a matrix's column space. It is at most the number of rows and of columns, an outer product has rank at most one, a product has rank at most the rank of either factor, and transposing does not change it.
- The read operator of a linear classifier has rank one, the composed read operator of a chain has rank at most either stage's, and the effective rank of a spectrum is the continuous version chapter 4 allocates bits by.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. OT-7 [demonstrated]. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and ... [geometric-observation/claims/LEDGER.md:30](#).

FIRST STATED

Chapter 0 section 0.2 and chapter 2 section 2.2 of *Data Mining as Observation*, with the rank bound of the chain rule in [geometric-observation/chapters/ch06_mathematical_preliminaries.md:10-27](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48 ; geometric-observation/chapters/ch06_mathematical_preliminaries.md:10-27
chapter 12 section 12.2	pullback composition and the rank bound	geometric-observation/chapters/ch06_mathematical_preliminaries.md:10-27 ; chapter 2 of this book

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Rank.lean](#), theorems `rank_le_width`, `rank_le_height`, `rank_outer_le_one`, `rank_mul_le`, `rank_zero`, `rank_transpose`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Pipeline.lean](#), theorems `quotient_inherited`, `quotient_inherited_chain`, `rank_comp_le_first`, `rank_comp_le_second`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[effective rank](#) (p. 123); [rank certificate](#) (p. 285); [read subspace](#) (p. 293); [Jacobian](#) (p. 192); [outer product](#) (p. 248).

SEE ALSO

Ledger rows that cite the entry's records without naming it: GO-1.

rank certificate

instrument. Entry id `rank-certificate`.

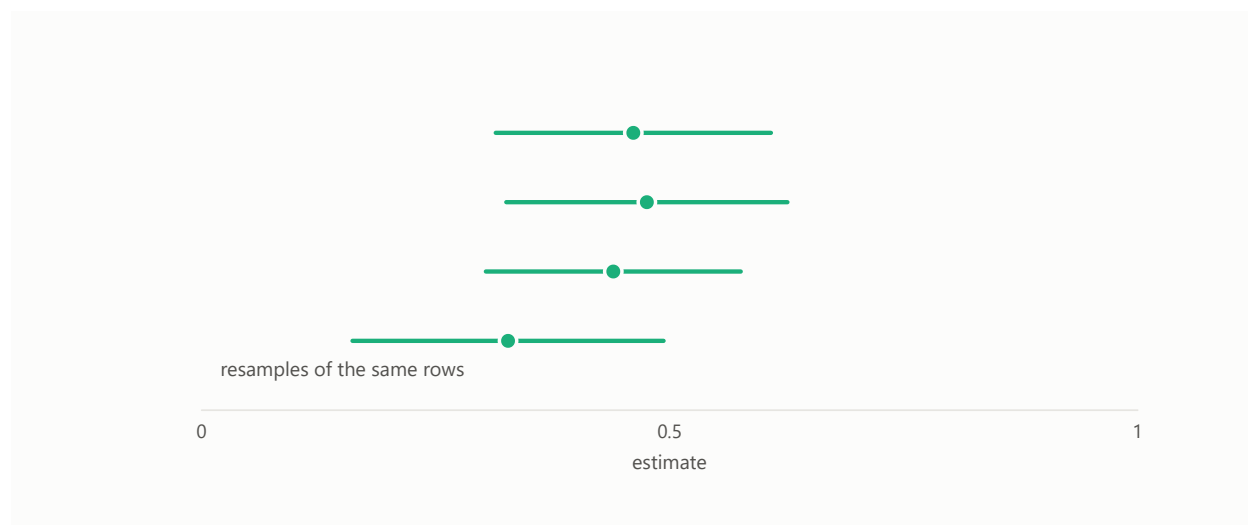


Figure 175: A bound on which neighbour rankings the compression preserved.

DEFINITION

A bound, computed from a compressed representation, that certifies which neighbour rankings the compression preserved for a consumer and which it did not. In its strict setting the floor is a guarantee, in its percentile setting an estimate. A vacuous result supplies no guarantee about the corpus, and falling back to exact reranking is the program's policy for it rather than a consequence. Chapter 11.

Example. A ratio spread κ of 1.0148 with μ of 0.0664 certifies Kendall tau at least 0.8671 across 19900 pairs.

EQUATION

none

CONDITIONS

- In the strict setting, the zeroth and hundredth percentiles of the distance ratio, the floor is a guarantee for the anchor sample, broken by a single collapsed pair.
- In the percentile setting the implementation computes the distortion on the retained central pairs and the concentration over all pairs, and the floor is a robust estimate, not a guarantee on either set.
- The floor is about the global ordering of the sampled pairwise distances. It does not directly certify a query's top-k recall, one query's neighbour order, or the trimmed tail.
- A floor on a sample of anchors is a statement about that sample. Carrying it to the corpus needs a sampling argument the certificate records the inputs for and does not supply.
- A vacuous certificate supplies no guarantee about the corpus. Reranking exactly is the policy the program adopts when the certificate is vacuous, and not something the certificate proves necessary.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

The compression program's certificate specification, `turboquant-pro/docs/CERTIFICATE_SPEC.md`, DOI 10.5281/zenodo.20660087, and chapter 11 section 11.3 of *Data Mining as Observation*.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- [turboquant-pro/turboquant_pro/rank_certificate.py:25-36](#) at 856c4cb. * **Strict** (`lo=0`, `hi=100`): kappa is the true worst-case distortion (max/min per-pair ratio). The bound is then unconditional – a genuine distribution-free floor over *all* pairs – but a single collapsed or near-duplicate pair (ratio $\rightarrow 0$ or huge) sends kappa $\rightarrow \infty$ and makes the certificate vacuous, so it is brittle to data-artifact pairs. * **Robust** (the `lo=2.5`, `hi=97.5` default): kappa is the percentile-robust distortion, trimming the most-distorted ~5% of pairs. This is the sensible default and matches the source paper's torus protocol, but the resulting floor is **conditional**: it holds for the central 95% of pairs, *not* unconditionally over all of them. The trimmed tail can invert arbitrarily, so the reported floor is a robust estimate, not a hard worst-case guarantee. Use the strict regime when you need the unconditional bound.

MACHINE CHECKED

[lean/DataMiningAsObservation/RankCertificate.lean](#), theorems `nn_preserved`, `nn_preserved_of_kappa_one`, `kappa_ge_one`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 11.

RELATED

[certificate](#) (p. 59); [vacuity threshold](#) (p. 371); [the flip](#) (p. 147); [hubness](#) (p. 176).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.2, 11.3.

Ledger rows that cite the entry's records without naming it: GO-3.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.5, chapter 8 section 8.9, chapter 9 section 9.3, chapter 11 section 11.2, chapter 11 section 11.3, chapter 11 section 11.5.

rank-faithful

concept. Entry id `rank-faithful`.

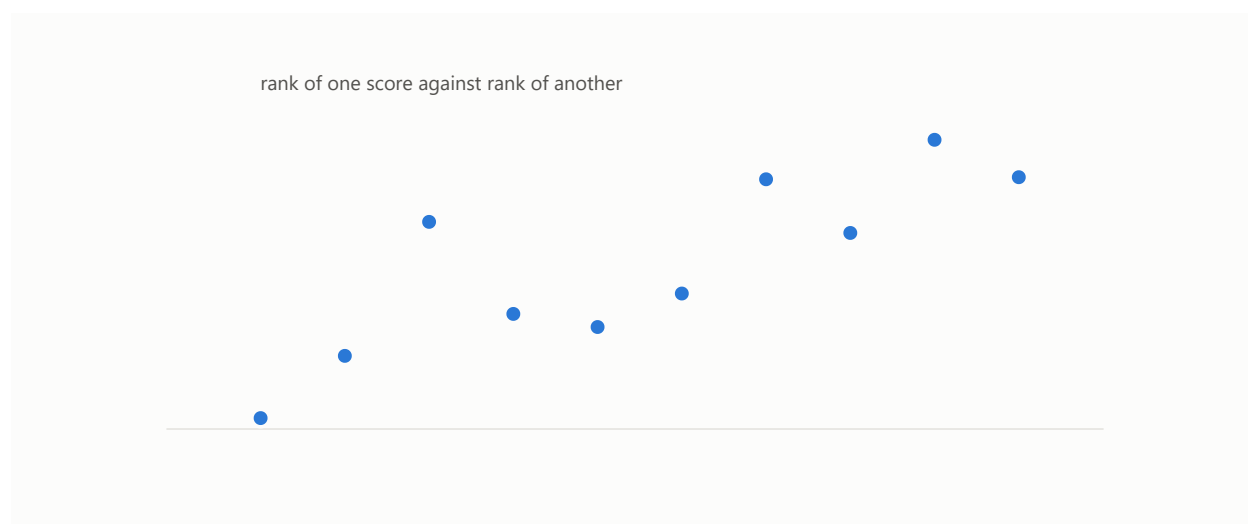


Figure 176: Compressed distances that keep the exact ranking.

DEFINITION

Of a map, preserving the ordering of distances without any bound on the stretch. Chapter 0 section 0.10.

Example. A compression under which every one of the 19900 pair rankings is kept is rank-faithful.

EQUATION

Book equation 0.21.

$$\frac{1}{K} d(x, y) \leq d'(f(x), f(y)) \leq K d(x, y) \quad \text{for all } x, y.$$

CONDITIONS

- A map is rank-faithful when it preserves the ordering of distances with no bound on the stretch. Every strictly increasing transform of the distance is rank-faithful, so a reader that ranks, the nearest-neighbour reader among them, returns the same answer under it.
- Rank-faithful and bi-Lipschitz are separate words. The square preserves every ordering and stretches large distances beyond any fixed factor, and the substrate table measures rank agreement, not stretch.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 9, `geometric-observation/chapters/ch09_legibility.md:10-41`, DOI 10.5281/zenodo.21776291, and chapter 0 section 0.10 and chapter 3 section 3.3 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	row normalization is the projection onto the read subspace of the geodesic-rank consumer	geometric-observation/chapters/ch09_legibility.md:31-41 ; geometric-observation/chapters/ch03_historical_precursors.md:98-110
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation/chapters/ch09_legibility.md:10-41 ; the-angular-observer/theorem.md:110-146

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/RankFaithful.lean](#), theorems `isRankFaithful_of_strictMono`, `closer_set_eq`, `near-est_eq`, `sq_rankFaithful_on_nonneg`, `sq_not_bi_lipschitz`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3.

RELATED

[bi-Lipschitz](#) (p. 42); [rank certificate](#) (p. 285); [recognizer](#) (p. 296); [quotient](#) (p. 279).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.2.

Ledger rows that cite the entry's records without naming it: NEG-1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.2, chapter 3 section 3.3, chapter 9 section 9.2, chapter 11 section 11.1.

read direction

concept. Entry id `read-direction`.

DEFINITION

An eigenvector of the read operator with a nonzero eigenvalue, a direction the consumer is sensitive to. Chapter 2 section 2.2.

Example. For the consumer $3x_1 + 4x_2$ the read direction is $(3, 4)$ over 5.

EQUATION

none

CONDITIONS

- An eigenvector of the read operator with a nonzero eigenvalue, a direction the consumer is sensitive to. An affine consumer has exactly one, its weight vector, the projection onto a read direction is idempotent, and its residual is orthogonal to it.
- The code that puts the whole error on the read direction is the worst, the control that destroys the read direction is confirmed worst, and no code of a given total error can be read as more than the total.

Conditions are curated in `entries.toml` rather than read from a record.

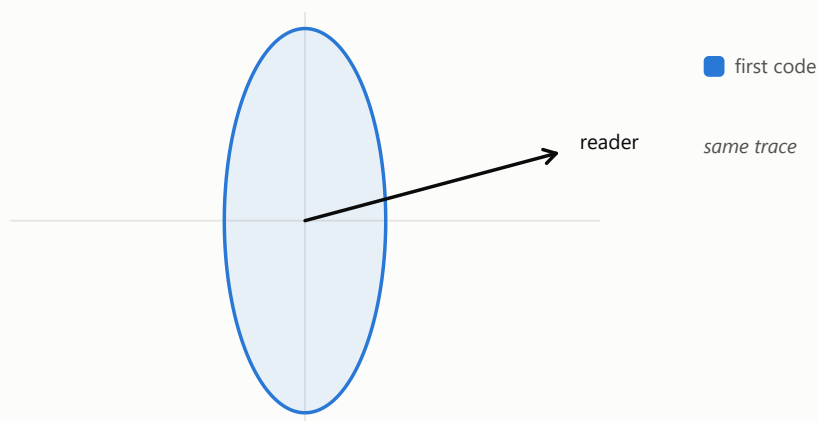


Figure 177: A direction the consumer is sensitive to.

LEDGER

none

FIRST STATED

Chapter 2 section 2.2 and chapter 4 section 4.3 of *Data Mining as Observation*, with the control that destroys the read direction in `geometric-observation/chapters/ch08_value.md:40-70`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Projection.lean](#), theorems `proj_proj`, `dot_sub_proj`, `proj_sq_le`, `proj_add_orth`, `proj_add`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2, 3, 4, 6, 8, 11, 12.

RELATED

[read subspace](#) (p. 293); [read operator](#) (p. 291); [sensitivity](#) (p. 328); [nuisance](#) (p. 240); [projection](#) (p. 272).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.9, 6.1, 4.2.

Ledger rows that cite the entry's records without naming it: GO-1, GO-2 (pos. half: consumer-projected covariance *controls*).

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 4 section 4.3, chapter 6 section 6.2, chapter 11 section 11.7.

read distortion

concept. Entry id read-distortion.

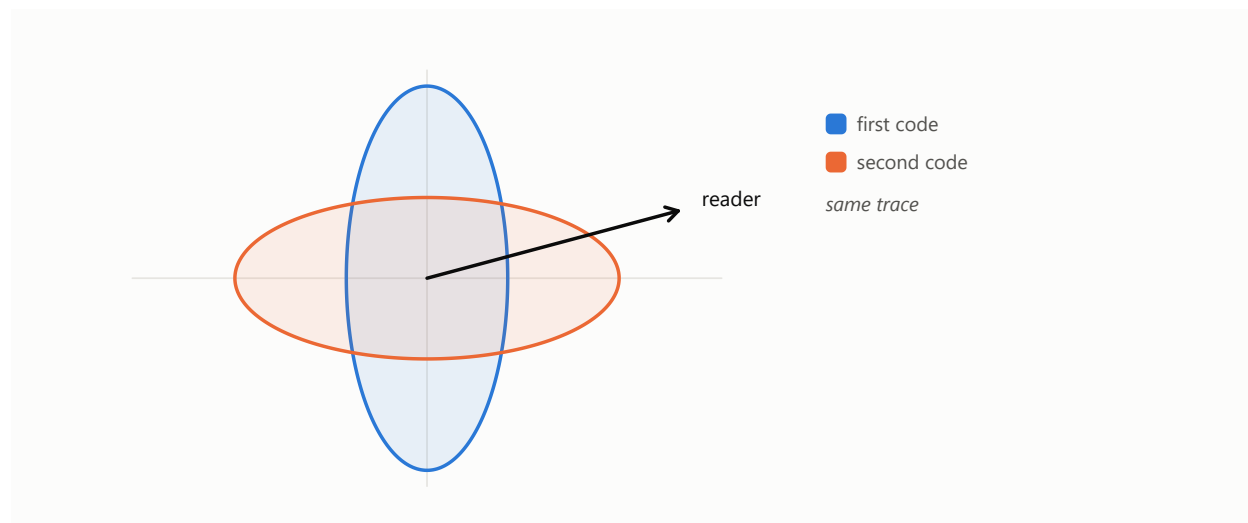


Figure 178: The error a code costs a reader, weighted by what it reads.

DEFINITION

The error as a consumer experiences it, the trace of the read operator times the error's second-moment matrix, which is its covariance when the error is centered. For the identity reader it is mean squared error. Equation 0.10.

Example. A reader with operator $\text{diag}(1, 0)$ charges a code with errors $(0.3, 1.7)$ only 0.3.

Known as, or related to prior art. The consumer-weighted squared error. Task-based quantization studies the same objective for a known task.

EQUATION

none

CONDITIONS

- Exact, with no centering, for a fixed read operator, which is the affine case. For a nonlinear consumer whose error depends on the row it is a first-order factorized surrogate, exact when the local operator and the error's outer product are uncorrelated across rows.
- The second-moment matrix of the error, not its covariance, unless the error is centered by construction.
- The trace of the second moment is the mean squared vector norm, which is the dimension times the per-coordinate mean squared error.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 5, `geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:49-75`, DOI 10.5281/zenodo.21776291, and chapter 2 of the same volume for the failure of observer-free measurement.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:49-75

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadDistortion.lean](#), theorems `read_distortion`, `identity_reader`, `quad_one`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 2, 4.

RELATED

[the flip](#) (p. 147); [alignment](#) (p. 15); [identity reader](#) (p. 180); [quotient](#) (p. 279).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.10, 1.2, 4.1.

Ledger rows that cite the entry's records without naming it: GO-2 (pos. half: consumer-projected covariance *controls*).

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.2, chapter 4 section 4.3.

read operator

concept. Entry id read-operator.

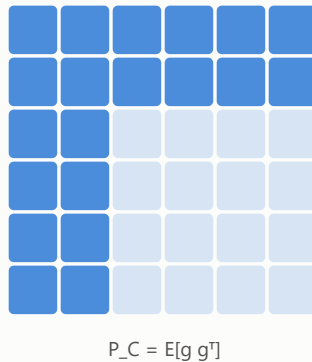


Figure 179: The averaged outer product of the sensitivity with itself.

DEFINITION

The average outer product of a consumer's sensitivity over a dataset, an average of local linearizations whose kernel is the set of directions unread at almost every row of the workload. Its range is the read subspace and its

kernel is the nuisance. Equations 0.9 and 0.11.

Example. The affine consumer $3x_1 + 4x_2$ has read operator with rows (9, 12) and (12, 16) at every row.

Known as, or related to prior art. In the scalar Euclidean-output case the read operator is the active-subspace matrix of Constantine and Gleich, the covariance of the gradient. What the program adds is the pullback through the output metric, the observer triple, the budget, and the audit discipline.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

CONDITIONS

- Each local operator is positive semidefinite, so the kernel of the workload average is the intersection of the local kernels up to sets of rows of measure zero, and an average-null direction is locally unread at almost every row of that workload.
- The average depends on the workload it was taken over and is written with that workload where the difference matters.
- Its off-diagonal entries are co-sensitivities, how two features' sensitivities vary together across rows, and not interactions, which are cross-partials and live in the Hessian.
- The metric in the vector-output form is the local geometry of the output metric where that metric has a local quadratic representation. A dataset-level loss has none, and the operator is then taken on the score with the identity geometry.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* GO-EC-3 [predicted]. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe ... [geometric-observation/claims/LEDGER.md:162](#).

FIRST STATED

Volume 14, chapters 4 and 5, [geometric-observation/chapters/ch04_the_observer_triple.md:9-60](#) and [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48](#), DOI 10.5281/zenodo.21776291.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.2	the observer triple, read subspace, nuisance, same read operator means same read geometry	geometric-observation/chapters/ch04_the_observer_triple.md:9-60 ; geometric-observation/OBSERVATION.md:1-10

FAILURES AND CORRECTIONS

- [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:42-47](#) at 787a933. **Operating-point dependence.** P_C is a *local* object — it depends on x_0 through J . For a linear consumer it is global; for a nonlinear one it varies over X , and the honest version of every result carries P_C as a field, not a constant. The blind probe of Chapter 10 recovers P_C at an operating point precisely because it is local; averaging it over a data distribution gives the $\bar{P}_C$ that enters the alignment law κ (Chapter 12).

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 7, 8, 11, 12, 13, 14.

RELATED

[read distortion](#) (p. 290); [quotient](#) (p. 279); [identity reader](#) (p. 180); [blind probe](#) (p. 45); [the flip](#) (p. 147).

SEE ALSO

Ledger rows that cite the entry's records without naming it: OT-7, GO-1.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 2 section 2.3, chapter 6 section 6.1, chapter 6 section 6.2, chapter 11 section 11.7.

read subspace

concept. Entry id `read-subspace`.

DEFINITION

The directions of the data a consumer can distinguish, spanned by the eigenvectors of its read operator with nonzero eigenvalue. Chapter 0 section 0.5.

Example. The consumer $3 \times 1 + 4 \times 2$ has a one-dimensional read subspace spanned by $(3, 4)$.

Known as, or related to prior art. The active subspace of Constantine and Gleich when the output is scalar and Euclidean.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 6.1.

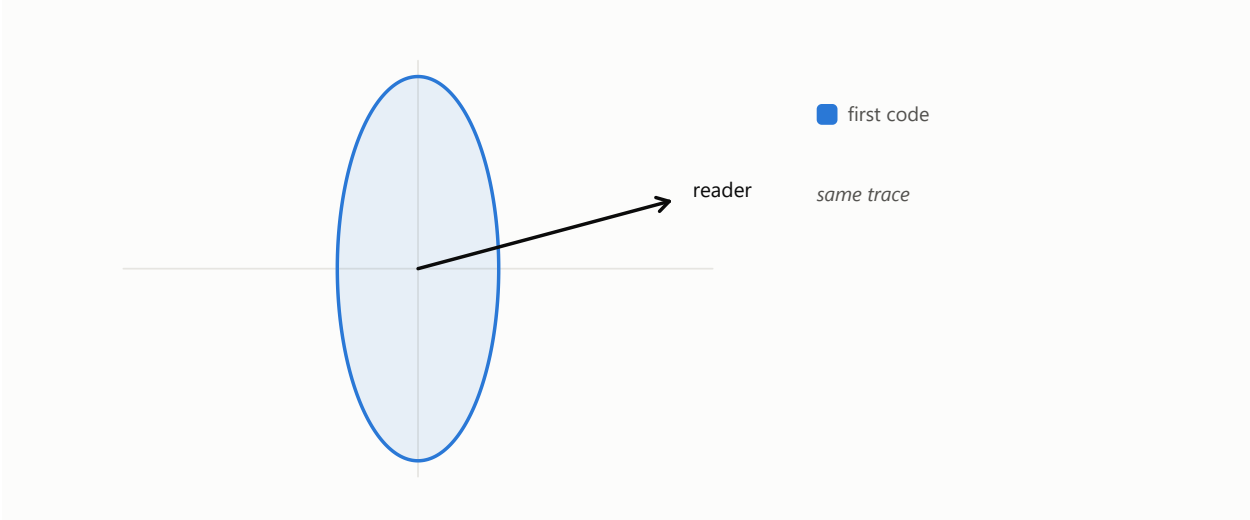


Figure 180: The directions the consumer reads.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

CONDITIONS

- The read subspace is spanned by the eigenvectors of the read operator with nonzero eigenvalue, and it is small. A linear classifier reads one direction, an attention head a handful, and the input has hundreds or thousands.
- Pointwise, a scalar-output consumer reads one direction at each row, and the averaged operator can still have full rank, as the consumer that reads the squared length of a row shows.
- Recovering it from a black box costs two consumer calls per input dimension per row and is a cliff at the full dimension.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 5, [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48](#), DOI [10.5281/zenodo.21776291](#), where the read subspace is the range of the read operator.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48 ; geometric-observation/chapters/ch06_mathematical_preliminaries.md:10-27

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 9, 10, 11, 12, 13.

RELATED

[nuisance](#) (p. 240); [read operator](#) (p. 291); [blind probe](#) (p. 45); [budget cliff](#) (p. 52).

SEE ALSO

Ledger rows that cite the entry's records without naming it: OT-3, NEG-12, GO-B-Llama, GO-B-Llama-rematch, GO-B-Llama-rematch.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 6 section 6.2, chapter 11 section 11.7.

recall at k

concept. Entry id `recall-at-k`.

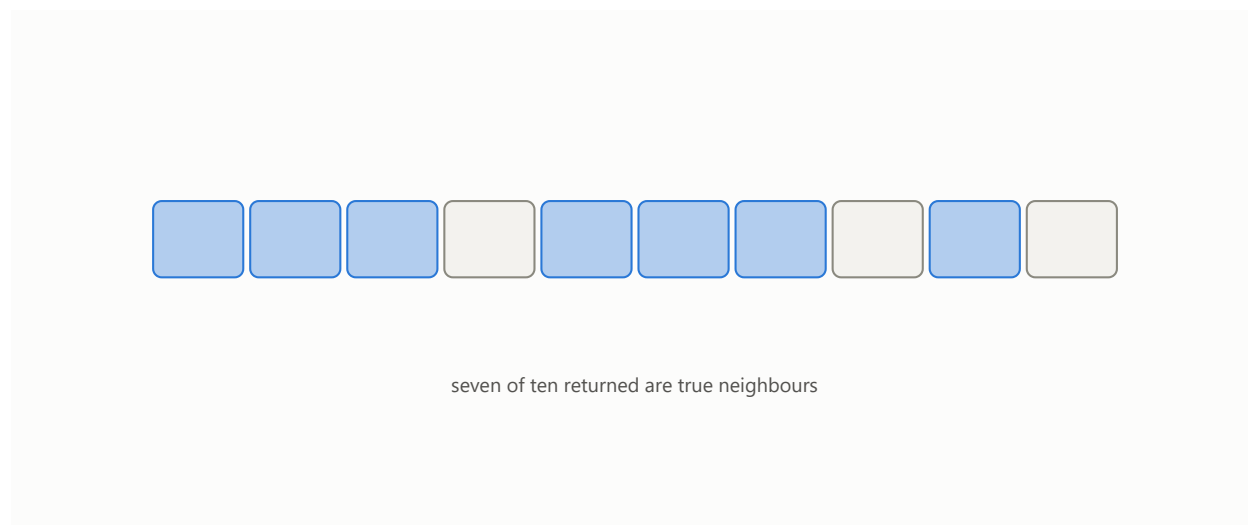


Figure 181: Of the k true neighbours, how many the k returned contain.

DEFINITION

The fraction of a query's true k nearest neighbours that an index returned. Equation 11.3.

Example. Returning 7 of a query's 10 true neighbours is recall at 10 of 0.7.

EQUATION

Book equation 11.3.

$$\text{recall}@k = \frac{|\text{returned top-}k \cap \text{true top-}k|}{k},$$

true top- k computed from the uncompressed vectors.

CONDITIONS

- The fraction of a query's true k nearest neighbours, computed from the uncompressed vectors, that the index returned. It lies in the unit interval and is one exactly when the returned list is the true list.
- The aggregate over queries is a weighted mean over strata, so a stratum that fails entirely moves it by no more than its weight. That is why the anti-hub stratum is reported on its own, and why aggregate recall is never an acceptance metric on its own.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 11 section 11.3 of *Data Mining as Observation*, with the program's stratified form in `turboquant-pro`, `turboquant-pro/docs/HUBNESS_PRIMER.md:86-131`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/RecallAtK.lean](#), theorems `recallAtK_mem_unit`, `recallAtK_eq_one_iff`, `aggregate_le_of_failing`, `aggregate_example`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 10, 11, 12.

RELATED

[anti-hub](#) (p. 20); [min-over-strata](#) (p. 226); [rank certificate](#) (p. 285); [hubness](#) (p. 176).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 10.7.

Ledger rows that cite the entry's records without naming it: NEG-14, GO-B-Llama, GO-B-Llama-rematch.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.5, chapter 10 section 10.3, chapter 11 section 11.6.

recognizer

instrument. Entry id `recognizer`.

DEFINITION

The instrument that decides whether a clustering certificate is right, wrong, or vacuous, by naming the manifold from the eigenvalue multiplets or certifying that none is present. Chapter 9.

Example. The ratios of the lowest eight eigenvalues matched the sphere template, and a shape matching no template was refused.



Figure 182: The spectrum read against a finite list of templates, and a refusal when none matches.

EQUATION

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

CONDITIONS

- The recognizer names a manifold from the low eigenvalue multiplets of a neighbourhood graph’s Laplacian by template match, or certifies that none is present, and it can only name shapes in its template set.
- It was held to a preregistered held-out battery, and the vacuity threshold below which it certifies nothing is derived from the problem rather than tuned.
- The limitation applies to every spectral reading in the book. A finite graph spectrum names a shape only against a finite list of templates, and the manifold, multiplet, and Weyl’s law entries carry the same scope.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 11, `geometric-observation/chapters/ch11_the_recognizer.md:1-95`, DOI 10.5281/zenodo.21776291, and the-angular-observer’s recognizer battery, `the-angular-observer/PREREG_RECOGNIZER_BATTERY.md:1-80`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 9 section 9.2	the recognizer’s mechanism, low multiplets and angular distances, dimension before shape by Weyl’s law, refusal, the growth gotcha	geometric-observation/chapters/ch11_the_recognizer.md:1-95

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Recognizer.lean](#), theorems `templateDist_self`, `templateDist_comm`, `templateDist_scale`, `templateDist_nonneg`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 9.

RELATED

[certificate](#) (p. 59); [vacuity threshold](#) (p. 371); [hubness](#) (p. 176).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 9.3, 0.32.

Ledger rows that cite the entry's records without naming it: GO-3.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.3, chapter 9 section 9.1, chapter 9 section 9.2, chapter 9 section 9.3, chapter 11 section 11.1.

reconstruction error

concept. Entry id `reconstruction-error`.

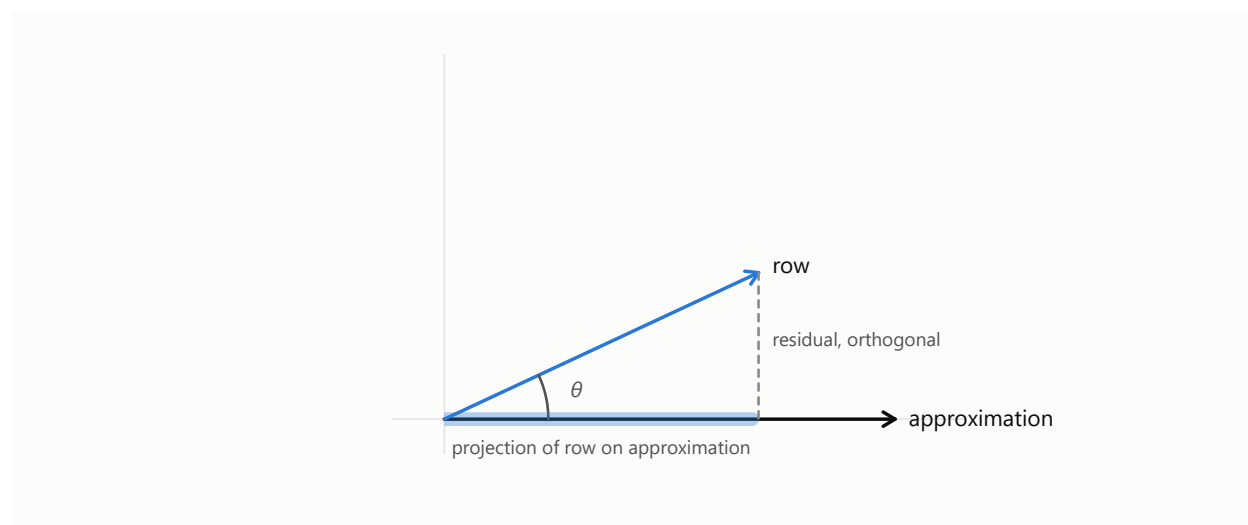


Figure 183: The squared length of the difference between a row and its approximation.

DEFINITION

The squared length of the difference between a row and its approximation, the identity reader's distortion. Chapter 1 section 1.4.

Example. Approximating $(3, 4)$ by $(3, 0)$ costs reconstruction error 16.

EQUATION

none

CONDITIONS

- The squared length of the difference between a row and its approximation, the identity reader's distortion. It is nonnegative, zero exactly when the approximation is the row, a sum of per-coordinate squared errors, and for a projection onto a unit direction it is the squared length less the squared component along it.
- A reconstruction cosine of 0.995 sat beside a perplexity of ten thousand, recalibration improved reconstruction and worsened the consumer, and at matched reconstruction the consumer-aware code won in twelve of twelve domains.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-2 (neg. half: not reconstruction) [demonstrated]. At matched bits, downstream preservation is not controlled by reconstruction error. [geometric-observation/claims/LEDGER.md:63](#).
- *measures*. GO-6 [demonstrated]. At matched rate, output coding $\leq$ surrogate $\leq$ reconstruction on the consumer metric; the output–reconstruction gap is governed by the ker P_C entropy share, and the surrogate–output gap vanishes as rate grows. [geometric-observation/claims/LEDGER.md:68](#).

FIRST STATED

Chapter 1 section 1.4 and chapter 4 section 4.1 of *Data Mining as Observation*, with the identity slice in `geometric-observation/chapters/ch02_failure_of_observer_free_measurement.md:1-40`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	geometric-observation/claims/LEDGER.md row GO-6; geometric-observation/chapters/ch07_cost.md

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReconstructionError.lean](#), theorems `recon_nonneg`, `recon_eq_zero_iff`, `recon_sum`, `recon_proj`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/ReadDistortion.lean](#), theorems `read_distortion`, `identity_reader`, `quad_one`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 4, 6, 7, 8, 11, 12, 13.

RELATED

[identity reader](#) (p. 180); [read distortion](#) (p. 290); [distortion](#) (p. 116); [principal component analysis](#) (p. 269).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 4.2, 1.1, 0.5.

Ledger rows that cite the entry's records without naming it: NEG-4.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 3 section 3.2.

refresh floor

correction. Entry id `refresh-floor`.

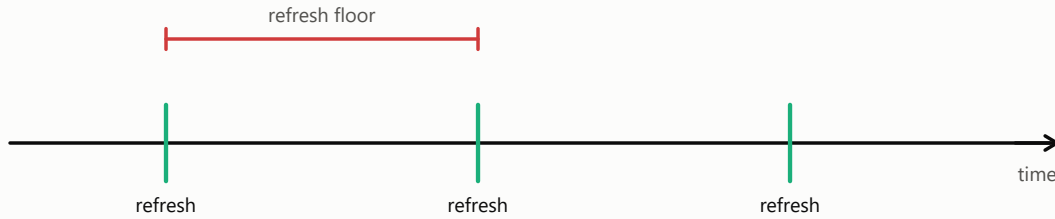


Figure 184: The shortest renewal that keeps a certificate within its error.

DEFINITION

A claimed lower bound on how often a certificate must be renewed. The proportional-to-coherence-time version was refuted and replaced by a sealed horizon. Chapter 13.

Example. At 10 hertz Doppler the sealed refresh floor was 4 slots of one millisecond.

EQUATION

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz, } 1 \text{ TTI at } \geq 50 \text{ Hz.}$

CONDITIONS

- The proportional law, a floor of about 0.177 times the coherence time, was fit at a relaxed threshold against a fresh baseline that never met the budget. It is refuted, the record is kept, and it is not cited.
- The sealed replacement calibrated the baseline to the budget first and found floors of a few transmission intervals at 10 hertz and one at 50 hertz and above, with the optimal linear predictor unable to beat the one-coherence-time wall.
- Volume 14's book outline still cites the refuted law, an open item.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects.* OT-4 [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [geometric-observation/claims/LEDGER.md:36](#).
- *refutes or corrects.* OT-11 [void]. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [geometric-observation/claims/LEDGER.md:48](#).

FIRST STATED

An unsealed exploration in the radio freshness track, `observation-theory-campaigns/analysis/csi/CSI-refreshfloor.json`, refuted and replaced by the sealed sweep `observation-theory-campaigns/analysis/csi/PREREG-XPROTO-CSI-SWEEP2.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.3	BGP 0.351, IS-IS 0.184, OSPF 0.083, second collector	observation-theory-campaigns/analysis/mongo/PREREG-XPROTO-MG.md:42 ; observation-theory-campaigns/experiments/ROUTING-TELEMETRY-TRACK.md:130-150 ; geometric-observation/BOOK-OUTLINE.md:75
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35
chapter 13 section 13.4	refuted fit 0.177 T_{coh} R^2 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	observation-theory-campaigns\analysis\csi\CSI-refreshfloor.json:2,111-114 ; observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-47
chapter 13 section 13.4	sealed correction, 0.5 dB calibration, floors 4/6/4, 2/2/3, 1, slopes 0.1008, 0.1398, 0.1086, R^2 0.7376, 0.9218, 0.6174, bars B1 to B4 and MC1 to MC4 all true, seeds 20260827 to 20260829, sealed 1d3de4a	observation-theory-campaigns/analysis/csi/PREREG-XPROTO-CSI-SWEEP2.md:1-60 ; observation-theory-campaigns/analysis/csi/XPROTO-CSI-SWEEP2-graded.json ; observation-theory-campaigns/experiments/SEALS.md:90
chapter 13 section 13.4	freshness sweep classes, sensing refuted at about 1.6 times	geometric-observation/BOOK-OUTLINE.md:70-90

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [geometric-observation/claims/LEDGER.md:36](#).
- OT-11, [void]. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [geometric-observation/claims/LEDGER.md:48](#).
- [observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-50](#) at f6d1fe2. **The age horizon** ([analysis/csi/](#), XPROTO-CSI-SWEEP2 sealed 2026-08-27, graded PASS): at the calibrated 0.10 budget the largest compliant report period is a few TTI at 10 Hz and 1 TTI at 50 Hz and above, so there is no proportionality law to fit. The earlier [csi-refreshfloor.*](#) claim of a floor $\approx 0.177 \cdot T_{\text{coh}}$ ($R^2=0.915$) was an **unsealed exploration** measured at a relaxed 0.15 threshold against the 0.10 target, with a fresh baseline that never met the budget. It is refuted; the record is kept, not cited. Separately, the optimal linear predictor cannot beat the one-coherence-time wall (Gaussian fading $\rightarrow$ Wiener optimal), and that wall sits well outside the budget horizon. The **RAN governor** ([analysis/ran](#)) is the reference O-RAN rApp core: observe $\rightarrow$ measure false-clear $\rightarrow$ refresh at the floor $\rightarrow$ escalate (diversity / re-route).

MACHINE CHECKED

[lean/DataMiningAsObservation/CoherenceTime.lean](#), theorems `clarke_to_three_decimals`, `examples`, `refuted_law_exceeds_sealed`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 13.

RELATED

[coherence time](#) (p. 67); [certificate](#) (p. 59); [false-clear rate](#) (p. 141).

SEE ALSO

none

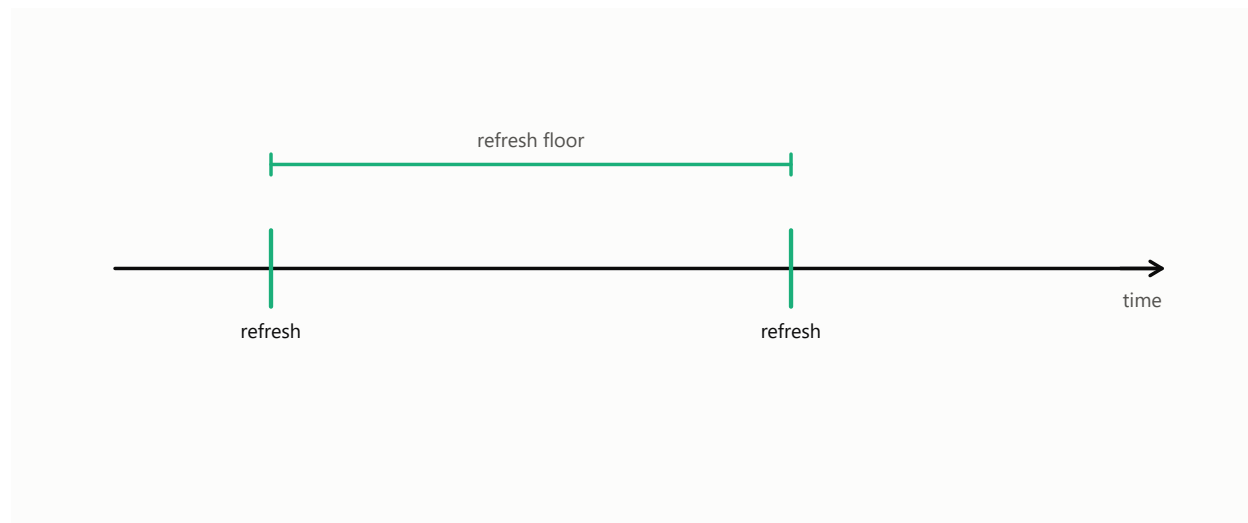
refresh interval*instrument.* Entry id refresh-interval.

Figure 185: The longest renewal period that keeps a certificate within its error.

DEFINITION

The longest renewal period that keeps a certificate within its promised error, a measured property of the system and not a convention. Chapter 13.

Example. A certificate that stays within its error for 40 milliseconds has refresh interval 40 milliseconds.

EQUATION

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz, } 1 \text{ TTI at } \geq 50 \text{ Hz.}$

CONDITIONS

- The longest renewal period that keeps a certificate within its promised error, a measured property of the system and not a convention. A certificate older than the interval is stale for every consumer whose floor is shorter, and the floors are counted in slots that add over consecutive durations.
- There is no proportionality law between the interval and the coherence time. The fit of the floor as 0.177 of the coherence time was refuted and is not to be cited, and the sealed floors were 4, 1, and 1 slots at 10, 50, and 400 hertz.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 13 section 13.2 of *Data Mining as Observation*, with the sealed floors in observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35 and the refuted fit in observation-theory-campaigns/analysis/csi/CSI-refreshfloor.json.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/CoherenceTime.lean](#), theorems `clarke_to_three_decimals`, `examples`, `refuted_law_exceeds_sealed`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/TTI.lean](#), theorems `slots_of_ms`, `slots_add`, `slots_mono`, `slots_coherence`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 13, 14.

RELATED

[refresh floor](#) (p. 299); [coherence time](#) (p. 67); [certificate](#) (p. 59); [false-clear rate](#) (p. 141); [transmission time interval](#) (p. 367).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.25, 0.26.

Ledger rows that cite the entry's records without naming it: OT-4.

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.3, chapter 13 section 13.4.

refusal

concept. Entry id refusal.

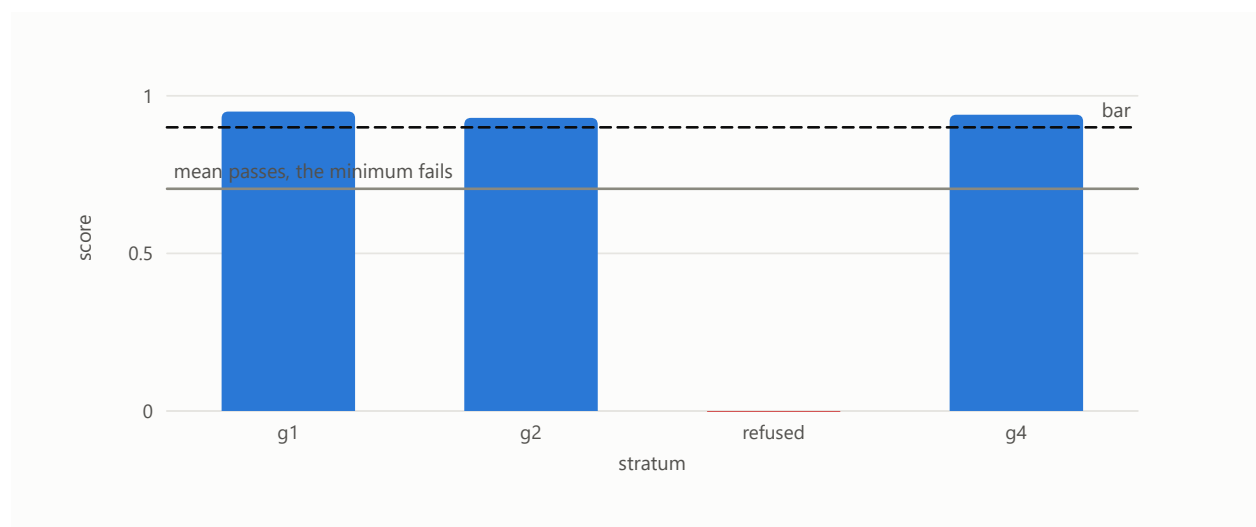


Figure 186: A number the instrument declines to report, counted as a verdict.

DEFINITION

An instrument's declining to report a number it cannot support. A refusal is a verdict, and it is counted. Chapter 2 section 2.4 and chapter 9 section 9.2.

Example. The probe refused to report when 16 samples in 128 dimensions could not resolve rank fifteen.

EQUATION

none

CONDITIONS

- An instrument's declining to report a number it cannot support, when samples do not exceed the dimension, when no template matches, or when a group is too thin. A refusal is a verdict, it is counted, and a mean never hides it.
- The recognizer refuses when the ratios match no stored template, the probe refuses to extrapolate past its fit range, and the strata design refuses rather than warns.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 2 section 2.4 and chapter 9 section 9.2 of *Data Mining as Observation*, with the refusal regimes in `readscope/readscope/regimes.py:1-60`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	readscope/readscope/regimes.py:1-60
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro/docs/STRATA_RFC.md:24-98

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Abstention.lean](#), theorems `abstain_not_passes`, `verdict_eq_none_iff`, `passes_verdict_iff`, `inf_le_of_subset`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2, 8, 9, 10, 11, 14.

RELATED

[abstention](#) (p. 12); [vacuity threshold](#) (p. 371); [recognizer](#) (p. 296); [verdict](#) (p. 375); [coverage](#) (p. 92).

SEE ALSO

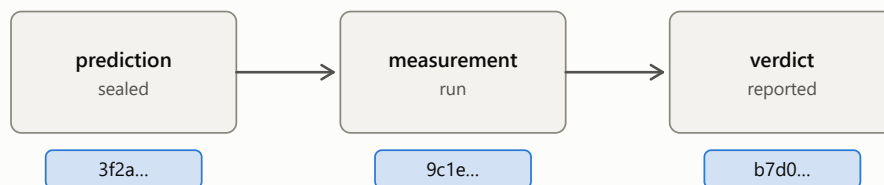
Book equations stated beside the entry's terms, not defining it: 10.7, 14.4, 9.3.

Ledger rows that cite the entry's records without naming it: GO-3, NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.6, chapter 6 section 6.1, chapter 10 section 10.4.

registered

concept. Entry id registered.



each file's digest is recorded before the next step, and a changed digest proves a changed file

Figure 187: Written into a sealed file before the measurement.

DEFINITION

Of a claim, bar, null, or budget, written into a sealed file before the measurement. Chapter 8 section 8.8.

Example. The bar of 0.90 for anti-hub recall was registered before the compressed index was scored.

EQUATION

none

CONDITIONS

- Of a claim, bar, null, or budget, written into a sealed file before the measurement, so that a changed digest would prove a changed prediction. A registered bar discriminates only when the null fails it and the real system passes it.
- A registered miss is kept as a miss, an identifier burned at pilot is not reused, and every number in the book was registered before it was measured or is labelled as not having been.
- Registration proves what was claimed and not when, and its chronology rests on the seal's public receipt, signed tag, or archival deposit.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.8 of *Data Mining as Observation*, with the template in `observation-theory-campaigns/experiments/PREREG-TEMPLATE.md:1-71` and the seal rule in `observation-theory-campaigns/experiments/SEALS.md:1-10`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Seal.lean](#), theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Bar.lean](#), theorems `passes_anti`, `passes_mono`, `discriminates_iff`, `no_bar_of_null_ge`, `exists_bar_of_lt`, `vacuous_of_null_passes`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[preregistration](#) (p. 267); [sealed](#) (p. 324); [bar](#) (p. 39); [ledger class](#) (p. 211); [declaration](#) (p. 103).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.1, 8.3.

Ledger rows that cite the entry's records without naming it: GO-1, GO-3, NEG-11.

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.8, chapter 9 section 9.2.

reliability weight

concept. Entry id reliability-weight.

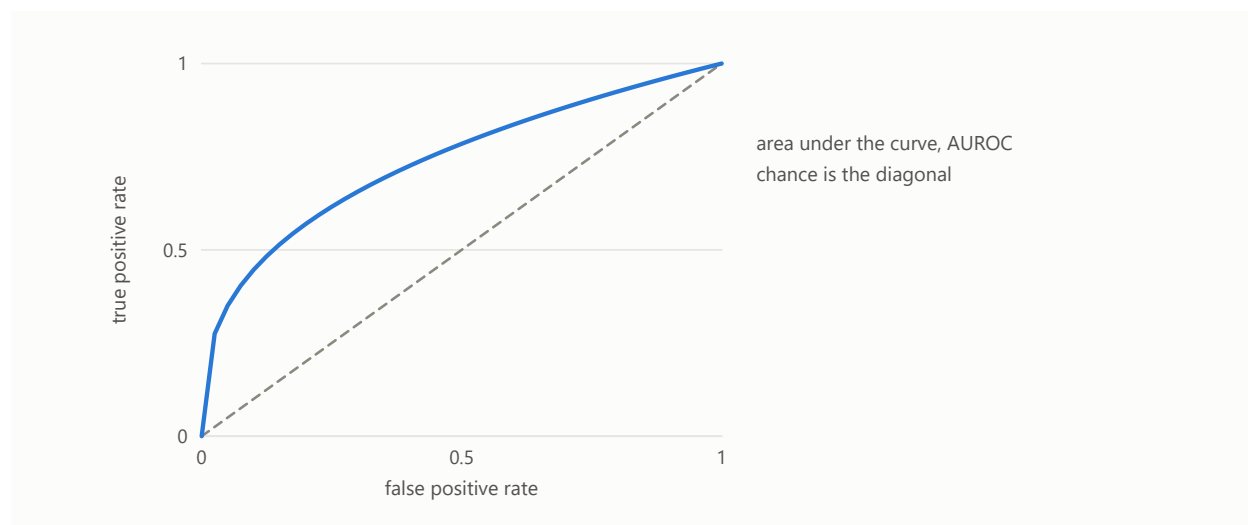


Figure 188: Twice the held-out AUROC minus one, or zero.

DEFINITION

Zero or twice a validated encoder's held-out AUROC minus one, whichever is larger, used as that encoder's authority. It is a discrimination weight, unchanged by increasing transforms of the score and not by decreasing ones, and the calibration evidence is the expected calibration error reported beside it. Equation 0.38.

Example. A held-out AUROC of 0.8035 gives reliability weight 0.607, and an AUROC of 0.45 gives zero.

EQUATION

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

CONDITIONS

- Zero or twice the held-out AUROC minus one, whichever is larger. It is a discrimination weight, unchanged by a strictly increasing transform of the encoder's score and not by a decreasing one, which sends A to 1 minus A and the weight to zero whenever A was above one half.
- The calibration evidence is the expected calibration error reported beside it, and the weight is computed on a held-out split under the cross-corpus gate, so it is one formula from one source.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

The moral-embedding program's calibrated authority, `gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63` and `xbse/README.md:175-195`, and chapter 12 section 12.5 and chapter 14 section 14.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.5	ECE 0.018 to 0.101 vs raw up to 0.223, reliability weight, audit binding	<code>xbse/README.md:175-195</code> ; gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63
chapter 14 section 14.2	reliability weights and calibration errors per axis, the collapsed family's mean weight 0.559 against the general valence channel's own 0.735, the three design rules, 0.048 to 0.049 and 0.089 to 0.101 at 696 pairs	gtc-prototype/docs/CALIBRATED_AUTHORITY.md:1-65

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReliabilityWeight.lean](#), theorems `pair_le_one`, `auROCNum_le`, `auROC_le_one`, `weight_mem_unit`, `weight_eq_zero_of_le_half`, `weight_comp`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 12, 14.

RELATED

[Monotone Invariance Theorem](#) (p. 227); [harness](#) (p. 168); [posited versus measured](#) (p. 263); [certificate](#) (p. 59).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.16.

replica

concept. Entry id `replica`.

DEFINITION

A copy of a database kept on another machine. Its lag is how far it trails the primary. Chapter 13.

Example. A replica 50 writes behind the primary returns the old value for those 50 rows.



Figure 189: A copy of a database on another machine, trailing the primary.

EQUATION

Book equation 0.26.

$$FC = \Pr [W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

CONDITIONS

- A copy of a database kept on another machine, trailing the primary by a lag. The keys stale under a lag grow with the lag and vanish at no lag. A naive certificate that clears every read has coverage one and a false-clear rate equal to the stale fraction, and a witness-backed certificate that clears only keys outside the lag has a false-clear rate of zero at the price of coverage.
- Which reads are stale is consumer-relative, since two readers of the same replica have different coherence times, and the measured false-clear rates on ZooKeeper, Postgres, and MongoDB are the ledger's numbers.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 13 section 13.3 of *Data Mining as Observation*, with the program's substrate measurements in `observation-theory-campaigns/analysis/mongo/PREREG-XPROTO-MG.md` and the database rows of the freshness track.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Replica.lean](#), theorems `stale_mono`, `stale_zero`, `naive_certificate`, `witnessed_certificate`, `witnessed_coverage`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 4, 13.

RELATED

[freshness](#) (p. 156); [certificate](#) (p. 59); [witness](#) (p. 382); [false-clear rate](#) (p. 141); [coverage](#) (p. 92).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 13.1.

Ledger rows that cite the entry's records without naming it: OT-11.

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.3.

rerank

instrument. Entry id rerank.

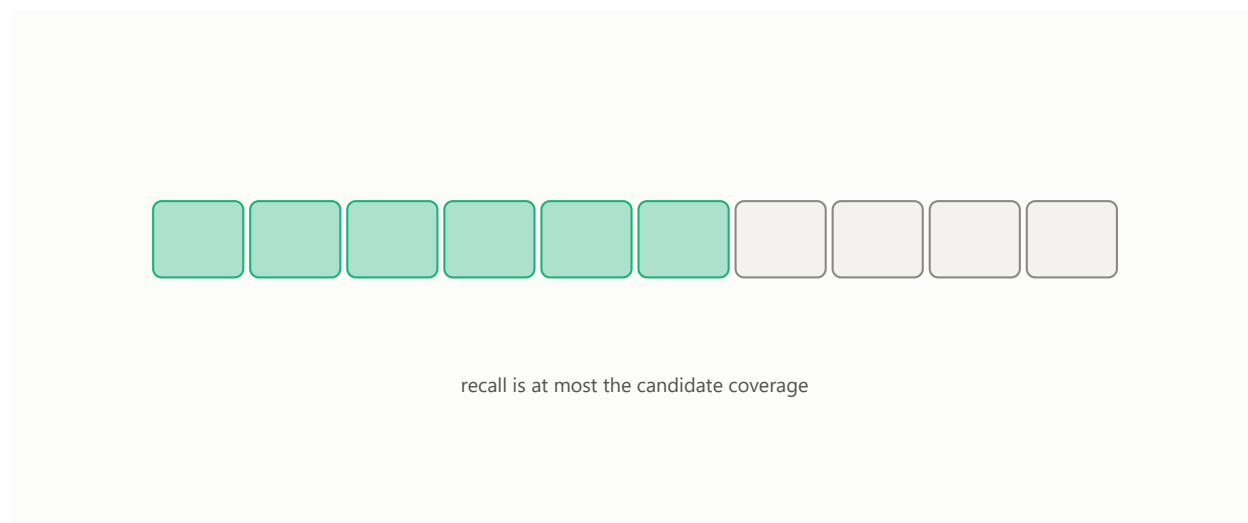


Figure 190: A second scorer reorders the candidate list and cannot add to it.

DEFINITION

Reordering a candidate list with a second scorer. Its recall is at most the candidate coverage. Chapter 10 section 10.5.

Example. Reranking 100 candidates that hold 70 of the true neighbours can reach recall 0.7 and no more.

EQUATION

none

CONDITIONS

- Reordering a candidate list with a second scorer. A rerank cannot return a row the list does not hold, so its recall is at most the candidate coverage, an oracle rerank reaches the coverage exactly, and a deeper list can only raise the coverage.
- Reranking the compressed candidates read 0.6627, identical to the unreranked recall, because candidate coverage was the wall, and the depth curve from 51 to 501 moved coverage from 0.696 to 0.926.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 10 section 10.5 of *Data Mining as Observation*, with the candidate-coverage finding in `turboquant-pro/docs/RESULTS_strata_phase23_gates.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 10 section 10.5	rerank identical 0.6627, candidate coverage, depth curve 51 to 501, coverage 0.696 to 0.926, fidelity 0.663 to 0.702, originals rerank 0.923, 4096 bytes on 148	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:45-110

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Rerank.lean](#), theorems `hits_le_coverage`, `recall_le_coverage`, `oracle_rerank`, `coverage_mono`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 4, 10, 11, 12.

RELATED

[recall at k](#) (p. 295); [anti-hub recall](#) (p. 22); [inverted file](#) (p. 185); [rank certificate](#) (p. 285); [retrieval-augmented pipeline](#) (p. 312).

SEE ALSO

Book equations stated beside the entry’s terms, not defining it: 11.3, 10.7, 12.3.
Ledger rows that cite the entry’s records without naming it: NEG-14, GO-B-Llama, GO-B-Llama-rematch.
Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.5, chapter 10 section 10.4, chapter 10 section 10.5, chapter 11 section 11.4, chapter 12 section 12.4.

residualization

instrument. Entry id residualization.

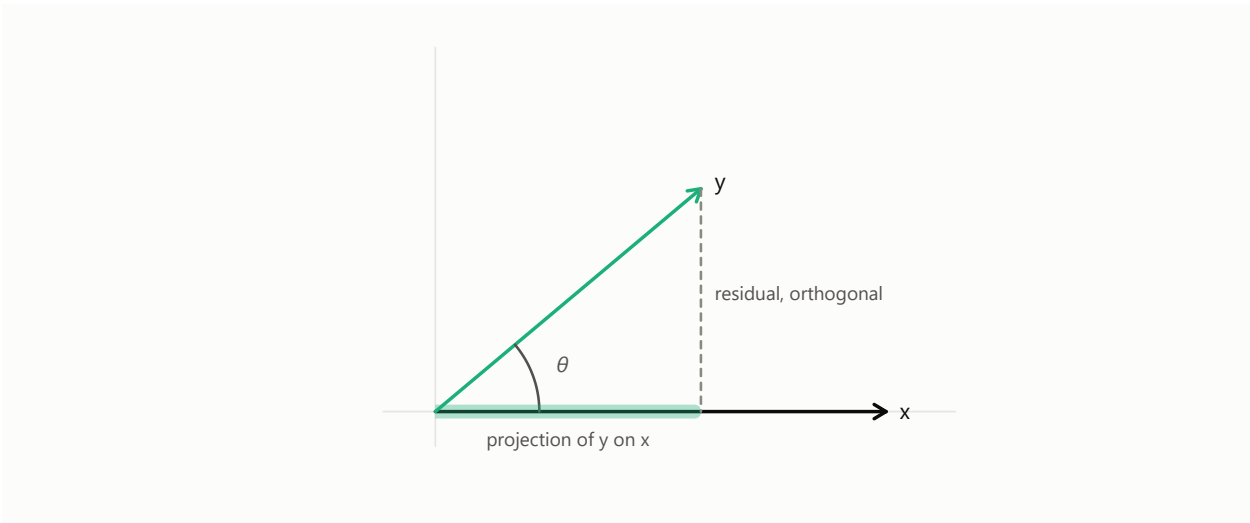


Figure 191: The least-squares multiple of one column removed from another.

DEFINITION

Removing from a column the least-squares multiple of another. The residual is orthogonal to the column removed. Chapter 8 section 8.6 and chapter 12 section 12.5.

Example. Removing the least-squares multiple of (1, 1) from (3, 1) leaves (1, -1), which is orthogonal to (1, 1).

EQUATION

none

CONDITIONS

- Removing from a column the least-squares multiple of another. The residual is orthogonal to the column removed, its squared length is the original less the squared dot product over the removed column's squared length, a column orthogonal to the removed one is unchanged, and a column residualized on itself vanishes.
- Whether named categories survive residualization on the general valence channel is chapter 12's test, and the book-length signal fell from 0.241 to a residual 0.093 at six and a half standard errors.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.6 and chapter 12 section 12.5 of *Data Mining as Observation*, with the bifactor readout in `xbse/docs/BIFACTOR_READOUT.md:14-100`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.6	books 0.241 at 17 sigma, residual 0.093, R2 0.009, z 6.5, p 5.7e-11, 85 percent, fiction 0.131 on 2250	<code>geometric-aesthetics/book/src/chapter-17-empirical-evidence-for-geometric-aesthetics.md:95-106</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

`lean/DataMiningAsObservation/Residual.lean`, theorems `residual_orth`, `residual_sq`, `residual_of_orth`, `residual_self`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 8, 12, 14.

RELATED

[correlation](#) (p. 86); [orthogonal](#) (p. 247); [confound](#) (p. 77); [contraction](#) (p. 80); [projection](#) (p. 272).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.1, 12.3, 14.2.

Ledger rows that cite the entry's records without naming it: GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.6, chapter 12 section 12.5, chapter 14 section 14.8.

retrieval-augmented pipeline

concept. Entry id retrieval-augmented-pipeline.

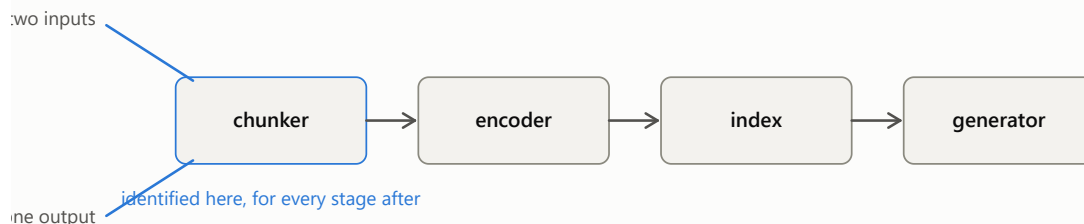


Figure 192: Chunker, encoder, index, and generator, each reading the one before.

DEFINITION

A system that answers a question by chunking documents, embedding the chunks, indexing them, retrieving the nearest to the embedded question, and handing them to a generator. A chain of observers. Chapter 12.

Example. Chunker, encoder, index, generator: a query becomes a vector, the vector a candidate list, and the list with the question an answer.

EQUATION

Book equation 12.1.

$$x \sim_{\text{stage}} x' \implies x \sim_{\text{pipeline}} x' \quad \text{for every stage, differentiable or not,}$$

$$\text{rank } P_{\text{run}}(x) \leq \min_{\text{stages in the run}} \text{rank } J_{\text{stage}}(x) \quad \text{on a differentiable run of stages, at each row.}$$

CONDITIONS

- A system that chunks documents, embeds the chunks, indexes them, retrieves the nearest to the embedded question, and hands them to a generator. Whatever one stage declares the same, every later stage and the pipeline declare the same, and for linear stages the rank of the composition is at most the rank of any stage, so the pipeline's read subspace is no larger than its narrowest stage's.
- Both hold for every stage, differentiable or not. The benchmark score of the whole is a certificate whose false-clear rate on deployment slices is the chapter's measurement.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 12 of *Data Mining as Observation*, with the program's streaming-retrieval staleness row OT-11 and the legal-citation retrieval flip.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Pipeline.lean](#), theorems `quotient_inherited`, `quotient_inherited_chain`, `rank_comp_le_first`, `rank_comp_le_second`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 8, 10, 11, 12, 13, 14.

RELATED

[quotient](#) (p. 279); [read subspace](#) (p. 293); [deployment mismatch](#) (p. 108); [freshness](#) (p. 156).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 13.1.

Ledger rows that cite the entry's records without naming it: OT-11, GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 12 section 12.3, chapter 12 section 12.6.

Robin Hood index

instrument. Entry id `robin-hood-index`.

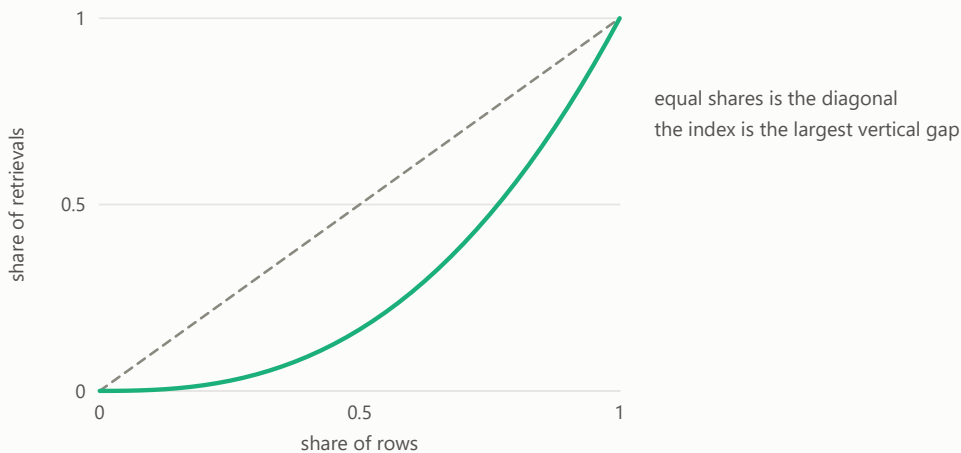


Figure 193: The share of retrievals that would have to move to equalize the rows.

DEFINITION

The fraction of a total count that would have to move from points above the mean to points below it to make every count equal. Equation 0.35.

Example. If 20 percent of the rows take 60 percent of the retrievals, the Robin Hood index is 0.40.

EQUATION

Book equation 0.35.

$$\text{RH} = \sum_i \max\left(0, \frac{N_i}{\sum_j N_j} - \frac{1}{n}\right).$$

CONDITIONS

- The fraction of the total count that would have to move from points above the mean to points below it to make every count equal, the excess above the mean over the total, which is half the total absolute deviation over the total. It is nonnegative, zero exactly at equality, at most one, and unchanged by a common scaling of the counts.
- The strata design reports it per area beside the skew, and the first run's ratio of 1.30 against a bar of 1.5 is the number the abstention rule was written around.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Hoover's index of concentration, applied to neighbour counts in turboquant-pro's strata design, turboquant-pro/docs/STRATA_RFC.md:24-98, and chapter 0 section 0.16 of *Data Mining as Observation*.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/RobinHood.lean](#), theorems `sum_dev`, `excess_eq_half_abs`, `robinHood_nonneg`, `robinHood_eq_zero_iff`, `robinHood_le_one`, `robinHood_scale`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 10.

RELATED

[hubness](#) (p. 176); [anti-hub](#) (p. 20); [Poisson ceiling](#) (p. 262); [abstention](#) (p. 12).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 10.4.

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.4.

ROC curve

instrument. Entry id `roc-curve`.

DEFINITION

The path of true positive rate against false positive rate as the threshold sweeps, whose area is the AUROC. Chapter 5 section 5.4.

Example. A score with true positive rate 0.75 at false positive rate 0.25 traces one point of the curve, and the sweep of thresholds traces the rest.

EQUATION

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$$J \leq 2A - 1 \text{ when the ROC curve is concave, and not in general.}$$



Figure 194: True positive rate against false positive rate as the threshold sweeps.

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

CONDITIONS

- The path of true positive rate against false positive rate as the threshold sweeps. Its area is the AUROC, which lies in the unit interval, is one half for a constant score, is one for a perfect ranking, and is unchanged by any strictly increasing transform of the score.
- The bound from AUROC to the best F1 through the Youden index holds for concave curves and not for every curve, with the counterexample at AUROC 0.75, and the book corrects the source on that point.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 5 section 5.4 of *Data Mining as Observation*, with the Youden ceiling and its correction to concave curves in `theory-radar/paper/astar_paper.tex:94-110`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Auroc.lean](#), theorems `pair_nonneg`, `auroc_nonneg`, `auroc_le_one`, `auroc_perfect`, `auroc_reversed`, `auroc_chance`, `auroc_monotone_invariant`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/YoudenF1.lean](#), theorems `f1_eq`, `f1_le_of_youden`, `auroc_eq`, `youden_eq`, `youden_exceeds_auroc_form`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 5, 6, 7, 8, 11, 12, 13, 14.

RELATED

[AUROC](#) (p. 32); [Youden index](#) (p. 387); [threshold](#) (p. 362); [precision, recall](#) (p. 266); [F1](#) (p. 140).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.28.

Sources-table rows that share a record with the entry without naming it: chapter 5 section 5.4, chapter 6 section 6.3.

rotary position embedding

concept. Entry id rotary-position-embedding.

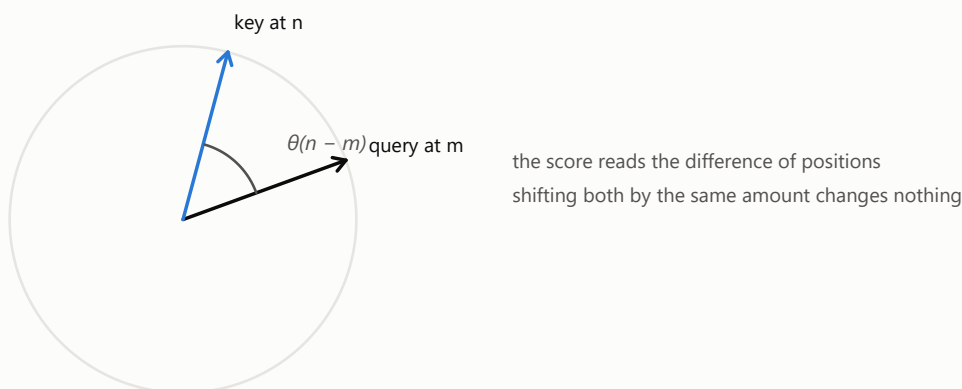


Figure 195: A rotation by position, under which a head reads relative position only.

DEFINITION

A rotation applied to queries and keys that encodes where each token sits in the sequence. Chapter 0 section 0.11.

Example. A query at position 5 and a key at position 3 score exactly as they would at positions 105 and 103.

EQUATION

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

CONDITIONS

- A rotation applied to queries and keys by an angle proportional to position. A rotation preserves length, and the score of a query at one position against a key at another depends only on the difference of positions, so a head reads relative position and shifting both positions by the same amount changes no score.
- A key quantizer that rounds the rotated key rounds its position as well as its content, which is part of why the direction-only quantizer moved the scores.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Su and others, RoFormer, 2021, as chapter 0 section 0.11 of *Data Mining as Observation* states it, with the program's key-side measurements in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro/docs/KV_KEYS_FINDING.md:61-86 ; the-angular-observer/README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Rope.lean](#), theorems `rot_length`, `rot_dot`, `relative_position`, `shift_invariant`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[attention](#) (p. 26); [query, key, value](#) (p. 276); [KV cache](#) (p. 200); [dot product](#) (p. 118).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.1.

Ledger rows that cite the entry's records without naming it: NEG-2.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 2 section 2.5, chapter 3 section 3.2, chapter 8 section 8.2.

ROUGE

concept. Entry id rouge.

DEFINITION

A family of scores for generated text against a reference. Chapter 0 section 0.11.

Example. A generated text that reproduces 6 of a reference's 8 unigrams has ROUGE-1 recall 0.75.

EQUATION

none

CONDITIONS

- A family of scores for generated text against a reference. ROUGE-1 recall counts the reference's words the candidate contains, with multiplicity, over the reference's length. It lies in the unit interval, is one for the reference itself, and is one for any rearrangement of the reference, since it reads bags and not order.

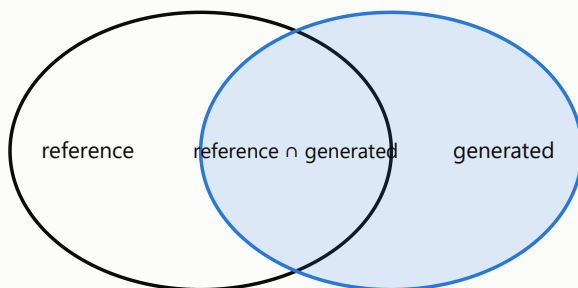


Figure 196: The overlap of the generated text's n-grams with the reference's.

- A ROUGE score therefore certifies less than it seems to. The 13.7 ROUGE-L difference at 512 tokens re-validated at negative 0.31 on forty documents, and the claims ledger carries both numbers.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Lin, ROUGE, 2004, as chapter 0 section 0.11 states it, with the program's 13.7 ROUGE-L number and its re-validation in `turboquant-pro/CLAIMS.md`: 63-81.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.9	13.7 ROUGE-L, 0.25/4.19/9.60/13.7 at 64/128/256/512, re-validation negative 0.31 n=40, 26.64 under symmetric nf4, _quant_nf4a_group unchanged since 289bdfc before 4f7baab	turboquant-pro/CLAIMS.md:63-81 ; turboquant-pro/benchmarks/kvquant_matrix/REVAL-2026-08-08.md

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Rouge.lean](#), theorems `overlap_le`, `rouge1_mem_unit`, `rouge1_self`, `rouge1_perm`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 8.

RELATED

[bag of words](#) (p. 36); [perplexity](#) (p. 258); [harness](#) (p. 168); [certificate](#) (p. 59).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.22.

Ledger rows that cite the entry's records without naming it: NEG-4.

S

safe pruning

concept. Entry id safe-pruning.

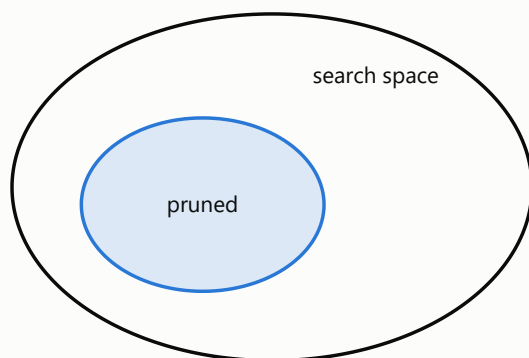


Figure 197: A branch dropped without loss because a bound says nothing in it can win.

DEFINITION

A rule for discarding candidates during a search that is guaranteed to discard no solution. The Apriori principle and the Monotone Invariance Theorem are the book's two instances. Chapter 5.

Example. A node whose Youden ceiling is 0.70 cannot beat the best F1 found so far, 0.80, so its subtree is skipped.

EQUATION

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$J \leq 2A - 1$ when the ROC curve is concave, and not in general.

CONDITIONS

- A pruning rule is safe when the pruned search returns the same answer as the unpruned one on every input. The Apriori principle and the Monotone Invariance Theorem are safe by proof.
- A learned or heuristic rule cannot earn that word from a finite test. It can be empirically lossless on the held-out cases, and the record says which cases.
- The AUROC form of the F1 ceiling is a safe pruning bound only for concave ROC curves. The Youden form is sound for every score at the leaf, and pruning a subtree by its root's ceiling is a heuristic in both forms.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 5 section 5.4 of *Data Mining as Observation*, with the program's two owned instances in theory-radar and arc-equivariant-search.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- [theory-radar/ERRATA.md:3-75](#) at 37c4e6c. 2026-09-03. The AUROC to F1 bound holds only for concave ROC curves **What was claimed.** Theorem “AUROC-F1 Bound” in `paper/astar_paper.tex`, the derivation in `src/symbolic_search/_auroc_proof.py`, and the pruning bounds in `run_astar_v2.py` and `src/symbolic_search/_heuristic_dag.py` all rested on the statement that any ROC curve with area A has maximum Youden index $J = \max(\text{TPR} - \text{FPR})$ at most $2A - 1$. **Why it is wrong.** The statement holds for concave ROC curves (equivalently, scores whose likelihood ratio is monotone), because a concave curve lies above the two chords through its Youden point and so has area at least $(1 + J)/2$. It fails for non-concave curves. Counterexample, found by an external reviewer of the companion textbook: rank three quarters of the positives above every negative and the remaining quarter below every negative. AUROC is 0.75, the maximum Youden index is 0.75, and at prevalence one half the best thresholded F1 is 0.857, above the 0.80 the AUROC form of the ceiling allows. The sound ceiling for that score, from $J = 0.75$, is 0.889, and the score sits under it. The `auroc_f1_bound` formula in `run_astar_v2.py`, $2A\pi/(A\pi + (1 - A)(1 - \pi))$, is not a valid upper bound either: at prevalence 0.1 the same score reaches F1 0.857 against a “bound” of 0.5. **Consequence for reported results.** The searches that used these bounds to prune could have discarded a candidate formula whose ROC curve was not concave and whose true best F1 exceeded the computed ceiling. The formulas and wins reported by those runs are therefore what a pruned search found, and are lower bounds on what an unpruned search would find. The Monotone Invariance Theorem and the results that do not depend on the AUROC bound are unaffected. **Correction.** The sound form of the ceiling substitutes the measured maximum Youden index of the candidate’s own scores for $2A - 1$. It holds for every score, costs the same sort as AUROC, and is what `max_f1_for_youden` and `youden_f1_bound` now compute. The AUROC form is retained under its original names with a docstring stating the concavity condition, and the pruning sites now use the measured index. A rerun of the published searches under the corrected bound was done on 2026-09-03 and 2026-09-04, `rerun_youden_2026_09_03.py`, results in `results/rerun_youden_2026_09_03.json`, run on the Atlas workstation, CPU only. **Rerun, part A, the pre-filter admissibility table (tab:auroc).** Depth-2 pairwise enumeration on the eight datasets, exhaustive against the AUROC pre-filter at the seven published thresholds, with the sound Youden ceiling beside it. The evaluation counts under the pre-filter reproduce the published table (Breast Cancer 8700 pairs, 2576 evaluated at 0.75; Wine 1560, 413; Circles 20, 4; Moons 20, 2). The pre-filter lost no optimum on any dataset at any threshold, which is what the paper reported. That was luck of the data, not the theorem: the retracted ceiling sat below the F1 a pair formula actually reach

MACHINE CHECKED

[lean/DataMiningAsObservation/SafePruning.lean](#), theorems `support_anti`, `apriori`, `subset_of_frequent`, at observation-data-mining 95af30c; what the check covers is stated in the book’s [appendix C](#).

[lean/DataMiningAsObservation/YoudenF1.lean](#), theorems `f1_eq`, `f1_le_of_youden`, `auroc_eq`, `youden_eq`, `youden_exceeds_auroc_form`, at observation-data-mining 95af30c; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 5.

RELATED

[Apriori principle](#) (p. 25); [Monotone Invariance Theorem](#) (p. 227); [Youden F1 bound](#) (p. 385); [formula search](#) (p. 154).

SEE ALSO

Book equations stated beside the entry’s terms, not defining it: 5.3.

Sources-table rows that share a record with the entry without naming it: chapter 5 section 5.4, chapter 6 section 6.3, chapter 6 section 6.4, chapter 6 section 6.5, chapter 7 section 7.3, chapter 8 section 8.5.

sampling

instrument. Entry id sampling.

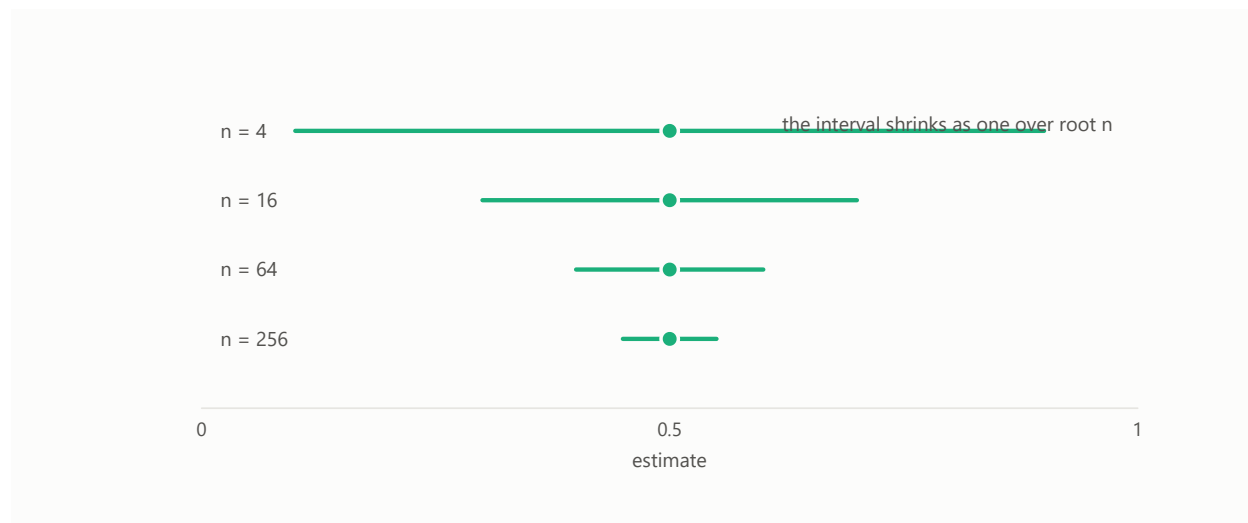


Figure 198: Which rows to read, and the standard error of the mean of those read.

DEFINITION

Choosing which rows to read. The standard error of a sample mean falls as one over the square root of the sample size. Chapter 2 section 2.7.

Example. Reading 350000 of 2391361 rows is a 15 percent sample, and its mean has standard error the spread over the square root of 350000.

EQUATION

none

CONDITIONS

- Choosing which rows to read. A sample's mean lies between its smallest and largest row, the standard error of the mean falls as one over the square root of the sample size, and a sample too thin to score is abstained on rather than averaged in.
- Sampling and aggregation are design choices that decide what the reader can see, and the first strata run read 350000 of 2391361 rows and abstained on the strata with two rows and one.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 2 section 2.7 of *Data Mining as Observation*, with the first strata run's sample in `turboquant-pro/docs/STRATA_RFC.md`: 24-98.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/StandardError.lean](#), theorems `se_pos`, `se_quarter`, `se_antitone`, `se_tendsto_zero`, `thousand_folds`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Ensemble.lean](#), theorems `average_between`, `sq_average_le`, `average_const`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14.

RELATED

[standard error](#) (p. 347); [aggregation](#) (p. 13); [bootstrap](#) (p. 49); [seed](#) (p. 326); [stratification](#) (p. 350).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.18, 8.2, 2.1.

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.7, chapter 8 section 8.1, chapter 10 section 10.4.

score

concept. Entry id score.

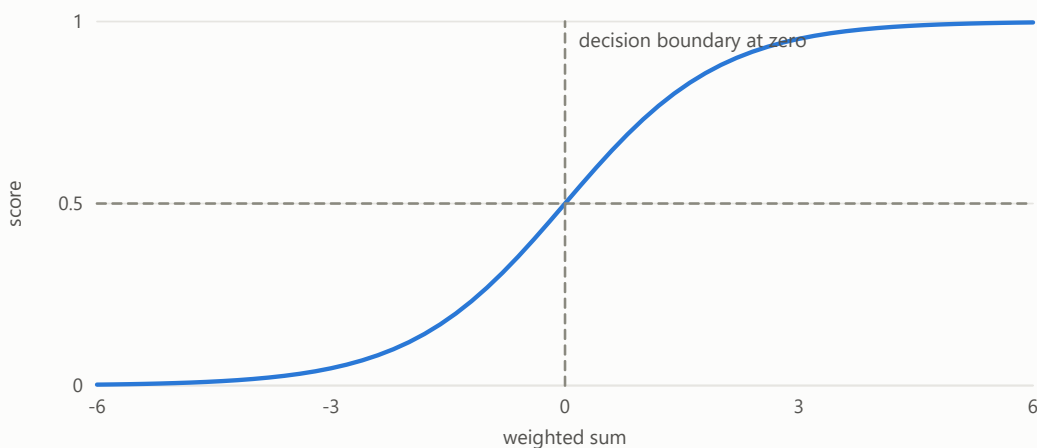


Figure 199: A number per row that a threshold turns into a decision.

DEFINITION

The number a classifier produces per row before a threshold turns it into a decision. Chapter 6.

Example. A classifier outputs 0.73 for a row; the threshold, not the score, makes the decision.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y\right) = \max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(f(X), \tau)], y\right)$$

for every strictly monotone g .

CONDITIONS

- The number a classifier produces per row before a threshold turns it into a decision. Every decision at every threshold is a function of the scores' ordering, so a strictly increasing recalibration of the scores with the matching recalibration of the threshold changes no decision.
- What a recalibration can change is calibration, which is why the reliability weight reads the ordering and the expected calibration error reads the values, and neither stands in for the other.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the program's threshold sweeps in theory-radar.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Threshold.lean](#), theorems `predicted_anti`, `tp_anti`, `fp_anti`, `decision_comp`, `predicted_extremes`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[classifier](#) (p. 65); [threshold](#) (p. 362); [Monotone Invariance Theorem](#) (p. 227); [calibration](#) (p. 56); [reliability weight](#) (p. 306).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 5 section 5.4, chapter 6 section 6.3.

sealed

instrument. Entry id sealed.

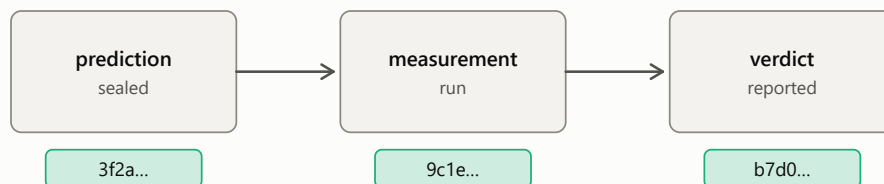
DEFINITION

Of a prediction, committed with its hash recorded before the measurement was run, so that anyone with the repository can verify it has not changed, and with a public push, a signed tag, or an archival deposit to fix when. Chapter 0 section 0.12.

Example. The prediction file's SHA-256 was recorded in a commit pushed before the run, and the seal names the commit.

EQUATION

none



each file's digest is recorded before the next step, and a changed digest proves a changed file

Figure 200: Committed with its hash recorded before the measurement.

CONDITIONS

- Of a prediction, committed with its hash recorded before the measurement, so that anyone with the repository can verify that the file has not changed. The hash proves the content. The time it existed is established by the public push, the signed tag, the transparency log, or the archival deposit that the seal names, since a commit's dates are supplied fields.
- A changed digest proves a changed file, an equal digest is evidence and not proof because a fixed-length digest of longer inputs cannot be injective, and every claim class in the ledger rests on a seal.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

The program's seal ledger, `observation-theory-campaigns/experiments/SEALS.md:1-10`, a seal being the registration id, the sealing commit, and the SHA-256 of the sealed file at that commit, and chapter 0 section 0.12 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.8	seal = commit plus SHA-256	observation-theory-campaigns/experiments/SEALS.md:1-10
chapter 8 section 8.8	declaration drafted 2026-08-17, sealed 2026-08-18, one count corrected, G2	geometric-observation/crucible/DECLARATION-V1.md:1-20 ; geometric-observation/crucible/OT-CRUCIBLE-4.md:31-35
chapter 12 section 12.6	XPROTO-LLM, benchmark 0.909, 0.920, 0.909, thirty slices, target 0.8, naive 0.333, aware 0.033, spread 0.380, deployment mean 0.736, six bars on three seeds, sealed 2026-08-25 at b61f7f1	observation-theory-campaigns/experiments/LLM-EVAL-TRACK.md:1-50 ; observation-theory-campaigns/analysis/llm/XPROTO-LLM-graded.json ; observation-theory-campaigns/experiments/SEALS.md:85

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.4	sealed correction, 0.5 dB calibration, floors 4/6/4, 2/2/3, 1, slopes 0.1008, 0.1398, 0.1086, R^2 0.7376, 0.9218, 0.6174, bars B1 to B4 and MC1 to MC4 all true, seeds 20260827 to 20260829, sealed 1d3de4a	observation-theory-campaigns/analysis/csi/PREREG-XPROTO-CSI-SWEEP2.md:1-60 ; observation-theory-campaigns/analysis/csi/XPROTO-CSI-SWEEP2-graded.json ; observation-theory-campaigns/experiments/SEALS.md:90

FAILURES AND CORRECTIONS

- [observation-theory-campaigns/ERRATA.md:111-120](#) at f6d1fe2. Standing note on records and runners Two failures this session shared one shape. A record was committed from a runner later found defective, and the correction was left uncommitted where nothing reading the repository could see it. The rule adopted in response is that a rerun writes to a new path and never over a committed record, and that the superseded record stays in the repository naming what replaced it. See the PF4-007 subsection of experiments/CAMPAIGN.md.

MACHINE CHECKED

[lean/DataMiningAsObservation/Seal.lean](#), theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[preregistration](#) (p. 267); [ledger class](#) (p. 211); [certificate](#) (p. 59); [refresh floor](#) (p. 299).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.1.

Ledger rows that cite the entry's records without naming it: OT-7, OT-11.

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.8.

seed

concept. Entry id seed.

DEFINITION

The number that fixes a run's pseudo-randomness. Two runs with the same seed agree only when code, data, environment, and execution are also fixed, and different seeds are necessary but not sufficient for an independent replication. Chapter 8.

Example. Seed 20260827 fixed the run, and seeds 20260828 and 20260829 were the two replications.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

CONDITIONS

- The number that fixes a run's pseudo-randomness. Two runs with the same seed reproduce each other only when the code, data, environment, and execution are also fixed, since nondeterministic execution can differ at the same seed, and runs with different seeds are necessary but not sufficient for an independent

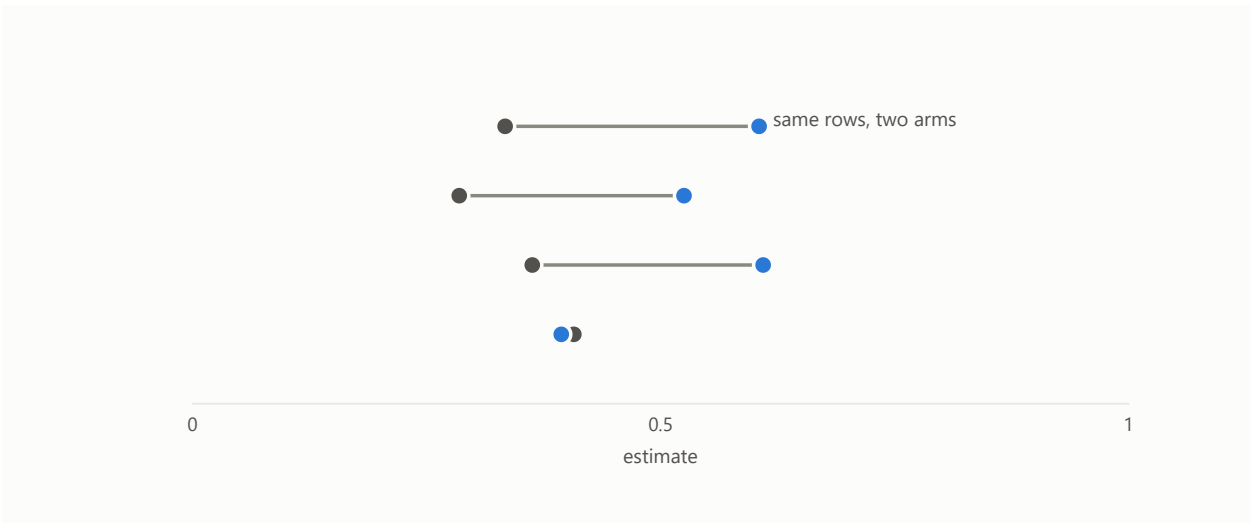


Figure 201: The number that fixes a run’s pseudo-randomness.

- replication.
- The standard error over n seeds falls as one over the square root of n , a paired comparison across seeds has smaller variance than an unpaired one exactly when the two arms covary, and a preregistration names its seeds before the run.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.1 of *Data Mining as Observation*, with the disjoint-seed rule of the database freshness track in `observation-theory-campaigns/experiments/DATABASE-FRESHNESS-TRACK.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation/claims/LEDGER.md row GO-4
chapter 13 section 13.3	ZooKeeper hot 0.99 cold 0.01 witnessed 0.0, Postgres 0.50 to 0.06, MongoDB 0.47 to 0.03, production Postgres 0.47 to 0.02, real substrates, disjoint seeds	observation-theory-campaigns/experiments/DATABASE-FRESHNESS-TRACK.md:1-45

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/StandardError.lean](#), theorems `se_pos`, `se_quarter`, `se_antitone`, `se_tendsto_zero`, `thousand_folds`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

[lean/DataMiningAsObservation/Bootstrap.lean](#), theorems `mean_sub`, `var_sub`, `paired_lt_iff`, `cov_comm`, at [observation-data-mining 95af30c](#); what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 4, 6, 7, 8, 9, 11, 12, 13.

RELATED

[harness](#) (p. 168); [standard error](#) (p. 347); [bootstrap](#) (p. 49); [preregistration](#) (p. 267); [sealed](#) (p. 324).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.18, 8.2.

Ledger rows that cite the entry's records without naming it: NEG-14.

sensitivity

concept. Entry id sensitivity.

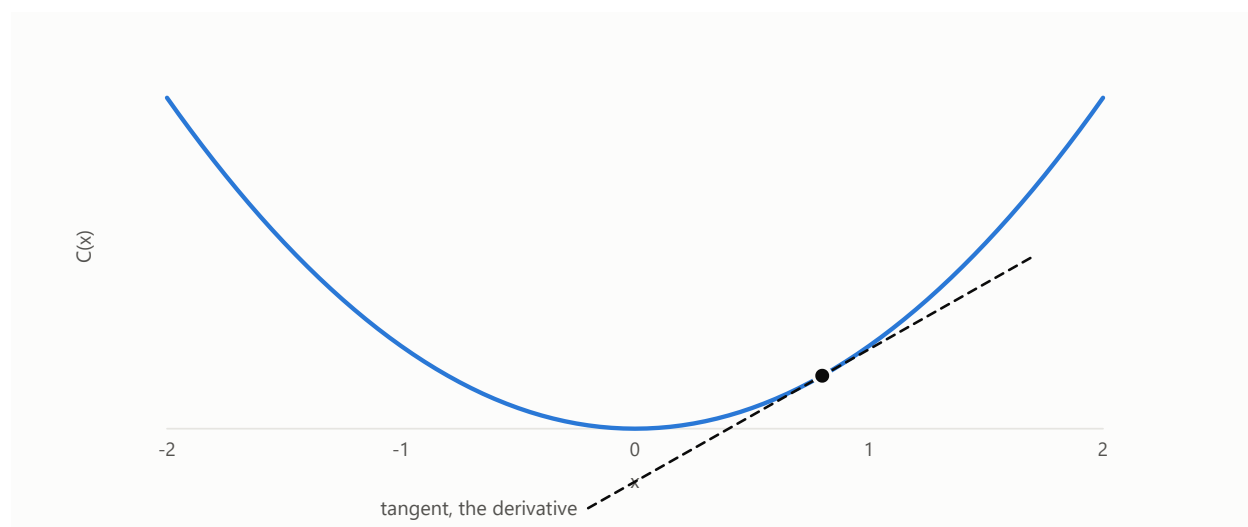


Figure 202: The gradient of the consumer at a row.

DEFINITION

The vector of partial derivatives of a consumer with respect to each coordinate of its input, at one point. Equation 0.8.

Example. For $C(x_1, x_2)$ equal to $3x_1 + 4x_2$ the sensitivity is $(3, 4)$ at every row.

Known as, or related to prior art. The gradient of the consumer at a row, the quantity active-subspace methods average.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

Book equation 14.4.

$$\text{attribution}_j(x) \approx g_j(x) \delta_j, \quad \overline{\text{importance}}_j \approx (P_C)_{jj} = \mathbb{E}[g_j^2].$$

CONDITIONS

- The sensitivity of a consumer at a row is the vector of partial derivatives of its output with respect to each coordinate, measurable without the formula by a central finite difference at two calls per coordinate.
- The read operator is the workload average of the sensitivity's outer product, so its diagonal is the expected squared sensitivity to each feature and its off-diagonal entries are co-sensitivities, not interactions.
- A gradient-based attribution estimates the sensitivity. Permutation importance, partial dependence, and Shapley values measure other things.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 5, `geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48`, DOI 10.5281/zenodo.21776291, as the gradient of the consumer at a row.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	readscope/readscope/regimes.py:1-60

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadOperator.lean](#), theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 7, 8, 11, 14.

RELATED

[read operator](#) (p. 291); [consumer](#) (p. 79); [blind probe](#) (p. 45); [read subspace](#) (p. 293).

SEE ALSO

Ledger rows that cite the entry's records without naming it: NEG-14, GO-EC-3.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2, chapter 2 section 2.3.

session deck

reference. Entry id `session-deck`.

where	numbers	source, file and lines
session 3	the four failure modes	session-outlines.md:38
session 6	formula or black box	session-outlines.md:117-136

every number names the file and lines it came from, at a commit

Figure 203: The course’s lecture deck, cited by line range.

DEFINITION

The lecture deck of ECE 514, cited by line range from the instructor’s session outlines, which are not public. Chapter 1 section 1.6.

Example. Session outline lines 117 to 136 state the transferable idea of chapter 5 as a learning outcome.

EQUATION

none

CONDITIONS

- The lecture deck of the course the book accompanies, ECE 514, cited by line range from the instructor’s session outlines. A row that names it names a file and lines like every other source, and a changed digest would prove a changed deck.
- The deck is not public. The book quotes what it says and marks each such row as instructor working documents, and the reader who wants the deck asks the instructor.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 1 section 1.6 of *Data Mining as Observation*, citing the instructor’s working documents, `ECE_514-01_FA26_session-outlines.md`, which are not public.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Seal.lean](#), theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 2, 3, 4, 5, 6, 8, 10, 11, 12, 14.

RELATED

[sources table](#) (p. 338); [commit hash](#) (p. 70); [harness](#) (p. 168); [preregistration](#) (p. 267).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 1.1.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.1, chapter 1 section 1.3, chapter 1 section 1.6, chapter 1 section 1.7, chapter 2 section 2.3, chapter 5 section 5.3, chapter 7 section 7.6.

shard

instrument. Entry id shard.



each shard returns its top k, and the global top k lies in the union

Figure 204: A partition searched separately, with the answers merged.

DEFINITION

One partition of an index that is searched separately, with the results merged. The global top k lies inside the union of the shards' top k. Chapter 13.

Example. Eight shards each returning their top 10 hand the merger 80 candidates, among which the global top 10 lies.

EQUATION

none

CONDITIONS

- One partition of an index that is searched separately, with the results merged. If each shard returns its own top k and the global top k is strict, the global top k lies inside the union of the shards' answers, so the merge loses nothing and reads at most the number of shards times k candidates.
- A placement study over 64 shards reported search times taken with the file cache already holding the shards, and a speedup of 63.7 times was fragmentation, so a sharded number names whether it is warm and what the shards were.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 13 section 13.1 of *Data Mining as Observation*, with the placement study in `turboquant-pro/docs/RESEARCH_ROADMAP.md:133-160` and the fleet run in `openvector-bench/README.md:250-263`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.1	22.30 s cold vs 0.835 s warm at eight probes, every prior number warm, fragmentation 63.7 bracketing 42 to 47, 64 shards, routing sparsity computed and discarded	turboquant-pro/docs/RESEARCH_ROADMAP.md:133-160

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Shard.lean](#), theorems `global_in_local`, `global_in_union`, `merge_cost`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 11, 13.

RELATED

[inverted file](#) (p. 185); [recall at k](#) (p. 295); [cold, warm](#) (p. 69); [cache](#) (p. 55); [budget](#) (p. 51).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.3, 11.4, 13.2.

Ledger rows that cite the entry's records without naming it: GO-12.

Sources-table rows that share a record with the entry without naming it: chapter 8 section 8.4, chapter 11 section 11.4, chapter 13 section 13.1.

silhouette

concept. Entry id silhouette.

DEFINITION

For a row, the difference between its mean distance to the nearest other cluster and its mean distance to its own cluster, divided by the larger of the two. Equation 0.30.

Example. A point with mean distance 1 to its own cluster and 4.5 to the nearest other has silhouette 3.5 over 4.5, or 0.778.

EQUATION

Book equation 0.30.

$$\text{SSE} = \sum_k \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad s_i = \frac{b_i - a_i}{\max(a_i, b_i)}.$$

CONDITIONS

- For a row, the difference between its mean distance to the nearest other cluster and its mean distance to its own cluster, over the larger of the two. It lies between minus one and one, is positive exactly when the row is closer to its own cluster, and is unchanged when every distance is scaled by the same factor.

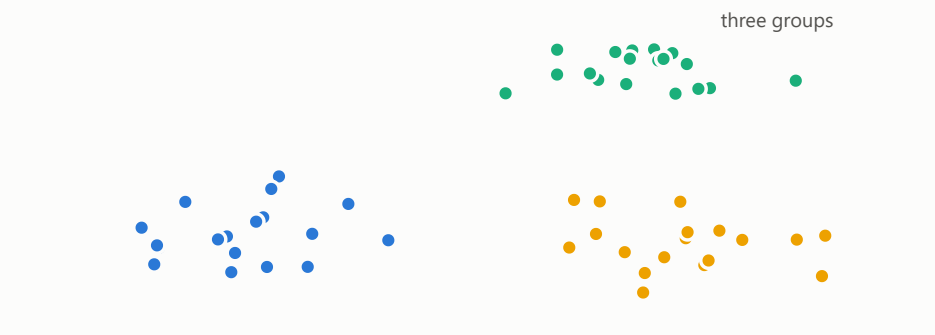


Figure 205: Distance to the nearest other cluster against distance within its own.

- It is computed under the identity reader on the distances it is given, so it validates a clustering for that reader and not for a consumer that reads other directions, which is why chapter 9 asks the recognizer to name the manifold instead.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Rousseeuw, silhouettes, 1987, as TSK chapter 7 presents it and chapter 0 section 0.15 states it, and chapter 9 of *Data Mining as Observation*, where it is one validity index among those the recognizer replaces.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Silhouette.lean](#), theorems `silhouette_mem`, `silhouette_pos_iff`, `silhouette_scale`, `silhouette_self`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 9, 10.

RELATED

[recognizer](#) (p. 296); [vacuity threshold](#) (p. 371); [identity reader](#) (p. 180); [quotient](#) (p. 279).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 9.1.

Simpson's paradox

concept. Entry id `simpsons-paradox`.

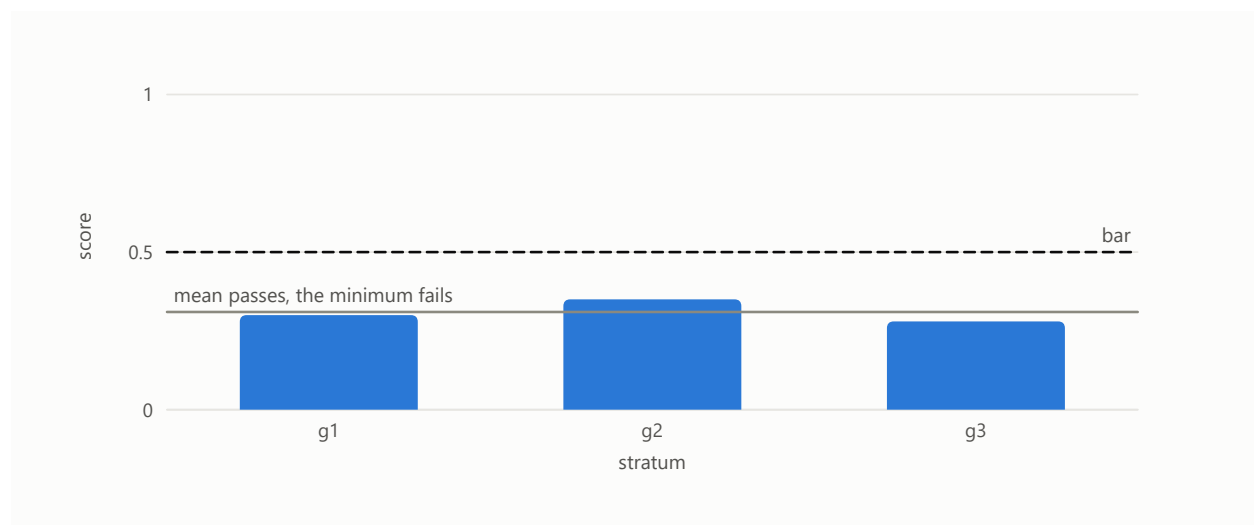


Figure 206: Every group shows one sign and the aggregate shows the other.

DEFINITION

A reversal of a comparison when groups are combined, which the book reads as an aggregation quotient. Chapter 5.

Example. A treatment better in both the young and the old can be worse overall when the two groups are mixed unequally.

EQUATION

none

CONDITIONS

- A pooled rate is the size-weighted mean of the group rates, so pooling is a quotient that declares the group labels irrelevant, and it can reverse a comparison that holds in every group when the groups have different sizes under the two arms.
- With equal group sizes under both arms the pooled comparison agrees with a comparison that holds in every group. The book's answer is the min-over-strata verdict, which never pools.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

TSK chapter 5 on objective measures, as chapter 5 section 5.5 of *Data Mining as Observation* reads it, an aggregation quotient.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Simpson.lean](#), theorems `pooled_eq_weighted`, `reversal`, `no_reversal_of_equal_sizes`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 5.

RELATED

[quotient](#) (p. 279); [min-over-strata](#) (p. 226); [abstention](#) (p. 12); [deployment mismatch](#) (p. 108).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 10.7.

skewness

concept. Entry id skewness.

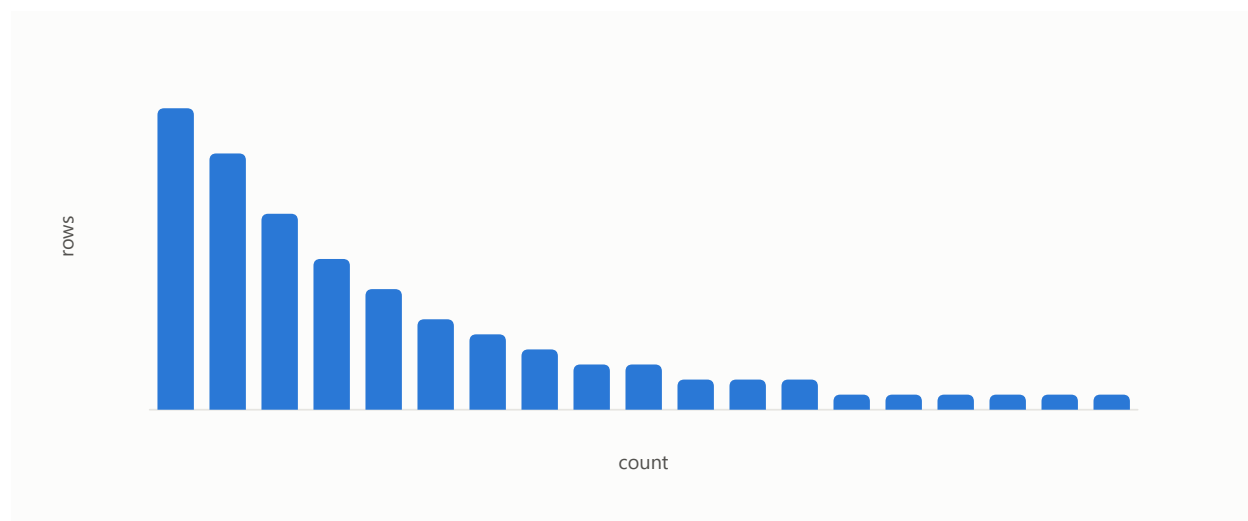


Figure 207: The mean cubed deviation, zero for a symmetric column.

DEFINITION

The mean cubed deviation of a column from its mean over the cube of its spread, zero for a symmetric column. The hubness gate reads it. Chapter 3 section 3.5.

Example. Counts of 1, 1, 1, 1, and 10 have a long right tail and positive skewness, and a symmetric column has zero.

EQUATION

none

CONDITIONS

- The weighted mean cubed deviation of a column from its mean over the cube of its spread. It is unchanged by a shift and by a positive rescaling, changes sign when the column is negated, and is zero for a column symmetric about its mean.
- The hubness gate reads the skew of the neighbour-count distribution, and the remedy that was to fix it moved the skew from 3.970 to 3.177 while the compressed path still failed every stratum.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 3 section 3.5 and chapter 10 section 10.2 of *Data Mining as Observation*, with the hubness gate's skew in `openvector-bench/results/R13_STAGE1_RESULT.md:75-110`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.5	Gate A design error, A prime skew 3.970 to 3.177, max 287 to 213, Robin Hood 0.372 to 0.261, fraction 0.117 to 0.079, compressed path 0.663 vs 0.90, seven strata, 0.62 to 0.69 vs 0.76 to 0.84	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:1-45

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Skewness.lean](#), theorems `skew_shift`, `skew_scale`, `skew_neg`, `skew_symmetric`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 3, 5, 10, 11.

RELATED

[hubness](#) (p. 176); [Robin Hood index](#) (p. 313); [variance](#) (p. 374); [gate](#) (p. 159); [percentile](#) (p. 255).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.35, 10.3, 0.17.

Ledger rows that cite the entry's records without naming it: NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.2, chapter 10 section 10.4, chapter 10 section 10.5.

softmax

concept. Entry id softmax.

DEFINITION

The function that exponentiates a list of numbers and divides by their sum, so that they are positive and add to one. Equation 0.23.

Example. Scores 2, 1, and 0 give weights 0.665, 0.245, and 0.090, which sum to one.

EQUATION

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

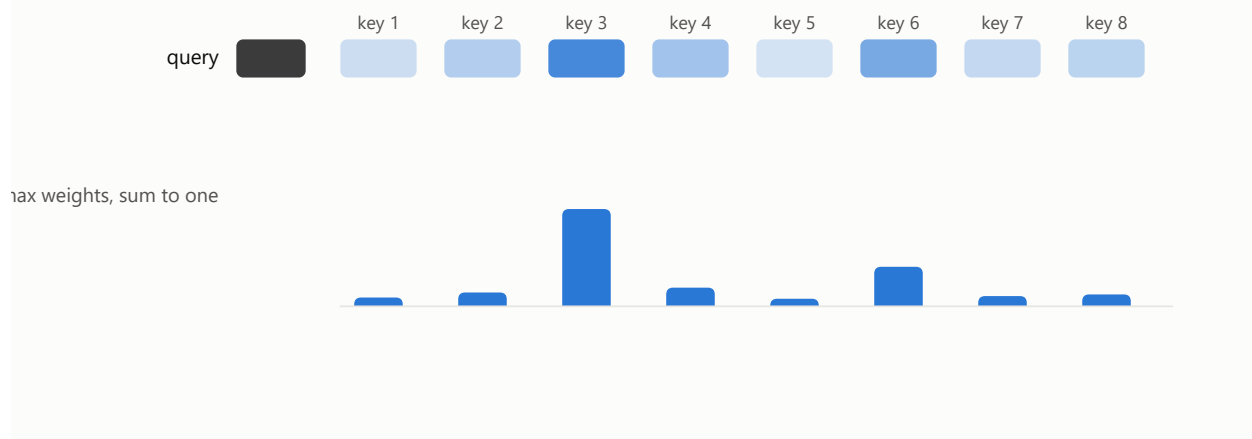


Figure 208: Exponentials normalized to sum to one.

CONDITIONS

- The softmax exponentiates a list of scores and divides by the sum, so the weights are positive and add to one. Adding the same constant to every score leaves it unchanged, and a larger score gets a larger weight.
- As a consumer it reads the scores through their differences, which is why the two proxies for softmax-KL in the ledger, the variance ratio and the projected-variance trace, were refuted.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects*. NEG-8 [refuted]. The Var-ratio `tang_qproj` is a ≥ 0.9 -Spearman rank proxy for softmax-KL under every consumer. [geometric-observation/claims/LEDGER.md:101](#).
- *refutes or corrects*. NEG-9 [refuted]. The projected-variance trace $\text{tr}(\mathbf{P}_C \Sigma \delta)$ is a complete rank statistic for softmax-KL across all arms. [geometric-observation/claims/LEDGER.md:102](#).

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's softmax reader in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro/docs/KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-8, [refuted]. The Var-ratio `tang_qproj` is a ≥ 0.9 -Spearman rank proxy for softmax-KL under every consumer. [geometric-observation/claims/LEDGER.md:101](#).
- NEG-9, [refuted]. The projected-variance trace $\text{tr}(\mathbf{P}_C \Sigma \delta)$ is a complete rank statistic for softmax-KL across all arms. [geometric-observation/claims/LEDGER.md:102](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/Softmax.lean](#), theorems `denom_pos`, `softmax_pos`, `softmax_sum`, `softmax_shift`, `softmax_lt_iff`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 8, 11.

RELATED

[attention](#) (p. 26); [KL divergence](#) (p. 196); [perplexity](#) (p. 258); [consumer](#) (p. 79).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.24.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.4, chapter 3 section 3.2, chapter 8 section 8.2, chapter 11 section 11.1, chapter 11 section 11.2.

sources table

instrument. Entry id `sources-table`.

where	numbers	source, file and lines
c	h	a
0	.	4
r	e	p

every number names the file and lines it came from, at a commit

Figure 209: For every number, the file and lines it came from.

DEFINITION

The table at the end of every chapter naming, for each number, the repository, file, and line range it came from. Chapter 0 section 0.21.

Example. The row for section 4.2 names `readscope/readscope/allocate.py` lines 1 to 100 for the water-filling formula.

EQUATION

none

CONDITIONS

- The table at the end of every chapter naming, for each number, the repository, file, and line range it came from, so that no number in the book has to be taken on trust. A row that names a commit hash binds the number to one state of the file, since a changed digest proves a changed file.
- The table is a citation and not a proof. The book's checker verifies that every cited path exists, and the encyclopedia's checker verifies the numbers of its hand-filled entries against the generated ones, but the numbers themselves are the records' and the reader who wants them opens the file.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.21 of *Data Mining as Observation*, with the program's claims ledger that the same discipline enforces in `turboquant-pro/CLAIMS.md:1-27`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Seal.lean](#), theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[commit hash](#) (p. 70); [sealed](#) (p. 324); [ledger class](#) (p. 211); [preregistration](#) (p. 267); [certificate](#) (p. 59).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.1.

Sources-table rows that share a record with the entry without naming it: chapter 11 section 11.5.

Spearman correlation

concept. Entry id `spearman-correlation`.

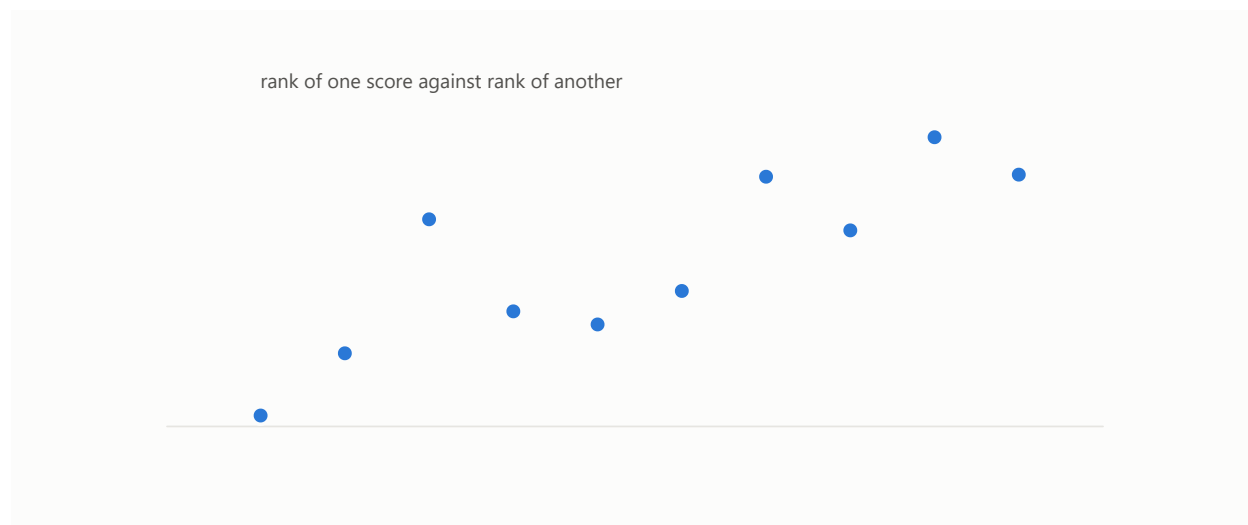


Figure 210: Pearson's correlation of the ranks.

DEFINITION

The ordinary correlation between two lists of ranks. Equation 0.14.

Example. Ranks (1, 2, 3) against (1, 3, 2) give Spearman correlation 0.5.

EQUATION

Book equation 0.14.

$$\rho_S = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)}, \quad d_i = \text{difference of the two ranks of item } i.$$

CONDITIONS

- The ordinary correlation between two lists of ranks, one minus six times the sum of squared rank differences over $n(n^2 - 1)$. It is at most one, one when the rankings agree, and minus one for three items in opposite orders, and since ranks depend on the ordering alone a strictly increasing transform of either score leaves it unchanged.
- A high Spearman between a proxy and a consumer's loss is not a certificate. The variance-ratio proxy with Spearman above 0.9 is a refuted ledger row, and the rank certificate's floor on Spearman is the strict setting's guarantee, not a measured correlation.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *refutes or corrects*. NEG-8 [refuted]. The Var-ratio `tang_qproj` is a ≥ 0.9 -Spearman rank proxy for softmax-KL under every consumer. [geometric-observation/claims/LEDGER.md:101](#).

FIRST STATED

Spearman, the proof and measurement of association between two things, 1904, as chapter 0 section 0.9 states it, and the rank statistic of the recognizer battery in the-angular-observer and of the rank certificate in readscape.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- NEG-8, [refuted]. The Var-ratio `tang_qproj` is a ≥ 0.9 -Spearman rank proxy for softmax-KL under every consumer. [geometric-observation/claims/LEDGER.md:101](#).

MACHINE CHECKED

[lean/DataMiningAsObservation/Spearman.lean](#), theorems `rho_le_one`, `rho_identical`, `rho_reversed_three`, `rank_comp`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 4, 7, 9, 11.

RELATED

[Kendall correlation](#) (p. 194); [rank certificate](#) (p. 285); [recognizer](#) (p. 296); [Monotone Invariance Theorem](#) (p. 227).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.2.

Ledger rows that cite the entry's records without naming it: GO-3.

spectral clustering

instrument. Entry id `spectral-clustering`.

DEFINITION

Clustering the rows of the row-normalized spectral embedding, which reads the angle and discards the density. Chapter 3 section 3.3 and chapter 9.

Example. Row-normalize the two lowest nontrivial eigenvectors of the Laplacian and run k-means on the angles.

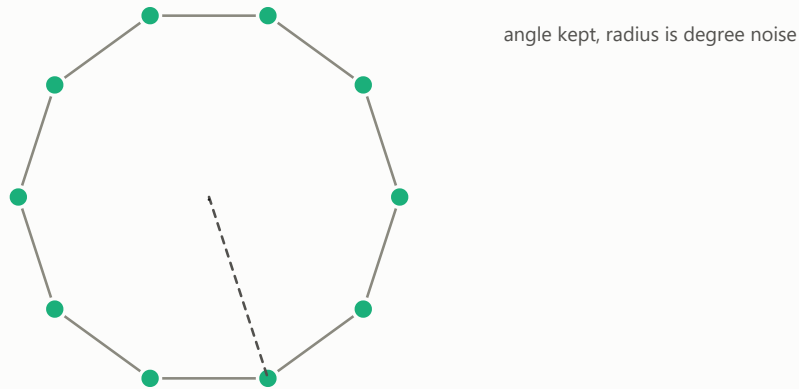


Figure 211: Clustering the row-normalized spectral embedding.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

CONDITIONS

- Clustering the rows of the row-normalized spectral embedding, which reads the angle and discards the density. Row normalization puts every row on the unit sphere, keeps the cosine between rows, and ignores a positive rescaling, and the Laplacian's quadratic form is nonnegative, zero on constants, and positive across any edge.
- Ng, Jordan, and Weiss normalized because it improved results, and chapter 3 says why. The Euclidean distance in the embedding converges to degree noise and the angle keeps the geodesics.
- The row normalization applies to eigenvectors of the normalized Laplacian, whose leading vector is proportional to the square root of the degree and not constant.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Ng, Jordan, and Weiss, on spectral clustering, 2002, as chapter 3 section 3.3 and chapter 9 section 9.1 of *Data Mining as Observation* read it, with the angular reading in the `-angular-observer/theorem.md:110-146`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation/chapters/ch09_legibility.md:10-41 ; the-angular-observer/theorem.md:110-146

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/SpectralEmbedding.lean](#), theorems `dot_self_nonneg`, `rowNormalize_unit`, `dot_rowNormalize`, `rowNormalize_smul`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Laplacian.lean](#), theorems `quad_eq`, `quad_nonneg`, `quad_const`, `quad_pos_of_edge`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 9.

RELATED

[spectral embedding](#) (p. 342); [Laplacian](#) (p. 207); [k-means](#) (p. 198); [recognizer](#) (p. 296); [geodesic distance](#) (p. 162).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.20.

Ledger rows that cite the entry's records without naming it: GO-3, NEG-1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.3, chapter 9 section 9.2.

spectral embedding

instrument. Entry id `spectral-embedding`.

DEFINITION

The coordinates a node receives from its values in the first few nontrivial eigenvectors of the graph Laplacian. Clustering those coordinates is spectral clustering. Chapter 3.

Example. A ring of ten nodes embeds on a circle, and the angle between two nodes' embeddings tracks their distance along the ring.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

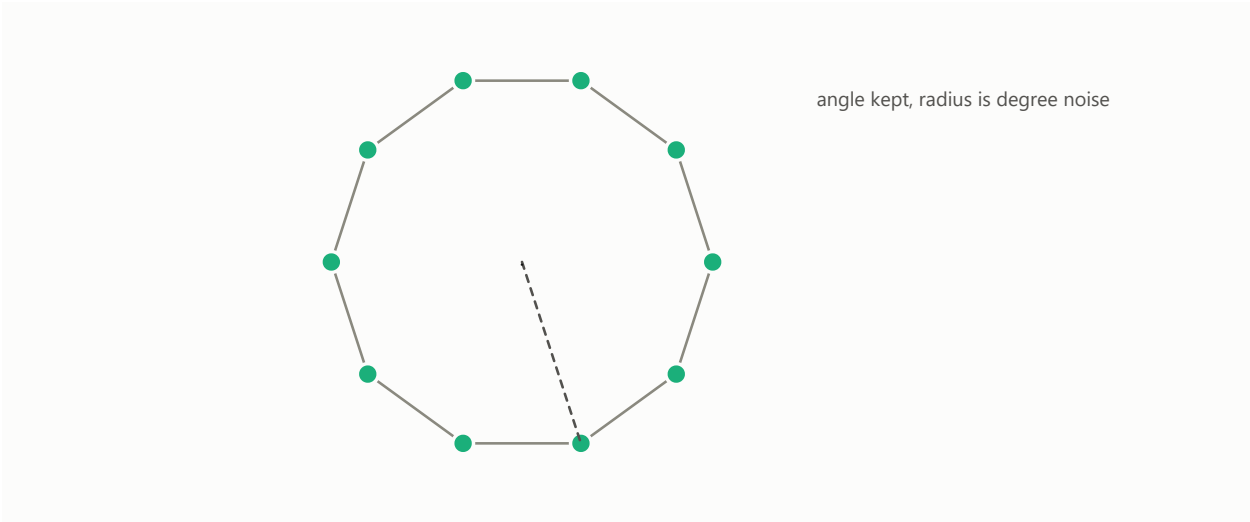


Figure 212: A node’s values in the low eigenvectors of the Laplacian.

CONDITIONS

- The coordinates a node receives from its values in the first few nontrivial eigenvectors of the graph Laplacian. Row normalization puts every embedded row on the unit sphere, leaves the cosine between rows unchanged, and ignores a positive rescaling of a row, so it is the projection onto the read subspace of the geodesic-rank consumer.
- Its magnitude converges to degree noise and its angle keeps the geodesics, so the cosine between embedded rows is the right similarity and the Euclidean distance the wrong one. Clustering the normalized coordinates is spectral clustering, and a density reader of the same embedding reads the complementary coordinate.
- The embedding of equation 0.20 scales each eigenvector entry by the eigenvalue and the degree. Its squared Euclidean distance is the resistance, commute time over volume, exactly for the full spectrum and approximately when only the low modes are kept.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Ng, Jordan, and Weiss, on spectral clustering, 2002, as chapter 3 section 3.3 of *Data Mining as Observation* reads it, with the program’s angular reading in `the-angular-observer/theorem.md:110-146`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 3 section 3.3	row normalization is the projection onto the read subspace of the geodesic-rank consumer	geometric-observation/chapters/ch09_legibility.md:31-41 ; geometric-observation/chapters/ch03_historical_precursors.md:98-110

chapter 9 section 9.1

the geodesic-rank reader discards the radius, row normalization is the angular projection

[geometric-observation/chapters/ch09_legibility.md:10-41](#); the-angular-observer/theorem.md:110-146

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/SpectralEmbedding.lean](#), theorems `dot_self_nonneg`, `rowNormalize_unit`, `dot_rowNormalize`, `rowNormalize_smul`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Laplacian.lean](#), theorems `quad_eq`, `quad_nonneg`, `quad_const`, `quad_pos_of_edge`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 9.

RELATED

[Laplacian](#) (p. 207); [commute time](#) (p. 72); [geodesic distance](#) (p. 162); [degree](#) (p. 105); [recognizer](#) (p. 296).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.20.

Ledger rows that cite the entry's records without naming it: NEG-1.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.3.

spectrum

concept. Entry id spectrum.

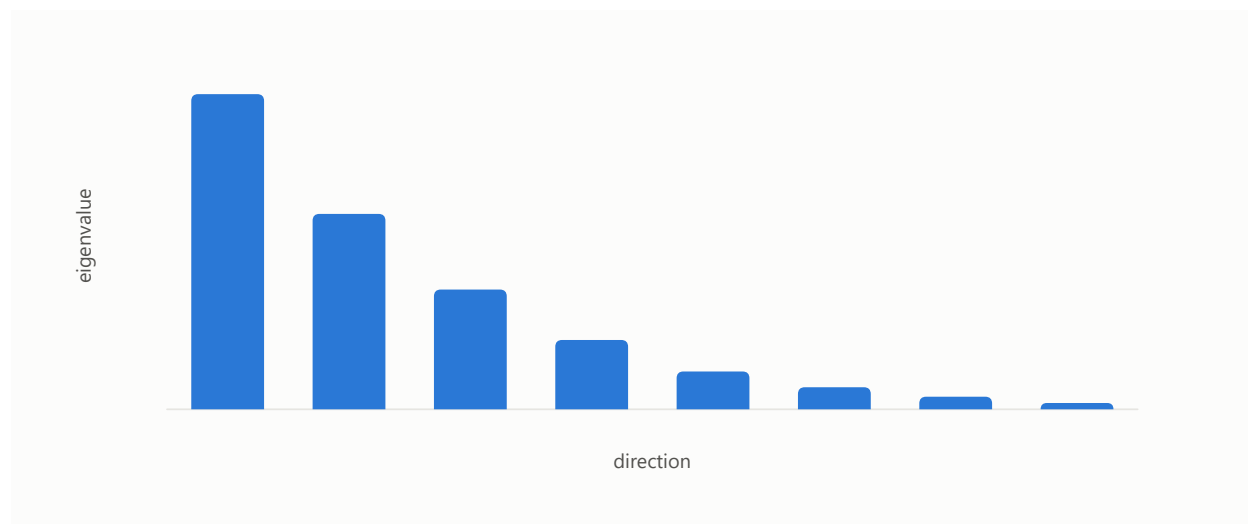


Figure 213: The eigenvalues of a matrix.

DEFINITION

The list of a matrix's eigenvalues. Chapter 0 section 0.4.

Example. `diag(0.3, 1.7)` has spectrum (0.3, 1.7), trace 2.0, and effective rank 1.34.

EQUATION

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

CONDITIONS

- The list of a matrix's eigenvalues. Its effective rank, the square of its sum over the sum of its squares, lies between one and the dimension, is the dimension for a flat spectrum, and is one for a single eigenvalue.
- The spectrum is not invariant under a change of basis of the input where the trace pairing and the rank are, and the Laplacian's spectrum is what the recognizer reads for dimension and shape.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. OT-7 [demonstrated]. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and ... [geometric-observation/claims/LEDGER.md:30](#).

FIRST STATED

Chapter 0 section 0.4 of *Data Mining as Observation*, with the program's spectrum records in `readscope`, `readscope/readscope/spectrum.py:35-70`, and the recognizer battery.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/EffectiveRank.lean](#), theorems `sq_sum_le`, `effRank_le`, `sum_sq_le_sq_sum`, `one_le_effRank`, `effRank_const`, `effRank_single`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 3, 4, 9, 11, 14.

RELATED

[eigenvalue](#), [eigenvector](#) (p. 125); [effective rank](#) (p. 123); [recognizer](#) (p. 296); [Laplacian](#) (p. 207); [isotropic](#) (p. 187).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 9.2.

Ledger rows that cite the entry's records without naming it: GO-3.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.2, chapter 9 section 9.4, chapter 14 section 14.3, chapter 14 section 14.5, chapter 14 section 14.7.

split

instrument. Entry id `split`.



Figure 214: The rows a model is fit on and the rows it is scored on.

DEFINITION

A partition of the rows into a part the model is fit on and a part it is scored on. A scorer that has read the test part scores it perfectly. Chapter 8 section 8.1.

Example. 900 rows for fitting and 100 held out, with the 100 never read until the score.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

CONDITIONS

- A partition of the rows into a part the model is fit on and a part it is scored on. The parts' sizes add to the row count, a row in the test part is not in the training part, and a scorer that has read the test part scores it perfectly, which is leakage.
- The virgin split of the non-oracle test raised the flip's margin to 0.796 against 0.780, and the split is fixed before any transform is fit, since an imputation fit on all rows reads the test part.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 8 section 8.1 of *Data Mining as Observation*, with the virgin split in `geometric-observation/chapters/ch10_the_blind_probe.md:100-118`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/CrossValidation.lean](#), theorems `sizes_sum`, `accuracy_weighted`, `accuracy_mean_of_equal`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Leakage.lean](#), theorems `errors_lookup_eq_zero`, `lookup_default`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[cross-validation](#) (p. 96); [leakage](#) (p. 210); [harness](#) (p. 168); [seed](#) (p. 326); [stratification](#) (p. 350).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.2.

Ledger rows that cite the entry's records without naming it: NEG-4, GO-B-legal (035→036).

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.7, chapter 6 section 6.4, chapter 8 section 8.1, chapter 8 section 8.5, chapter 12 section 12.3.

standard error

concept. Entry id standard-error.

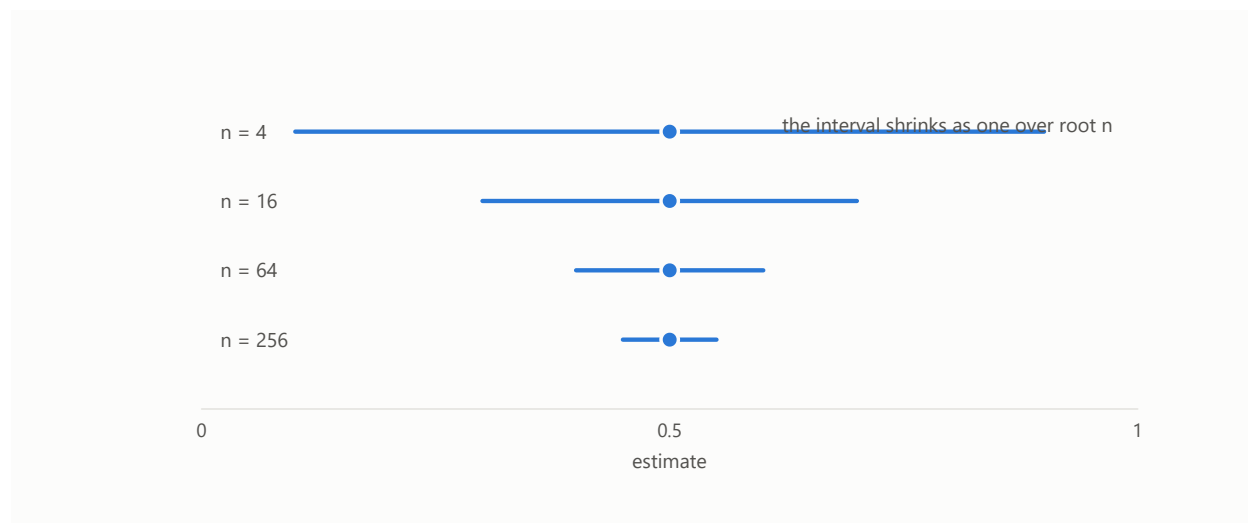


Figure 215: The spread of an estimate over repetitions, falling as one over root n .

DEFINITION

The spread of an estimate across repeated samples. Chapter 0 section 0.9.

Example. A spread of 0.4 over 16 seeds gives standard error 0.1, and over 64 seeds 0.05.

EQUATION

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

CONDITIONS

- The spread of an estimate across repeated samples, the spread of one draw over the square root of the number of independent draws. It is positive, falls as the draws grow, halves only when the draws quadruple, and tends to zero.
- Dependent draws do not shrink it this way. A thousand repeated folds shrink the naive standard error by a factor near 31.6 and the corrected one by 15.84 less, which is the Nadeau and Bengio correction.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.9 of *Data Mining as Observation*, with the program's own inflation in `constraint-gap/review/FINDINGS.md:1-35`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_eff 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	<code>constraint-gap/review/FINDINGS.md:1-35</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/StandardError.lean](#), theorems `se_pos`, `se_quarter`, `se_antitone`, `se_tendsto_zero`, `thousand_folds`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 6, 7, 8, 13, 14.

RELATED

[Nadeau and Bengio correction](#) (p. 233); [harness](#) (p. 168); [multiple comparisons](#) (p. 229); [preregistration](#) (p. 267).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.18.

Sources-table rows that share a record with the entry without naming it: chapter 6 section 6.4.

standardization

instrument. Entry id standardization.

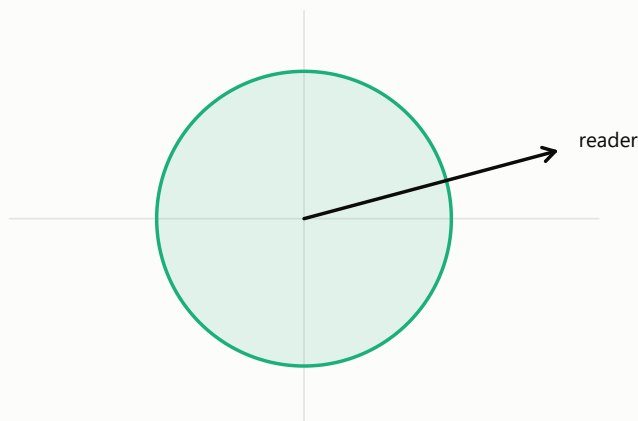


Figure 216: Each column centred and divided by its spread.

DEFINITION

Subtracting a column's mean and dividing by its spread, so that it has mean zero and variance one. It is invertible and preserves every ordering, so it changes the reader and not the data. Chapter 2.

Example. Values 2, 4, and 6 with mean 4 and spread 1.633 become -1.22, 0, and 1.22.

EQUATION

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 0.6.

$$x_w = \Sigma^{-1/2}(x - \mu), \quad \Sigma^{-1/2} = \sum_i \lambda_i^{-1/2} v_i v_i^\top.$$

CONDITIONS

- Subtracting a column's mean and dividing by its spread. The standardized column has weighted mean zero and weighted variance one, the transform is invertible, and it preserves every ordering of the rows, so it changes the reader's geometry and not what the table holds.
- An invertible transform identifies no two rows and forms no quotient. For a consumer that reads distances across columns the declaration that scale is irrelevant is almost always right, and for a consumer that reads one column it does nothing.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 2 section 2.5 of *Data Mining as Observation*, with the whitened code's record in `geometric-observation/chapters/ch08_value.md:84-97`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Standardization.lean](#), theorems `mean_zero`, `variance_one`, `standardize_inv`, `standardize_lt_iff`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 9, 10, 11.

RELATED

[whitening](#) (p. 381); [imputation](#) (p. 182); [Euclidean distance](#) (p. 133); [quotient](#) (p. 279); [outlier](#) (p. 250).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 3.1.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.1, chapter 2 section 2.5, chapter 4 section 4.3.

stratification

instrument. Entry id stratification.

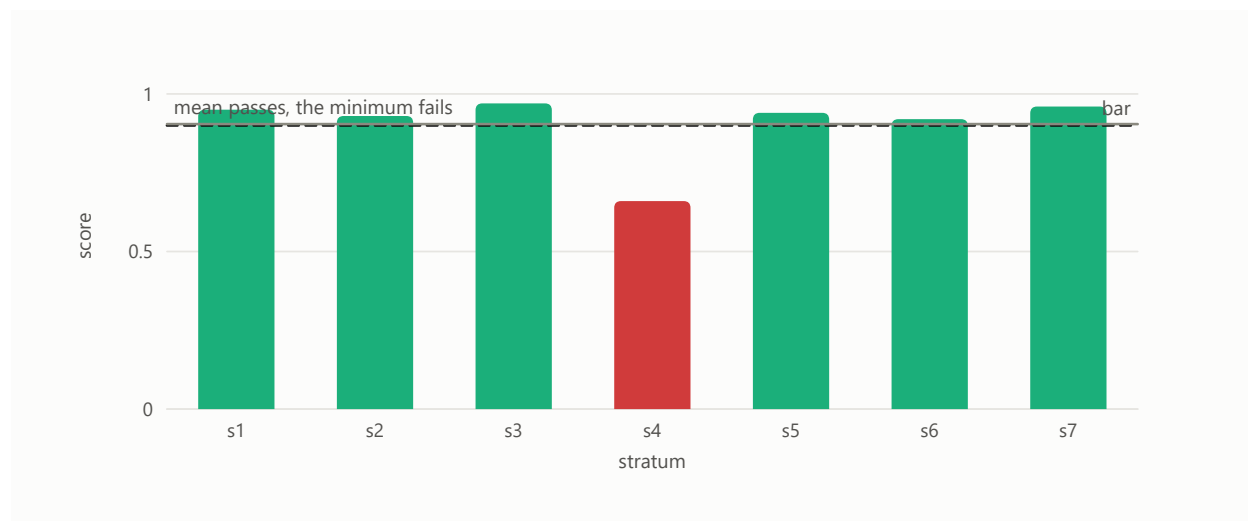


Figure 217: Rows scored in strata, so the minimum can be reported.

DEFINITION

Splitting rows into strata, by difficulty or by group, and scoring each separately, so that the report can take the minimum over strata. Chapter 10 and chapter 12.

Example. Fourteen strata by difficulty, seven eligible, two abstained for having 2 rows and 1.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

CONDITIONS

- Splitting rows into strata, by difficulty or by group, and scoring each separately, so that the report can take the minimum over strata and count the strata too thin to score. A weighted aggregate over strata is at most the pass bar when one stratum fails by enough, and the folds of a stratified split test every row once.
- The first strata run scored seven of fourteen eligible strata and abstained on those with two rows and one, and the second prediction inverted, which the record keeps.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 10 section 10.4 of *Data Mining as Observation*, with the strata design in `turboquant-pro/docs/STRATA_RFC.md:24-98`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.5	Gate A design error, A prime skew 3.970 to 3.177, max 287 to 213, Robin Hood 0.372 to 0.261, fraction 0.117 to 0.079, compressed path 0.663 vs 0.90, seven strata, 0.62 to 0.69 vs 0.76 to 0.84	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:1-45

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

`lean/DataMiningAsObservation/Certificate.lean`, theorems `falseClear_mul_coverage`, `coverage_empty`, `falseClear_mem_unit`, `minOverStrata_passes_iff`, `minOverStrata_le_weighted_mean`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

`lean/DataMiningAsObservation/CrossValidation.lean`, theorems `sizes_sum`, `accuracy_weighted`, `accuracy_mean_of_equal`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 5, 8, 10, 11, 12.

RELATED

[min-over-strata](#) (p. 226); [abstention](#) (p. 12); [anti-hub recall](#) (p. 22); [cross-validation](#) (p. 96); [harness](#) (p. 168).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 14.5, 8.2.

Ledger rows that cite the entry's records without naming it: NEG-14, GO-B-Llama, GO-B-Llama-rematch.

Sources-table rows that share a record with the entry without naming it: chapter 10 section 10.4, chapter 10 section 10.5, chapter 12 section 12.4.

sum of squared errors

concept. Entry id `sum-of-squared-errors`.

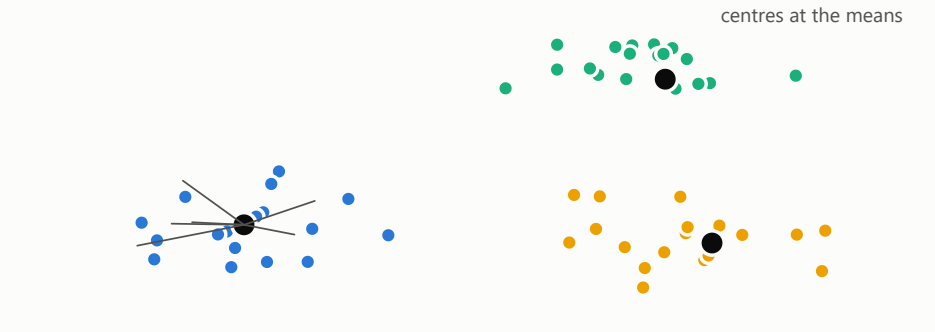


Figure 218: The squared distance to the cluster centre, summed.

DEFINITION

The sum over rows of the squared distance to the row's cluster centre, the validity index k-means minimizes. Chapter 0 section 0.15.

Example. Points 1, 2, and 6 with centres 1.5 and 6 have sum of squared errors $0.25 + 0.25 + 0 = 0.5$.

EQUATION

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

Book equation 0.30.

$$\text{SSE} = \sum_k \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad s_i = \frac{b_i - a_i}{\max(a_i, b_i)}.$$

CONDITIONS

- The sum over rows of the squared distance to the row's cluster centre, the validity index k-means minimizes. The error about any centre is the error about the mean plus the count times the squared distance between the two, so the mean minimizes it, and it is the identity reader's distortion summed within clusters.
- Two clusterings of the same rows with the same sum of squared errors can be read oppositely by a consumer that weighs directions unequally.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.15 and chapter 9 section 9.1 of *Data Mining as Observation*.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/KMeans.lean](#), theorems `sse_decomposition`, `mean_minimizes`, `assign_nearest`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/ReconstructionError.lean](#), theorems `recon_nonneg`, `recon_eq_zero_iff`, `recon_sum`, `recon_proj`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 9.

RELATED

[k-means](#) (p. 198); [validity index](#) (p. 372); [reconstruction error](#) (p. 298); [identity reader](#) (p. 180); [silhouette](#) (p. 332).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 4.2.

Ledger rows that cite the entry's records without naming it: GO-3.

Sources-table rows that share a record with the entry without naming it: chapter 9 section 9.4, chapter 10 section 10.1.

support

concept. Entry id support.

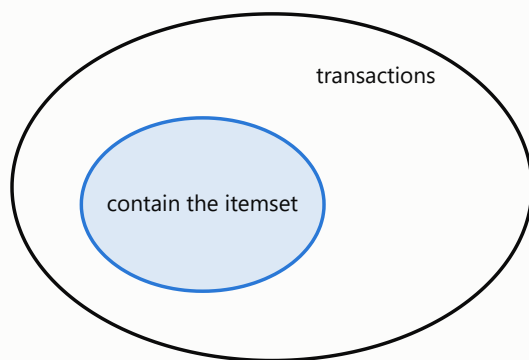


Figure 219: The fraction of transactions that contain the itemset.

DEFINITION

Of an itemset, the fraction of transactions that contain it. Chapter 5.

Example. An itemset in 30 of 200 baskets has support 0.15.

EQUATION

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

CONDITIONS

- The fraction of transactions that contain an itemset. Support cannot increase when an itemset grows, so an infrequent itemset has only infrequent supersets and every subset of a frequent itemset is frequent, which is the license to prune the lattice.
- The count of candidates rather than the count of transactions decides whether a run finishes, and a rare consequent lets lift reach the transaction count on a single co-occurrence.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

TSK chapter 5, as chapter 5 section 5.1 of *Data Mining as Observation* states it, with anti-monotonicity as the Apriori license.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/SafePruning.lean](#), theorems `support_anti`, `apriori`, `subset_of_frequent`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 2, 5, 6, 7, 8.

RELATED

[itemset](#) (p. 188); [Apriori principle](#) (p. 25); [lift](#) (p. 214); [cross-support ratio](#) (p. 95).

SEE ALSO

none

surrogate

instrument. Entry id surrogate.

DEFINITION

A quadratic stand-in for the consumer's output metric, the read distortion, used to allocate bits when the output itself cannot be. Chapter 4 section 4.2.

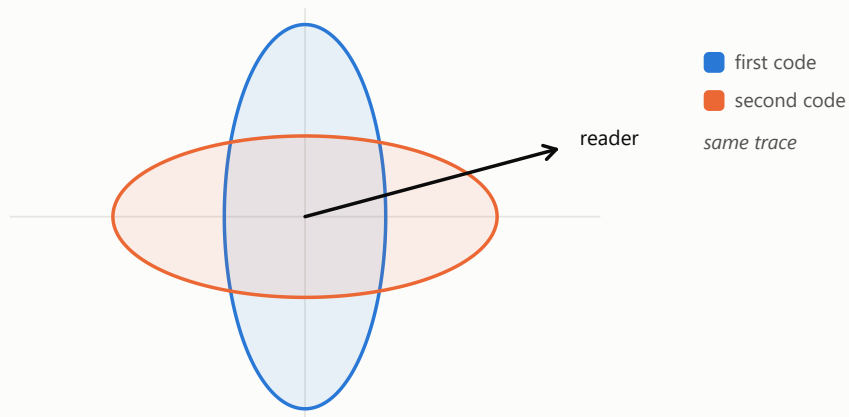


Figure 220: A quadratic stand-in for the consumer's output metric.

Example. The read distortion 0.3 stands in for the consumer's own loss when bits are allocated.

EQUATION

none

CONDITIONS

- A quadratic stand-in for the consumer's output metric, the read distortion, used to allocate bits when the output itself cannot be. The read distortion is a quadratic form in the read operator, and two readers at different angles rank the same two errors differently.
- At every rate the output coder is at or below the surrogate, which is at or below reconstruction, and the surrogate-to-output gap shrinks from 0.41 to 0.005 as the rate grows, while the read distortion is a control and not a complete rank statistic.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures*. GO-6 [demonstrated]. At matched rate, output coding $\leq$ surrogate $\leq$ reconstruction on the consumer metric; the output-reconstruction gap is governed by the ker P_C entropy share, and the surrogate-output gap vanishes as rate grows. [geometric-observation/claims/LEDGER.md:68](#).

FIRST STATED

Chapter 4 section 4.2 of *Data Mining as Observation*, with the surrogate ordering in [geometric-observation/claims/LEDGER.md](#) row GO-6.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	water-filling formula, directions below the water get no bits, the surrogate caveat	readscope/readscope/allocate.py:1-100
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	geometric-observation/claims/LEDGER.md row GO-6; geometric-observation/chapters/ch07_cost.md

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ReadDistortion.lean](#), theorems `read_distortion`, `identity_reader`, `quad_one`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Isotropy.lean](#), theorems `isotropic_reads_same`, `isotropic_no_flip`, `anisotropic_readers_differ`, `flip_iff_anisotropic`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 4, 8.

RELATED

[read distortion](#) (p. 290); [distortion](#) (p. 116); [output metric](#) (p. 251); [identity reader](#) (p. 180).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 4.2, 4.6, 0.11.

Ledger rows that cite the entry's records without naming it: NEG-9.

sweep

instrument. Entry id sweep.

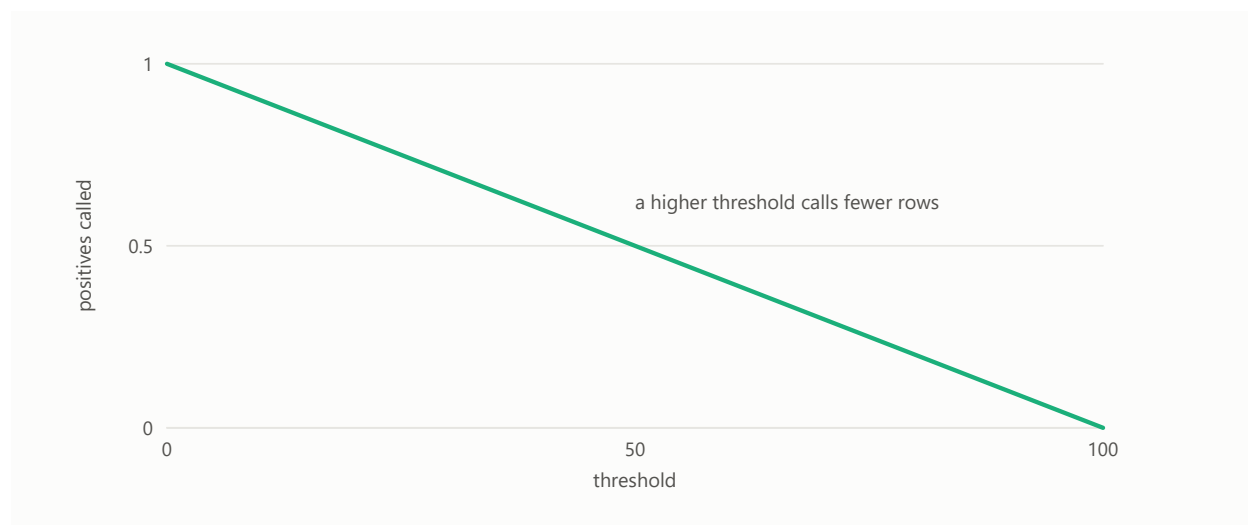


Figure 221: A measurement run across a range of one parameter.

DEFINITION

Running a measurement across a range of one parameter and reading the curve. Chapter 5 section 5.4 and chapter 9 section 9.3.

Example. Thresholds from 0.1 to 0.9 in steps of 0.1 trace nine points of the ROC curve.

EQUATION

none

CONDITIONS

- Running a measurement across a range of one parameter, a threshold, a budget, or a degradation, and reading the curve. As a threshold rises the set called positive can only shrink, so true and false positives fall together and the ROC curve is traced.
- Fourteen corpora were swept through degradation until retrieval died, the budget sweep found the cliff at k equal to d , and the threshold sweep found the optimal F1 by sort.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 5 section 5.4 and chapter 9 section 9.3 of *Data Mining as Observation*, with the degradation sweep in `openvector-bench/README.md:60-96`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Threshold.lean](#), theorems `predicted_anti`, `tp_anti`, `fp_anti`, `decision_comp`, `predicted_extremes`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 3, 4, 5, 6, 7, 8, 9, 10, 13.

RELATED

[threshold](#) (p. 362); [ROC curve](#) (p. 314); [budget cliff](#) (p. 52); [sensitivity](#) (p. 328); [harness](#) (p. 168).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 6.2, 0.28, 9.3.

Ledger rows that cite the entry's records without naming it: GO-3, GO-4.

Sources-table rows that share a record with the entry without naming it: chapter 5 section 5.4, chapter 11 section 11.4, chapter 11 section 11.7, chapter 12 section 12.2.

T

teacher forcing

concept. Entry id teacher-forcing.

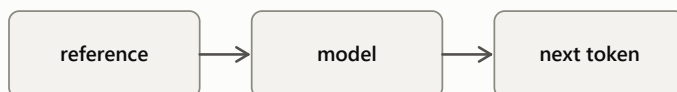


Figure 222: The reference token fed in at each step, so an error cannot compound.

DEFINITION

Feeding a language model the correct text as context rather than its own earlier outputs. Chapter 0 section 0.11.

Example. At step 5 the model is fed the reference's fifth token, not its own fourth guess.

EQUATION

none

CONDITIONS

- Feeding a model the correct text as context rather than its own earlier outputs. The forced and free runs agree exactly as long as the forced predictions match the reference, and they part at the first token the model gets wrong, after which the free run's context is no longer the reference's.
- Perplexity is measured under teacher forcing, so a good perplexity does not certify a free run. The C-12 record's thirteen-point difference vanished under teacher forcing, and the operator-drift claim it was meant to support is refuted in the ledger.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's C-12 record in `readscope`, `readscope/CALIBRATION.md`:600-660.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/TeacherForcing.lean](#), theorems `free_eq_take`, `free_length`, `free_ne_of_error`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 11.

RELATED

[perplexity](#) (p. 258); [drift](#) (p. 119); [harness](#) (p. 168); [deployment mismatch](#) (p. 108).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.22, 11.1.

Ledger rows that cite the entry's records without naming it: OT-4.

template match

instrument. Entry id `template-match`.

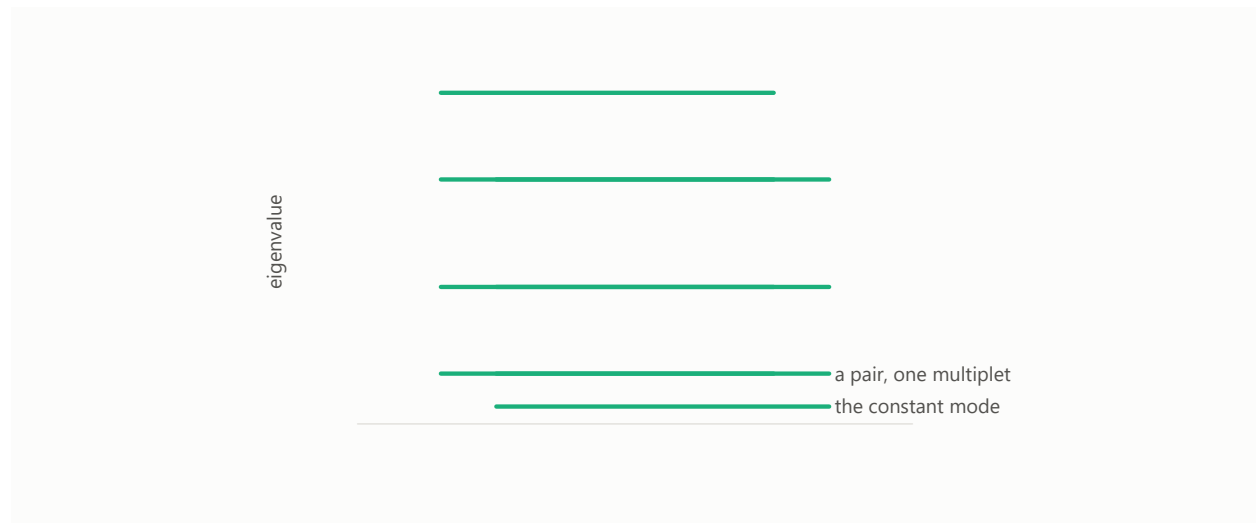


Figure 223: The spectrum's ratios against a stored template.

DEFINITION

Comparing a measured list of numbers, such as eigenvalue ratios, to a stored list for each candidate shape and picking the smallest distance between the logarithms. Equation 0.32.

Example. The lowest eight eigenvalue ratios matched the sphere template within tolerance, and the torus template did not.

EQUATION

Book equation 0.32.

$$\text{dist}(m, t) = \sqrt{\frac{1}{8} \sum_{k=1}^8 (\ln m_k - \ln t_k)^2}.$$

CONDITIONS

- Comparing a measured list of numbers, the eigenvalue ratios, to a stored list for each candidate shape and picking the smallest root-mean-square distance between the logarithms. The distance is zero between a

list and itself, symmetric, and unchanged when both lists are scaled by the same factor, so the match reads shape and not size.

- The templates are frozen before the battery is run, with their code hash recorded, and the match can only name shapes in its template set, which is why the recognizer also certifies that none is present.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 11, `geometric-observation/chapters/ch11_the_recognizer.md:1-95`, DOI 10.5281/zenodo.21776291, and the recognizer battery in the-angular-observer.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	<code>the-angular-observer/PREREG_RECOGNIZER_BATTERY.md:1-80</code>

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Recognizer.lean](#), theorems `templateDist_self`, `templateDist_comm`, `templateDist_scale`, `templateDist_nonneg`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 8, 9, 14.

RELATED

[recognizer](#) (p. 296); [vacuity threshold](#) (p. 371); [intrinsic dimension](#) (p. 184); [Laplacian](#) (p. 207).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 9.3.

Ledger rows that cite the entry's records without naming it: GO-3.

Sources-table rows that share a record with the entry without naming it: chapter 3 section 3.3, chapter 9 section 9.1, chapter 9 section 9.2, chapter 11 section 11.1.

TF-IDF

concept. Entry id `tf-idf`.

DEFINITION

A weighting of term counts by how rare the term is across the collection, so that a term in every document carries no weight. Equation 0.36.

Example. A term appearing 5 times in a 100-word document and in 1 of 100 documents weighs 0.05 times the log of 100.

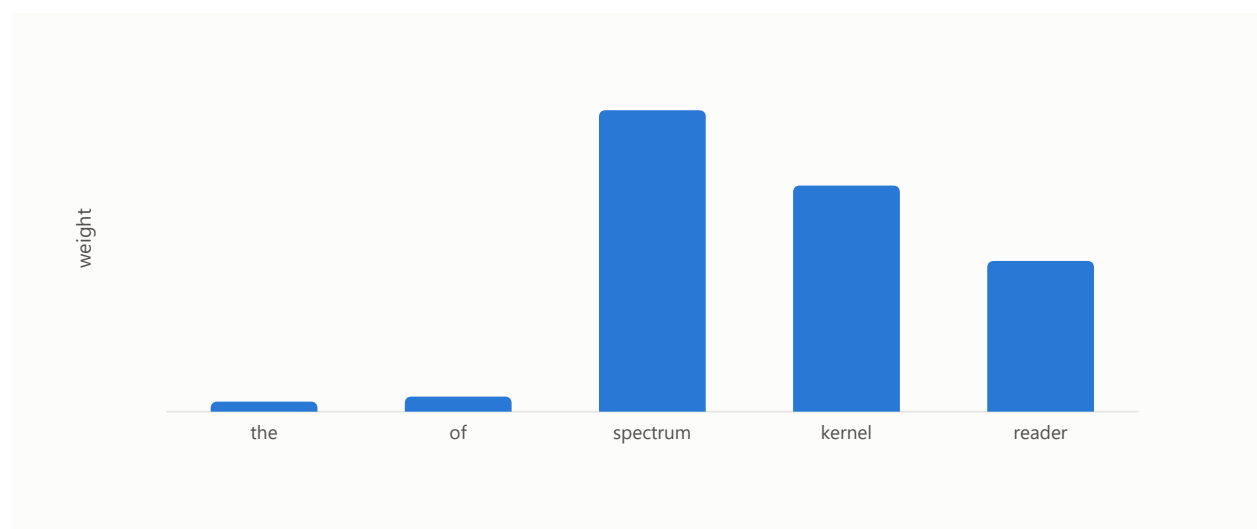


Figure 224: Term frequency scaled down by how many documents carry the term.

EQUATION

Book equation 0.36.

$$w_{t,d} = \text{tf}_{t,d} \cdot \ln \frac{N}{\text{df}_t}, \quad \text{tf}_{t,d} = \frac{\text{count of } t \text{ in } d}{\text{length of } d}, \quad \text{df}_t = \text{documents containing } t.$$

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

CONDITIONS

- A term's weight is its frequency in the document times the logarithm of the document count over the number containing it. A term in every document carries no weight, the weight is nonnegative and falls as the term spreads, and the frequency lies in the unit interval.
- It is a reader that reads rarity. The legal-citation flip used a frozen TF-IDF to SVD baseline trained on the training split only, and the flip tied while its magnitude overshot, which the ledger carries as partial.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Spärck Jones, a statistical interpretation of term specificity, 1972, as chapter 0 section 0.17 states it, with the program's frozen LSA baseline in ledger row GO-B-legal.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/TFIDF.lean](#), theorems `weight_everywhere`, `weight_nonneg`, `weight_antitone`, `tf_mem_unit`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 12.

RELATED

[bag of words](#) (p. 36); [retrieval-augmented pipeline](#) (p. 312); [cross-corpus gate](#) (p. 94); [quotient](#) (p. 279).

SEE ALSO

Ledger rows that cite the entry's records without naming it: GO-B-legal (035→036).

threshold

concept. Entry id threshold.

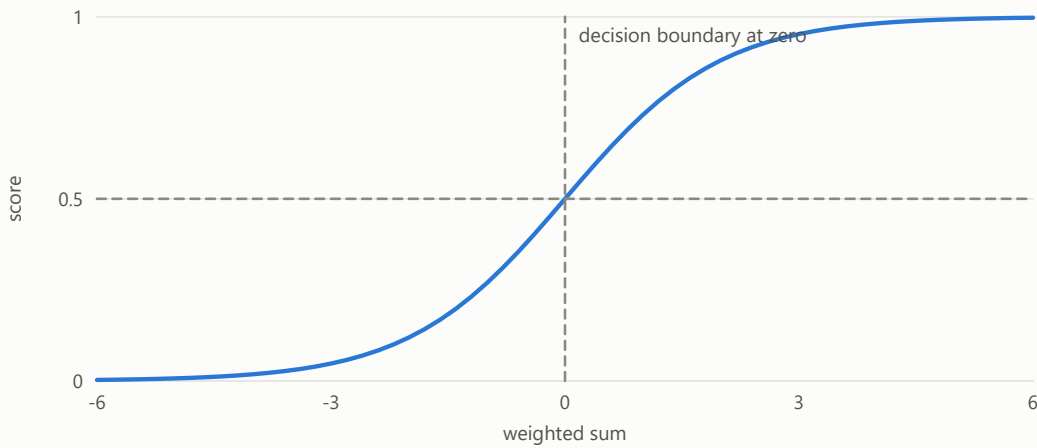


Figure 225: The value that turns a score into a decision.

DEFINITION

The value at which a score becomes a positive decision. Chapter 6.

Example. Scores 0.3, 0.6, and 0.9 at threshold 0.5 give one negative and two positives; at 0.7, two negatives and one positive.

EQUATION

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y) = \max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(f(X), \tau)], y)$$

for every strictly monotone g .

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

CONDITIONS

- The value at which a score becomes a positive decision. Raising it can only shrink the predicted set, so true positives and false positives can only fall, which is why the ROC curve is traced by a single sweep, and below every score everything is positive while above every score nothing is.

- A strictly increasing recalibration of the score with the matching recalibration of the threshold leaves every decision unchanged, which is the Monotone Invariance Theorem at the level of one decision.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the program's threshold sweeps in theory-radar.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Threshold.lean](#), theorems `predicted_anti`, `tp_anti`, `fp_anti`, `decision_comp`, `predicted_extremes`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

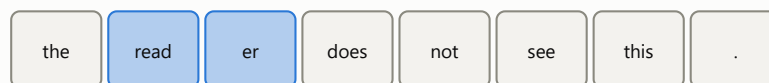
[Monotone Invariance Theorem](#) (p. 227); [Youden F1 bound](#) (p. 385); [reliability weight](#) (p. 306); [calibration](#) (p. 56).

SEE ALSO

none

token

concept. Entry id token.



two tokens for one word

Figure 226: A piece of text, roughly a word, the unit a language model reads.

DEFINITION

A piece of text, roughly a word, the unit a language model reads and writes. Chapter 0 section 0.11.

Example. The word unbelievable may be three tokens, un, believ, and able.

EQUATION

Book equation 0.22.

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(w_t \mid w_{<t}).$$

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

CONDITIONS

- A piece of text, roughly a word, the unit a language model reads and writes. A model reads a sequence of tokens and outputs a probability for the next one, and perplexity is two to the power of the average bits per token, so it is at least one and equals the vocabulary size for a uniform guess.
- The key-value cache stores one key and one value per token per head, and a generation of 512 tokens under a wrong codebook was where the retracted degradation curve was measured. Token counts are the budget in every serving row of chapter 13.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's serving measurements in `geometric-observation/experiments/GO-kv-serving-flip-NOTES.md`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Perplexity.lean](#), theorems `bits_nonneg`, `one_le_perplexity`, `perplexity_uniform`, `perplexity_mono`, `bits_of_finding`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 4, 8, 11, 12, 13.

RELATED

[perplexity](#) (p. 258); [attention](#) (p. 26); [KV cache](#) (p. 200); [rotary position embedding](#) (p. 316); [teacher forcing](#) (p. 358).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 11.1.

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.6.

trace

concept. Entry id trace.

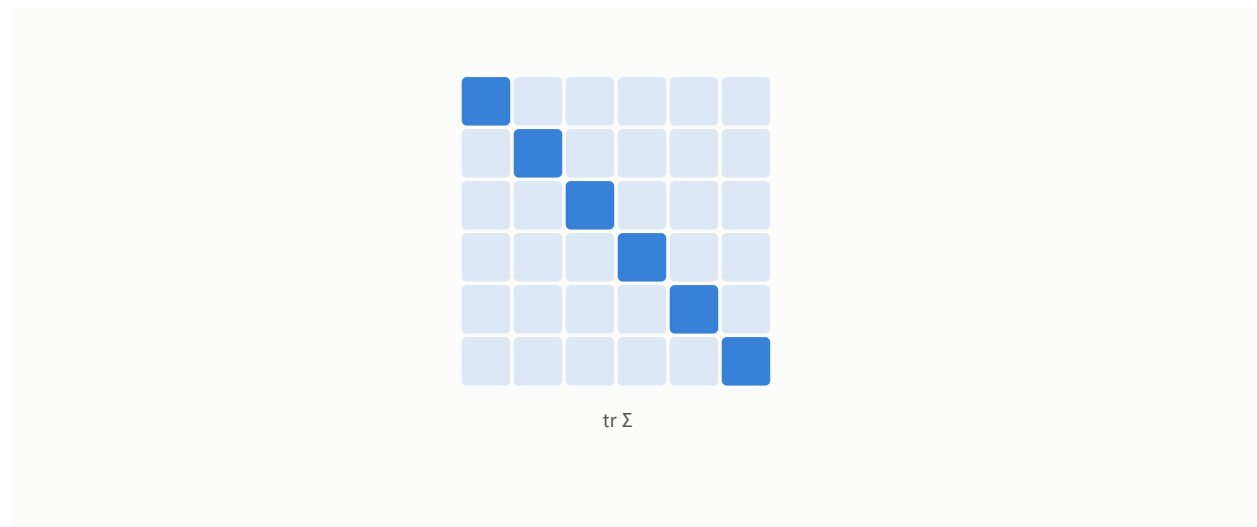


Figure 227: The sum of the diagonal, the total variance for a covariance.

DEFINITION

The sum of a matrix's diagonal, which for a covariance is the total variance. Equation 0.3.

Example. $\text{diag}(0.3, 1.7)$ has trace 2.0, the total variance.

EQUATION

Book equation 0.3.

$$\Sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)], \quad \text{tr } \Sigma = \sum_i \Sigma_{ii}.$$

Book equation 0.10.

$$d_O = \text{tr}(P_C M_\delta) = \mathbb{E}[\delta^\top P_C \delta], \quad M_\delta = \mathbb{E}[\delta \delta^\top], \quad P_C = I \Rightarrow d_O = \text{tr } M_\delta.$$

CONDITIONS

- The sum of a matrix's diagonal. It is linear, the trace of the identity is the dimension, the trace of an outer product is the vector's squared length, the trace of a product does not depend on the order, and the trace of a read operator times a rank-one error is the read distortion of that error.
- For a covariance the trace is the total variance, and the trace pairing of a read operator with a second moment is the invariant the ledger's row OT-7 names, unchanged by a change of basis where the spectrum and effective rank are not.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* OT-7 [demonstrated]. The damage form, trace pairing, rank, and loading covariance are $\text{GL}(d)$ -invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and ... [geometric-observation/claims/LEDGER.md:30](#).

FIRST STATED

Chapter 0 section 0.2 of *Data Mining as Observation*, with the trace pairing of Volume 14's invariance row OT-7.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Trace.lean](#), theorems `trace_add`, `trace_smul`, `trace_one`, `trace_vecMulVec`, `trace_mul_comm`, `trace_read`, `trace_identity_read`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 3, 4, 13.

RELATED

[covariance matrix](#) (p. 91); [read distortion](#) (p. 290); [identity reader](#) (p. 180); [effective rank](#) (p. 123).

SEE ALSO

Ledger rows that cite the entry's records without naming it: OT-2.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.2.

transaction

concept. Entry id transaction.

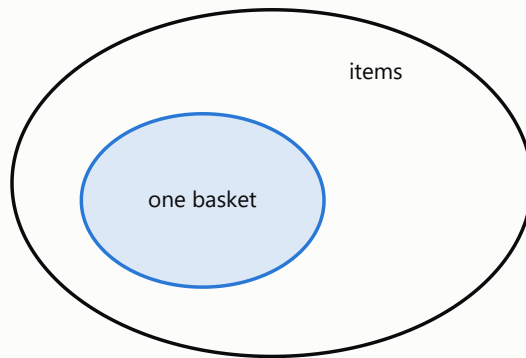


Figure 228: One row of a market-basket table.

DEFINITION

One row of a market-basket table, the set of items bought together. Chapter 5 section 5.1.

Example. Bread, butter, and jam bought together is one transaction, one basket.

EQUATION

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

CONDITIONS

- One row of a market-basket table, the set of items bought together. The support of an itemset counts the transactions that contain it, it cannot rise as the itemset grows, and the support fraction lies in the unit interval.
- A transaction is the unit the consumer reads in chapter 5, and which items count as the same is the quotient the pattern report has to name.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Agrawal, Imieliński, and Swami, mining association rules, 1993, as chapter 5 section 5.1 of *Data Mining as Observation* reads it.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/ItemSet.lean](#), theorems `supportFrac_mem_unit`, `supportFrac_empty`, `supportFrac_anti`, `supportFrac_union_le`, `closed_of_maximal`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/SafePruning.lean](#), theorems `support_anti`, `apriori`, `subset_of_frequent`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 1, 5, 13.

RELATED

[itemset](#) (p. 188); [support](#) (p. 353); [confidence](#) (p. 75); [anti-monotonicity](#) (p. 24).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 5 section 5.1 to 5.3, 5.5.

transmission time interval

concept. Entry id `transmission-time-interval`.

DEFINITION

The one-millisecond slot in which a 5G tower schedules a transmission, abbreviated TTI. Chapter 13.

Example. A coherence time of 20 milliseconds spans 20 slots.

EQUATION

Book equation 13.2.

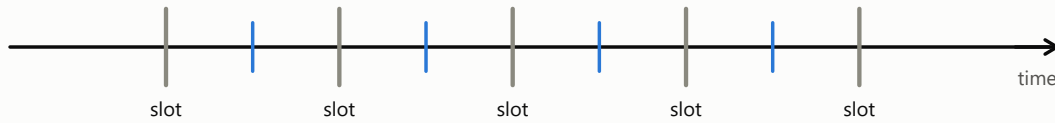


Figure 229: The one-millisecond slot a tower schedules in.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz, } 1 \text{ TTI at } \geq 50 \text{ Hz.}$

CONDITIONS

- The one-millisecond slot in which a 5G tower schedules a transmission, abbreviated TTI. A duration of m milliseconds spans m slots, slots add over consecutive durations, and a longer duration spans more of them, which is the arithmetic the refresh floors are counted in.
- The sealed floors were 4, 1, and 1 slots at Doppler frequencies of 10, 50, and 400 hertz, and the earlier fit of the floor as a fixed fraction of the coherence time was refuted and is not to be cited.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.13 of *Data Mining as Observation*, with the program's sealed floors in `observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/TTI.lean](#), theorems `slots_of_ms`, `slots_add`, `slots_mono`, `slots_coherence`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 13.

RELATED

[block error rate](#) (p. 46); [coherence time](#) (p. 67); [refresh floor](#) (p. 299); [freshness](#) (p. 156).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.25.

Sources-table rows that share a record with the entry without naming it: chapter 13 section 13.3, chapter 13 section 13.4.

truncation

instrument. Entry id truncation.

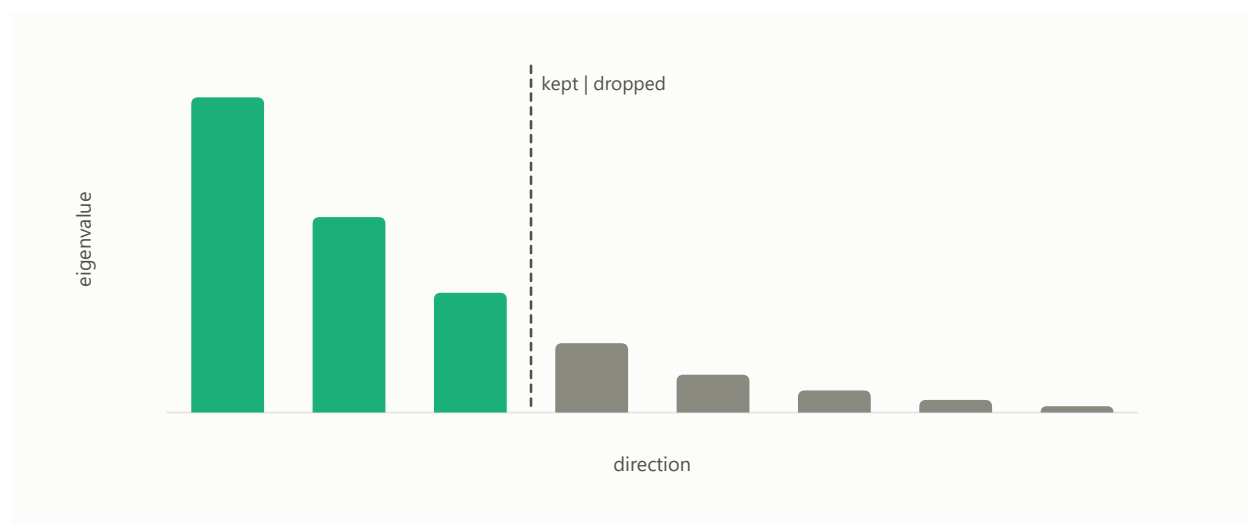


Figure 230: The first k components kept and the rest dropped.

DEFINITION

Keeping the first k components of a spectrum and dropping the rest, a bet that the consumer reads the top of the spectrum. Chapter 4 section 4.5.

Example. Keeping 64 of 300 GloVe components kept 73 percent of the variance and lost 0.685 against 0.862 downstream.

EQUATION

none

CONDITIONS

- Keeping the first k components of a spectrum and dropping the rest. The variance kept grows with k and reaches one at full rank, and the error is the sum of the dropped eigenvalues, so truncation is a bet that the consumer reads the top of the spectrum.
- Truncation to 64 components kept 73 percent of GloVe's variance and lost 0.685 against 0.862 downstream, is reproducible, and loses on compact sets.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 4 section 4.5 of *Data Mining as Observation*, with the truncation claim in `turboquant-pro/CLAIMS.md:28-49` and the GloVe table in `turboquant-pro/benchmarks/RESULTS_glove.md:1-40`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.5	truncation claim reproducible, loses on compact sets	turboquant-pro/CLAIMS.md:28-49
chapter 11 section 11.5	9.6x at recall 0.999 CI-gated on GloVe 1.18M; 32x at 0.9993 on private 199k LaBSE, ties OPQ, beats RaBitQ, 20x build; 27.7x and 114x reported; PCA truncation loses on compact sets; 20x at 199k and 4x at 1M over OPQ; RaBitQ builds in under a second	turboquant-pro/CLAIMS.md:28-49

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/PCA.lean](#), theorems `varAlong_basis`, `varAlong_le`, `varAlong_ge`, `dropped_eq`, `dropped_nonneg`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/ExplainedVariance.lean](#), theorems `explained_mem_unit`, `explained_mono`, `explained_full`, `retained_identity`, `retained_example`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 4, 11.

RELATED

[principal component analysis](#) (p. 269); [explained variance](#) (p. 137); [budget](#) (p. 51); [spectrum](#) (p. 344); [concentrated](#) (p. 73).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 4.2, 0.7, 11.4.

Ledger rows that cite the entry's records without naming it: GO-4.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.5.

V

vacuity threshold

result. Entry id vacuity-threshold.

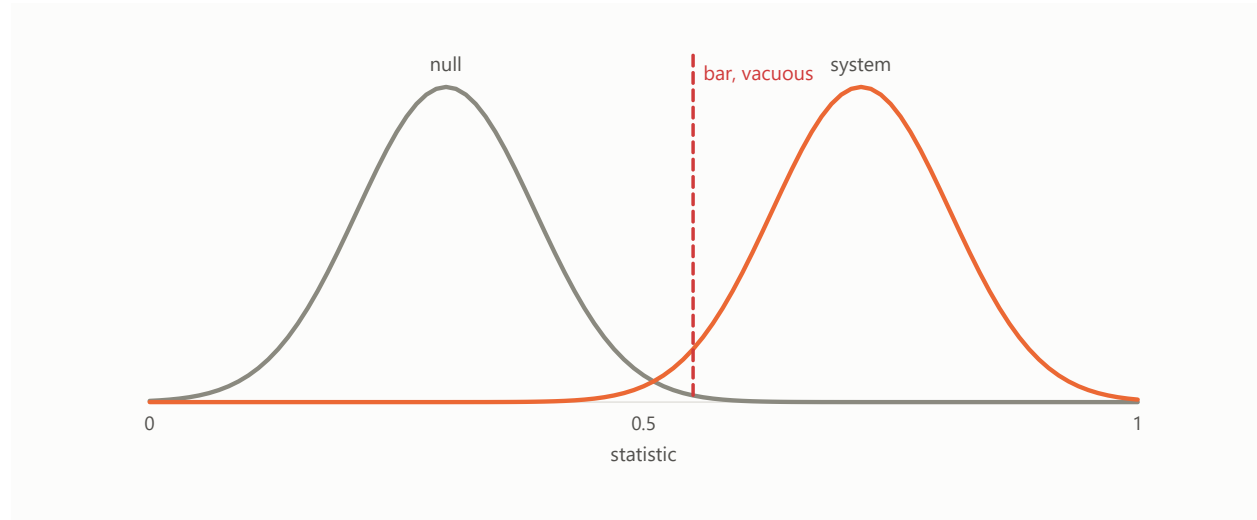


Figure 231: The statistic below which the certificate says nothing.

DEFINITION

The value of a certificate's statistic below which the certificate is uninformative, derived from the problem rather than tuned to the data. Chapter 9.

Example. A certificate statistic at the maximum certifiable value, 1.83, is vacuous, and the derived threshold says so before the data is read.

EQUATION

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E} \left[\max_{N-1} \mathcal{N}(0, 1) \right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous.}$$

CONDITIONS

- The threshold is derived from the problem, the expected maximum of the distractors' margins, and is not tuned to the data.
- A certificate at or below the threshold proves nothing about the object. It has not shown that no structure exists, only that this test could not certify one at this resolution.
- Tested under seal on retrieval, which is recognition with one candidate per query, and the transition has a finite width.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

- *measures.* GO-3 [demonstrated]. The certificate's vacuity threshold predicts where single-stage retrieval dies. [geometric-observation/claims/LEDGER.md:65](#).

FIRST STATED

Volume 14, the GO-3 registration and its notes, [geometric-observation/experiments/G03-certificate-vacuity-v3-NOTES.md:1-60](#), DOI 10.5281/zenodo.21776291, and chapter 9 section 9.3 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.3	the margin certificate, mu crit as the expected maximum of N minus 1 standard normals, rho, death at 0.948 within 6 percent, Spearman 0.991 vs 0.873, fourteen corpora, six gates, the v1 to v3 path, the standing correction	geometric-observation/experiments/G03-certificate-vacuity-v3-NOTES.md:1-60 ; geometric-observation/claims/LEDGER.md row GO-3

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Vacuity.lean](#), theorems `maxOver_mono`, `rho_eq_one_iff`, `rho_lt_one_of_lt`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 8, 9.

RELATED

[certificate](#) (p. 59); [rank certificate](#) (p. 285); [recognizer](#) (p. 296).

SEE ALSO

none

validity index

concept. Entry id `validity-index`.

DEFINITION

A score for a clustering computed without labels. Chapter 9.

Example. The silhouette of a clustering with two well separated groups is near 0.8, and near 0 the clustering says nothing.

EQUATION

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

Book equation 0.30.

$$\text{SSE} = \sum_k \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad s_i = \frac{b_i - a_i}{\max(a_i, b_i)}.$$

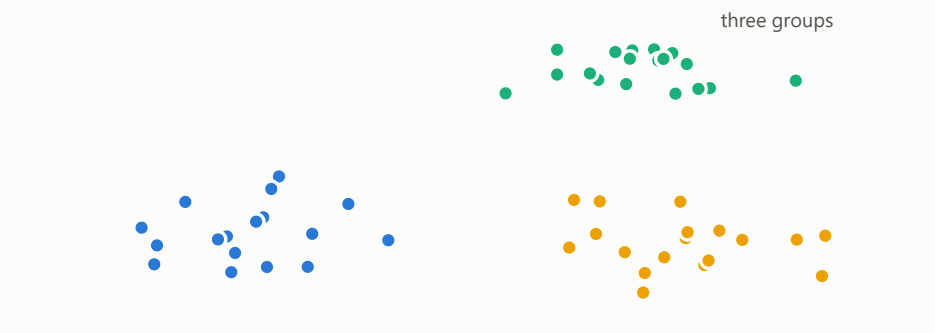


Figure 232: A score for a clustering without labels.

CONDITIONS

- A score for a clustering computed without labels. The silhouette is the case, in the interval from minus one to one, positive exactly when a row is closer to its own cluster, and unchanged by a common rescaling of the distances.
- A validity index is an output metric and needs a null, since it is computed under the identity reader on the distances it is given, which is why chapter 9 asks the recognizer to name the manifold or certify that none is present.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

TSK chapter 7 on cluster validity, as chapter 9 section 9.1 of *Data Mining as Observation* reads it, an output metric that needs a null.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Silhouette.lean](#), theorems `silhouette_mem`, `silhouette_pos_iff`, `silhouette_scale`, `silhouette_self`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[silhouette](#) (p. 332); [recognizer](#) (p. 296); [vacuity threshold](#) (p. 371); [null model](#) (p. 242).

SEE ALSO

Ledger rows that cite the entry's records without naming it: GO-3.

variance

concept. Entry id variance.

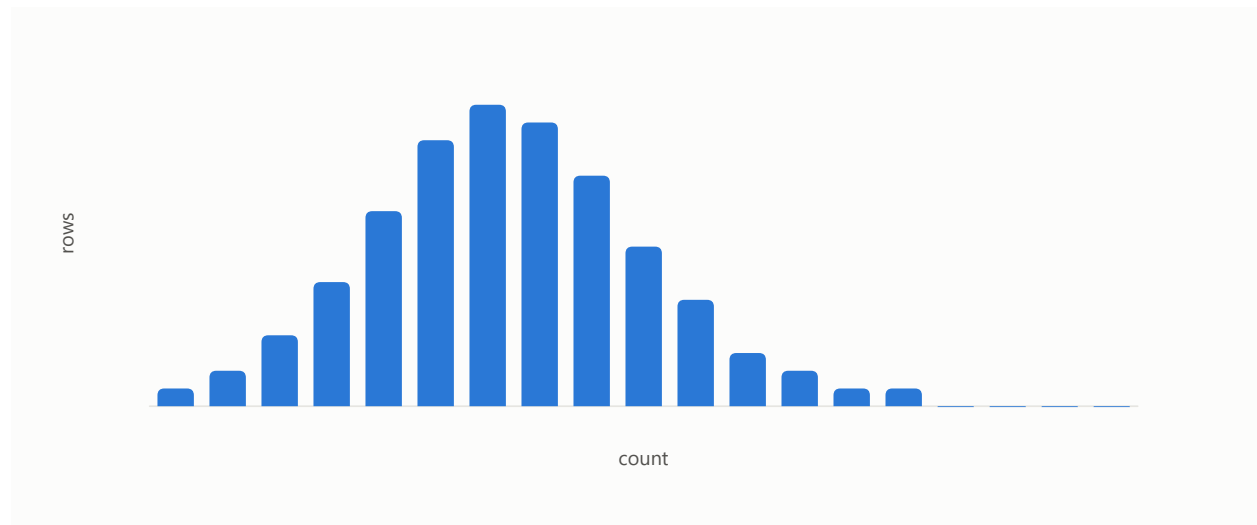


Figure 233: The mean squared deviation from the mean.

DEFINITION

The mean squared deviation of a column from its mean, the identity reader's measure of spread. Chapter 0 section 0.3.

Example. Values 2, 4, and 6 have mean 4 and variance $(4 + 0 + 4)$ over 3, or 2.667.

EQUATION

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

CONDITIONS

- The weighted mean of the squared deviation of a column from its weighted mean. It is nonnegative, equals the mean square less the squared mean, is zero for a constant column, scales with the square of a rescaling, and with positive weights is zero only for a constant column.
- Variance is the identity reader's measure of spread. A direction with large variance and no group structure pulls principal components and k-means centres alike, and whether a consumer reads that direction is a separate question.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.3 of *Data Mining as Observation*, with the program's spectrum tools in `readscope/readscope/spectrum.py:35-70`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Variance.lean](#), theorems `variance_nonneg`, `variance_eq`, `variance_const`, `variance_smul`, `eq_mean_of_variance_eq_zero`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 12, 14.

RELATED

[covariance matrix](#) (p. 91); [explained variance](#) (p. 137); [standardization](#) (p. 348); [spectrum](#) (p. 344); [effective rank](#) (p. 123).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.5, 0.7.

Ledger rows that cite the entry's records without naming it: OT-7.

Sources-table rows that share a record with the entry without naming it: chapter 4 section 4.2, chapter 9 section 9.4.

verdict

concept. Entry id verdict.

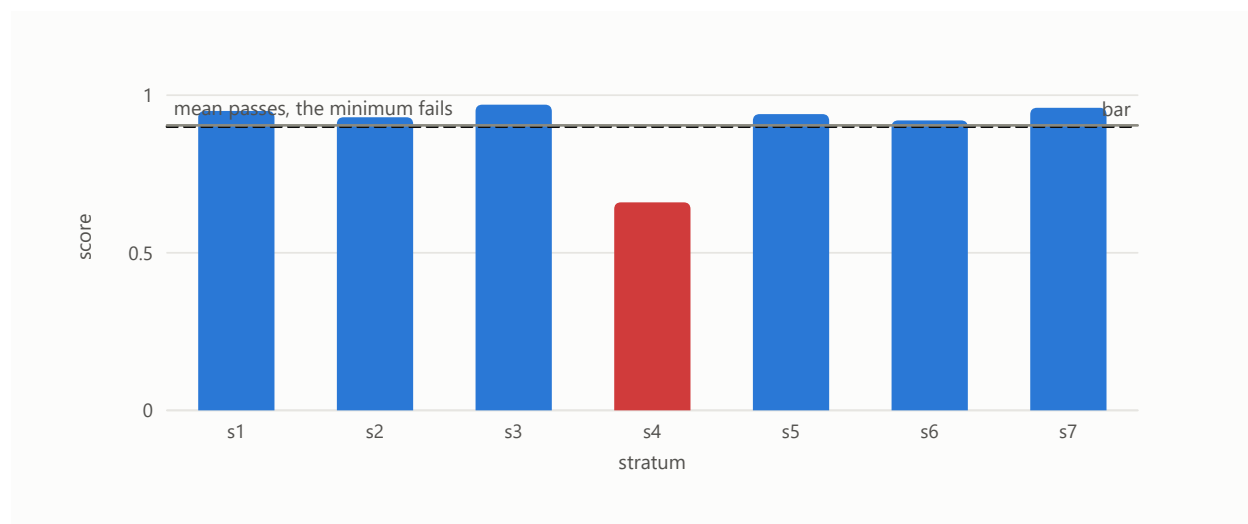


Figure 234: Pass, fail, or abstain, taken as the worst group.

DEFINITION

The outcome of a registered test, pass, fail, or abstain, taken as the worst group with the groups too thin to score counted. Chapter 1 section 1.5 and chapter 10 section 10.4.

Example. Strata at 0.95, 0.93, 0.97, and 0.66 against a bar of 0.90 give the verdict fail, with the failing stratum named.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 14.5.

$$\text{FC}_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} \text{FC}_g, \quad \text{ABSTAIN otherwise.}$$

CONDITIONS

- The outcome of a registered test, pass, fail, or abstain, taken as the worst group with the groups too thin to score counted. A verdict at one bar implies the verdict at every lower bar, and an abstention is reported as a verdict and not dropped.
- A verdict is relative to a budget and does not transfer to another, and the same two codes get opposite verdicts from two output metrics, so a verdict names its bar, its null, its budget, and its consumer.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 1 section 1.5 and chapter 10 section 10.4 of *Data Mining as Observation*, with the six classes in `geometric-observation/PROTOCOL.md:58-75`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer/experiments/manifold-recovery/battery_result.json

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Bar.lean](#), theorems `passes_anti`, `passes_mono`, `discriminates_iff`, `no_bar_of_null_ge`, `exists_bar_of_lt`, `vacuous_of_null_passes`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

[lean/DataMiningAsObservation/Abstention.lean](#), theorems `abstain_not_passes`, `verdict_eq_none_iff`, `passes_verdict_iff`, `inf_le_of_subset`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[bar](#) (p. 39); [abstention](#) (p. 12); [ledger class](#) (p. 211); [min-over-strata](#) (p. 226); [certificate](#) (p. 59).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 8.3.

Ledger rows that cite the entry's records without naming it: GO-3, NEG-14.

Sources-table rows that share a record with the entry without naming it: chapter 1 section 1.5, chapter 3 section 3.5, chapter 8 section 8.10, chapter 10 section 10.2, chapter 11 section 11.6.

W

water-filling

concept. Entry id water-filling.

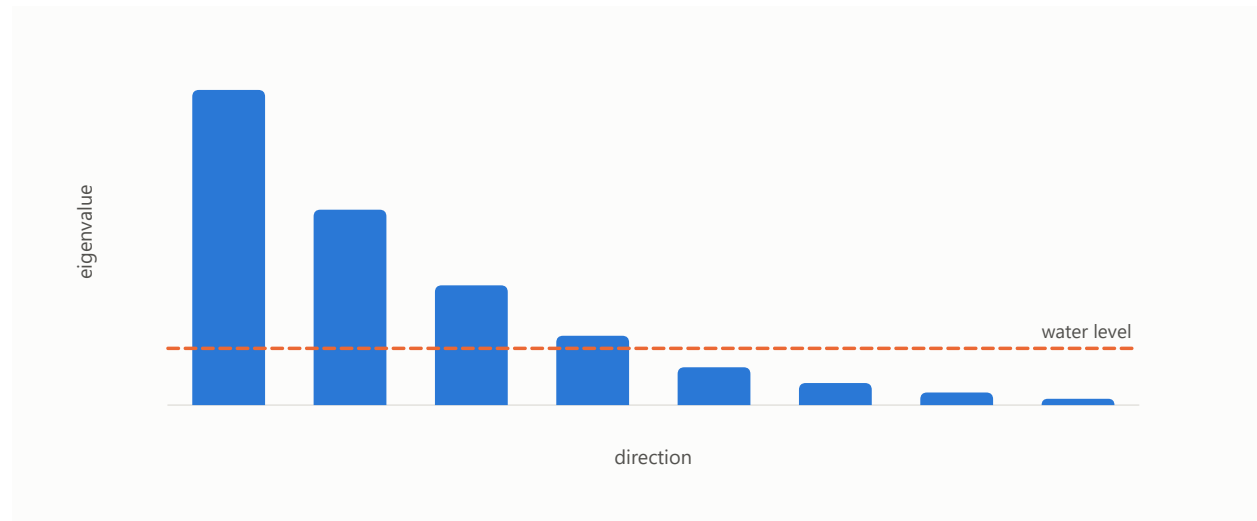


Figure 235: Directions below the water get no bits.

DEFINITION

The allocation of a bit budget across directions that gives each direction half the log of its sensitivity-weighted variance over a common water level, and nothing to directions under the water. Equation 0.13.

Example. Directions with variances 4, 1, and 0.1 and a water level of 0.5 get bits in proportion to the log of 8, the log of 2, and nothing.

Known as, or related to prior art. Reverse water-filling of rate-distortion theory, per Cover and Thomas, applied to the read-weighted spectrum.

EQUATION

none

CONDITIONS

- The closed form assumes a uniform quantizer at high rate, so that each bit quarters the squared error, and directions taken as the eigenvectors of the covariance so that their errors add.
- The sensitivity of each direction is read in that same eigenbasis, as the quadratic form of the read operator along the eigenvector, and pairing the eigenvalues of the two matrices is valid only when they share an eigenbasis.
- Coarse quantizers, unequal codebook steps, and errors not spread evenly depart from the formula, and chapter 4 reports where a measured allocation did.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Shannon's power allocation across channels, applied with a consumer's sensitivity in place of a signal's power in readscape and turboquant-pro, and chapter 0 section 0.7 and chapter 4 section 4.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	water-filling formula, directions below the water get no bits, the surrogate caveat	readscope/readscope/allocate.py:1-100

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/WaterFilling.lean](#), theorems `contribution_eq_water`, `terms_mul`, `two_sqrt_le`, `distortion2_ge`, `distortion2_eq_of_equal`, at observation-data-mining 95af30c; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 4.

RELATED

[read operator](#) (p. 291); [read distortion](#) (p. 290); [alignment](#) (p. 15); [the flip](#) (p. 147).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.13, 4.2.

Sources-table rows that share a record with the entry without naming it: chapter 2 section 2.3, chapter 2 section 2.4, chapter 2 section 2.6, chapter 3 section 3.4, chapter 4 section 4.2, chapter 4 section 4.4, chapter 4 section 4.5, chapter 6 section 6.1, chapter 8 section 8.3, chapter 8 section 8.4, chapter 11 section 11.7, chapter 11 section 11.9.

Weyl's law

concept. Entry id `weyls-law`.

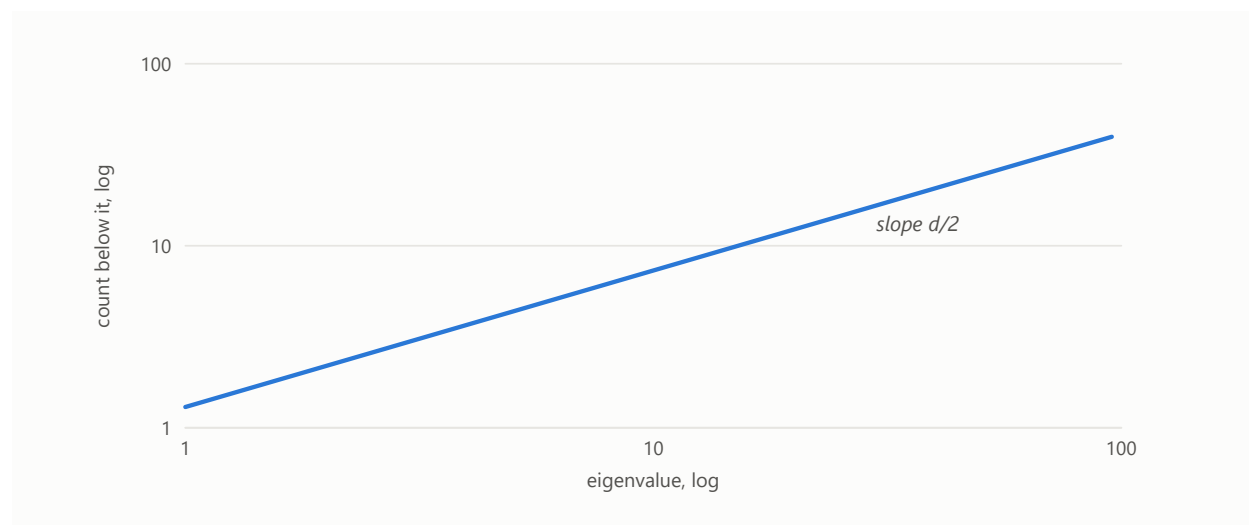


Figure 236: The eigenvalue count grows like the value to half the dimension.

DEFINITION

The asymptotic rule for how the number of eigenvalues of a continuum Laplacian below a value grows with that value, whose exponent reveals the dimension of the space. Reading it from a finite graph needs the conditions under which the graph Laplacian converges. Equation 0.31.

Example. On a surface, dimension two, the eigenvalue count grows linearly with the value; on a curve, as the square root.

EQUATION

Book equation 0.31.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

CONDITIONS

- An asymptotic statement about the continuum Laplacian. The number of eigenvalues below a value grows like that value to the power of half the dimension, so the dimension is in the exponent and is read from the slope of the count against the value on log axes. The Lean file checks that arithmetic and not the law.
- Carrying it to a finite neighbourhood graph needs sampling, graph-construction, normalization, and scaling conditions under which the graph Laplacian converges, and only the low part of the spectrum is read. The recognizer's battery recovered dimensions of 1.81, 2.80, and 1.74 on three sealed templates under those conditions and against a finite list.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Weyl, 1911, as chapter 0 section 0.15 of *Data Mining as Observation* states it, with the program's dimension reading in `geometric-observation/chapters/ch11_the_recognizer.md:1-95`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.2	the recognizer's mechanism, low multiplets and angular distances, dimension before shape by Weyl's law, refusal, the growth gotcha	geometric-observation/chapters/ch11_the_recognizer.md:1-95

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/IntrinsicDimension.lean](#), theorems `log_weyl`, `dimension_from_slope`, `weyl_double`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 9.

RELATED

[intrinsic dimension](#) (p. 184); [manifold](#) (p. 220); [Laplacian](#) (p. 207); [multiplet](#) (p. 230); [recognizer](#) (p. 296).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.19.

Ledger rows that cite the entry's records without naming it: GO-3.

Sources-table rows that share a record with the entry without naming it: chapter 9 section 9.2.

whitening

concept. Entry id whitening.

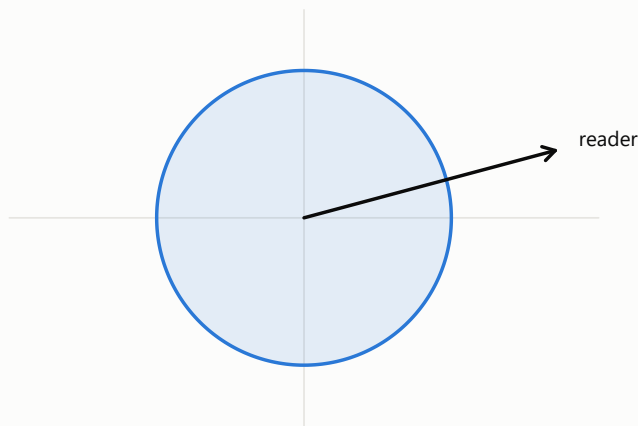


Figure 237: Rescaling so that the covariance becomes the identity.

DEFINITION

Rescaling centred data by the inverse square root of a positive-definite covariance, or a pseudoinverse on its support, so that the covariance becomes the identity. Euclidean distance after whitening is Mahalanobis distance before it. Equation 0.6.

Example. With covariance $\text{diag}(1, 4)$, the row $(3, 4)$ whitens to $(3, 2)$.

EQUATION

Book equation 0.6.

$$x_w = \Sigma^{-1/2}(x - \mu), \quad \Sigma^{-1/2} = \sum_i \lambda_i^{-1/2} v_i v_i^\top.$$

CONDITIONS

- Rescaling centred data by the inverse square root of its covariance, so that the covariance becomes the identity. It needs a positive-definite covariance, or a pseudoinverse restricted to the covariance's support, and after it every direction has variance one.
- Euclidean distance after whitening equals Mahalanobis distance before it, so whitening changes which reader the Euclidean reader is and not what the data holds. The whitened code won at every budget on the whale corpus, 0.83, 0.85, 0.97 against 0.41, 0.80, 0.89.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Chapter 0 section 0.2 of *Data Mining as Observation*, with the program's whitened code in Volume 14 chapter 8, [geometric-observation/chapters/ch08_value.md:84-97](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.5	whitened code wins at every budget	geometric-observation/chapters/ch08_value.md:84-97
chapter 4 section 4.3	whitened code on whale 0.83, 0.85, 0.97 vs 0.41, 0.80, 0.89	geometric-observation/chapters/ch08_value.md:84-97

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/Whitening.lean](#), theorems `whiten_unit_variance`, `whiten_sq_sum`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 2, 4, 10.

RELATED

[Mahalanobis distance](#) (p. 219); [identity reader](#) (p. 180); [read operator](#) (p. 291); [water-filling](#) (p. 378).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.5.

Ledger rows that cite the entry's records without naming it: GO-2 (neg. half: not reconstruction).

witness

concept. Entry id witness.

DEFINITION

An independent measurement of whether a certificate's claim was true. Chapter 13.

Example. The witness read the replica's actual position where the certificate had said fresh, and that disagreement is one false clear.

EQUATION

Book equation 0.26.

$$\text{FC} = \Pr [W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

Book equation 13.1.

$$\text{FC} = \Pr [W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr} (P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

CONDITIONS

- A witness is independent of the certificate it grades. A certificate that grades itself has no witness.

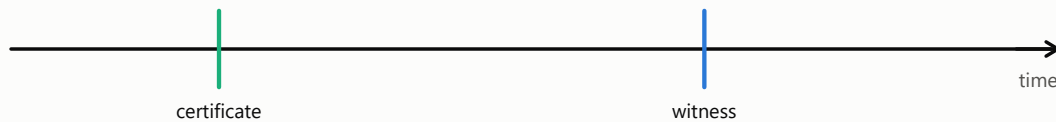


Figure 238: An independent measurement of whether the certificate’s claim was true.

- The witness is the consumer’s own outcome or a measurement of it, so the same certificate has a different witness for each consumer, as the hot and cold readers of one replica show.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Volume 14, chapter 19, `geometric-observation/chapters/ch19_the_certificate_that_ages.md:61-76`, and the freshness program’s umbrella document `observation-theory-campaigns/experiments/FRESHNESS-PROGRAM.md`.

MEASUREMENTS

From `observation-theory-campaigns/experiments/DATABASE-FRESHNESS-TRACK.md` at `f6d1fe2`.

- line 23. XPROTO-PG (analysis/pgrep); Postgres, `recovery_min_apply_delay`; WAL LSN; $\sim 0.50 \rightarrow \sim 0.06$
- line 24. XPROTO-MG (analysis/mongo); MongoDB delayed secondary; `oplog ts`; $\sim 0.47 \rightarrow \sim 0.03$
- line 25. XPROTO-PGX (analysis/pgx); production PG, netem lag; WAL LSN, `pg_stat_statements` footprint; $\sim 0.47 \rightarrow \sim 0.02$
- line 27. XPROTO-ZK (analysis/zk); ZooKeeper 3.9 ensemble; `zxid, sync()`; hot 0.99 / cold 0.01, witnessed 0.0

From `observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md` at `f6d1fe2`.

- line 25. XPROTO-CSI (analysis/csi); CQI $\rightarrow$ MCS; HARQ; $0.34-0.37 \rightarrow 0.10$ (OLLA); pass 08-23
- line 26. XPROTO-BEAM (analysis/beam); mmWave beam index; HARQ; $0.31 \rightarrow 0.02$ (BFR); pass 08-23
- line 27. XPROTO-AICSI (analysis/aicsi); neural-CSI recon (turboquant bridge); precoder/HARQ; recon wins yet $0.28 \rightarrow 0.13$; pass 08-23, scope-corrected 08-25 warning
- line 28. XPROTO-HO (analysis/ho); RSRP $\rightarrow$ serving cell; RLF; $0.31-0.44 \rightarrow 0.09-0.12$; pass 08-23
- line 30. XPROTO-PHY (analysis/phy); PMI / RI / TA; HARQ; $0.27-0.42 \rightarrow 0.055-0.13$; pass 08-24

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

`lean/DataMiningAsObservation/Certificate.lean`, theorems `falseClear_mul_coverage`, `coverage_empty`, `falseClear_mem_unit`, `minOverStrata_passes_iff`, `minOverStrata_le_weighted_mean`, at `observation-data-mining 95af30c`; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 0, 12, 13, 14.

RELATED

[certificate](#) (p. 59); [false-clear rate](#) (p. 141); [coverage](#) (p. 92).

SEE ALSO

Sources-table rows that share a record with the entry without naming it: chapter 12 section 12.6, chapter 13 section 13.3, chapter 13 section 13.4.

Y

Youden F1 bound

correction. Entry id youden-f1-bound.

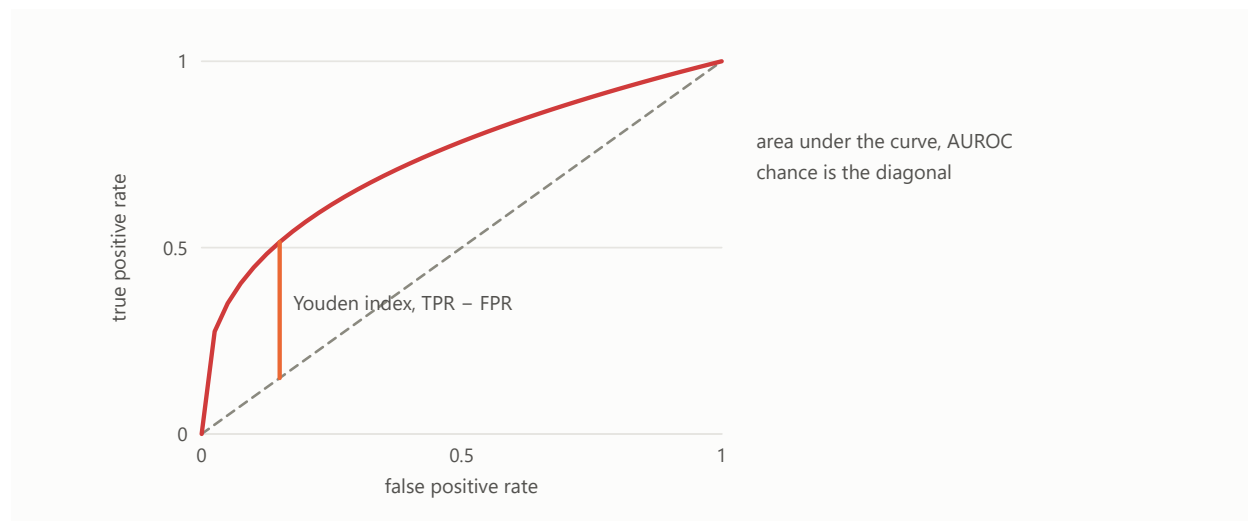


Figure 239: The Youden index bounds the best F1, for every score.

DEFINITION

The bound on the best F1 a score can reach at any threshold, from the Youden index, true positive rate minus false positive rate, maximized over thresholds. The bound holds for every score. The form that reads the ceiling from the AUROC holds only for concave ROC curves, with a counterexample at AUROC 0.75.

Example. A Youden index of 0.5 at the best threshold bounds the F1 the score can reach at any threshold, for every score.

EQUATION

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$$J \leq 2A - 1 \text{ when the ROC curve is concave, and not in general.}$$

CONDITIONS

- The Youden form holds for every score with a finite sample.
- The AUROC form holds for concave ROC curves, and as a pruning bound it is safe only there.
- Inside a tree search either form bounds the formula itself and not its descendants, so pruning a subtree by its root's ceiling is a heuristic in both cases.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

The AUROC form, theorem “AUROC-F1 Bound” in `theory-radar/paper/astar_paper.tex` and the derivation in `theory-radar/src/symbolic_search/_auroc_proof.py`, repository DOI 10.5281/zenodo.20660206. The corrected

form at theory-radar 1b3d105.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	Monotone Invariance and AUROC invariance theorems, the AUROC to F1 bound via the Youden index, stated in the source for every ROC curve and corrected in the book to concave curves, with the counterexample at AUROC 0.75 and F1 0.857	theory-radar/paper/astar_paper.tex:94-170 ; theory-radar/paper/theory_radar_paper.tex:290-304
chapter 5 section 5.4	seven thresholds up to 0.75, eight datasets, 56 conditions, zero admissibility violations, reductions 70 to 99.1 percent	theory-radar/paper/astar_paper.tex:170-200
chapter 5 section 5.4	the rerun of 2026-09-03 and 2026-09-04, pre-filter counts 2576 of 8700 and 413 of 1560, ceiling violations 812 and 412, Youden pruning 199 and 38, A* optima 0.958 and 0.694 in strict and fast modes, Youden mode short by 0.006 and 0.002	theory-radar/results/rerun_youden_2026_09_03.json ; theory-radar/ERRATA.md
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory-radar/paper/theory_radar_paper.tex:290-304 ; theory-radar/paper/astar_paper.tex:94-110

Rerun of the A* paper's pruning tables by `theory-radar/rerun_youden_2026_09_03.py`. Part A is depth-2 pairwise enumeration on eight datasets, exhaustive against the AUROC pre-filter at seven thresholds and against Youden pruning. Part B is the depth-3 A* search with 50,000 expansions in strict mode (no pre-filter), the published fast mode (AUROC pre-filter at 0.52), and the youden mode.

Part A, recorded at commit aedc053 of theory-radar:

Dataset, depth 2 pairs	Pairs	Exhaustive F1
Circles (2)	20	0.9960
Moons (2)	20	0.8330
Breast Cancer (30)	8700	0.9542
Wine (13), class 0 vs rest	1560	0.9204
Synthetic (10)	900	0.9556
Synthetic (20)	3800	0.9460
Synthetic (30)	8700	0.9637
Synthetic (40)	15600	0.9425

Part B.

Dataset	strict F1 and formula	fast F1
Circles (2)	0.9965, (x1 hypot x2)	0.9965
Moons (2)	0.8927, (neg(x1) max x2)	0.8927
Synthetic (20)	1.0000, (x0 hypot x1)	1.0000

Dataset	strict F1 and formula	fast F1
Breast Cancer (10)	0.9578, $((f1 \text{ min } f2) + f7)$	0.9578
Diabetes (8)	0.6937, $((d5 \text{ min } d7) + d1)$	0.6937

FAILURES AND CORRECTIONS

- [theory-radar/ERRATA.md:3-75](#) at 37c4e6c. 2026-09-03. The AUROC to F1 bound holds only for concave ROC curves **What was claimed.** Theorem “AUROC-F1 Bound” in `paper/astar_paper.tex`, the derivation in `src/symbolic_search/_auroc_proof.py`, and the pruning bounds in `run_astar_v2.py` and `src/symbolic_search/_heuristic_dag.py` all rested on the statement that any ROC curve with area A has maximum Youden index $J = \max(\text{TPR} - \text{FPR})$ at most $2A - 1$. **Why it is wrong.** The statement holds for concave ROC curves (equivalently, scores whose likelihood ratio is monotone), because a concave curve lies above the two chords through its Youden point and so has area at least $(1 + J)/2$. It fails for non-concave curves. Counterexample, found by an external reviewer of the companion textbook: rank three quarters of the positives above every negative and the remaining quarter below every negative. AUROC is 0.75, the maximum Youden index is 0.75, and at prevalence one half the best thresholded F1 is 0.857, above the 0.80 the AUROC form of the ceiling allows. The sound ceiling for that score, from $J = 0.75$, is 0.889, and the score sits under it. The `auroc_f1_bound` formula in `run_astar_v2.py`, $2A\pi/(A\pi + (1 - A)(1 - \pi))$, is not a valid upper bound either: at prevalence 0.1 the same score reaches F1 0.857 against a “bound” of 0.5. **Consequence for reported results.** The searches that used these bounds to prune could have discarded a candidate formula whose ROC curve was not concave and whose true best F1 exceeded the computed ceiling. The formulas and wins reported by those runs are therefore what a pruned search found, and are lower bounds on what an unpruned search would find. The Monotone Invariance Theorem and the results that do not depend on the AUROC bound are unaffected. **Correction.** The sound form of the ceiling substitutes the measured maximum Youden index of the candidate’s own scores for $2A - 1$. It holds for every score, costs the same sort as AUROC, and is what `max_f1_for_youden` and `youden_f1_bound` now compute. The AUROC form is retained under its original names with a docstring stating the concavity condition, and the pruning sites now use the measured index. A rerun of the published searches under the corrected bound was done on 2026-09-03 and 2026-09-04, `rerun_youden_2026_09_03.py`, results in `results/rerun_youden_2026_09_03.json`, run on the Atlas workstation, CPU only. **Rerun, part A, the pre-filter admissibility table (tab: auroc).** Depth-2 pairwise enumeration on the eight datasets, exhaustive against the AUROC pre-filter at the seven published thresholds, with the sound Youden ceiling beside it. The evaluation counts under the pre-filter reproduce the published table (Breast Cancer 8700 pairs, 2576 evaluated at 0.75; Wine 1560, 413; Circles 20, 4; Moons 20, 2). The pre-filter lost no optimum on any dataset at any threshold, which is what the paper reported. That was luck of the data, not the theorem: the retracted ceiling sat below the F1 a pair formula actually reach

MACHINE CHECKED

[lean/DataMiningAsObservation/YoudenF1.lean](#), theorems `f1_eq`, `f1_le_of_youden`, `auroc_eq`, `youden_eq`, `youden_exceeds_auroc_form`, at observation-data-mining 95af30c; what the check covers is stated in the book’s [appendix C](#).

USED IN

Data Mining as Observation chapters 5.

RELATED

[Monotone Invariance Theorem](#) (p. 227); [formula search](#) (p. 154); [safe pruning](#) (p. 320); [Apriori principle](#) (p. 25).

SEE ALSO

none

Youden index

instrument. Entry id youden-index.



Figure 240: True positive rate minus false positive rate at a threshold.

DEFINITION

True positive rate minus false positive rate at a threshold, whose maximum over thresholds is the pointwise ceiling that prunes the formula search. Chapter 5 section 5.4.

Example. True positive rate 0.75 and false positive rate 0.25 give Youden index 0.5.

EQUATION

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$$J \leq 2A - 1 \text{ when the ROC curve is concave, and not in general.}$$

CONDITIONS

- True positive rate minus false positive rate at a threshold, whose maximum over thresholds bounds the best F1 the score can reach at any threshold. The bound is exact for concave ROC curves and fails for a curve at AUROC 0.75.
- The pointwise ceiling on a node of the formula search is proved, and using it to prune the node's descendants is a heuristic that missed the optimum on two of five datasets in the rerun.

Conditions are curated in `entries.toml` rather than read from a record.

LEDGER

none

FIRST STATED

Youden, index for rating diagnostic tests, 1950, as chapter 5 section 5.4 of *Data Mining as Observation* reads it, with the pointwise ceiling in `theory-radar/paper/astar_paper.tex:94-110`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

MACHINE CHECKED

[lean/DataMiningAsObservation/YoudenF1.lean](#), theorems `f1_eq`, `f1_le_of_youden`, `auroc_eq`, `youden_eq`, `youden_exceeds_auroc_form`, at `observation-data-mining 95af30c`; what the check covers is stated in the book's [appendix C](#).

USED IN

Data Mining as Observation chapters 5.

RELATED

[ROC curve](#) (p. 314); [AUROC](#) (p. 32); [F1](#) (p. 140); [safe pruning](#) (p. 320).

SEE ALSO

Book equations stated beside the entry's terms, not defining it: 0.28, 0.38.

Sources-table rows that share a record with the entry without naming it: chapter 5 section 5.4, chapter 6 section 6.3.

Entries by kind

Concepts, 159. abstention (p. 12); aggregation (p. 13); alignment (p. 15); anisotropic (p. 18); anti-hub (p. 20); anti-monotonicity (p. 24); attention (p. 26); attribute type (p. 28); AUROC (p. 32); bag of words (p. 36); balanced accuracy (p. 37); bi-Lipschitz (p. 42); bit (p. 43); block error rate (p. 46); budget (p. 51); cache (p. 55); calibration (p. 56); capacity (p. 57); certificate (p. 59); chance level (p. 62); chunk (p. 63); classifier (p. 65); codebook (p. 66); coherence time (p. 67); cold, warm (p. 69); commute time (p. 72); concentrated (p. 73); confidence (p. 75); confidence interval (p. 76); confound (p. 77); consumer (p. 79); contrastive objective (p. 82); correction (p. 85); correlation (p. 86); cosine (p. 87); coupling null (p. 89); covariance matrix (p. 91); coverage (p. 92); cross-support ratio (p. 95); curvature (p. 98); decision boundary (p. 101); degree (p. 105); density (p. 106); deployment mismatch (p. 108); disparate impact ratio (p. 114); distance concentration (p. 115); distortion (p. 116); dot product (p. 118); drift (p. 119); effective rank (p. 123); eigenvalue, eigenvector (p. 125); embedding (p. 126); encoder (p. 127); equalized odds (p. 130); escalation (p. 132); Euclidean distance (p. 133); eviction (p. 134); explained variance (p. 137); F1 (p. 140); false-clear rate (p. 141); floor, ceiling (p. 151); footprint (p. 153); freshness (p. 156); Gaussian (p. 160); geodesic distance (p. 162); geometric decision cost (p. 164); graph (p. 166); harness (p. 168); head (p. 170); Hessian (p. 172); hub (p. 175); identity reader (p. 180); intrinsic dimension (p. 184); isotropic (p. 187); itemset (p. 188); Jaccard (p. 191); Jacobian (p. 192); Kendall correlation (p. 194); kernel (p. 195); KL divergence (p. 196); KV cache (p. 200); lag (p. 202); language model (p. 206); Laplacian (p. 207); leakage (p. 210); ledger class (p. 211); licensed, not licensed (p. 213); lift (p. 214); Mahalanobis distance (p. 219); manifold (p. 220); margin (p. 221); metric (p. 224); min-over-strata (p. 226); multiple comparisons (p. 229); multiplet (p. 230); nat (p. 236); nearest neighbour (p. 238); neighbourhood graph (p. 239); nuisance (p. 240); observer (p. 244); operating point (p. 245); orthogonal (p. 247); outer product (p. 248); outlier (p. 250); output metric (p. 251); paraphrase class (p. 254); percentile (p. 255); perplexity (p. 258); posited versus measured (p. 263); positive semidefinite (p. 265); precision, recall (p. 266); projection (p. 272); p-value (p. 273); quantization (p. 275); query, key, value (p. 276); query set (p. 278); quotient (p. 279); rank (p. 283); rank-faithful (p. 286); read direction (p. 288); read distortion (p. 290); read operator (p. 291); read subspace (p. 293); recall at k (p. 295); reconstruction error (p. 298); refusal (p. 303); registered (p. 305); reliability weight (p. 306); replica (p. 307);

retrieval-augmented pipeline (p. 312); rotary position embedding (p. 316); ROUGE (p. 317); safe pruning (p. 320); score (p. 323); seed (p. 326); sensitivity (p. 328); silhouette (p. 332); Simpson's paradox (p. 333); skewness (p. 335); softmax (p. 336); Spearman correlation (p. 339); spectrum (p. 344); standard error (p. 347); sum of squared errors (p. 351); support (p. 353); teacher forcing (p. 358); TF-IDF (p. 360); threshold (p. 362); token (p. 363); trace (p. 365); transaction (p. 366); transmission time interval (p. 367); validity index (p. 372); variance (p. 374); verdict (p. 375); water-filling (p. 378); Weyl's law (p. 379); whitening (p. 381); witness (p. 382).

Instruments, 71. allocation (p. 16); anti arm (p. 19); anti-hub recall (p. 22); attribution (p. 29); audit (p. 31); bagging (p. 35); bar (p. 39); baseline (p. 40); blind probe (p. 45); boosting (p. 48); bootstrap (p. 49); challenge set (p. 61); commit hash (p. 70); contraction (p. 80); control (p. 83); cross-corpus gate (p. 94); cross-validation (p. 96); DBSCAN (p. 100); decision tree (p. 102); declaration (p. 103); detector (p. 109); direction-only quantizer (p. 111); discretization (p. 112); early stopping (p. 122); ensemble (p. 129); expected calibration error (p. 136); fine-tuning (p. 144); finite difference (p. 146); formula search (p. 154); gate (p. 159); hash (p. 169); hierarchical clustering (p. 173); importance (p. 181); imputation (p. 182); inverted file (p. 185); k-means (p. 198); latent semantic analysis (p. 209); linear classifier (p. 216); logistic regression (p. 217); matched bits (p. 223); naive Bayes (p. 235); null model (p. 242); paired (p. 253); per-channel quantizer (p. 257); planted (p. 260); Poisson ceiling (p. 262); preregistration (p. 267); principal component analysis (p. 269); product quantization (p. 271); random forest (p. 282); rank certificate (p. 285); recognizer (p. 296); refresh interval (p. 302); rerank (p. 309); residualization (p. 310); Robin Hood index (p. 313); ROC curve (p. 314); sampling (p. 322); sealed (p. 324); shard (p. 331); sources table (p. 338); spectral clustering (p. 340); spectral embedding (p. 342); split (p. 346); standardization (p. 348); stratification (p. 350); surrogate (p. 354); sweep (p. 356); template match (p. 359); truncation (p. 369); Youden index (p. 387).

Results, 7. Apriori principle (p. 25); budget cliff (p. 52); the flip (p. 147); hubness (p. 176); Landauer's principle (p. 203); Monotone Invariance Theorem (p. 227); vacuity threshold (p. 371).

Corrections, 3. Nadeau and Bengio correction (p. 233); refresh floor (p. 299); Youden F1 bound (p. 385).

References, 1. session deck (p. 329).

Entries by chapter of the book

The chapter of *Data Mining as Observation* in whose body the entry's terms appear.

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Provenance

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